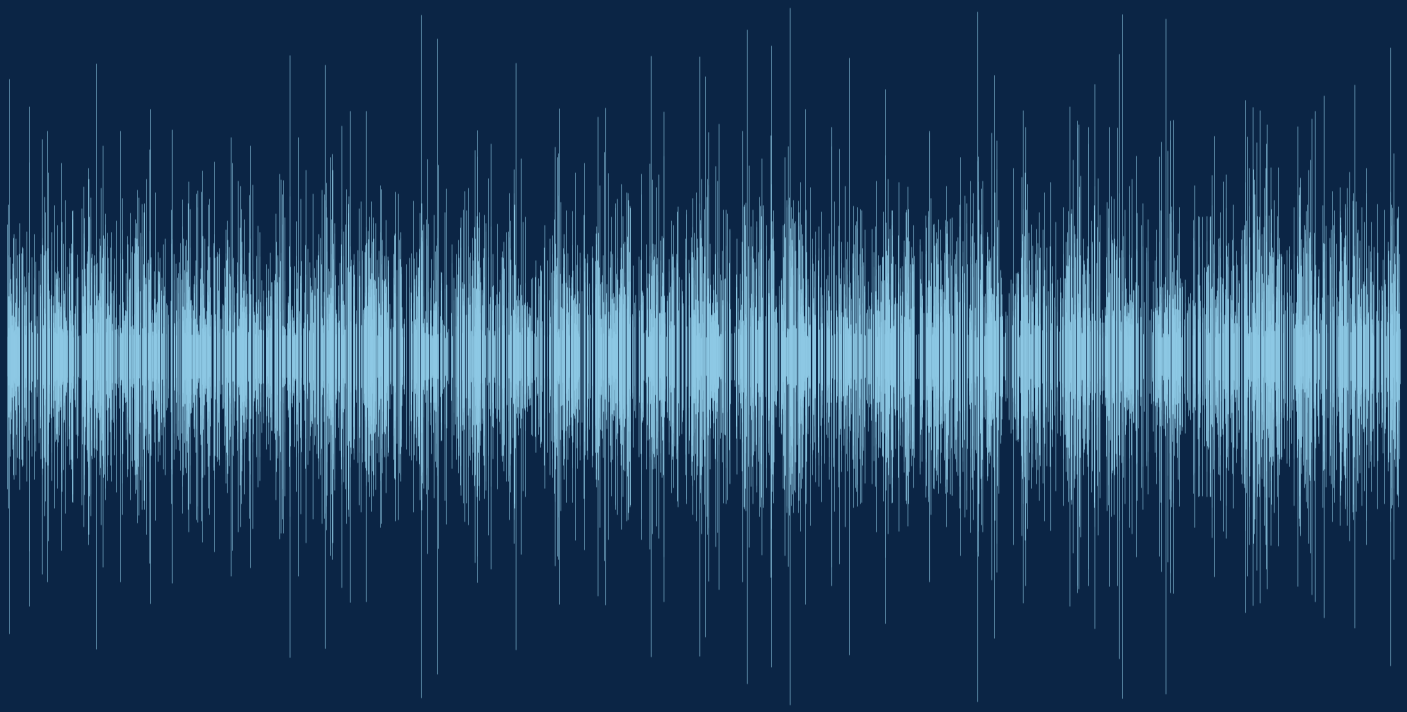


Netuno User Manual

Simulation of rainwater harvesting systems in buildings: the method, the tutorial, the guides and the references, in one volume.

COMPLETE EDITION · SIX CHAPTERS AND TEN APPENDICES

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When citing results obtained with the program, also state the version used and the source of the rainfall series, following the list in section 4.10.

About this translation

This is the English translation of the Portuguese manual, chapter by chapter, with the same section numbers, figures, tables and equations. The Portuguese text is the reference: where the two differ, the Portuguese one prevails, and the difference is a defect to be reported (section 6.4).

Netuno comes in two editions built from the same code (section 1.2). The **Brazil Edition (BRL)** ships with Brazilian data: the CLIMBra climate database, the catalogue of municipal water and electricity tariffs, the grid emission factors and the Brazilian certifications. The **International Edition (INT)** ships without them, and the user supplies the rainfall series, the tariff, the currency and the emission factors. Both editions calculate in the same way. Where a section describes a feature that only the Brazil Edition offers, it says so at the top.

The figures are shared with the Portuguese manual, and the labels drawn inside some of them are in Portuguese; each caption, in English, says what the figure shows. Numbers follow the English convention (a point for decimals, a comma for thousands). Brazilian standards are cited by their Portuguese titles, with an English description where needed.

How to use this manual

It was not written to be read from cover to cover. Section 1.3 tells you where to enter, depending on what you want to do.

You want to	Read
learn to use the program	chapter 4, the tutorial
solve a case	chapter 3, the guides
publish a study	chapter 2, the method; and section 4.10
review a study	chapter 5, and appendices A and B
avoid mistakes	section 5.1

The manual assumes an engineering reader who does not yet know the subject: concepts are introduced before they are used, and the vocabulary is in appendix G. Two kinds of box appear throughout the text:

The box	When it appears
with a **blue** rule	a warning or remark that changes what you do
with an **orange** rule, marked *In depth*	the technical detail behind it; can be skipped on a first reading

> **Every number printed in this manual is calculated by the program.** The tables and figures were not typed: they are produced by running Netuno with the input data each section declares. Each published value has a matching automatic check, so that a fix in the program cannot leave the manual lying. Section 5.3 explains how this is done.

Contents

1 Start here

1.1 What Netuno does, and what it does not

1.2 Installation and first run

1.3 Where to start, depending on your goal

2 The method

2.1 The daily water balance

2.2 Ideal volume and the sweep

2.3 Efficiency indicators

2.4 Rainfall indices

2.5 Sizing the treatment line

2.6 Pumping energy

2.7 Economic analysis and tariff regimes

2.8 Life cycle assessment

2.9 Certifications

2.10 Multi-criteria decision

3 Guides available in the program

3.1 Comparing cities

3.2 Using a future climate scenario

3.3 Adjusting the tariff of your municipality

3.4 Modulating the demand by calendar

3.5 Simulating schools, hospitals and shops

3.6 Comparing many cases at once

3.7 Sensitivity analysis

3.8 Importing a Netuno 4 project

3.9 Exporting to CAD, BIM and spreadsheets

4 Tutorial: one example from start to finish

4.1 The project and the rain

4.2 The building and the demand

4.3 The first simulation

4.4 Choosing the volume

4.5 Is an upper tank needed?

4.6 What quality the use requires

4.7 What it costs and how fast it pays back

4.8 How much carbon it avoids

4.9 Which volume to choose, after all

4.10 The report, and what to declare in a paper

5 Mistakes, limitations and reliability

5.1 Ten mistakes the program does not prevent

5.2 Known limitations

5.3 How the program is verified

6 Fixes, bugs and versioning

6.1 Comparison with Netuno 4

6.2 Error messages and what to do

6.3 Version history and what changed

6.4 How to report a problem

Appendices

Appendix A -- The project parameters

Appendix B -- The program's processes

Appendix C -- Emission factors

Appendix D -- Tariff regimes

Appendix E -- Climate database

Appendix F -- Files, folders and formats

Appendix G -- Glossary

Appendix H -- References

Appendix I -- One-page summaries

Appendix J -- Example files

List of figures

Figure 1 -- Components of a building rainwater harvesting system and the path of the water, from collection to the point of use

Figure 2 -- The same system seen in section

Figure 3 -- Destination of the precipitation of one day

Figure 4 -- Sequence of operations of the daily balance, in the order the program runs them, in the notation of the equations of the daily balance section

Figure 5 -- The six dimensions of analysis and the question assessed in each one

Figure 6 -- The four reading paths of the manual

Figure 7 -- Which part of the total consumption rainwater can supply, depending on the uses the project intends

Figure 8 -- The twelve steps of the program, grouped by nature

Figure 9 -- The ten days of the worked case: (a) collected volume, in blue, and overflowed volume, in orange, by day -- on day 9 the overflow almost equals the collection; (b) volume stored in the lower tank at the end of each day, after consumption, with the 2,000 L capacity dashed; on days 6, 7 and 9 the tank reaches capacity, overflows and ends the day with 1,920 L

Figure 10 -- Sequence of operations of the daily balance: (a) the order adopted by the program, with the overflow calculated before consumption; (b) the reversed order (YBS), with consumption withdrawn before the overflow

Figure 11 -- Saving potential curve of the same building, swept with a 250 L step and with a 5,000 L step

Figure 12 -- Sweep of the tutorial project read by three criteria: (a) saving potential and (b) net present value of the same set of volumes

Figure 13 -- The three efficiency indicators and what each one counts

Figure 14 -- Daily supply classes: full (V_con,t = D_pl,t), partial (0 < V_con,t < D_pl,t) and none (V_con,t = 0). The filled height represents the fraction of the day's rainwater demand supplied by rainwater

Figure 15 -- Four of the nine indices marked over the year 1995 in Florianópolis: PRCPTOT and SDII, at the top; RX1day, the largest daily value of the year; CDD, the longest sequence of dry days, in the shaded band; and R20mm, the days with 20 mm or more, marked at the bottom

Figure 16 -- Walsh and Lawler (1981) seasonality index in the four capitals of the guide, with the classes of the index in the background and the mean annual precipitation under the name of each city

Figure 17 -- Total annual precipitation (PRCPTOT) of the Florianópolis series, with the mean of the period as a dashed line, in the indices tab

Figure 18 -- Longest annual sequence of dry days (CDD) and of wet days (CWD) of the Florianópolis series, in the indices tab

Figure 19 -- Components of the treatment line sized by the program, in the order of the flow, and destination of the material retained

Figure 20 -- The three disinfection technologies, the residual effect of each one and the make-up of the annual running cost (OPEX). The residual effect distinguishes chlorination from the others: it protects the stored water, while ultraviolet radiation and ozone act only on the water passing through the equipment

Figure 21 -- Energy chain of pumping

Figure 22 -- (a) Manometric head of a building rising main: geometric difference in level between the water surfaces of the cistern and the upper tank, plus the head losses; (b) energy line along a stretch of pipe, with the

continuous friction loss (h_f) and the local losses at a valve and an elbow (h_loc)

Figure 23 -- Minimum efficiency of two-pole single-phase induction motors, according to Table 13 of ABNT NBR 17094-2:2016

Figure 24 -- Monthly water and sewage bill of the tutorial project, broken down into parts, without and with the system

Figure 25 -- The six tariff block codes and the effect of each one on the monthly water and sewage bill

Figure 26 -- Distribution of the 652 municipalities of the catalogue among the six tariff regimes

Figure 27 -- Initial investment broken down into the seven items the program adds up, for three volumes of polyethylene tank

Figure 28 -- Cash flow of the tutorial project over twenty years: annual saving on the water and sewage bill and annual costs, with the initial investment in year zero, in nominal values; and cumulative balance discounted at the minimum attractive rate of return, which crosses zero at the payback time of 16.3 years

Figure 29 -- The seven stages of the inventory, grouped by the nature of the emission

Figure 30 -- Boundary of the program's environmental assessment: (a) processes included -- embodied carbon of the tanks, the piping and the pump set, operational emissions of pumping and of treatment consumables, and the credit for the potable water avoided; (b) processes excluded -- civil works, maintenance and replacement, end of life and impact categories beyond global warming

Figure 31 -- The thresholds of reduction in potable water consumption with which the program places the project in LEED v4 (credits WEc1 and WEc2), LEED v5 (credit WEc2, options 1, 2 and 4), AQUA-HQE Residential and EDGE, on the same scale

Figure 32 -- TOPSIS score of each volume of the sweep of the tutorial project, from 1 to 20 m³, with a polyethylene cistern and equal weights among the seven criteria; the star marks the volume with the highest score

Figure 33 -- First-rank acceptability (b^1) of each candidate volume, obtained with five thousand weight vectors drawn on the simplex

Figure 34 -- (a) Mean annual precipitation and (b) potable water saving potential of the tutorial project, with 150 m² of catchment, 600 L/day of demand, a 28% suppliable fraction and a 4,000 L cistern, in four state capitals

Figure 35 -- Mean monthly precipitation in (a) Florianópolis, (b) São Paulo, (c) Brasília and (d) Manaus, on the same scale; the dashed line is the monthly mean of each series

Figure 36 -- The shape of the two emission pathways available in the database

Figure 37 -- Water part of the monthly bill as a function of consumption, in three minimum-charge regimes

Figure 38 -- The four modulation modes, what each one asks of the user and the situation in which each one is justified

Figure 39 -- The three demand profiles used in the comparison, expressed as the ratio between the month's demand and the mean demand

Figure 40 -- (a) Monthly demand profiles of the school and the beach hotel over the mean monthly precipitation of Florianópolis; (b) saving potential of each profile with the same 4,000 L tank

Figure 41 -- The eight available building types, grouped by the tariff category the utility charges

Figure 42 -- Plausible range of variation of each assumption in a sensitivity analysis, with the origin of the uncertainty

Figure 43 -- The variation of the net present value when each assumption is changed by 30% up and down, keeping the others

Figure 44 -- Florianópolis precipitation series used throughout the tutorial: (a) total precipitation of each year, with the thirty-year mean as a solid line and the years above the mean in blue and below it in orange; (b) frequency distribution of daily precipitation on wet days, on a logarithmic scale, which shows how few days account for the largest volumes

Figure 45 -- The mean seasonal cycle of the series, calculated and drawn by the program as soon as the precipitation is loaded

Figure 46 -- The runoff coefficient the program gives to each roof material, with the values of the study of materials published in the journal MIX Sustentável (2020). The dashed line marks 0.85, the collection factor of ABNT NBR 15527 adopted in the tutorial

Figure 47 -- The usual ranges of per capita consumption by building type

Figure 48 -- Typical split of the consumption of a house by end use

Figure 49 -- States of the suppliable-fraction field in the interface

Figure 50 -- The level of the cistern over the first year of the series, with the daily rain in the background

Figure 51 -- (a) Saving potential as a function of the tank volume, approaching the 28% ceiling without crossing it, with the ideal volume marked; (b) marginal gain, on a logarithmic scale, with the 1 %/m³ threshold adopted in the tutorial

Figure 52 -- The first chart of the results tab: the potable water saving potential as a function of the lower tank volume

Figure 53 -- The second chart: the marginal gain, in percentage points per added cubic metre

Figure 54 -- The third chart: the overflowed volume and the rainwater volume consumed, both annual, on the same scale

Figure 55 -- The fourth chart: the split of the days of the period into fully supplied, partly supplied and not supplied

Figure 56 -- Use of the lower tank capacity in the tutorial project, with a rainwater demand of 168 L/day: (a) duration curves of the stored autonomy at the end of each day, in days of rainwater demand, for 2, 4, 10 and 20 m³; (b) annual turnovers of the volume, given by the ratio between the annual rainwater consumption and the capacity, with the fraction of days on which the tank ends empty (up to 5% of the capacity) shown under each volume

Figure 57 -- The program's four quality classes and the normative status of each one

Figure 58 -- The chart of the economic tab: the net present value of each volume of the sweep, with a concrete cistern, with the economic maximum and the technical optimum marked

Figure 59 -- The chart of the environmental tab for the same project, with a concrete cistern: the parts of the carbon balance over twenty years

Figure 60 -- Cumulative net emissions of the 4,000 L cistern over the service life, for polyethylene and concrete

Figure 61 -- Net emissions over twenty years of the 4,000 L cistern, by material

Figure 62 -- Net emissions over twenty years against the cistern volume, for (a) polyethylene and (b) concrete, in the tutorial project

Figure 63 -- The chart of the decision tab: the closeness-to-ideal score of each candidate volume, with the winner highlighted

Figure 64 -- The second chart of the decision tab: how close the chosen volume comes to the best possible value on each of the seven criteria

Figure 65 -- Catchment area as the horizontal projection of the roof, according to item 3.6 of ABNT NBR 15527:2019

Figure 66 -- Number of tests and of files in each family of the verification suite, as collected in this version

Figure 67 -- The five verification layers of the program, from the unit tests, numerous and aimed at local errors, to the manual check, occasional and aimed at model errors

Figure 68 -- The mean divergence relative to Netuno 4, broken down

Figure 69 -- How much of each cubic metre produced actually reaches the building, by Brazilian macro-region

Figure 70 -- The distribution of the 652 catalogued municipalities across the Brazilian states, with colours identifying the macro-region

List of tables

Table 1 -- System requirements to install and run the program

Table 2 -- What changes between the Brazil Edition and the International Edition

Table 3 -- File formats the program saves and the content of each one

Table 4 -- Example files distributed with the program

Table 5 -- Recommended reading path depending on the reader's goal

Table 6 -- Choice between fixed volume and sweep depending on the question of the study

Table 7 -- Hydraulic criteria to decide on the upper tank

Table 8 -- Input data of the ten-day worked case

Table 9 -- Precipitation series of the ten days of the worked case

Table 10 -- Daily water balance of the worked case, day by day, with collection, consumption, overflow and storage

Table 11 -- Check of the mass balance closure over the ten days

Table 12 -- Indicators of the ten-day worked case

Table 13 -- Variables of Equation 1

Table 14 -- Variables of Equation 2

Table 15 -- Variables of Equation 5

Table 16 -- Withdrawal of the day and volume stored at the end of the day under strict YAS, strict YBS and in Netuno

Table 17 -- Variables of Equation 7

Table 18 -- Variables of Equation 8

Table 19 -- Variables of Equation 9

Table 20 -- Variables of Equation 10

Table 21 -- Variables of Equation 11

Table 22 -- Variables of Equation 12

Table 23 -- Volumes given by the methods of Annex A of ABNT NBR 15527:2007 for the tutorial project, rounded to 250 L, and the potential each one yields in the daily simulation

Table 24 -- Storage fraction of the sweep volumes and the simulation time step Fewkes and Butler (2000) recommend for each range

Table 25 -- Variables of Equation 13

Table 26 -- **Table 99** -- Representative series length, in years, depending on the suppliable fraction

Table 27 -- Variables of Equation 14

Table 28 -- Variables of Equation 15

Table 29 -- Variables of Equation 16

Table 30 -- Variables of Equation 17

Table 31 -- Variables of Equation 18

Table 32 -- Variables of Equation 19

Table 33 -- Variables of Equation 20

Table 34 -- Summary of the indicators, with the denominator and the question each one answers

Table 35 -- Variables of Equation 21

Table 36 -- Variables of Equation 22

Table 37 -- Variables of Equation 23

Table 38 -- Variables of Equation 24

Table 39 -- Variables of Equation 25

Table 40 -- Variables of Equation 26

Table 41 -- Variables of Equation 27

Table 42 -- Variables of Equation 28

Table 43 -- Variables of Equation 29

Table 44 -- Mechanical inlet filter sized by the roof area

Table 45 -- Variables of Equation 30

Table 46 -- Polishing filter recommendation by target quality class

Table 47 -- Sizing and pricing of the first flush in the tutorial project

Table 48 -- Variables of Equation 31

Table 49 -- The two efficiencies of the pumping chain and what each one converts

Table 50 -- Variables of Equation 32

Table 51 -- Variables of Equation 33

Table 52 -- Variables of Equation 34

Table 53 -- Variables of Equation 35

Table 54 -- Variables of Equation 36

Table 55 -- Fallback values adopted when the project does not declare them

Table 56 -- Variables of Equation 37

Table 57 -- Annual energy of six catalogue pressure pumps in the tutorial's direct pressurisation, with the toilet flow limiting the pump flow

Table 58 -- Variables of Equation 38

Table 59 -- Variables of Equation 39

Table 60 -- Variables of Equation 40

Table 61 -- Coding and application of each minimum-charge regime in the calculation of the bill

Table 62 -- Variables of Equation 41

Table 63 -- Parts of the annual running cost and the origin of each one

Table 64 -- Variables of Equation 42

Table 65 -- Variables of Equation 44

Table 66 -- Variables of Equation 45

Table 67 -- EN 15978 life cycle modules and what the program's assessment covers in each one

Table 68 -- Variables of Equation 46

Table 69 -- Mass per useful cubic metre of each tank material

Table 70 -- Carbon inventory of the final tutorial project, item by item, with the share of each component in the embodied carbon; negative annual values are credits

Table 71 -- Variables of Equation 47

Table 72 -- Variables of Equation 48

Table 73 -- Variables of Equation 49

Table 74 -- Variables of Equation 50

Table 75 -- Variables of Equation 51

Table 76 -- Options of LEED v5 credit WEc2 to which rainwater harvesting contributes

Table 77 -- Items of the water consumption theme of AQUA-HQE Residential related to rainwater harvesting

Table 78 -- Score of criterion 3.6 of Selo Casa Azul + CAIXA according to the scope of use

Table 79 -- The seven decision criteria, the direction of preference and the group of each one

Table 80 -- Variables of Equation 52

Table 81 -- TOPSIS closeness score with equal weights across the dimensions and first-rank acceptability in SMAA-VIKOR, with 5,000 random weight vectors, for the tutorial volumes

Table 82 -- The same tutorial project simulated in four Brazilian state capitals

Table 83 -- Ratio between the wettest and the driest month in the four capitals

Table 84 -- Content of the national climate database distributed with the program

Table 85 -- The six tariff block codes and the effect of each one on the bill

Table 86 -- The six tariff regimes and what happens when saving in each one

Table 87 -- Effect of the water saving on the bill in two opposite tariff regimes

Table 88 -- Exponent n of the CORSAN Exponent Table, residential category

Table 89 -- Monthly bill in Alegrete (RS) by the exponential formula, effective 01/07/2026

Table 90 -- The three ways of comparing alternatives depending on what varies between them

Table 91 -- Subcommands of the command line and the purpose of each one

Table 92 -- Potential, net present value and payback of the final tutorial project with each assumption varied alone, from the low to the high value

Table 93 -- Reading the SMAA-VIKOR acceptability of the winning volume

Table 94 -- Netuno 4 data recovered by the import

Table 95 -- Export formats and the purpose of each one

Table 96 -- Spreadsheet export options and when to use each one

Table 97 -- Characteristics of the Florianópolis precipitation series used in the tutorial

Table 98 -- The three ways of applying the first flush

Table 99 -- State of the tutorial project at the end of defining the rain and the catchment

Table 100 -- State of the tutorial project at the end of defining the demand

Table 101 -- Comparison between the annual rainwater supply and the suppliable demand

Table 102 -- Result of the first tutorial simulation, with a 3,000 L tank

Table 103 -- Classification of each day of the simulation by the degree of supply

Table 104 -- State of the tutorial project after the first simulation

Table 105 -- Volume sweep of the tutorial project, with the marginal gain of each added cubic metre

Table 106 -- Marginal saving gain for each cubic metre added to the tank

Table 107 -- Indicators of the ideal volume obtained with the 1 %/m³ threshold adopted in the tutorial

Table 108 -- Ideal volume obtained with each marginal-gain threshold, for the same project

Table 109 -- Cost and performance compared between the ideal volume and the largest volume swept

Table 110 -- State of the tutorial project after choosing the volume

Table 111 -- The three supply arrangements and what each one adds to the project

Table 112 -- Single tank compared with two tanks, keeping the same total storage

Table 113 -- Effect of the pump trigger level on the saving potential and on the number of starts

Table 114 -- State of the tutorial project after the decision on the upper tank

Table 115 -- Cost of the treatment line according to the disinfection technology adopted

Table 116 -- Fine-filtration recommendation and cost of the line by quality class

Table 117 -- Mechanical inlet filter sized by the roof area

Table 118 -- State of the tutorial project after defining the quality class

Table 119 -- Make-up of the water and sewage bill before and after the rainwater system

Table 120 -- Economic indicators of the tutorial project over twenty years of analysis

Table 121 -- The six tariff regimes and the effect of each one on the saving

Table 122 -- Make-up of the initial investment of the tutorial project

Table 123 -- Economic indicators of the tutorial project over twenty years

Table 124 -- Assumptions of the economic projection and the weight of each one in the result

Table 125 -- State of the tutorial project after the economic analysis

Table 126 -- Carbon balance of the tutorial project with a concrete cistern

Table 127 -- Net carbon emissions of the tutorial project according to the tank material

Table 128 -- Effect of the concrete cistern volume on the net carbon emissions

Table 129 -- Effect of switching from concrete to polyethylene on the three dimensions of analysis

Table 130 -- State of the tutorial project after the environmental analysis

Table 131 -- Multi-criteria decision matrix: each sweep volume assessed in the three dimensions of analysis

Table 132 -- Winning volume under each criterion taken alone

Table 133 -- The seven multi-criteria criteria and the direction of preference of each one

Table 134 -- Volume chosen by TOPSIS according to the dimension given the dominant weight, and the runner-up

Table 135 -- Summary of the tutorial project, with all the parameters and results

Table 136 -- The sets of reference cases exported from Netuno 4

Table 137 -- Divergence measured between the two versions, by set of cases

Table 138 -- Divergence in three successively more controlled configurations

Table 139 -- Attribution of the divergence to its three sources

Table 140 -- Variables of Equation 53

Table 141 -- Physical input validation messages and the action for each one

Table 142 -- Suppliable fraction messages and the action for each one

Table 143 -- Precipitation series error messages and the action for each one

Table 144 -- Volume sweep error messages

Table 145 -- Chlorination parameter error messages

Table 146 -- Economic analysis error messages and the action for each one

Table 147 -- Multi-criteria decision error messages and the action for each one

Table 148 -- Pump set parameter error messages

Table 149 -- Error situations when reading and saving files

Table 150 -- Warnings that do not stop the calculation and must be stated in the study

Table 151 -- **Table 99** -- Overview of the published versions

Table 152 -- Fixes that alter the results of studies done with an earlier version

Table 153 -- Behaviour fixes that do not alter any calculated result

Table 154 -- Project parameters: identification

Table 155 -- Project parameters: rainfall

Table 156 -- Project parameters: catchment

Table 157 -- Project parameters: building

Table 158 -- Project parameters: demand

Table 159 -- Project parameters: irrigation

Table 160 -- Project parameters: water quality

Table 161 -- Project parameters: tanks

Table 162 -- Project parameters: pump set

Table 163 -- Project parameters: piping

Table 164 -- Project parameters: investment

Table 165 -- Project parameters: operation

Table 166 -- Project parameters: tariffs

Table 167 -- Project parameters: energy

Table 168 -- Project parameters: incentives

Table 169 -- Project parameters: financial

Table 170 -- Project parameters: certifications

Table 171 -- Project parameters: other

Table 172 -- Emission factors of the tank materials, with mass and boundary

Table 173 -- Emission factors of the treatment consumables

Table 174 -- Manufacturing emission of the pump set by power rating, with the catalogue mass

Table 175 -- Provenance of the hypochlorite emission factor

Table 176 -- Distribution of the catalogue municipalities across the six tariff regimes

Table 177 -- Confidence grade of the tariff records in the catalogue

Table 178 -- Coverage of the tariff catalogue: municipalities by category and status of the data

Table 179 -- The tariff structures of the catalogue, numbered for the price-by-block tables that follow

Table 180 -- Price of water by monthly consumption interval, in R$/m³, by municipality and category (part 1 of 3: 0-5 to 20-25 m³)

Table 181 -- Price of water by monthly consumption interval, in R$/m³, by municipality and category (part 2 of 3: 25-30 to 70-75 m³)

Table 182 -- Price of water by monthly consumption interval, in R$/m³, by municipality and category (part 3 of 3: 75-80 to 5,000 m³)

Table 183 -- Tariff categories not yet collected, by provider

Table 184 -- Effective date and last check of the source of each tariff structure, with the status of the data

Table 185 -- The CORSAN exponential formula in each tariff table and collected category

Table 186 -- Contents of the national climate database

Table 187 -- Historical precipitation series distributed with the program

Table 188 -- Formats the program reads and where each one is loaded

Table 189 -- Formats the program writes and where each one is generated

Table 190 -- The seven sections of the project file and what each one holds

Table 191 -- Location of the program's files and of the user's files

Table 192 -- Compatibility between versions of the project file

Table 193 -- Terminology for rainwater and stormwater, in Portuguese and English

Table 194 -- Fundamental quantities of the water balance and relations between fraction and potential

Table 195 -- Reference sheet: the ten steps of a complete study and the section that deals with each one

Table 196 -- Reference sheet: the ten design pitfalls and where each one is dealt with

Table 197 -- Reference sheet: the six tariff regimes and the effect of saving

Table 198 -- Reference sheet: the indicators and the question each one answers

Table 199 -- The nine example cases and what each one demonstrates

Table 200 -- Result each example case produces when reread and simulated

List of equations

Equation 1 -- Mass balance closure

Equation 2 -- Volume collected in the day

Equation 3 -- Volume available before consumption

Equation 4 -- Overflow volume

Equation 5 -- Volume stored before consumption

Equation 6 -- Rainwater volume consumed

Equation 7 -- Volume left for the next day

Equation 8 -- Potable mains top-up

Equation 9 -- Classification of the day by supply

Equation 10 -- The three parts of the daily demand

Equation 11 -- Marginal gain per added cubic metre

Equation 12 -- Ideal volume by the marginal-gain criterion

Equation 13 -- Storage fraction

Equation 14 -- Potable water saving potential

Equation 15 -- Volumetric reliability

Equation 16 -- Relation between potential and reliability

Equation 17 -- Mean daily overflow

Equation 18 -- Efficiency of use of the collected water

Equation 19 -- Annual potable water volume saved

Equation 20 -- Mean daily unmet demand

Equation 21 -- Total annual precipitation, PRCPTOT

Equation 22 -- Precipitation on very wet days, R95p

Equation 23 -- Maximum one-day precipitation, RX1day

Equation 24 -- Maximum five-day precipitation, RX5day

Equation 25 -- Simple daily intensity index, SDII

Equation 26 -- Days of heavy precipitation, R20mm

Equation 27 -- Consecutive dry days, CDD

Equation 28 -- Consecutive wet days, CWD

Equation 29 -- Seasonality index, SI

Equation 30 -- Mass of chlorine product per cubic metre

Equation 31 -- Incremental investment of the treatment line

Equation 32 -- Specific pumping energy

Equation 33 -- Specific energy from the flow rate

Equation 34 -- Pumping flow rate deduced from the power

Equation 35 -- Energy used at start-ups

Equation 36 -- Annual pump energy

Equation 37 -- Annual energy of the pressure pump

Equation 38 -- Monthly bill saving

Equation 39 -- CORSAN exponential formula

Equation 40 -- Sewage part of the bill

Equation 41 -- Total initial investment

Equation 42 -- Cumulative inflation factor

Equation 43 -- Number of tariff adjustments applied

Equation 44 -- Cumulative tariff adjustment factor

Equation 45 -- Net present value

Equation 46 -- Embodied emission of a component

Equation 47 -- Annual emission of the pumping energy

Equation 48 -- Net annual emission balance

Equation 49 -- Environmental payback time

Equation 50 -- Carbon balance at the end of service life

Equation 51 -- Specific emission per litre delivered

Equation 52 -- TOPSIS closeness coefficient

Equation 53 -- Series transformation to emulate Netuno 4

1. Start here

1.1. What Netuno does, and what it does not

Netuno simulates rainwater harvesting systems in buildings. Given the rainfall regime of a place, the roof of a building and the consumption of the people who use it, it answers **how much potable water the system replaces**, and at what cost, with what quality and with what carbon impact.

This section starts from the beginning, so the first two subsections deal with the object itself before dealing with the program. Readers who already design building water systems will find them elementary and can go straight to the six dimensions of analysis, which is where Netuno appears; readers who are new to the subject will find there the vocabulary that the rest of the manual uses without explaining again.

1.1.1. What a rainwater harvesting system is

It is a system that **collects the rain falling on the roof and uses it instead of mains water**, for the uses that do not need potable water.

The idea is old and simple. What makes it difficult is one thing only: rain does not fall when you need it. That is why the system needs a tank, and that is why the size of that tank is the central question of every project -- and of this manual. Figure 1 presents the components of the system and the path of the water, from collection to the point of use.

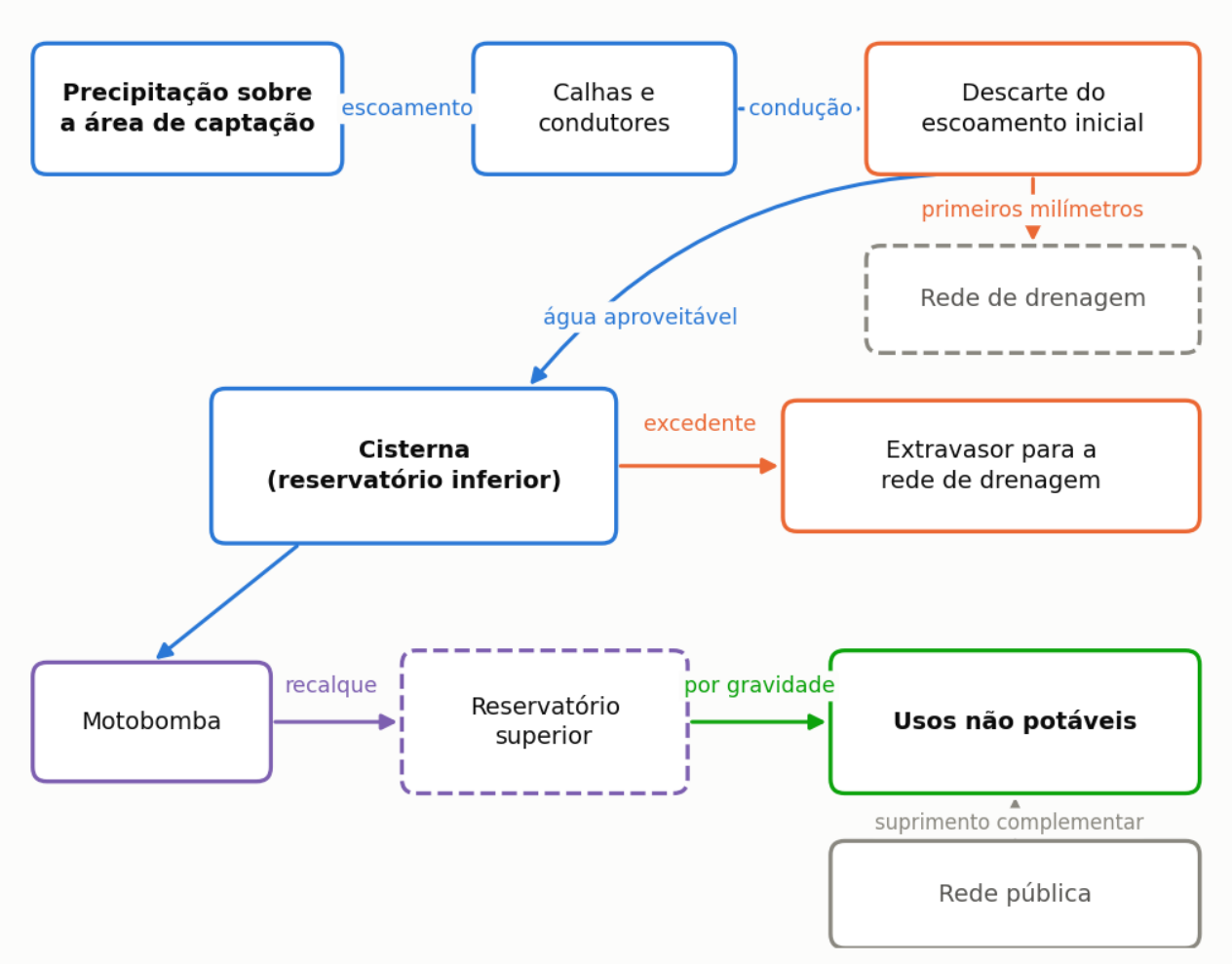


Figure 1 -- Components of a building rainwater harvesting system and the path of the water, from collection to the point of use. The upper tank is dashed because it is optional: in a single-storey building the pump can pressurise the points of use directly, and the upper tank changes when water is supplied, not the volume available.

Following the drawing, from left to right:

The roof is the catchment area. What counts is the **horizontal projection** of the roof, not the sloping area, as item 3.6 of ABNT NBR 15527:2019 defines it -- it is the roof's shadow on the ground that receives the rain. Getting this wrong is the first of the ten mistakes in section 5.1.

The gutters and downpipes carry the water to the cistern. The program does not size them; that is hydraulic design, and in Brazil it follows ABNT NBR 10844.

The first-flush device diverts the first millimetres of each event, which carry the dust, leaves and bird droppings accumulated on the roof. It is the cheapest quality control measure, and section 4.6 shows how much it changes the water that reaches the tank.

The cistern, or lower tank, stores the collected water. When the inflow exceeds the capacity, the surplus **overflows** to the drainage system. This overflow is not waste through carelessness: it is the unavoidable consequence of storing a resource that arrives concentrated.

The pump and the upper tank distribute the water. In a single-storey building the pump can pressurise the points of use directly. In buildings of two or more storeys an elevated tank is usually adopted: the water is pumped up to it and distributed by gravity.

The non-potable uses are the destination: toilets and urinals, floor and vehicle washing, landscape irrigation. When rainwater is not enough, the **mains** supplies the rest, and supply is never interrupted.

The diagram of Figure 1 shows how the parts are linked, which is what matters to follow the calculation. It lacks, however, the information that settles half of the design questions: where each part sits in relation to the others, and at what height. The section of Figure 2 restores that geometry, because the difference in level is what determines whether water flows by gravity or needs a pump, and because the position of the cistern relative to the ground is a construction decision, not a calculation one.

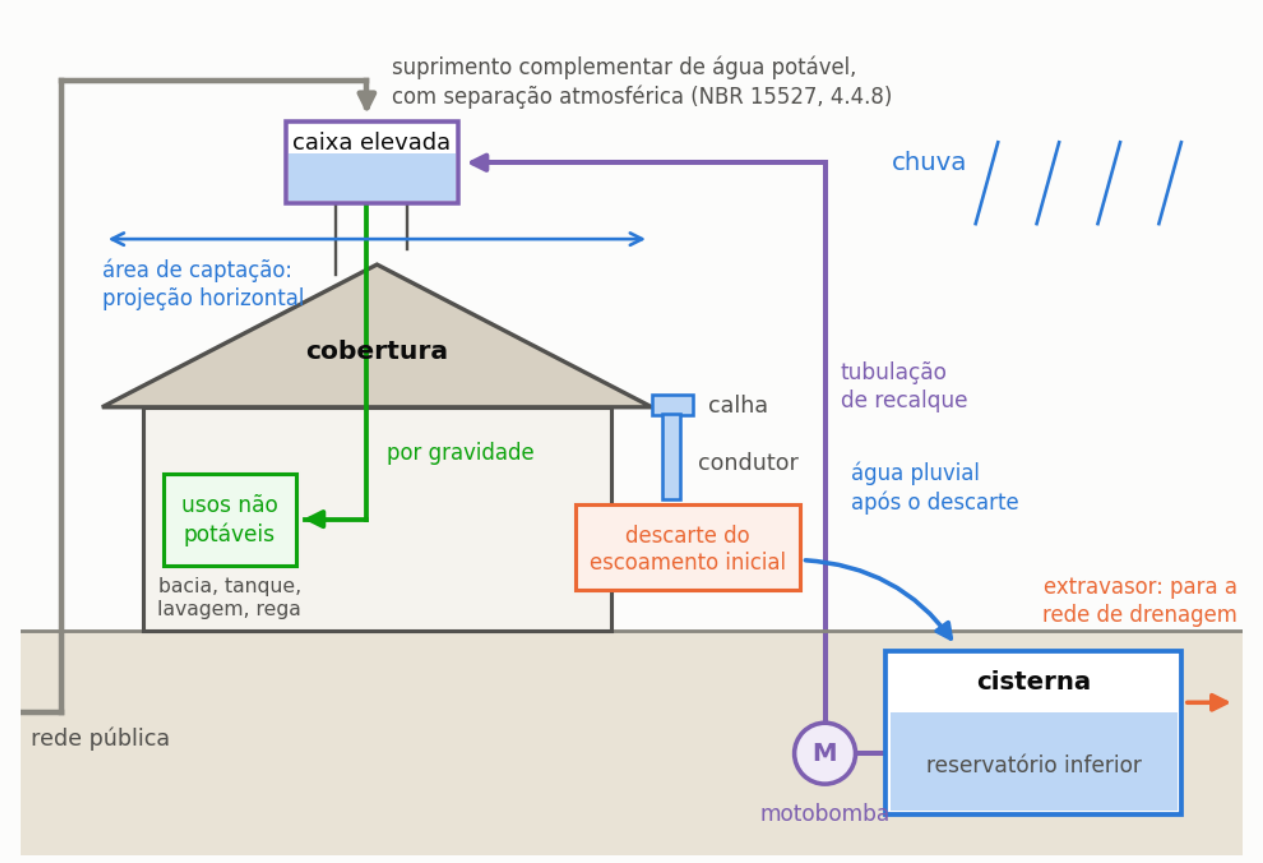


Figure 2 -- The same system seen in section. The catchment area is the horizontal projection of the roof, not the sloping area of the roof. The elevated tank receives the water pumped from the cistern and returns it to the points of use by gravity; the mains feeds it, through an air gap, supplying the volume that rainwater does not cover. The cistern overflow carries the surplus to the drainage system.

> **Vocabulary the manual uses all the time.** *Cistern* and *lower tank* designate the same component. *Overflow* is the surplus volume that the overflow pipe carries to the drainage system. *Rainwater demand* is the part of the consumption that rainwater can supply; *total demand* is the whole consumption of the building. Appendix G has the complete glossary.

1.1.2. What the program calculates, in one day

The program does an inflow and outflow account, every day, for each day of the rainfall series. Figure 3 shows how the precipitation of one day is split, and the account is this:

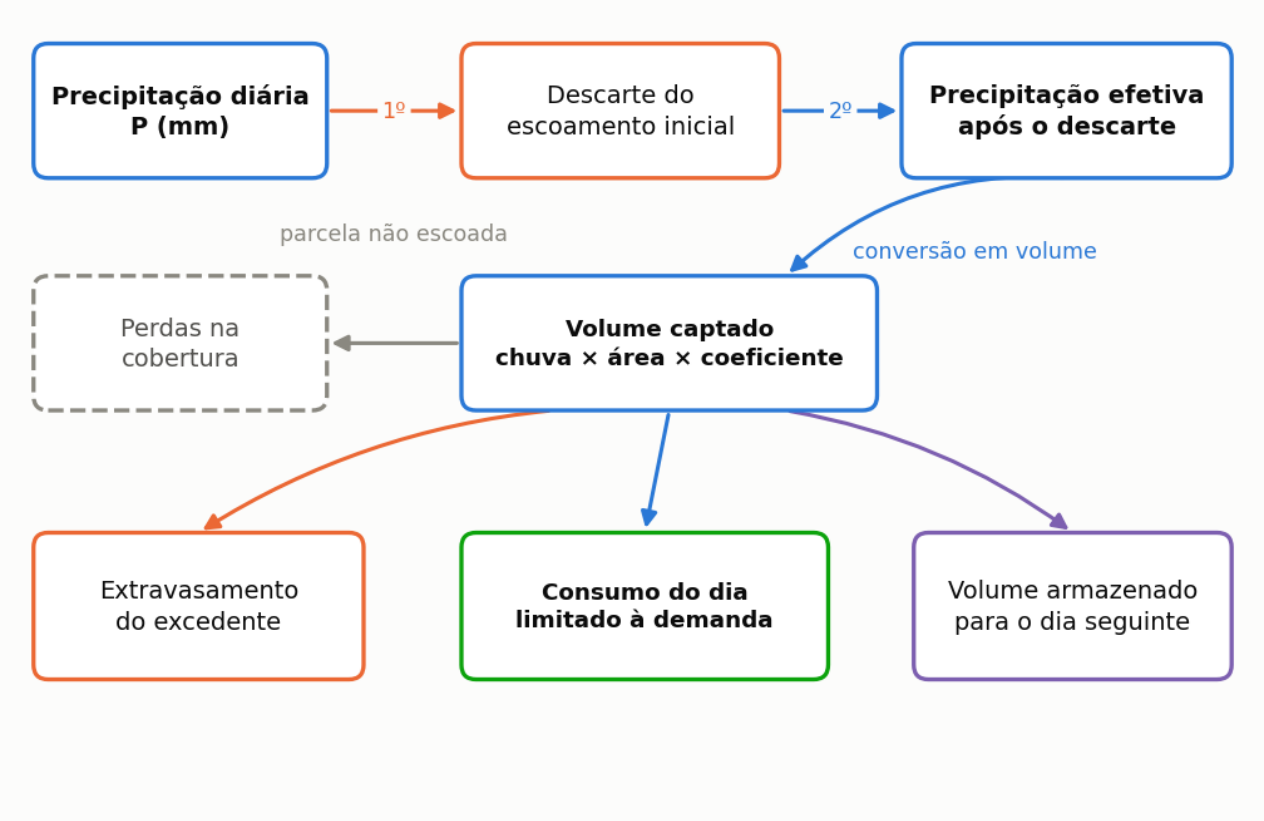


Figure 3 -- Destination of the precipitation of one day. Each box is a part of the balance, and the sum of the overflow, the consumption and the stored volume equals the collected volume. The overflow is calculated before the day's consumption, and the day's rain can already supply the day's consumption: an order between the YAS and YBS rules, explained in the section on the daily balance.

Reading the drawing:

1. **The first flush is deducted.** With 12 mm of precipitation and a 2 mm first flush, 10 mm of effective precipitation remain.
2. **It is converted into volume.** Ten millimetres on 100 m² of roof, with a runoff coefficient of 0.85, give 850 litres. The calculation is `mm x m² x coefficient`, and it comes out directly in litres, because one millimetre on one square metre is exactly one litre.
3. **It is added to the volume stored in the cistern**, and whatever exceeds the capacity **overflows** to the drainage system.
4. **The day's consumption is withdrawn**, up to the rainwater demand.
5. **The remaining volume is carried to the next day** -- and it is this dependence between consecutive days that makes a daily simulation necessary.

The five steps form a cycle, not a list: the volume left at the end of the fifth step is exactly the volume from which the third step starts on the next day. This link between one day and the next is the one thing that a calculation on the

annual mean cannot reproduce, and it is the reason the simulation is daily. Figure 4 represents this sequence in the notation of the daily balance equations.

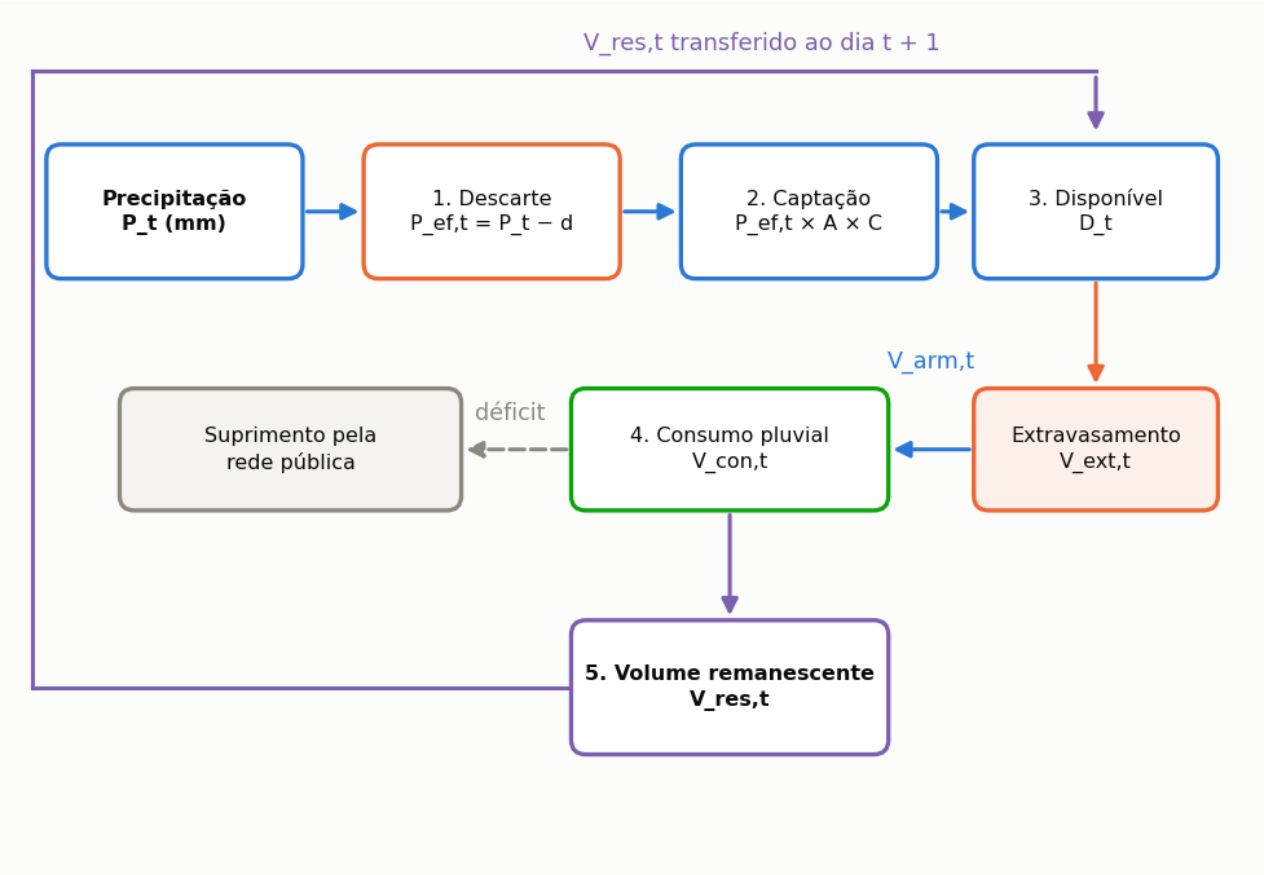


Figure 4 -- Sequence of operations of the daily balance, in the order the program runs them, in the notation of the equations of the daily balance section. The purple link around the drawing represents the dependence between consecutive days: the remaining volume V_res,t is the starting volume of the next day.

> **IN DEPTH . WHY THE ANNUAL MEAN IS NOT ENOUGH**
>
> It is tempting to divide the annual rain by 365 and do the calculation in one go. The result is always optimistic, and sometimes very much so. The reason is that rain arrives **concentrated**: it fills the cistern in one day, overflows, and then is missing for two weeks. The mean represents neither the overflow nor the deficit, and both reduce the yield. Section 2.1 shows this happening in a ten-day table that you can redo in a spreadsheet, and section 4.4 shows the size of the error.

Repeated over thirty years of series, this calculation produces the program's central answer: the **potable water saving potential**, the fraction of the building's total consumption that rain managed to replace.

1.1.3. The six dimensions

The volume of water supplied is the first question of a project, but not the only one. The program assesses the system in six dimensions, which generally lead to different optimal volumes (Figure 5).



Figure 5 -- The six dimensions of analysis and the question assessed in each one. The program goes through them in this order, and each one is treated in a section of its own in the method chapter.

Each dimension has its own treatment in the method chapter: the technical one in sections 2.1 to 2.4, quality in 2.5, economics in 2.7, the environment in 2.8, certification in 2.9 and the decision in 2.10.

The sixth dimension is what sets the program apart from a spreadsheet. The first five produce **diverging** results: the volume with the largest water saving is rarely the one with the best financial return, and neither is usually the one with the best environmental performance. The multi-criteria decision makes explicit the weights given to each criterion.

1.1.4. What the program does not do

Starting with the limits is unusual, and deliberate. It saves the time of anyone who opened the program expecting something else.

It does not size the plumbing. It does not calculate the routing, the diameter of each run or the slope of the network. The pump wizard estimates the diameter and the head loss of the rising main to select the equipment, but it does not replace the plumbing design; the diagram exported in DXF is a diagram, and the construction design belongs to the designer.

It does not detail the first-flush device. The program calculates the volume of the first-flush chamber, equal to the catchment area times the depth, and prices it with the leaf separator; the arrangement of the fittings, the purge and the connection to the drainage are separate hydraulic design.

It does not replace water quality testing. It sizes treatment equipment from assumptions. The constants of the roof wash-off model are not calibrated against measurements, and the resulting quality is an assumption, not a result.

It does not model an hourly sub-step. A 40 mm rain in two hours and the same depth over twenty-four hours give the same result.

It does not do a complete life cycle inventory. The assessment is a **screening** one, in a single impact category -- carbon. There is no eutrophication, acidification, land use or toxicity.

It does not certify. It shows where the project would fit. Certification is decided by the certification body, with the complete documentation of the development.

It does not know the site. It does not know whether the volume fits on the plot, whether the water table is high, or whether the structure can carry the elevated tank.

Section 5.2 has the complete list of known limitations, with the consequences of each one.

1.1.5. Who it is for

The manual serves four reader profiles, with reading paths independent of one another (Figure 6).

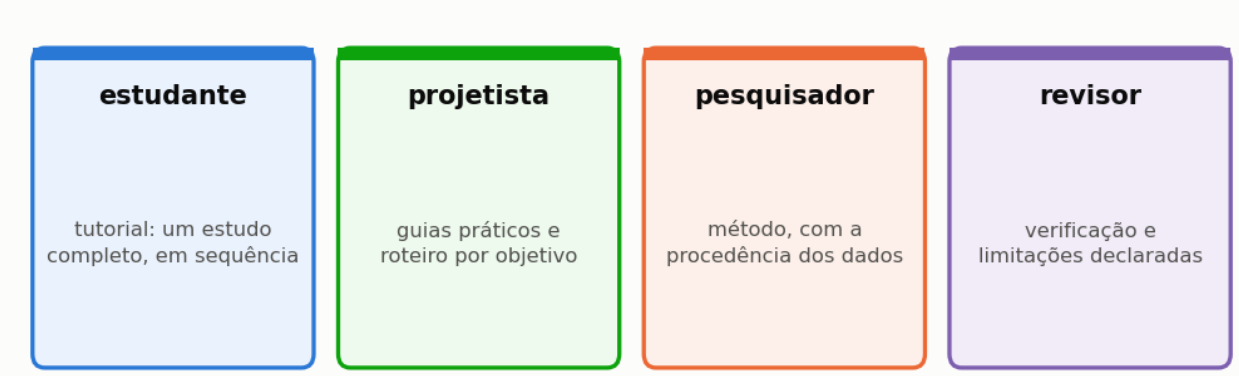


Figure 6 -- The four reading paths of the manual. They are independent of one another, and the manual was organised so that each one can be followed on its own.

In section numbers: the student follows chapter 4, which is the tutorial from start to finish; the designer uses the guides of chapter 3 and the route of section 1.3; the researcher goes to the method of chapter 2; and the reviewer finds in chapter 5 the verification and the declared limits.

Anyone starting now should go through the whole of chapter 4, in the order in which it is written. It builds a project from scratch and follows it to the final decision, and each section ends by showing where the project has got to, so that the reader sees the study taking shape instead of receiving it finished.

1.1.6. What changed from Netuno 4

Netuno 4 is the predecessor, published in 2014 by the same group. Netuno 5 keeps the conceptual model -- a daily behavioural balance -- and adds six dimensions of analysis that did not exist.

> **IN DEPTH . THE TWO DELIBERATE DIFFERENCES OF METHOD**
>
> Anyone comparing results of the two versions must declare them.
>
> **The treatment of the first flush.** Netuno 4 treats the depth as an **event threshold**: below it nothing is collected, above it the whole rain is collected. Netuno 5 **subtracts** the depth, which is physically more faithful.
>
> **The optimal volume criterion.** The implementations and the search grids differ.
>
> Together, the two account for 91% of the divergence measured between the programs. Over 4,050 reference cases, Netuno 5 estimates 2.3 to 2.7 percentage points less saving. **With both neutralised, the engine reproduces Netuno 4 with a mean error of 0.25 percentage points and 100% of the cases within 5 points.** Section 6.1 has the complete breakdown.

1.1.7. A word about results

Every number the program produces is **conditional on the declared assumptions**. It does not discover truths; it calculates consequences.

That is why the manual insists so much on declaration. Section 4.10 has the list of what must appear in a paper, section 5.1 has the ten mistakes the program does not prevent, and section 5.2 has what it does not cover.

A study that declares its assumptions and its limits is more useful -- and more defensible -- than one that presents a number on its own.

1.2. Installation and first run

> By the end of this section you will have the program installed, open, and a simulation finished with a result you can check against this manual.

1.2.1. What you need

Table 1 -- System requirements to install and run the program

Requirement	What is needed
system	Windows 10 or 11, 64-bit
disk space	about 1.5 GB for the Brazil Edition, with the national climate database; about 0.7 GB for the International Edition
memory	4 GB is enough; 8 GB makes the sweep more comfortable
privileges	administrator, to install for all users; without them, the installation goes to the user's folder

Nothing else needs to be installed. The program is distributed already assembled, with all the components it depends on to calculate, and does not require any prior installation of interpreters, libraries or auxiliary packages. Nor does it need an internet connection to work: everything it consults during a simulation is on the computer itself.

1.2.2. The two editions

The program comes in two editions, with their own installers, which can be installed side by side. The green (BRL) or blue (INT) badge at the top of the sidebar, the window title and the footer of every report and export say which edition the study came from.

Table 2 -- What changes between the Brazil Edition and the International Edition

	Brazil Edition (BRL)	International Edition (INT)
installer	`Netuno_Setup_5_0_0.exe`	`Netuno_INT_Setup_5_0_0.exe`
language on opening	Portuguese	English
rainfall	CLIMBra database (652 cities) or your own series	your own series
water and sewage tariffs	catalogue of 652 municipalities, or typed in	typed in
electricity	ANEEL tariff or typed in; tariff flags	typed in
currency and costs	Brazilian reais, with reference prices	currency chosen; costs typed in
grid and potable water emission factors	Brazilian databases (MCTI, routes by region)	from the project's country, mandatory
certifications and incentives	LEED, EDGE, AQUA-HQE, Selo Casa Azul, IPTU Verde	LEED, EDGE

Without the country's emission factors, the International Edition does not calculate the environmental assessment: a Brazilian factor applied to another country would give a wrong result without any warning.

Both editions calculate in exactly the same way: the daily balance, the first flush, the runoff coefficients, the economic analysis and the life cycle assessment are the same code. What the International Edition leaves out is Brazilian **data**, not method. The differences between the Brazilian standards the method follows and the European standard EN 16941-1:2024 are listed, item by item, in the repository document `docs/normas_nbr_x_en16941.md`.

1.2.3. Installation

1. Run `Netuno_Setup_5_0_0.exe` (Brazil Edition) or `Netuno_INT_Setup_5_0_0.exe` (International Edition). The installer asks whether the installation is for all users, which requires administrator rights, or only for you.
2. Choose the installation language and accept the licence, the GPL version 3.
3. Choose the installation type. The **standard** one brings the program, the manual, the cases, the examples and all the languages. The **custom** one lets you untick the Spanish and Chinese language packs. In the Brazil Edition, the national climate database always comes along, embedded in the program.
4. Accept the suggested folder, `C:\Program Files\Netuno 5` (Brazil Edition) or `C:\Program Files\Netuno 5 International` (International Edition), or choose another.
5. Tick, if you wish, **create a desktop icon**. The option **open .netuno files with Netuno 5** comes ticked in the Brazil Edition: with it, double-clicking a project opens it directly in the program. In the International Edition it comes unticked, so as not to take over the files of a Brazil Edition installed on the same computer.
6. When it finishes, the installer offers to open the program.

After that, the Start menu has two shortcuts: the program, and its manual in PDF -- this English manual in the International Edition, the Portuguese one in the Brazil Edition. To uninstall, use Windows *Apps and features*, as with any program; your projects stay where they were saved.

> **What the installer puts on the computer.** The program, the four historical rainfall series that come with the installation, the example files and, in the Brazil Edition, the tariff database of the 652 municipalities and the national climate database. Nothing is downloaded during use: Netuno works without an internet connection.

1.2.4. Where your files go

The program never decides for you where to save. Every time you save a project, a report or an export, it asks for the folder.

Table 3 -- File formats the program saves and the content of each one

What you save	Extension	Contains
project	`.netuno`	all the input data **and the rainfall series**
report	`.pdf`	the complete document of the study
results	`.xlsx`, `.csv`	the tables, for a spreadsheet
diagram	`.dxf`	the schematic drawing, for CAD

Among these formats, `.netuno` is the one that really matters to keep. It is self-contained: it carries the whole rainfall series inside it, and therefore opens on any other computer even without the files it came from. It is also the file to attach to a paper so that someone else can reproduce the study, and this property is what separates a verifiable piece of work from one that is merely reported.

1.2.5. The climate database

The program brings four long historical series ready to use -- Florianópolis, Brasília, Manaus and São Paulo -- and accepts any series of your own in CSV.

> **Brazil Edition.** The national climate database ships only with the Brazil Edition. In the International Edition, the rainfall series is always a file of your own: daily precipitation in millimetres, one value per day, as appendix F describes.

Besides them, the Brazil Edition brings the **national climate database**: 652 cities, 19 CMIP6 climate models and 2 future scenarios, SSP2-4.5 and SSP5-8.5, extracted from the CLIMBra dataset. The original files add up to about 6.5 GB. The program brings them compressed into a single database of about 800 MB, embedded in the installer. There is nothing to download or copy: the database appears on the first screen as a list of cities.

Appendix E describes the database, and section 3.2 shows how to use a future scenario.

1.2.6. The example files

The manual comes with ready-made projects that reproduce the results printed here.

Table 4 -- Example files distributed with the program

File	What it is for
`casos\01_tutorial_florianopolis.netuno`	the project that chapter 4 builds, ready
`casos\03_dez_dias.netuno`	the ten-day teaching case of section 2.1
`casos\05_manaus.netuno` and the following	the same project in other state capitals
`exemplos\*.netuno`	three complete studies: house, office and school

They are in the program folder. Appendix J lists the nine cases of the manual, with the chapter of each one and the result they produce.

It is worth starting with the ten-day case. Open it, run the simulation and compare the result with the table printed in section 2.1. If the two match, the installation corresponds exactly to this manual; if they differ, the manual and the program are of different versions, and section 6.3 explains what to do in that case.

1.2.7. The first simulation, in five minutes

The program is organised in **steps**, in the left sidebar, in the order in which you work. For the first simulation four of them are enough.

1. **Project & Rainfall.** Choose the city in the climate database, or load the file `Florianopolis1.csv` that comes with the program. Enter a catchment area of 150 m², a runoff coefficient of 0.85 and a first flush of 2.0 mm.
2. **Building & Demand.** Choose a single-family house, 4 residents, 150 litres per person per day, and a rainwater-suppliable fraction of 28%.
3. **Reservoir & Costs.** Enter a fixed volume of 4,000 L.
4. Run it, and go to **Technical Results**.

You should get a saving potential of about **27%**.

Section 4.3 explains what this number means, and section 4.4 asks the next question: is 4,000 L the right volume?

1.2.8. If something goes wrong

The program does not open. Check that the installation finished without errors and reinstall over it; reinstalling keeps your files, because they are not in the program folder.

The list of cities is empty. The national climate database comes only with the Brazil Edition. In the International Edition, use a rainfall file of your own.

Some part of the program is missing, such as the 3D volumetry. Run, in the Command Prompt, `"C:\Program Files\Netuno 5\Netuno.exe" --diagnostico` (in the International Edition, `"C:\Program Files\Netuno 5 International\Netuno_INT.exe" --diagnostico`). The program lists what the installation contains: the 3D viewer, the climate database, the tariff database and the manual. Each item is marked OK or FALTA (missing). The list is also saved in `netuno_diagnostico.txt`, in the Windows temporary folder. Send that file with the problem report.

The result does not match the manual. Check, in this order: the rainfall series loaded, the area, the coefficient, the first-flush depth, the demand and the rainwater-suppliable fraction. A difference in result is almost always a difference in input, and section 5.1 lists the most common mistakes.

An error message appeared. Section 6.2 has the program's messages, what each one means and what to do.

1.3. Where to start, depending on your goal

This manual brings together forty-nine sections in seven chapters, and it was not designed to be read from cover to cover. It works best as a reference work, in which each reader enters at the point that matches what they need to do, and this section exists precisely to show that entry point.

1.3.1. Route by goal

The table below lists the five most common purposes of people who open the program, with the reading path for each one.

Table 5 -- Recommended reading path depending on the reader's goal

You want to	Read
learn to use the program	the whole of chapter 4, sections 4.1 to 4.10, in order
solve a case today	the three questions below, and then the guide of chapter 3 that fits
publish a study	chapter 2, which has the method, and section 4.10, with the declaration list
review someone else's study	chapter 5, and appendices A and B
migrate from Netuno 4	sections 3.8 and 6.1
avoid mistakes	section 5.1, with the ten most common mistakes

Anyone with an hour and no previous familiarity should read the tutorial of chapter 4. It builds a complete study around a single project, and each section ends by showing where that project has got to, so that the reader follows the construction instead of receiving the finished result.

1.3.2. The three questions that define a study

Even before opening the program, it is worth answering three questions, because they determine a good part of the later decisions and because answering them afterwards usually means redoing work.

1.3.2.1. Which uses will rainwater supply?

The answer defines the **rainwater-suppliable fraction**, and that fraction is the ceiling of everything else: a system whose fraction is 28% will never save more than 28% of the total consumption, however large the cistern and however much it rains. No later design decision changes this limit, which is why it deserves to be set before anything else. Figure 7 brings together the usual ranges of the suppliable fraction depending on the intended uses.

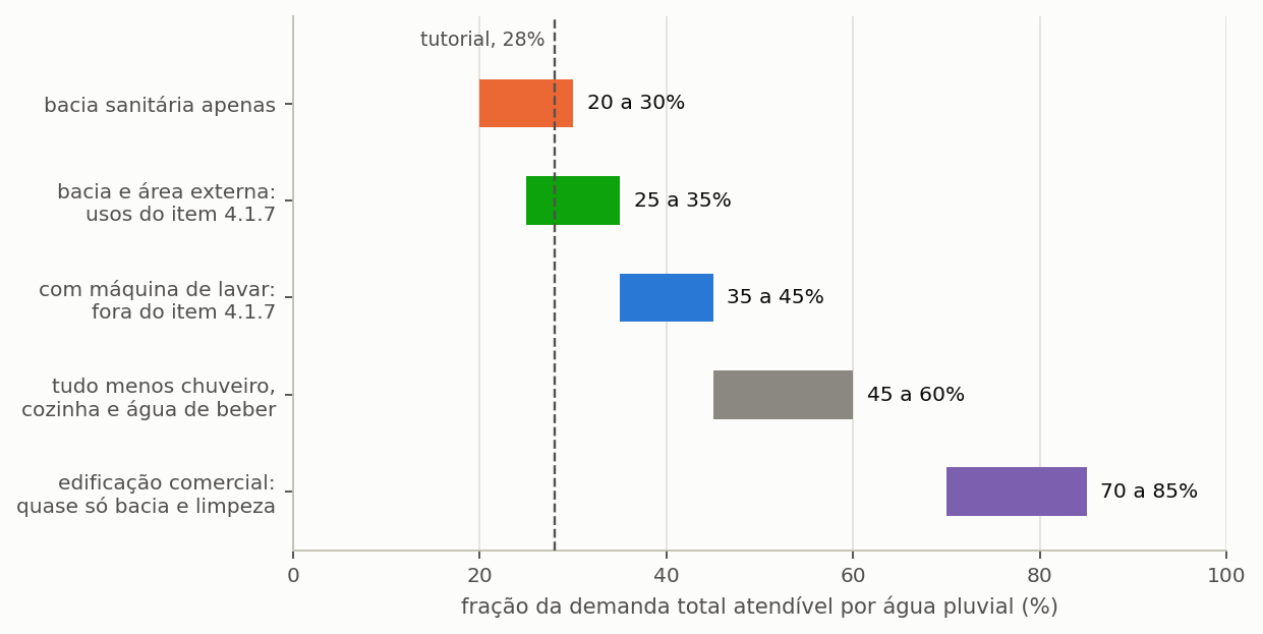


Figure 7 -- Which part of the total consumption rainwater can supply, depending on the uses the project intends. It is the ceiling of the saving potential: no tank, however large, exceeds this fraction.

The residential cases are dealt with in section 4.2, and the commercial case in section 3.5. The dashed line marks the 28% of the tutorial, which keeps to the uses that item 4.1.7 of ABNT NBR 15527:2019 admits.

1.3.2.2. Will you compare volumes or assess just one?

The distinction between assessing a known volume and looking for the right volume determines the simulation mode and, with it, the computing time and the richness of the result.

Table 6 -- Choice between fixed volume and sweep depending on the question of the study

If you want to	Use	Where it is dealt with
assess a cistern that already exists	fixed volume	section 4.3
find out which volume to buy	**sweep**	sections 4.4 and 2.2
choose between criteria that disagree	sweep and multi-criteria	sections 4.9 and 2.10

The sweep is almost always the right answer. It costs more computing time, in the order of seconds, and it produces the curve of diminishing returns, which is the strongest argument any study can have when justifying a volume.

1.3.2.3. Is an upper tank needed?

The third question is hydraulic, and the answer depends on the height of the building and on the pressure required at the points of use.

Table 7 -- Hydraulic criteria to decide on the upper tank

If the building	Then	Where it is dealt with
is single-storey and the points are close to the cistern	direct pressurisation, no tank	section 4.5
is a two-storey house or a multi-storey building	elevated tank, for **pressure**	section 4.5
needs a reserve against power cuts	elevated tank	section 4.5

> The elevated tank **does not increase** the water saving. Keeping the same total storage, it delivers slightly less and consumes pumping energy. The decision to adopt it must be hydraulic, and never based on the expectation of a larger yield.

1.3.3. The order of work in the program

The program's sidebar is numbered, and the numbering matches the order in which you work. The first four steps are data input, the next four show results, and the last four produce outputs or support the study (Figure 8).



Figure 8 -- The twelve steps of the program, grouped by nature. The first four are input and must be filled in order, because each one uses what the previous one defined; the next four are for reading results; the last four produce outputs or give support, and can be used at any time.

Where this manual deals with each one: step 1 in section 4.1, step 2 in 4.2, step 3 in 4.6, step 4 in sections 4.3, 4.4 and 4.7, step 5 in 4.3, step 6 in 4.7, step 7 in 4.8, step 8 in 4.9, step 9 in 4.10, step 10 in 3.9 and step 12 in 3.3. Step 11 is the documentation built into the program itself, and has no corresponding section here. In the International Edition, step 12, the Tariff Manager, does not appear, because the Brazilian tariff catalogue does not come with it.

It is worth saving the project at the end of each step. The parameters of appendix A are saved and read back in full, and the resulting file is the documentary evidence of the study, both for whoever will redo it in the future and for whoever will assess it.

1.3.4. Before publishing

Three readings are mandatory for anyone who intends to publish results, and none of them is long. Section 4.10 has the list of what to declare in a paper, together with a method paragraph ready to adapt. Section 5.1 brings together the ten design mistakes that the program has no way of preventing. And section 5.2 lists the known limitations -- of which it is worth reading not all, but those that affect the case in hand.

Appendix I condenses the three into one-page summaries, for quick reference.

2. The method

2.1. The daily water balance

> By the end of this section you will be able to redo, in a spreadsheet, exactly what Netuno does on each day of the series, and to say why the program reached the number it reached.

> **Before you start.** Chapter 2 is the technical layer of the manual: it shows the calculations from the inside. You do not need it to use the program -- chapter 4 is enough -- but you do need it to **defend** a result, whether before an examining board, in a review or in front of a client who asked where the number came from.

This is the section on which all the others depend. The saving potential, the cash flow, the avoided emission and the multi-criteria ranking all come from the numbers the daily balance produces. Whoever understands this section understands the program.

Netuno uses a **daily-step behavioural model**: for each day of the precipitation series, the program works out what enters the tank, what leaves it, what overflows and what stays stored. There is no probability distribution and no annual mean. There is the series, day after day, and the same calculation repeated.

2.1.1. The order of operations

For each day `t`, the program always runs, in this order:

1. **decides whether to divert** the first flush, according to the chosen mode;
2. **converts the depth into collected volume**, with the area and the coefficient;
3. **adds it to what was left from yesterday** and checks the overflow;
4. **supplies the rainwater demand** with what is stored;
5. **splits** the supply between indoor use and irrigation;
6. **tops up with mains water** what the rain did not supply;
7. **classifies** the day as one of full, partial or no supply.

The order is not arbitrary, and swapping two steps changes the result. Item 2.1.4 returns to this.

2.1.2. The worked case

Ten days, one house, no upper tank. The numbers were chosen to fit in a spreadsheet.

Table 8 -- Input data of the ten-day worked case

Input	Value
catchment area	100 m²
runoff coefficient	0.85
first-flush depth	2.0 mm, on every event
total demand	200 L/day
fraction suppliable by rainwater	40% = 80 L/day
lower tank volume	2,000 L
upper tank	none

The rainfall series of the ten days, in millimetres:

Table 9 -- Precipitation series of the ten days of the worked case

Day	1	2	3	4	5	6	7	8	9	10
Rain	0.0	12.4	1.6	0.0	0.0	28.0	5.2	0.0	41.0	0.0

Without outdoor demand, the daily rainwater demand is the indoor demand times the indoor non-potable fraction, which here coincides with the total suppliable fraction: 200 x 0.40 = **80 L/day**. That is the volume the system tries to deliver every day.

2.1.2.1. The table, day by day

Table 10 -- Daily water balance of the worked case, day by day, with collection, consumption, overflow and storage

Day	Rain	First flush	Effective rain	Collected	Rainwater demand	Consumed	Overflowed	In the tank	Supply
	mm	mm	mm	L	L	L	L	L	
1	0.0	0.0	0.0	0	80	0	0	0	none
2	12.4	2.0	10.4	884	80	80	0	804	full
3	1.6	1.6	0.0	0	80	80	0	724	full
4	0.0	0.0	0.0	0	80	80	0	644	full
5	0.0	0.0	0.0	0	80	80	0	564	full
6	28.0	2.0	26.0	2,210	80	80	774	1,920	full
7	5.2	2.0	3.2	272	80	80	192	1,920	full
8	0.0	0.0	0.0	0	80	80	0	1,840	full
9	41.0	2.0	39.0	3,315	80	80	3,155	1,920	full
10	0.0	0.0	0.0	0	80	80	0	1,840	full
sum				**6,681**		**720**	**4,121**	1,840	

Table 11 -- Check of the mass balance closure over the ten days

Closure check	Litres
collected in the period	6,681
minus the overflow	-4,121
minus the consumption	-720
minus what stayed in the tank	-1,840
residual	**0**

Table 12 -- Indicators of the ten-day worked case

Ten-day indicator	Value
potable water saving potential	36.0%
daily mean of rainwater consumed	72.0 L/day
daily mean of potable water consumed	128.0 L/day
daily mean overflow	412.1 L/day
days of full supply	90%
days of partial supply	0%
days of no supply	10%

2.1.3. Reading the table

Five things happen in these ten days, and each one teaches a rule of the program.

2.1.3.1. Day 1: empty tank

The simulation starts with the tank **empty**. There is no rain on day 1, nothing is stored, and the 80 L of rainwater demand are not met by a single drop: the day is one of **no** supply, and the 200 L of the whole day come from the mains.

This is not a defect of the chosen series. It is the initial condition of the model, and it matters: in short series, the first days penalise the result. With ten years of series the effect disappears; with ten days, it accounts for the whole gap between the potential obtained, 36.0%, and the suppliable fraction of 40%.

2.1.3.2. Day 3: the rain that is lost entirely

It rained 1.6 mm. The first-flush depth is 2.0 mm. The program does not subtract 1.6 and collect zero: it **diverts the whole day**.

> **Warning.** When the first-flush rule applies and the day's rain does not exceed the depth, the **whole** day is lost, not the difference. The calculation is `P_ef = P - d_f` when `P > d_f`, and **zero** otherwise.

The reason is physical: the diversion device has to fill before the water reaches the cistern. If the rain does not fill the device, nothing passes. In climates of light, frequent rain, this weighs far more than intuition suggests, and that is why the first-flush mode deserves attention in section 4.1.

> **Difference from Netuno 4.** Netuno 4 treats `d_f` as an **event threshold**: below it nothing is collected, above it the **whole** rain is collected, without subtraction. Netuno 5 subtracts the depth. The difference is deliberate and is quantified in section 6.1: Netuno 5 estimates, on average, 2.3 to 2.7 percentage points less saving over 4,050 reference cases, and **half** of that difference comes from here. Studies comparing the two versions must declare it.

2.1.3.3. Day 6: overflow comes before consumption

It rained 28 mm. After the 2 mm first flush, 26 mm of effective rain remain, which on 100 m² at a coefficient of 0.85 give 2,210 L collected.

The tank held 564 L. Added together, 2,774 L. The tank holds 2,000 L, so **774 L overflow**. Only after that does the program withdraw the 80 L of demand, leaving 1,920 L for the next day.

> **Warning.** This order defines the model. The overflow is calculated **before** the demand is withdrawn, as in the **YAS** rule (*yield after spillage*), but the day's own rain can already supply the day's demand, as in the **YBS** rule. Netuno therefore sits between the two, and the subsection «YAS, YBS and Netuno's order», further on, shows why.
>
> If consumption came before the overflow, the tank would accept water that would physically have overflowed already, and the program would overestimate the yield. The difference shows up precisely on days of heavy rain, which are those that weigh most in the annual total.

2.1.3.4. Day 9: a tank too small for the event

It rained 41 mm, and the system collected 3,315 L. The tank held 1,840 L and holds 2,000. **3,155 L overflowed** -- more than the whole tank can hold.

In a single day, the system threw away the equivalent of thirty-nine days of rainwater demand. It is the portrait of the mismatch between the rain, which arrives concentrated, and the demand, which is constant. Enlarging the tank recovers part of it, and that is exactly the question the volume sweep of section 2.2 answers. Figure 9 sums up the ten days.

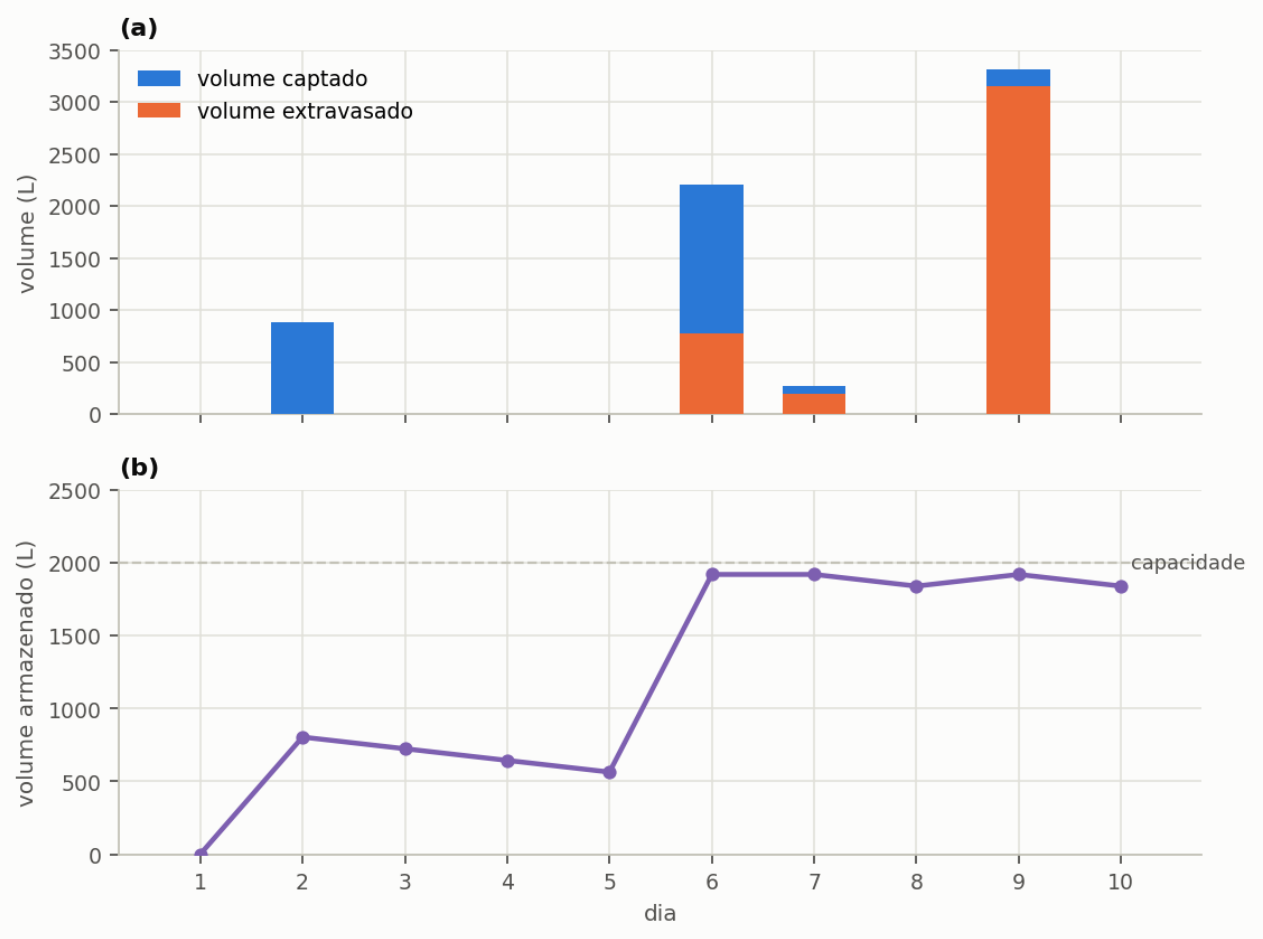


Figure 9 -- The ten days of the worked case: (a) collected volume, in blue, and overflowed volume, in orange, by day -- on day 9 the overflow almost equals the collection; (b) volume stored in the lower tank at the end of each day, after consumption, with the 2,000 L capacity dashed; on days 6, 7 and 9 the tank reaches capacity, overflows and ends the day with 1,920 L.

2.1.3.5. The sum: the balance closes

The last row of the table and the check table show the closure:

$$V_{\text{cap}} = V_{\text{ext}} + V_{\text{con}} + V_{\text{res}} \tag{1}$$

Table 13 -- Variables of Equation 1

Symbol	Meaning	Unit
`V_cap`	total volume collected in the period	L
`V_ext`	total volume overflowed in the period	L
`V_con`	total rainwater volume consumed in the period	L
`V_res`	volume left in the tank at the end of the period	L

In the ten days of the worked case, Equation 1 holds with 6,681 L collected, 4,121 L overflowed, 720 L consumed and 1,840 L left.

Zero residual. The program neither creates nor loses water, and this check is made automatically by the test suite at every change of the code, on ten-year series, with a residual of the order of 10-¹■ per cent of the collection.

2.1.4. The equations

The daily balance is written in five chained equations, which the program solves in this order for each day of the series. The notation follows that of section 4.1 for the collection part, and the index `t` always designates the current day of the simulation, so that `t-1` is the day immediately before.

2.1.4.1. The collected volume

Converting the precipitated depth into volume is the first operation of the day, and the simplest of all: the effective rain is multiplied by the catchment area and by the surface runoff coefficient.

$$V_{\mathrm{cap,\,t}} = P_{\mathrm{ef,\,t}} \times A \times C \tag{2}$$

Table 14 -- Variables of Equation 2

Symbol	Meaning	Unit
`V_cap,t`	volume collected on day `t`	L
`P_ef,t`	effective rain on day `t`, after deducting the first-flush depth	mm
`A`	catchment area, in horizontal projection	m²
`C`	surface runoff coefficient of the roof	dimensionless

The result of Equation 2 comes out directly in litres, with no conversion factor, because one millimetre of depth over one square metre of surface is exactly one litre of water. It is the only dimensional simplification of the model, and it is exact.

2.1.4.2. The overflow and the storage

The volume available before any consumption is the sum of what was left in the tank at the end of the previous day and what was collected today, as in Equation 3. Two parts follow from that sum, calculated in this order: the surplus that does not fit in the tank and therefore overflows, and the volume actually stored.

$$D_{\mathrm{t}} = V_{\mathrm{res,\,t-1}} + V_{\mathrm{cap,\,t}} \tag{3}$$

$$V_{\mathrm{ext,\,t}} = \max\left(0;\, D_{\mathrm{t}} - V_{\max}\right) \tag{4}$$

$$V_{\mathrm{arm,\,t}} = \min\left(D_{\mathrm{t}};\, V_{\max}\right) \tag{5}$$

Table 15 -- Variables of Equation 5

Symbol	Meaning	Unit
`D_t`	volume available on day `t`, before consumption	L
`V_res,t-1`	volume left in the tank at the end of the previous day	L
`V_cap,t`	volume collected on day `t`, from Equation 2	L
`V_ext,t`	volume overflowed on day `t`	L
`V_arm,t`	volume stored on day `t`, before consumption	L
`V_max`	useful capacity of the lower tank	L

The order in which Equations 4 and 5 appear defines the model. The overflow is calculated **before** the day's consumption is withdrawn, and that is why the program is usually described by the YAS rule, from *yield after spillage*. The description is approximate, and the next subsection says where it falls short. Reversing the order would make the tank accept, as consumption, water that would physically have overflowed already, and would overestimate the yield precisely on days of intense rain, which are those that weigh most in the annual total.

The difference between the two orders is invisible on most days and decisive on the few that matter. While the tank does not fill, overflowing before or after consuming gives exactly the same result, because there is no overflow at all; on days of intense rain, when the tank reaches capacity, the order determines whether a part of the water is counted as consumed or as lost. Those are precisely the days that weigh most in the annual total, and that is why a choice that looks like a formality shifts the whole result. Figure 10 compares the two sequences in the notation of the equations of this section.

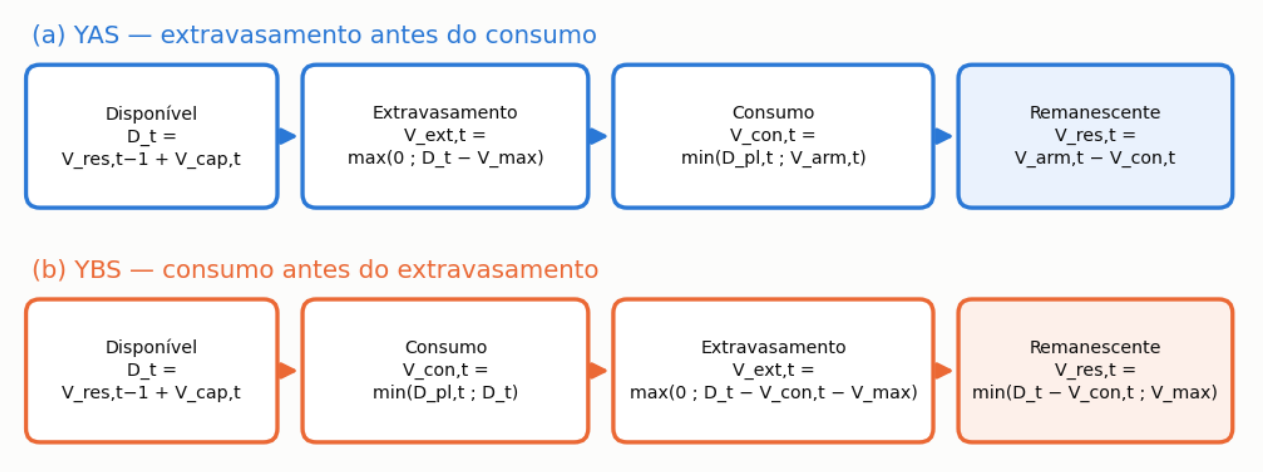


Figure 10 -- Sequence of operations of the daily balance: (a) the order adopted by the program, with the overflow calculated before consumption; (b) the reversed order (YBS), with consumption withdrawn before the overflow. The two sequences only diverge on days when the tank reaches capacity, and YBS assumes a consumption before the day's rain, which overestimates the yield.

2.1.4.3. YAS, YBS and Netuno's order

In the literature, YAS and YBS have a precise definition (Fewkes and Butler, 2000). With `Y_t` the volume withdrawn in the day, `Q_t` the collected volume, `S_t` the volume stored at the end of the day and `V_max` the capacity:

Table 16 -- Withdrawal of the day and volume stored at the end of the day under strict YAS, strict YBS and in Netuno

Rule	Withdrawal of the day	Stored at the end of the day
strict YAS	`Y_t = min(D_pl,t ; S_t-1)`	`S_t = min(S_t-1 + Q_t - Y_t ; V_max - Y_t)`
strict YBS	`Y_t = min(D_pl,t ; S_t-1 + Q_t)`	`S_t = min(S_t-1 + Q_t - Y_t ; V_max)`
Netuno	`Y_t = min(D_pl,t ; min(S_t-1 + Q_t ; V_max))`	`S_t = min(S_t-1 + Q_t ; V_max) - Y_t`

Under strict YAS, the day's withdrawal comes only from what was stored the day before: today's rain only serves tomorrow. Under strict YBS, today's rain supplies the demand before the tank overflows. Netuno does the overflow first, as YAS does, and then lets the day's rain, up to the capacity, supply the day's demand.

An inequality follows that holds for any series: the volume consumed by Netuno is never smaller than that of strict YAS nor larger than that of strict YBS. The result lies **between the two**, closer to YAS for large tanks, where the previous day almost always suffices, and further from it for small ones, where the day's rain weighs. Anyone comparing Netuno with a program or a standard that uses strict YAS, such as the detailed method of EN

16941-1:2024 (Annex A), should expect from Netuno a slightly higher saving potential for the same volume.

The program keeps this order by decision: it reproduces Netuno 4 and the studies of the research group that used it, and a change here would shift every published result.

2.1.4.4. Supplying the demand

Once it is established how much is in the tank, the program withdraws the smaller of the day's rainwater demand and the available volume, and passes on to the next day what is left.

$$V_{\mathrm{con,\,t}} = \min\left(D_{\mathrm{pl,\,t}}; V_{\mathrm{arm,\,t}}\right) \tag{6}$$

$$V_{\mathrm{res,\,t}} = V_{\mathrm{arm,\,t}} - V_{\mathrm{con,\,t}} \tag{7}$$

Table 17 -- Variables of Equation 7

Symbol	Meaning	Unit
`V_con,t`	rainwater volume consumed on day `t`	L
`D_pl,t`	rainwater demand of day `t`, built as in section 4.2	L
`V_arm,t`	volume stored on day `t`, from Equation 5	L
`V_res,t`	remaining volume, carried to the next day	L

Equation 7 is what makes the daily simulation necessary and irreplaceable: it links each day to the next and makes the tank a memory of what happened before. No calculation based on monthly or annual means can represent this chaining, and the whole mismatch between the concentrated supply and the spread-out demand follows from it.

2.1.4.5. The mains top-up

What the rain did not supply is bought from the utility, and that part brings together three different origins.

$$V_{\mathrm{pot,\,t}} = D_{\mathrm{pot,\,t}} + \Delta D_{\mathrm{int,\,t}} + \Delta D_{\mathrm{irr,\,t}} \tag{8}$$

Table 18 -- Variables of Equation 8

Symbol	Meaning	Unit
`V_pot,t`	volume of potable water consumed from the mains on day `t`	L
`D_pot,t`	demand that requires potable water by the nature of the use	L
`deltaD_int,t`	deficit of the indoor non-potable use, not met by the rain	L
`deltaD_irr,t`	irrigation deficit, **only** if the backup supply is on	L

The third part of Equation 8 deserves attention because it depends on a user choice. With the backup supply on, the mains comes to the rescue of irrigation on days when the cistern runs dry, and the deficit is met with potable water; with it off, the garden is simply not watered, the building does not buy that water, and the corresponding volume is counted as **unmet demand**, which is the third part of the demand balance dealt with later in this section.

2.1.4.6. Classifying the day

Finally, each day is classified by the degree of supply achieved, and it is from this classification that the percentages of days of full, partial and no supply reported in section 2.3 are calculated.

$$\mathrm{pleno}: V_{\mathrm{con,\,t}} = D_{\mathrm{pl,\,t}}; \mathrm{parcial}: 0 < V_{\mathrm{con,\,t}} < D_{\mathrm{pl,\,t}}; \mathrm{nulo}: V_{\mathrm{con,\,t}} = 0 \tag{9}$$

Table 19 -- Variables of Equation 9

Symbol	Meaning	Unit
`V_con,t`	rainwater volume consumed on day `t`	L
`D_pl,t`	rainwater demand of day `t`	L

In the equation, «pleno», «parcial» and «nulo» are the program's labels for full, partial and no supply. The comparison of Equation 9 is made with a tolerance of 10-■ L, not by exact equality. The reason is arithmetic, not conceptual: the balance works with floating-point numbers, and a difference of the order of the rounding error would classify as partial a day that was fully supplied.

2.1.5. The demand balance has three parts

The total demand of a day splits into three, not two:

$$D_{\mathrm{t}} = V_{\mathrm{con,\,t}} + V_{\mathrm{pot,\,t}} + V_{\mathrm{nao,\,t}} \tag{10}$$

Table 20 -- Variables of Equation 10

Symbol	Meaning	Unit
`D_t`	total demand of the building on day `t`	L
`V_con,t`	part supplied with rainwater	L
`V_pot,t`	part bought from the mains	L
`V_nao,t`	part nobody supplied, that is, unmet demand	L

The three parts are the rainwater consumption, the mains consumption and the **unmet demand**. The third is zero in the default configuration, in which the mains comes to the rescue of irrigation on dry days. It stops being zero when the user switches that rescue off, and then the program must show it: without it, the balance would close by construction, hiding the water nobody supplied.

2.1.6. With an upper tank

When the project has an upper water tank, step 4 changes. Instead of consuming directly from the lower tank, the program:

1. consumes from the upper tank what lies **above the trigger level**;
2. if something is still missing and there is water below, **starts the pump**: it transfers the smaller of what is in the lower tank and the free space above, and counts **one cycle**;
3. consumes again from what has just arrived;
4. repeats until the demand is met or the water runs out.

The trigger level is the reserve volume that is not consumed, and it serves to keep the pump from starting for every litre. It is capped at the volume of the upper tank: entering a trigger level larger than the tank does not make the program refuse it, it makes it adopt the full tank.

> **The upper tank does not increase the water saving.** It changes **when** the water is delivered, not **how much**. What it adds is cost: the investment in the tank and the pump, the pumping energy and the associated emission. Section 4.5 shows this with numbers.

The number of cycles and the volume pumped come from here and feed the energy calculation of section 2.6.

2.1.7. What the model does not represent

- **There is no hourly sub-step.** A 40 mm rain concentrated in two hours and the same depth spread over twenty-four hours produce the same result. For tanks very small relative to the catchment area, this underestimates the overflow.
- **There is no loss by evaporation or leakage** in the tank.
- **The initial condition is an empty tank.** In series shorter than one year, this penalises the result noticeably.
- **The runoff coefficient is single and constant.** It does not vary with the intensity of the rain or with the time since the last event.
- **In the balance, the first flush is a depth, not a device with its own dynamics.** The first-flush chamber is sized and priced by the volume `A x d_f` (section 2.5), but its emptying between events is not simulated: the depth is applied according to the chosen first-flush mode.

Section 5.2 brings together these and the other known limitations.

2.1.8. To check

The worked case of this section is in the file `03_dez_dias.netuno`, in the cases folder that comes with the program. Open it, run it, and compare with the printed table.

If any number differs, it is the manual that is out of date relative to the program. And in that case the automatic check will already have warned, because every number in this table has a corresponding assertion in the test suite described in section 5.3.

2.2. Ideal volume and the sweep

> This section formalises what section 4.4 did in practice: how the program builds the grid of volumes and how it applies the ideal-volume criterion.

2.2.1. The discrete grid

The sweep simulates a **finite** set of volumes, and building that set follows three rules.

The step is rounded up to a multiple of 250 litres. It is the commercial quantum: tanks are sold in standard volumes, and an optimum at 1,300 L cannot be bought. When the step asked for is not a multiple of 250, the program adjusts it and records the adjustment in `step_adjusted`, so that the report can declare the effective step.

The number of positive volumes is limited to 40. A sweep from 0 to 50,000 L with a 100 L step would be 500 thirty-year simulations, and the gain in information does not pay the cost. When the limit is reached, the effective step grows.

The maximum volume entered is always kept as the last point of the grid. Only the last interval can therefore be smaller than the effective step.

Volume **zero** enters the grid as the reference alternative: it is the no-system scenario, against which the others are compared.

The practical consequence appears in Figure 11. With a fine step, the curve shows the knee where the return starts to fall off; with a coarse step, the knee falls between two points of the grid and the criterion picks the first tabulated volume after it, which may lie several cubic metres beyond the volume that really matters.

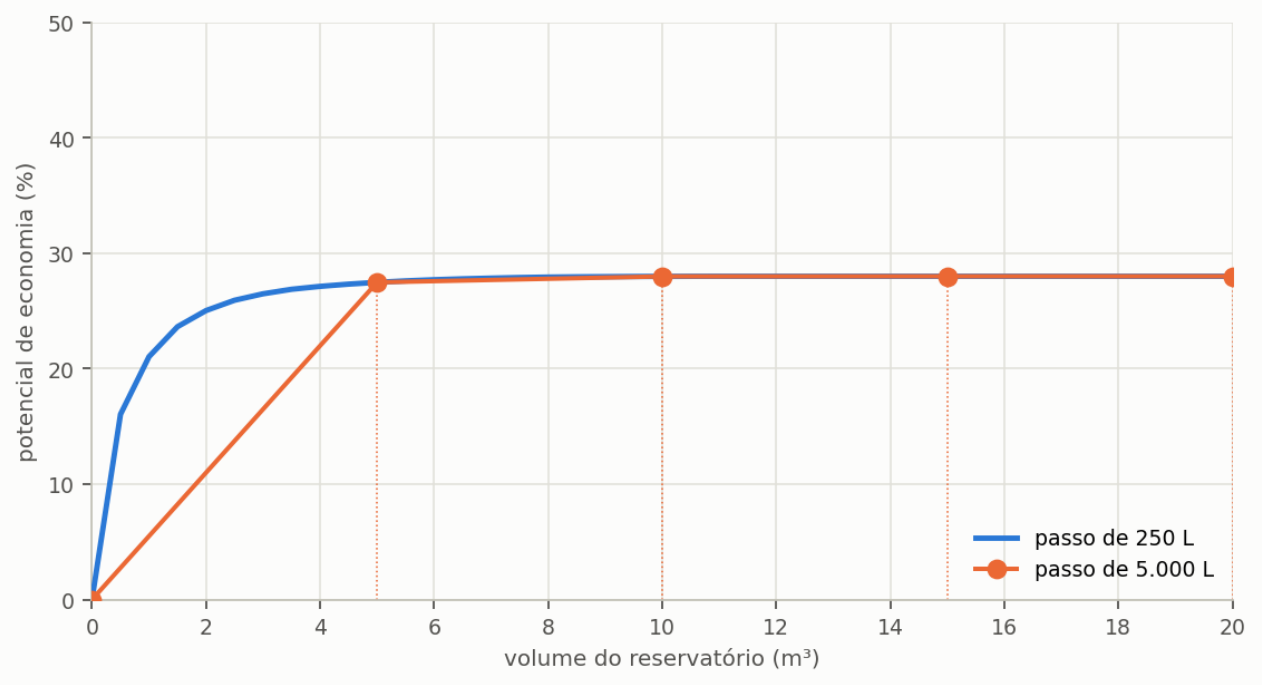


Figure 11 -- Saving potential curve of the same building, swept with a 250 L step and with a 5,000 L step. The coarse grid calculates each point it contains correctly, but it does not contain the stretch of greatest curvature, between 1 and 5 m³.

2.2.2. The marginal gain

The curve of saving potential against volume is monotonic and concave: each cubic metre added to the tank yields less than the previous one, and the yield tends to zero as the system approaches the ceiling set by the suppliable fraction. The program quantifies this diminishing yield by the discrete derivative of the curve, calculated for each volume of the grid from the second one on.

$$\Delta(V_i) = [\mathrm{Epot}(V_i) - \mathrm{Epot}(V_{i-1})] \div [(V_i - V_{i-1}) \div 1000] \tag{11}$$

Table 21 -- Variables of Equation 11

Symbol	Meaning	Unit
`delta(V_i)`	marginal gain of volume `V_i` relative to the previous one	%/m³
`Epot(V_i)`	potable water saving potential obtained with volume `V_i`	%
`Epot(V_i-1)`	saving potential obtained with the next smaller volume	%
`V_i`	tank volume at position `i` of the grid	L
`V_i-1`	next smaller volume of the grid	L
`1000`	conversion from litres to cubic metres	L/m³

The division by one thousand in the denominator of Equation 11 is there so that the result comes out in percentage points per added cubic metre, which is the unit in which the decision is made -- tanks are bought and priced in cubic metres, not in litres.

Being a discrete derivative, the marginal gain depends on the grid step. A fine grid produces a more local value, close to the true derivative at that point; a coarse grid produces the mean yield over a larger interval. The subsection on sensitivity to the step, further on, measures the size of that effect.

2.2.3. The ideal-volume criterion

The ideal volume is defined as the smallest positive volume of the grid whose marginal gain has already fallen below a threshold declared by the user.

$$V_{\mathrm{ideal}} = \min \left\{ V_i > 0 : \Delta(V_i) \leq \Delta\mathrm{lim} \right\} \tag{12}$$

Table 22 -- Variables of Equation 12

Symbol	Meaning	Unit
`V_ideal`	ideal volume by the marginal-gain criterion	L
`V_i`	tank volume at position `i` of the grid	L
`delta(V_i)`	marginal gain of volume `V_i`, from Equation 11	%/m³
`delta_lim`	marginal-gain threshold declared by the user; the program opens with 5.0, and the tutorial adopts 1.0	%/m³

When no volume of the grid meets the condition of Equation 12 -- which happens when the swept range ends before the curve flattens -- the program adopts the largest volume of the grid and records that circumstance in the report, so that the reader knows that the optimum found is a limit of the search and not a turning point of the curve. The Netuno 4 manual deals with the same case in its seventh chapter, and the conduct adopted here is the same.

2.2.3.1. An example, step by step

The sweep table of the tutorial, in section 4.4, lets you redo the calculation by hand. Between 3,000 L and 4,000 L the potential rises from 26.48% to 27.12%, that is, 0.64 percentage points for one more cubic metre. Since 0.64 is below the 1.0 %/m³ threshold, and the previous gain, between 2,000 L and 3,000 L, was 1.45 %/m³, the first volume to cross the threshold is 4,000 L, and it is the ideal volume.

Notice what happens right after. Between 6,000 L and 8,000 L the potential rises by 0.23 points, the same as in the previous interval, from 5,000 to 6,000 L, and yet the gain **per cubic metre** halves, from 0.23 to 0.11 %/m³, because the interval spans two cubic metres. That is why the criterion uses the gain per cubic metre, and not the gain between consecutive points of the grid: the latter depends on the step, and the former much less.

> **The criterion is hydraulic and only hydraulic.** It ignores cost, tariff and carbon. An ideal volume by this criterion is not a design recommendation; it is the point at which the yield curve stops paying off **in water**. Section 2.10 deals with the choice with the other criteria, which often point to different volumes.

This is one criterion among others equally defensible, and it is worth making that explicit. A designer who wants maximum water performance will choose the largest volume of the sweep; one answering for a budget will choose the volume with the largest net present value; the marginal-gain criterion lies between the two, as the one that identifies where the curve stops paying off. The three read the same sweep and need not coincide: in this project they point to three different volumes, and the maximum-performance one is the largest of the three. None of the three is wrong -- what is wrong is not saying which one was used. Figure 12 marks the volume chosen by each criterion, with the value noted in the chart legend.

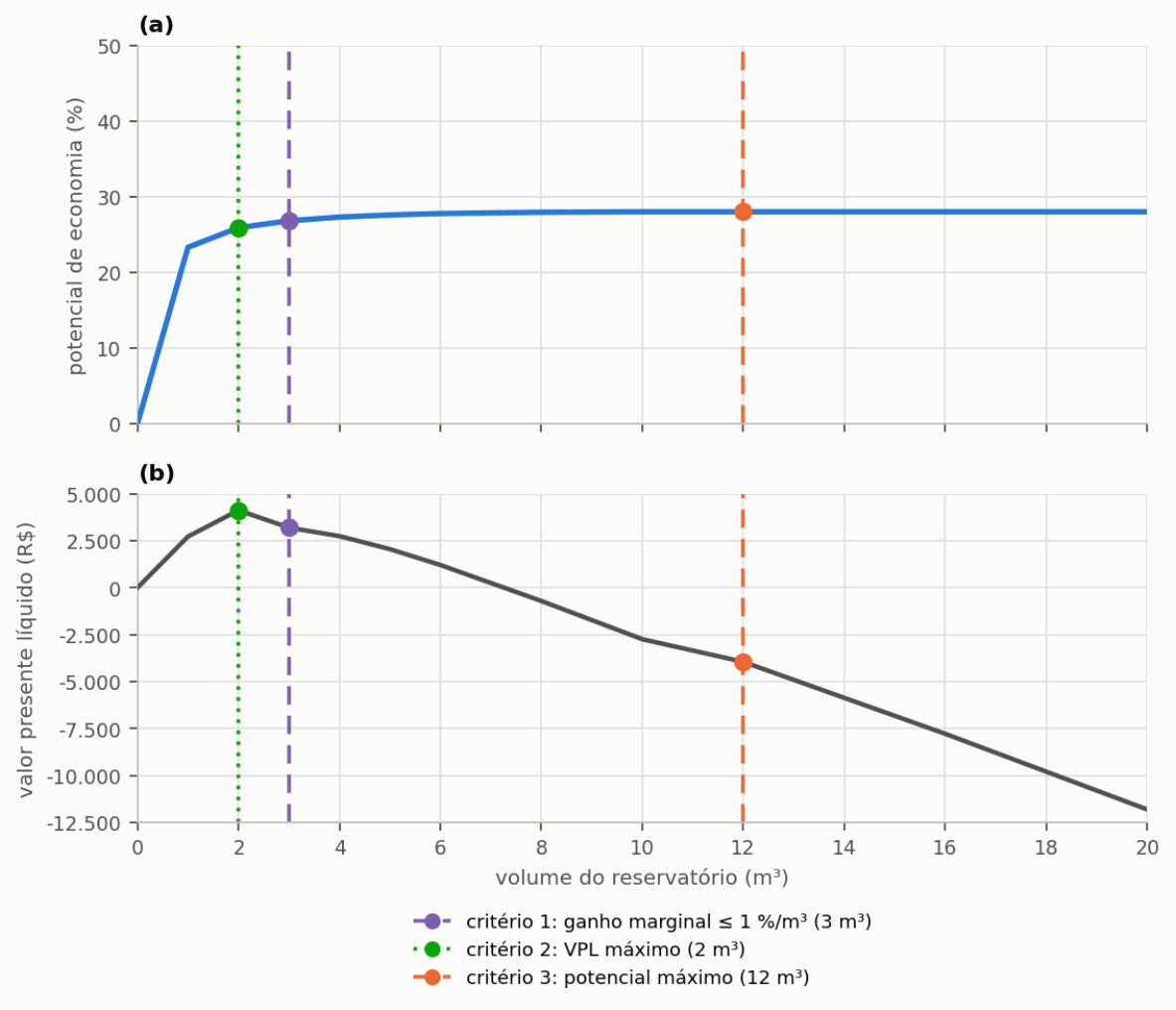


Figure 12 -- Sweep of the tutorial project read by three criteria: (a) saving potential and (b) net present value of the same set of volumes. The vertical lines mark the volume chosen by marginal gain below 1 %/m³, by maximum NPV and by maximum potential; the volume of each is noted in the legend, because it depends on the swept grid and on the economic assumptions in force.

2.2.4. Why the program simulates: the standard's empirical methods

The first edition of the Brazilian rainwater harvesting standard, ABNT NBR 15527:2007, had in its informative Annex A six methods to size the tank: Rippl's, simulation, Azevedo Neto's and three practical methods, the German, the English and the Australian. They are still in common use in projects and academic work, and it is worth understanding what each one answers before comparing them with the program's sweep. The European standard EN 16941-1:2024 has a simplified method of the same kind -- a number of days of the smaller of annual yield and demand -- next to the detailed daily simulation.

Rippl's method is the mass diagram applied to the tank: it accumulates, period by period, the difference between demand and usable rain, and adopts as the volume the largest cumulative deficit of the series. The resulting volume is the one that would meet **all** the demand even in the worst dry spell on record, and so it grows with the length of the series and with its resolution: with daily data, dry sequences appear whole; with monthly data, part of them is diluted in the monthly mean.

Azevedo Neto's method multiplies the mean annual precipitation `P`, in millimetres, the collection area `A`, in square metres, and the number of months of little rain `T` by the coefficient 0.042, and gives the volume in litres. The result depends entirely on `T`, and in cities without a defined dry season, such as Florianópolis, the choice of `T` is arbitrary.

The practical methods are rules of thumb. The German one takes 6% of the smaller of annual usable rain and annual demand; the English one takes 5% of the product `P x A`; the Australian one does a monthly balance by trial

until the desired confidence is reached. Neither of the first two uses the distribution of rain over time, and the English one does not even use the demand.

The table below applies five of these methods to the tutorial project, rounds each volume to a multiple of 250 L and simulates, in the program itself, the potential each one produces.

Table 23 -- Volumes given by the methods of Annex A of ABNT NBR 15527:2007 for the tutorial project, rounded to 250 L, and the potential each one yields in the daily simulation

Method	Rule	Volume	Simulated potential
German practical	6% of the smaller of annual supply and demand	3,750 L	27.00%
Rippl, monthly series	largest cumulative deficit, months	4,500 L	27.32%
Azevedo Neto, T = 1 month	0.042 x P x A x T	11,250 L	28.00%
Rippl, daily series	largest cumulative deficit, days	7,000 L	27.83%
English practical	0.05 x P x A	13,500 L	28.00%
daily simulation, marginal gain of 1 %/m³	sweep	**4,000 L**	**27.12%**

Table 24 -- Storage fraction of the sweep volumes and the simulation time step Fewkes and Butler (2000) recommend for each range

Volume	Storage fraction S/(A.R)	Time step recommended by Fewkes and Butler
1,000 L	0.0037	hourly
2,000 L	0.0074	hourly
3,000 L	0.0111	daily
4,000 L	0.0149	daily
8,000 L	0.0297	daily
12,000 L	0.0446	daily
20,000 L	0.0743	daily

The first table shows what the concave curve of section 4.4 already announced. The methods that size for the worst dry spell, such as Rippl with a daily series, and those that ignore demand, such as the English one, lead to tanks of 7 to 14 m³, which deliver less than one percentage point more than the 4,000 L of the simulation at two to three times the volume. The German method and Rippl with a monthly series get the order of magnitude right in this case, but by coincidence: the German method has no way of knowing that the Florianópolis rain is well distributed, and would give the same volume for a city with six dry months.

The essential difference is that the empirical methods answer a fixed question, and the simulation answers the question the designer declares. Rippl asks which volume never fails; the English method asks what fraction of the rain to store; the sweep with marginal gain asks from which volume on the next cubic metre stops paying off. Ghisi (2010) showed, for three cities in the state of São Paulo, that the right volume varies greatly with local rain, roof area, number of residents and demands, and concluded that sizing tanks by local tradition, without simulation, can lead to low-efficiency systems.

2.2.5. The time step and the storage fraction

The program simulates day by day, and that choice has a known range of validity. Fewkes and Butler (2000) compared behavioural models with steps of one minute, one hour, one day and one month, took the hourly model with the YAS rule (section 2.1) as the reference, and showed that the right step depends on a dimensionless quantity, the **storage fraction**.

$$F_{\mathrm{S}} = S \div (A \times R) \tag{13}$$

Table 25 -- Variables of Equation 13

Symbol	Meaning	Unit
F_S	storage fraction	dimensionless
S	tank capacity	m³
A	catchment area in horizontal projection	m²
R	mean annual precipitation	m

According to those authors' results, the hourly model is the one indicated for fractions up to 0.01; the daily one, for fractions between 0.01 and 0.125; and the monthly one, only above 0.125. The physical reason is simple: the smaller the tank relative to the rain it receives, the more it fills and overflows **within** a single rainy day, while the demand is withdrawn over the hours; a daily step adds up in a single inflow what, hour by hour, would have come in and gone out little by little.

The second table calculates the fraction for the volumes of the tutorial sweep. The chosen 4,000 L cistern has a fraction of 0.015, within the range of the daily model; the 1,000 and 2,000 L tanks fall below 0.01, and for them the daily result should be read as an approximation. Campisano and Modica (2015), who simulated a residential system with daily and sub-daily steps from measured discharges, reached a complementary conclusion: the daily step is reliable for estimating **water saving**, but steps of at most one hour are needed when the objective is **stormwater retention**, above all in small tanks with high demand.

> **In practice.** For potable water saving in buildings, the program's daily step is adequate in the usual design range. If the storage fraction of your study falls below 0.01, declare the limitation; and if the objective is to attenuate peak flows in urban drainage, the program is not the right tool.

2.2.6. What the literature says about the ideal volume

There is no single definition of ideal volume, and the systematic review by Semaan et al. (2020) helps to place the program's criterion. Examining optimisation studies of domestic systems, the authors observed that sizing the tank is the most important optimisation objective, but that cost is the most used criterion; that undersizing produces systems unable to supply a reliable volume; and that oversizing raises the investment for a small marginal benefit and may even compromise water quality, as the water is stored for longer.

Campisano et al. (2017), in a broad review of urban rainwater harvesting systems, bring together criteria from three families: **water efficiency**, such as saving potential and reliability; **economic**, such as net present value and payback; and **drainage performance**, such as retention. The program covers the first two families and declares that it does not cover the third.

The marginal-gain criterion belongs to the first family and has an advantage over fixed reliability, as in Rippl's: it does not require the designer to choose an acceptable failure level, which rarely has a technical justification, but a level of diminishing return, which is read directly in cubic metres of tank. The disadvantage is symmetrical: the 1 %/m³ threshold, or the 5 %/m³ one, is also a choice, and must be declared.

2.2.7. Sensitivity to the step

The ideal volume depends on the sweep step, and it is worth knowing by how much.

A coarse step jumps over the knee of the curve: with a 5,000 L step, the first point assessed after zero already has the mean marginal gain of the whole interval, which is high, and the second may already be below the threshold. The ideal is shifted upwards.

A fine step resolves the knee better, at the cost of time.

Practical recommendation: a step between 500 and 1,000 L for residential tanks, and between 2,000 and 5,000 L for large buildings, always declaring the step adopted together with the ideal volume.

2.2.8. Why the curve has this shape

The rainwater yield curve is concave and asymptotic, always. Two reasons:

Small events are the majority. The first litres of storage capture the frequent, small rains, which account for most of the days. The last litres only serve the few extreme events.

The suppliable fraction is a ceiling. The potential cannot exceed the fraction of the demand that rain supplies, defined in section 4.2. The curve approaches that ceiling and never crosses it -- an invariant the test suite checks up to volumes of 100 m³.

2.2.9. Is the series long enough?

Every result of this section depends on the rainfall series loaded, and the question the reviewer asks first is whether it is enough. Geraldi and Ghisi answered it in three papers: sub-series are extracted from a long series, each one is simulated, and it is counted how often they reproduce the result of the whole series. The representative length is the shortest that reproduces the result in most windows, and that **keeps reproducing** it at longer lengths -- an isolated hit in a short sub-series is not worth a recommendation.

The program has this analysis in the sweep step, in the «is the rainfall series long enough?» button. It uses the same grid of volumes and the same marginal-gain threshold as the sweep, because the answer depends on them, and it returns the verdict in one sentence, with the frequency table by length.

The answer belongs to this project, not to the city. It is the point the papers do not isolate and that the analysis makes visible: the series needed grows with the suppliable fraction. In the four thirty-year series that come with the program, with a 100 m² roof, four residents and 150 L per person per day:

Table 26 -- **Table 99** -- Representative series length, in years, depending on the suppliable fraction

Suppliable fraction	Florianópolis	São Paulo	Brasília	Manaus
20%	2	8	9	16
30%	8	18	14	20
40%	9	9	20	30
50%	24	25	29	30

With 20% of the demand met by rain, two years of series are enough in Florianópolis; with 50%, 24 are needed. The reason is direct: the larger the share to be covered, the more the cistern depends on rare dry spells, and the longer the series that contains them must be. A short series does not get the mean potential wrong -- it gets the volume wrong, which is what is bought.

> **Warning: a short series does not become a long one by averaging.** If the analysis says the project needs more years than the series has, the ideal volume calculated is provisional, and that is what is declared in the work. Repeating the simulation with another window of the same data does not solve it: what is missing is record, not processing. EN 16941-1:2024 asks for at least five years of daily data for its detailed method; Netuno keeps the Geraldi and Ghisi analysis, which answers for each project instead of fixing a minimum.

2.2.10. What the sweep does not answer

- **Which volume to buy.** Tanks exist in commercial volumes; the ideal calculated must be rounded to the nearest available volume, and the sweep already helps by working in multiples of 250 L.
- **Whether the system is worth it.** That is section 2.7.
- **Whether the volume fits on the plot.** The program does not know the site.
- **How much the cistern attenuates the drainage.** The daily step does not represent peak flows, and retention needs an hourly step or less, as Campisano and Modica (2015) show.

2.3. Efficiency indicators

> This section formally defines each indicator the program reports, with its unit, and says **what each one does not measure**.

A misread indicator is worse than none, and those of this section are often confused in the literature. The first two start from the same numerator -- the rainwater actually consumed -- and differ only in the denominator: a small arithmetic difference and a large difference in meaning, because each denominator asks a different question. The third is not a ratio between volumes but a count of days, and it answers a question neither of the other two reaches: how often the system can be counted on (Figure 13).

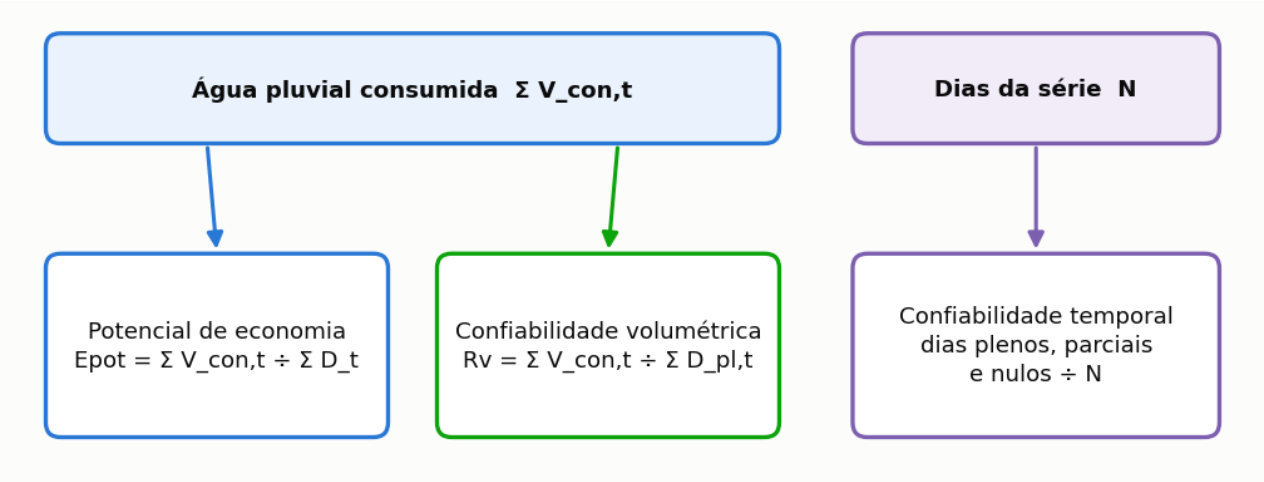


Figure 13 -- The three efficiency indicators and what each one counts. The saving potential and the volumetric reliability share the numerator and are related by the identity Epot = Rv x f, where f is the suppliable fraction; the time reliability counts days, not litres.

2.3.1. Potable water saving potential

The potable water saving potential is the program's central indicator and the one that appears in the summary of any study. It expresses the fraction of the **total** consumption of the building that rainwater managed to replace over the whole simulated series, not only of the part rain could supply.

$$\mathrm{Epot} = \left(\sum V_{\mathrm{con,\,t}} \div \sum D_{\mathrm{t}}\right) \times 100 \tag{14}$$

Table 27 -- Variables of Equation 14

Symbol	Meaning	Unit
`Epot`	potable water saving potential	%
`V_con,t`	rainwater volume consumed on day `t`	L
`D_t`	total demand of the building on day `t`	L
`soma`	sum over all the days of the simulated series	--

The denominator of Equation 14 is the **total** demand, and that choice is the origin of the most common confusion when reading results. A system that fully met all the demand assigned to it, without failing on a single day, would still show a saving potential equal to the declared suppliable fraction, not one hundred per cent. The ceiling, therefore, is not a limitation of the system: it is an arithmetic consequence of the definition of the indicator.

2.3.2. Volumetric reliability

Volumetric reliability answers a different, complementary question: of the water the system **should** supply, how much did it actually supply? The denominator here is the rainwater demand, not the total demand.

$$\mathrm{Rv} = \left(\sum V_{\mathrm{con,\,t}} \div \sum D_{\mathrm{pl,\,t}}\right) \times 100 \tag{15}$$

Table 28 -- Variables of Equation 15

Symbol	Meaning	Unit
`Rv`	volumetric reliability of the system	%
`V_con,t`	rainwater volume consumed on day `t`	L
`D_pl,t`	rainwater demand of day `t`, that is, the part rain can supply	L
`soma`	sum over all the days of the simulated series	--

Equations 14 and 15 differ only in the denominator, and that difference is enough for them to be related by an exact identity, useful as a quick check of any result.

$$\mathrm{Epot} = \mathrm{Rv} \times f \tag{16}$$

Table 29 -- Variables of Equation 16

Symbol	Meaning	Unit
`Epot`	potable water saving potential	%
`Rv`	volumetric reliability	%
`f`	fraction of the total demand suppliable by rainwater	dimensionless

A system with 97% reliability and a 28% suppliable fraction saves 27.1% of the total consumption, and the three numbers say the same thing in three ways. That is exactly the case of the tutorial's 4,000 L cistern, before the upper tank: the volumetric reliability is 96.86%, and 96.86% x 0.28 = 27.12%, the potential printed in section 4.4. Reporting only one of them, without declaring the suppliable fraction adopted, makes the result impossible to compare with that of another study -- and it is the most frequent cause of improper comparisons in the literature of the field.

The literature uses different names for the same ratio. Fewkes and Butler (2000) call **water saving efficiency** the quotient between the rainwater volume supplied and the demand the system should meet, which is the volumetric reliability of this section. Other authors call efficiency what here is saving potential; EN 16941-1:2024 calls it the coverage rate. When comparing with another work, check the denominator, not the name.

2.3.3. Time reliability

Time reliability classifies each day of the series into one of three mutually exclusive classes, according to the relation between the rainwater volume consumed and the day's rainwater demand, defined in Equation 9. The day is **full** when rainwater fully meets the rainwater demand; **partial**, when it meets part of it and the mains supplies the rest; and **none**, when the tank is empty and the mains supplies all the demand (Figure 14). The three percentages add up to 100%. The classification counts days, not volume: a day on which one litre was missing to meet the demand is partial, not almost full, and that is why time reliability does not convert directly into volume saved.

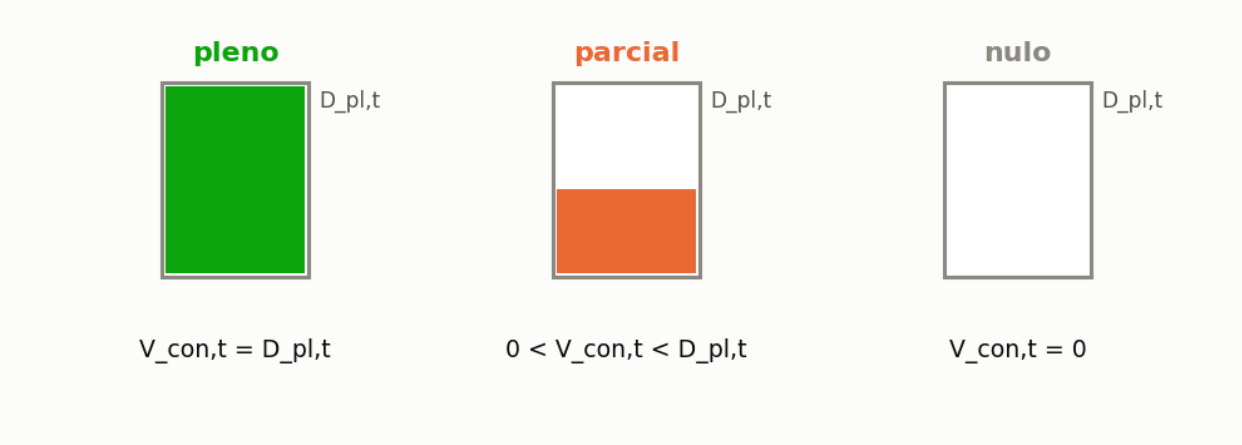


Figure 14 -- Daily supply classes: full (V_con,t = D_pl,t), partial (0 < V_con,t < D_pl,t) and none (V_con,t = 0). The filled height represents the fraction of the day's rainwater demand supplied by rainwater.

Formally, the day is **full** when `V_con,t = D_pl,t`, with a tolerance of 10-■ L; **partial** when `0 < V_con,t < D_pl,t`; and **none** when `V_con,t = 0`.

In the tutorial, the 4,000 L cistern has 96.6% of full days, 0.6% of partial days and 2.8% of days with no supply. Notice that the time reliability, 96.6%, is lower than the volumetric one, 96.86%. There is no contradiction: partial days do not count as full, but they contribute volume, and it is that water the volumetric reliability adds and the time reliability does not. The difference between the two is therefore an indirect measure of how far the failure days are partial failures.

What it measures: how often the user can count on the system.

What it does not measure: volume. A system can have 95% full days and a low potential if the rainwater demand is small, and the opposite also happens.

> Time reliability is the indicator that matters most for the **plumbing design**. Many days with no supply require a well-sized automatic backup supply, and the user notices the switching between the two sources.

2.3.4. Overflow

$$V_{\text{ext, med}} = \left(\sum V_{\text{ext, t}}\right) \div n \tag{17}$$

Table 30 -- Variables of Equation 17

Symbol	Meaning	Unit
`V_ext,med`	mean daily overflow	L/day
`V_ext,t`	volume overflowed on day `t`	L
`n`	number of days of the simulated series	day

What it measures: how small the cistern is relative to the rain events.

What it does not measure: waste. The water that overflows goes to the urban drainage, as it would without the system. There is no loss compared with the no-system scenario -- only an opportunity not taken.

> The overflow **never reaches zero** in a real system. It tends to a floor that is the difference between what the roof collects and what the demand consumes, and in the tutorial that floor is more than half of the water collected. Anyone who wants flood retention needs a detention tank, which is another project with another logic.

2.3.5. Efficiency of use of the collected water

The program does not print this indicator, but it follows directly from the previous ones and helps to understand the overflow. It is the fraction of the water that reached the cistern, after the first flush, that ended up consumed.

$$\eta\mathrm{cap} = \left(\sum V_{\mathrm{con,\,t}} \div \sum V_{\mathrm{cap,\,t}}\right) \times 100 \tag{18}$$

Table 31 -- Variables of Equation 18

Symbol	Meaning	Unit
`eta_cap`	efficiency of use of the collected water	%
`V_con,t`	rainwater volume consumed on day `t`	L
`V_cap,t`	volume collected on day `t`, after the first flush	L
`soma`	sum over all the days of the simulated series	--

In the tutorial's 4,000 L cistern, the roof delivers 187.4 m³ per year to the cistern; the system consumes 59.4 m³ and overflows 127.9 m³, and the two parts close the mass balance. The efficiency of use of the collected water is therefore 31.7%. The number explains why enlarging the cistern yields so little beyond a certain point: water is not missing on the roof, **demand** to consume it is. When supply greatly exceeds the rainwater demand, as in Florianópolis, the efficiency of use of the collected water is necessarily low, whatever the volume.

2.3.6. Duration and intensity of failures

The previous indicators say how much and how often the system fails, but not **how** it fails. Hashimoto, Stedinger and Loucks (1982) proposed, for water resource systems in general, three complementary criteria that are now used in rainwater harvesting too: **reliability**, which is the probability of the system being in a satisfactory state; **resilience**, which measures how quickly it recovers from a failure; and **vulnerability**, which measures the severity of failures when they occur.

The program reports the first and not the other two, but both can be calculated from the detailed daily series, exported to a spreadsheet. For the tutorial's 4,000 L cistern, over thirty years:

- there were 60 sequences of days that were not full, with a mean duration of 6.3 days and a maximum of 23 days. The resilience, taken as the inverse of the mean duration of a failure, is 0.16 per day;
- on failure days, a mean of 153 L was missing from a rainwater demand of 168 L, that is, about 90%. The typical failure, therefore, is not a small deficit: it is an empty cistern.

The design reading is direct. Long, almost total failures mean that the backup supply from the mains must be automatic and sized for the full flow of the rainwater points of use, not just to top up small shortfalls.

2.3.7. Annual volume saved

$$V_{\text{poup}} = (V_{\text{con, med}} \times 365,25) \div 1000 \tag{19}$$

Table 32 -- Variables of Equation 19

Symbol	Meaning	Unit
`V_poup`	volume of potable water saved per year	m³/year
`V_con,med`	mean daily rainwater consumption	L/day
`365,25`	length of the mean year, including leap years (365.25)	day/year
`1000`	conversion from litres to cubic metres	L/m³

What it measures: the basis of calculation of the economic analysis and the life cycle assessment.

> **IN DEPTH . THE 365.25 IS NOT CARELESSNESS**
> The conversion uses the mean year, including leap years, because long series contain them. Using 365 would introduce a systematic bias of 0.07% -- small, but for nothing.

2.3.8. Marginal gain

The formal definition of the marginal gain, with the grid of volumes made explicit, is in Equation 11, in section 2.2.

Defined in section 2.2. It is the indicator that decides the ideal volume, and it **only exists in a sweep** -- a single-volume simulation does not produce it.

2.3.9. Unmet demand

$$V_{\text{nao, med}} = (\sum V_{\text{nao, t}}) \div n \tag{20}$$

Table 33 -- Variables of Equation 20

Symbol	Meaning	Unit
`V_nao,med`	mean daily demand met by no source	L/day
`V_nao,t`	unmet demand on day `t`	L
`n`	number of days of the simulated series	day

The third part of the demand balance. It is zero in the default configuration and stops being zero when the irrigation backup supply is off.

Why it exists: without it, the balance would close by construction, attributing to the mains all the demand rain did not meet. Section 2.1 gives the details.

2.3.10. Summary table

Table 34 -- Summary of the indicators, with the denominator and the question each one answers

Indicator	Unit	Denominator	Answers
saving potential	%	total demand	how much potable consumption falls
volumetric reliability	%	rainwater demand	how much of what was intended was supplied
full, partial, no-supply days	%	days of the series	how often it can be counted on
overflow	L/day	--	is the tank small?
volume saved	m³/year	--	basis of the money and the carbon
marginal gain	%/m³	--	is it worth enlarging?
unmet demand	L/day	--	what nobody supplied

2.4. Rainfall indices

> This section gives the equation of each of the nine indices the program calculates, and declares three divergences from the ETCCDI standard.

The rainfall indices do not take part in the water balance and do not change any simulation result. Their role is to characterise the precipitation series -- and an honest characterisation is what makes it possible to compare studies of different cities.

2.4.1. The nine indices

The indices are calculated **year by year** over the loaded series, and the program reports the mean and the spread of the set of years. Throughout this section, `P_i` designates the precipitation of day `i` of the year, and a **wet day** is one on which `P_i >= 1 mm` -- an ETCCDI convention, adopted because depths below one millimetre are of the order of the measurement error of conventional rain gauges.

2.4.1.1. Total annual precipitation, PRCPTOT

Sum of the precipitation of all the wet days of the year. It differs from the gross annual total by excluding days of negligible depth, which keeps instrument noise from accumulating in the total.

$$\mathrm{PRCPTOT} = \sum P_\mathrm{i},\ \text{para}\ P_\mathrm{i} \geq 1\mathrm{mm} \tag{21}$$

Table 35 -- Variables of Equation 21

Symbol	Meaning	Unit
`PRCPTOT`	total precipitation of the wet days of the year	mm
`P_i`	precipitation of day `i`	mm
`soma`	sum over the days of the year that meet the condition	--

2.4.1.2. Precipitation on very wet days, R95p

Sum of the precipitation of the days whose depth exceeds the 95th percentile of the wet days. It measures how much of the annual total arrives in extreme events, and it is the index most directly linked to cistern overflow.

$$\mathrm{R95p} = \sum P_\mathrm{i},\ \text{para}\ P_\mathrm{i} > q_{95} \tag{22}$$

Table 36 -- Variables of Equation 22

Symbol	Meaning	Unit
`R95p`	precipitation accumulated on very wet days	mm
`P_i`	precipitation of day `i`	mm
`q_95`	95th percentile of the precipitation of the wet days of the series	mm

2.4.1.3. Maximum one-day precipitation, RX1day

The largest daily value recorded in the year. It is the index that sizes the overflow and the gutter, not the tank.

$$\mathrm{RX1day} = \max\left(P_{\mathrm{i}}\right) \tag{23}$$

Table 37 -- Variables of Equation 23

Symbol	Meaning	Unit
`RX1day`	largest daily precipitation of the year	mm
`P_i`	precipitation of day `i`	mm
`max`	maximum value over all the days of the year	--

2.4.1.4. Maximum five-day precipitation, RX5day

The largest five-consecutive-day total of the year. It matters more to rainwater harvesting than RX1day, because a tank fills over a prolonged event and not only at the daily peak.

$$\mathrm{RX5day} = \max\left(\sum P_{\mathrm{j}}\right),\ \text{para } j\text{de}iai + 4 \tag{24}$$

Table 38 -- Variables of Equation 24

Symbol	Meaning	Unit
`RX5day`	largest five-consecutive-day total of the year	mm
`P_j`	precipitation of day `j` of the window	mm
`i`	first day of each five-day window	--
`max`	maximum value over all the windows of the year	--

2.4.1.5. Simple daily intensity index, SDII

Mean precipitation of the wet days. It separates two regimes the annual total confuses: much rain on few days and little rain on many days can produce the same PRCPTOT and very different SDII.

$$\mathrm{SDII} = \mathrm{PRCPTOT} \div n_{\text{úmidos}} \tag{25}$$

Table 39 -- Variables of Equation 25

Symbol	Meaning	Unit
`SDII`	mean intensity of the wet days	mm/day
`PRCPTOT`	total precipitation of the wet days, from Equation 21	mm
`n_úmidos`	number of wet days of the year	day

2.4.1.6. Days of heavy precipitation, R20mm

Count of the days on which the depth reaches or exceeds twenty millimetres. In a rainwater harvesting system, these are typically the overflow days.

$$\mathrm{R20mm} = \#\left\{i : P_{\mathrm{i}} \geq 20\mathrm{mm}\right\} \tag{26}$$

Table 40 -- Variables of Equation 26

Symbol	Meaning	Unit
`R20mm`	number of days with heavy precipitation in the year	day
`P_i`	precipitation of day `i`	mm
`# { }`	count of the days that meet the condition	--

2.4.1.7. Consecutive dry days, CDD

Longest sequence of consecutive days with precipitation below one millimetre. It is the index that best predicts the performance of a rainwater system, because the longest dry spell is what empties the cistern.

$$\mathrm{CDD} = \max\left(L\right), \mathrm{com} P_{\mathrm{i}} < 1\mathrm{mm} \mathrm{em\ toda}\ a\ \mathrm{sequência}\ L \tag{27}$$

Table 41 -- Variables of Equation 27

Symbol	Meaning	Unit
`CDD`	longest sequence of consecutive dry days in the year	day
`L`	length of a sequence of consecutive dry days	day
`P_i`	precipitation of day `i`	mm

2.4.1.8. Consecutive wet days, CWD

Longest sequence of consecutive days with precipitation equal to or above one millimetre. It characterises the persistence of rain, and long sequences favour the complete filling of the tank.

$$\mathrm{CWD} = \max\left(L\right), \mathrm{com} P_{\mathrm{i}} \geq 1\mathrm{mm} \mathrm{em\ toda}\ a\ \mathrm{sequência}\ L \tag{28}$$

Table 42 -- Variables of Equation 28

Symbol	Meaning	Unit
`CWD`	longest sequence of consecutive wet days in the year	day
`L`	length of a sequence of consecutive wet days	day
`P_i`	precipitation of day `i`	mm

2.4.1.9. Seasonality index, SI

It measures the departure of the actual monthly distribution of the rain from a perfectly uniform distribution over the year. Unlike the previous eight, it does not belong to the ETCCDI set: it was proposed by Walsh and Lawler in 1981.

$$\mathrm{SI} = \left(1 \div R\right) \times \sum \left| X_{\mathrm{m}} - R \div 12 \right| \tag{29}$$

Table 43 -- Variables of Equation 29

Symbol	Meaning	Unit
`SI`	Walsh and Lawler seasonality index	dimensionless
`R`	total precipitation of the year	mm
`X_m`	total precipitation of month `m`, with `m` from 1 to 12	mm
`R ÷ 12`	monthly precipitation there would be if the rain were uniform	mm
`soma`	sum over the twelve months of the year	--
`\`	\	` absolute value of the difference --

The index of Equation 29 is zero when all months receive the same depth and tends to 1.83 when all the rain of the year falls in a single month. Between these extremes, Walsh and Lawler (1981) classify the regime as very equable (below 0.20), equable with a definite wetter season (0.20 to 0.39), rather seasonal with a short drier season (0.40 to 0.59), seasonal (0.60 to 0.79), markedly seasonal with a long drier season (0.80 to 0.99), most rain in three months or less (1.00 to 1.19) and extreme (1.20 or more).

The seasonality index is the one that matters most to rainwater harvesting, and section 3.1 shows why: it is the distribution of rain over the year, not the total precipitated, that decides the yield of the system.

The nine indices remain abstract while they stay in table form, and Figure 15 places them over one year of the series. Placed over a real year of series, each one points to an identifiable stretch of the record: the most intense day, the days above the threshold, the longest dry spell, the accumulated total.

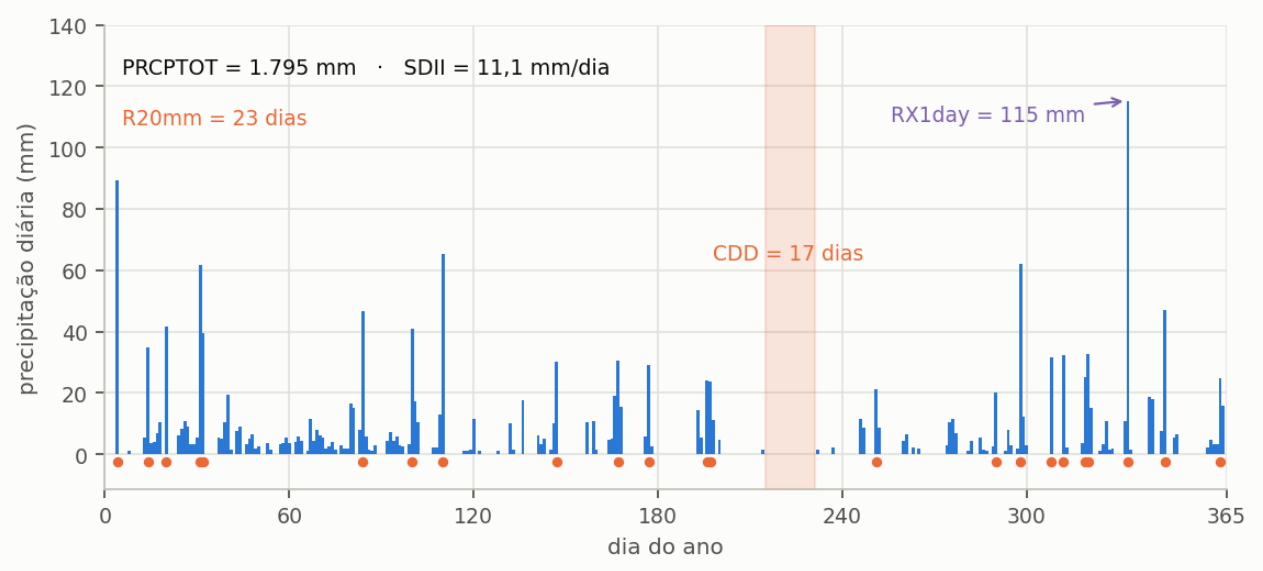


Figure 15 -- Four of the nine indices marked over the year 1995 in Florianópolis: PRCPTOT and SDII, at the top; RX1day, the largest daily value of the year; CDD, the longest sequence of dry days, in the shaded band; and R20mm, the days with 20 mm or more, marked at the bottom.

Seasonality is the index least like the others, because it describes neither intensity nor frequency: it describes how much the rain of the year concentrates in part of it. It is also the index that most separates the Brazilian state capitals from one another, and the one that best predicts whether a tank will have to get through a long dry spell (Figure 16).

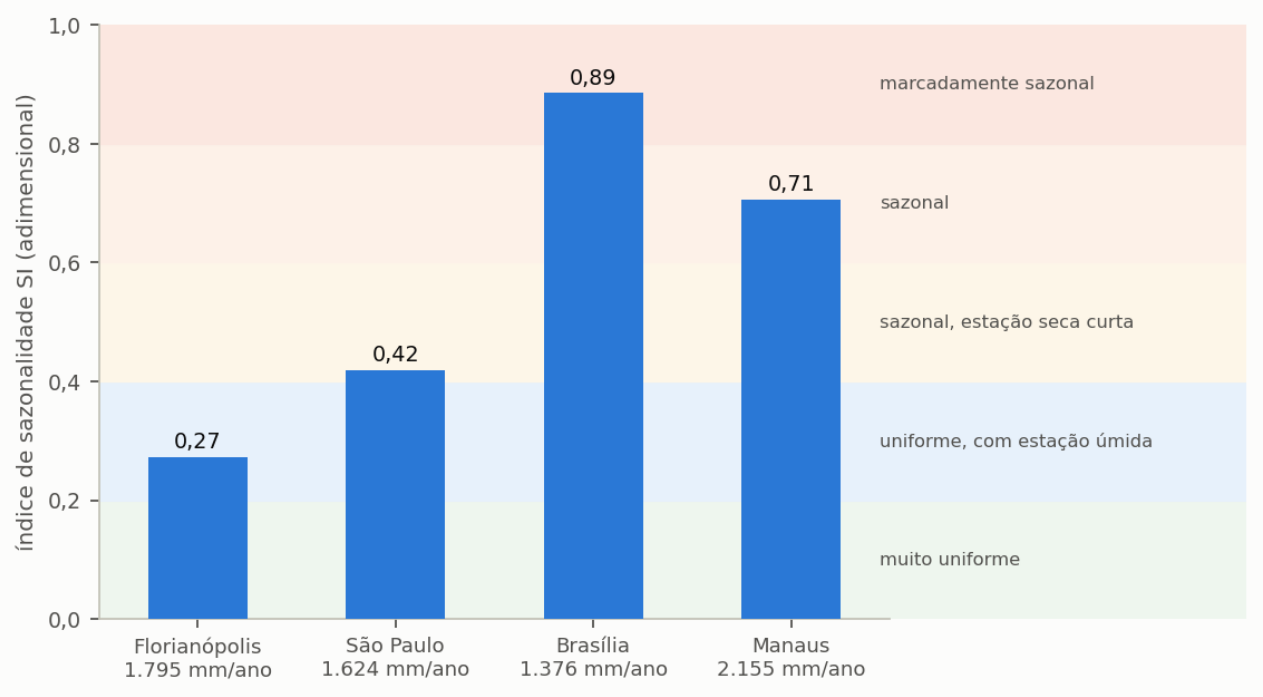


Figure 16 -- Walsh and Lawler (1981) seasonality index in the four capitals of the guide, with the classes of the index in the background and the mean annual precipitation under the name of each city. High annual precipitation and an even distribution over the year are independent properties.

2.4.2. Three divergences that must be declared

The program calls the set «ETCCDI plus seasonality», and the precision of the expression matters.

First: the set is not pure ETCCDI. The seasonality index is Walsh and Lawler's, and does not belong to the set of the *Expert Team on Climate Change Detection and Indices*. Conversely, **R10mm**, which is ETCCDI, is **not** implemented.

Eight of the nine indices are ETCCDI; the ninth is not; and one that would be is missing.

Second: the base period of the percentile. R95p requires a 95th percentile of the wet days. ETCCDI defines the fixed base period of 1961 to 1990. The program calculates the percentile over **the whole loaded series**.

The consequence is that the R95p of two series of different lengths is **not directly comparable**, because each one has its own threshold. For strict comparison with ETCCDI literature, trim the series to the base period.

The program calculates the percentile **only over the wet days**, which is correct: including dry days would shift the percentile down in seasonal series.

Third: the incomplete RX5day windows. The five-day moving sum uses partial windows at the start of each year, so that the first windows have one to four days. The effect is small and always in the direction of underestimating the RX5day of the first days of the year.

The program's indices tab presents each indicator year by year: the total annual precipitation (Figure 17) and the longest sequences of dry and wet days (Figure 18).

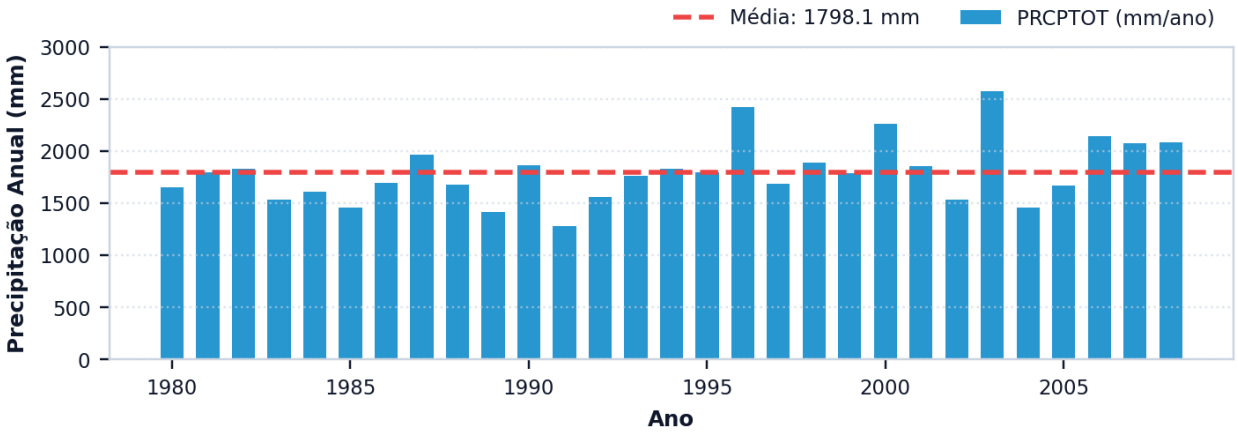


Figure 17 -- Total annual precipitation (PRCPTOT) of the Florianópolis series, with the mean of the period as a dashed line, in the indices tab. Figure generated by the program itself.

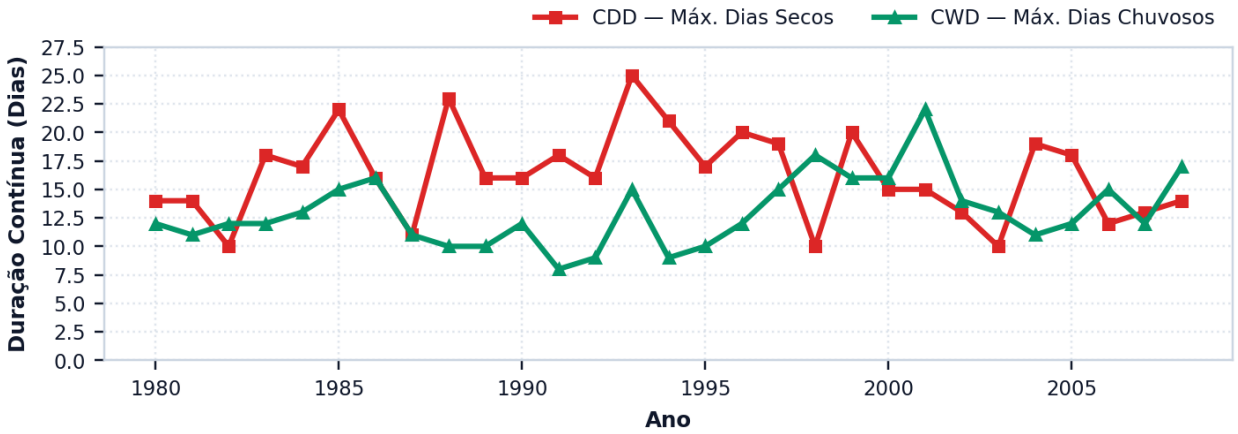


Figure 18 -- Longest annual sequence of dry days (CDD) and of wet days (CWD) of the Florianópolis series, in the indices tab. They are the two indices most directly linked to the tank volume the rainfall regime requires. Figure generated by the program itself.

2.4.3. What they are for in the study

Characterising the series in the method of the paper. PRCPTOT, SDII and SI describe the regime better than the annual mean alone.

Explaining the result. A high CDD explains a low-performing system despite abundant rain -- the case of Brasília in section 3.1.

Comparing climate scenarios. Between the historical and the projected series, what usually changes first is SDII and CDD, not PRCPTOT. Reporting only the annual total hides the change that most affects the system.

2.4.4. What the indices do not do

They do not enter the balance. They do not change the sizing. They do not replace the daily simulation -- which is precisely the program's argument against methods based on mean values.

An index is a summary. The balance uses the whole series.

2.5. Sizing the treatment line

> This section formalises how the program sizes the rainwater treatment plant and explains the **incremental rule**, which keeps treatment from being counted twice in the budget and in the cash flow.

The water running off a roof carries dust, leaves, soot and bird droppings, and no use -- not even toilet flushing -- admits it reaching the point of use as it left the roof. The rainwater treatment plant, called ETAP in the program (from the Portuguese *estação de tratamento de água pluvial*), is the set of equipment that does this conditioning, and the program sizes it from three inputs: the quality class the use requires, the roof area and the annual volume of rainwater actually consumed, the latter a result of the daily balance presented in section 2.1.

The treatment line sized by the program has four components in series. The **first flush** diverts the first depth of each rain, which carries the dirt accumulated on the roof; the **mechanical inlet filter** removes leaves and coarse solids before they reach the tank; **disinfection** is the mandatory stage, on which the microbiological safety of the stored water depends; and the **polishing filter** is optional, and meant for uses where appearance or the absence of dissolved residue matters.

The treatment line is a sequence, and the order of the stages is not arbitrary: each one removes what would make the next one ineffective. It is worth having the whole chain in view before discussing each part (Figure 19). The order is not interchangeable: disinfecting before filtering consumes the reagent in suspended organic matter and leaves micro-organisms protected in the particles. Turbidity is therefore a condition of the effectiveness of disinfection, not just an aesthetic parameter.

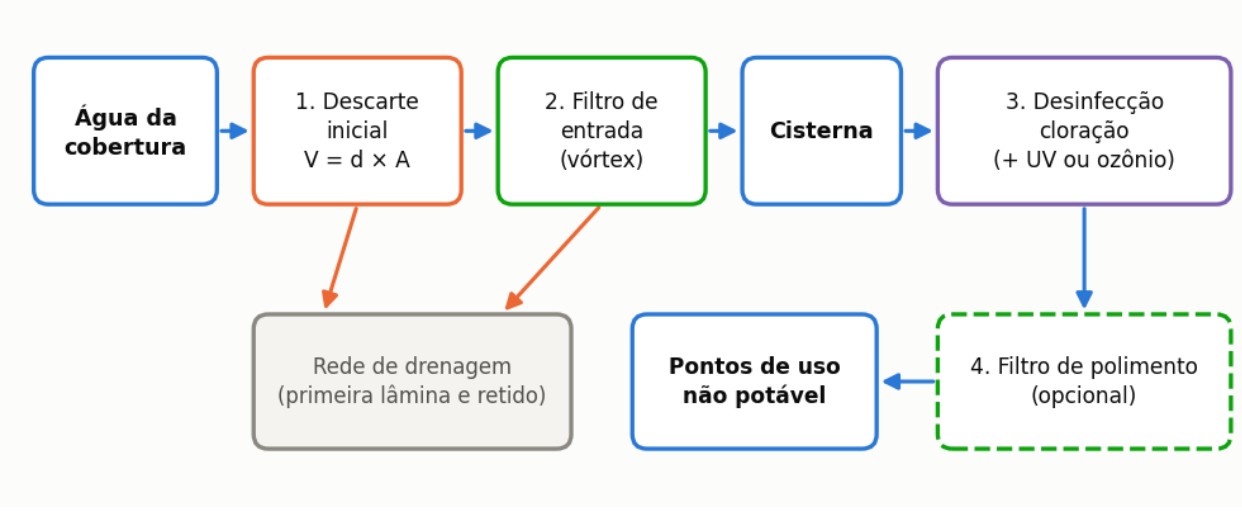


Figure 19 -- Components of the treatment line sized by the program, in the order of the flow, and destination of the material retained. The first flush and the inlet filter act before the cistern; disinfection acts after them because the reagent would be consumed by the suspended organic matter the previous stages remove; the polishing filter is optional.

2.5.1. The inlet filter

The inlet filter is sized by the roof area, not by the intended quality class, because what it has to cope with is the peak flow the roof produces -- a quantity that does not depend on the final destination of the water. The three ranges adopted by the program correspond to the capacities manufacturers themselves declare for the equipment common on the Brazilian market.

Table 44 -- Mechanical inlet filter sized by the roof area

Roof	Equipment	Mesh
up to 200 m²	self-cleaning vortex DN 100	350 µm
up to 500 m²	self-cleaning vortex DN 150	400 µm
above 500 m²	basket or cyclonic DN 200	500 µm

> The range limits follow the area the product itself declares it serves, not an arbitrary margin over it. That is why a 180 m² roof receives the DN 100 filter: the manufacturer declares the equipment enough for up to 200 m², and imposing that of the next range would make the installation more expensive without technical grounds.

2.5.2. Disinfection

Disinfection is the stage the program never dispenses with, and the reason lies in the concept of residual. Among the three technologies available, only chlorination leaves in the water a disinfecting agent that remains active after treatment; ultraviolet radiation and ozonation act at the moment of passage and leave nothing behind (Figure 20).

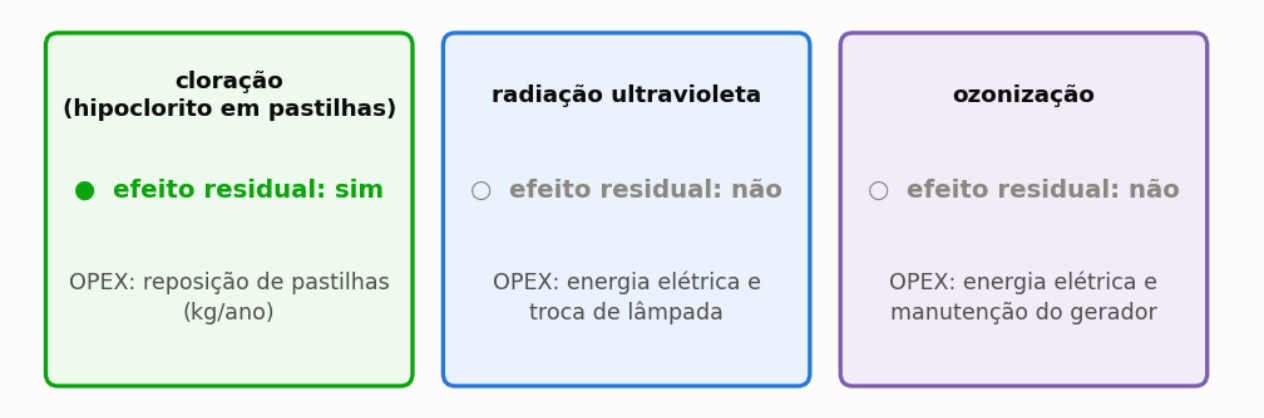


Figure 20 -- The three disinfection technologies, the residual effect of each one and the make-up of the annual running cost (OPEX). The residual effect distinguishes chlorination from the others: it protects the stored water, while ultraviolet radiation and ozone act only on the water passing through the equipment.

Ultraviolet and ozone come in as a barrier additional to chlorination, not as substitutes for it. Without a residual, the treated water becomes contaminated again on the way to the point of use, and the right practice -- in line with item 4.6.5 of ABNT NBR 15527:2019, which works with free residual chlorine -- keeps chlorination to protect the tank and the internal network. The economic consequence is direct: both options add the cost of chlorine to their own, and choosing either of them never reduces the running cost.

2.5.2.1. The mass of chlorine

The chlorination consumable is sized by the mass of commercial product consumed per cubic metre treated, which follows from the dose applied and the purity of the product bought.

$$m_{\text{prod}} = (C_{\text{Cl}} \div 1000) \div p \tag{30}$$

Table 45 -- Variables of Equation 30

Symbol	Meaning	Unit
`m_prod`	mass of commercial product consumed per cubic metre treated	kg/m³
`C_Cl`	dose of active chlorine applied to the water	mg Cl2/L
`1000`	conversion from milligrams per litre to kilograms per cubic metre	--
`p`	purity of the product, in mass of active chlorine per mass of product	dimensionless

> **What is bought is the product, not pure chlorine.** The calcium hypochlorite tablet contains about 65% active chlorine, and the price per kilogram refers to the commercial product, not to the active ingredient. Calculating the cost by the mass of active chlorine would underestimate it by the inverse of the purity, that is, by about 35% -- and it is a common mistake in budget spreadsheets.

2.5.3. The polishing filter

The polishing filter is optional and is the only point of the sizing where the target quality class influences the choice of equipment. For the more demanding classes, the program recommends including it, and the recommendation also varies in the type of filter medium: laundry, because it involves contact with fabric and skin, requires block activated carbon, whereas the other uses are satisfied with a sand filter or a twenty-micrometre cartridge.

Table 46 -- Polishing filter recommendation by target quality class

Class	Polishing recommended?	Equipment when ticked
1 -- NBR 15527:2019	no	sand filter or 20 µm cartridge
2 -- irrigation and washing	yes	likewise
3 -- cooling towers and boilers	yes	likewise
4 -- laundry	yes	**block activated carbon**

The recommendation, however, is not an imposition: the filter only enters the sizing if the user actually ticks it. When the class recommends it and the user declines, the program records the refusal in a field of its own and the report shows it, so that the decision is documented instead of disappearing.

> **Classes 2 to 4 are adaptations made by the program, not by NBR 15527.** The matrix of parameters by end use is a feature of the 2007 version of the standard and did not survive the 2019 revision. In a published study, present them as a criterion adopted by the author, never as a normative requirement.

2.5.4. The first flush

The first flush enters the program in two ways. In the water balance, it is the depth subtracted from each day's precipitation, described in section 2.1. In the budget, when the physical sizing of the first flush is enabled in step **3. Quality & Treatment** -- the default in the interface --, it is a subsystem with its own investment, running cost and embodied carbon, made up of the leaf separator and the first-flush chamber.

The volume of the first-flush chamber is the product of the first-flush depth and the catchment area, `V_desc = d x A`, in litres, with `d` in millimetres and `A` in square metres. Up to 250 L, the program prices a vertical PVC diversion column, at R$ 2.20 per litre plus R$ 120 in fittings and a slow purge valve; above 250 L, a rotomoulded polyethylene tank, with the price interpolated in the cost database plus R$ 150 for a sealing float and purge device. The leaf separator costs R$ 220 for up to 200 m² of catchment, R$ 440 for up to 500 m² and R$ 220 per unit, every 250 m², above that. The annual running cost of the subsystem is R$ 40, and all values are multiplied by the regional

factor. These are Brazilian reference prices; in the International Edition the costs are typed in.

Table 47 -- Sizing and pricing of the first flush in the tutorial project

Item	Value
first-flush depth	2 mm
catchment area	150 m²
volume of the first-flush chamber	300 L
polyethylene first-flush chamber	R$ 396.90
DN 100 leaf separator	R$ 220.00
investment of the subsystem	R$ 616.90
annual running cost	R$ 40.00

The investment of the subsystem is part of the investment of the treatment line and therefore enters the cash flow through the incremental rule below.

2.5.5. The incremental rule

The incremental rule is the part of the calculation most often got wrong in home-made implementations, and the error is always the same: counting treatment twice. It applies on three different fronts, and on each it takes its own form.

In the investment. The budget of a rainwater harvesting system already sets aside a generic treatment allowance within the accessories item, because no designer prices a system without some form of filtering. Adding the detailed line on top of that allowance would charge treatment twice, and so the program deducts the generic allowance and adds only the increment.

$$\mathrm{CAPEX}_{\mathrm{inc}} = \mathrm{CAPEX}_{\mathrm{linha}} - V_{\mathrm{gen}} \tag{31}$$

Table 48 -- Variables of Equation 31

Symbol	Meaning	Unit
`CAPEX_inc`	incremental investment of the treatment line	R$
`CAPEX_linha`	total investment in the equipment of the sized line	R$
`V_gen`	generic treatment allowance already included in the accessories item	R$

In the running cost. Chlorination is already paid in the chemical consumables line of the cash flow, so the incremental running cost of the treatment line deducts that part and the hypochlorite is paid once, not twice.

In the life cycle assessment. The treatment line **replaces** the chlorination term instead of adding to it, and the different treatment between the cash flow and the emission inventory follows from the two accounting for different quantities.

> The rule lives in a single module, consumed by **both** economic engines of the program. Two copies of the same rule diverge over time, and the divergence between them is invisible in the result: each engine responds consistently with itself.

2.5.6. Switching the treatment line off

The sizing of the treatment line can be switched off entirely, and in that case neither the investment nor the running cost of treatment enters the economic analysis. The purpose of the switch is to allow the study of the hypothesis of treatment already existing in the building -- in which case the cost should not be attributed to the rainwater system -- and to compare scenarios with and without the line. Switching it off does not mean the water needs no treatment: it means the treatment is not being priced here.

2.5.7. What the program does not size

The scope of the sizing ends at the equipment of the line, and four related items remain the designer's responsibility. The **hydraulic arrangement of the first flush** -- fittings, purge device and connection to the drainage -- is not designed: the program calculates only the volume of the first-flush chamber and its cost. The **piping** of the treatment line and the head loss it imposes are not calculated either. The **maintenance regime** -- changing the filter element, cleaning the vortex, replacing tablets -- is not planned by the program, although it determines a good part of the real running cost. And **quality control** remains entirely outside the scope: the program sizes equipment from assumptions, and does not replace laboratory testing.

ABNT NBR 15527 deals with control and maintenance, and the corresponding plan is the designer's responsibility. EN 16941-1:2024, in its Annex E, suggests maintenance intervals of six months for pre-treatment, gutters, filters and pumps, and twelve months for the tank.

2.6. Pumping energy

> This section gives the complete chain from the shaft to the socket, and explains why the nameplate power is **not** what the meter records.

Three versions of this calculation have existed in the program, and the first two made the mistakes described in the item «Why the complete chain matters»: treating the nameplate power as electrical power and deducting the pump efficiency twice. Anyone reading a study made with an old version needs to know this.

2.6.1. Nameplate power is shaft power

ABNT NBR 17094-2:2016 requires the **efficiency** to be marked on the motor's nameplate, as a quantity separate from the power. That only makes sense if the nameplate power is the mechanical power **at the shaft**.

Hence the chain has two efficiencies, not one:

Table 49 -- The two efficiencies of the pumping chain and what each one converts

Efficiency	What it converts
of the pump, `eta_b`	shaft power into hydraulic power delivered to the water
of the motor, `eta_m`	electrical power from the socket into mechanical power at the shaft

The program asks for both, in separate fields.

2.6.2. The equations

Specific pumping energy, per cubic metre lifted:

$$e = (\rho \times g \times H) \div (\eta b \times \eta m) \tag{32}$$

Table 50 -- Variables of Equation 32

Symbol	Meaning	Unit
`e`	specific energy consumption per volume pumped	kWh/m³
`rho`	density of water, taken as 1,000	kg/m³
`g`	acceleration of gravity, taken as 9.81	m/s²
`H`	total manometric head of the pumping	m head
`eta_b`	hydraulic efficiency of the pump	dimensionless
`eta_m`	efficiency of the electric motor	dimensionless

Equation 32 does not depend on the flow rate or the installed power. Lifting one cubic metre of water to a given height costs the same energy with a large or a small pump, because the work done is the same; what changes with the power is the time taken to do it, not the specific consumption.

When the operating flow rate is declared, the equivalent form:

$$e = P \div (Q \times \eta m) \tag{33}$$

Table 51 -- Variables of Equation 33

Symbol	Meaning	Unit
`e`	specific energy consumption per volume pumped	kWh/m³
`P`	nameplate power of the pump set, which is shaft power	kW
`Q`	pumping flow rate entered or deduced	m³/h
`eta_m`	efficiency of the electric motor	dimensionless

The hydraulic efficiency of the pump does not appear in Equation 33, and the absence is deliberate: it is already built into the fact that a motor of `P` kilowatts at the shaft makes the pump deliver `Q` cubic metres per hour at that head. Dividing by it again would count the hydraulic loss twice.

Flow rate, when not declared:

$$Q = (P \times \eta b) \div (\rho \times g \times H) \tag{34}$$

Table 52 -- Variables of Equation 34

Symbol	Meaning	Unit
`Q`	deduced pumping flow rate	m³/h
`P`	nameplate power of the pump set, which is shaft power	kW
`eta_b`	hydraulic efficiency of the pump	dimensionless
`rho`	density of water, taken as 1,000	kg/m³
`g`	acceleration of gravity, taken as 9.81	m/s²
`H`	total manometric head of the pumping	m head

Energy of the start-ups:

$$E_{\text{part}} = (P \times t_{\text{part}}) \div \eta m \tag{35}$$

Table 53 -- Variables of Equation 35

Symbol	Meaning	Unit
`E_part`	energy consumed by all the start-ups of the period	kWh
`P`	nameplate power of the pump set, which is shaft power	kW
`t_part`	total start-up time accumulated in the period	h
`eta_m`	efficiency of the electric motor	dimensionless

The pump efficiency is absent from Equation 35 for the physical reason that distinguishes it from the previous ones: during start-up there is no useful flow, and therefore no hydraulic loss to deduct. The motor draws energy from the socket, and that is all the utility's meter records.

The annual energy of the pump set is the sum of the two parts: that of the actual pumping, which depends on the volume lifted, and that of the start-ups, which depends on the number of starts.

$$E_{\text{ano}} = e \times V_{\text{rec}} + (P \times t_{\text{part}}) \div \eta m \tag{36}$$

Table 54 -- Variables of Equation 36

Symbol	Meaning	Unit
`E_ano`	electrical energy consumed by the pump set in the year	kWh/year
`e`	specific energy consumption per volume pumped	kWh/m³
`V_rec`	volume pumped in the year	m³/year
`P`	nameplate power of the pump set, which is shaft power	kW
`t_part`	total start-up time accumulated in the year	h/year
`eta_m`	efficiency of the electric motor	dimensionless

The nameplate power of a pump set is the mechanical power at the shaft, not the electrical power recorded by the meter (Figure 21). A 0.5 hp (0.37 kW) pump set with a motor of 60.5% efficiency draws about 0.61 kW from the mains: calculating the energy with the nameplate power would underestimate consumption by about 40%, and the energy running cost is precisely the part that reduces the saving of the system.

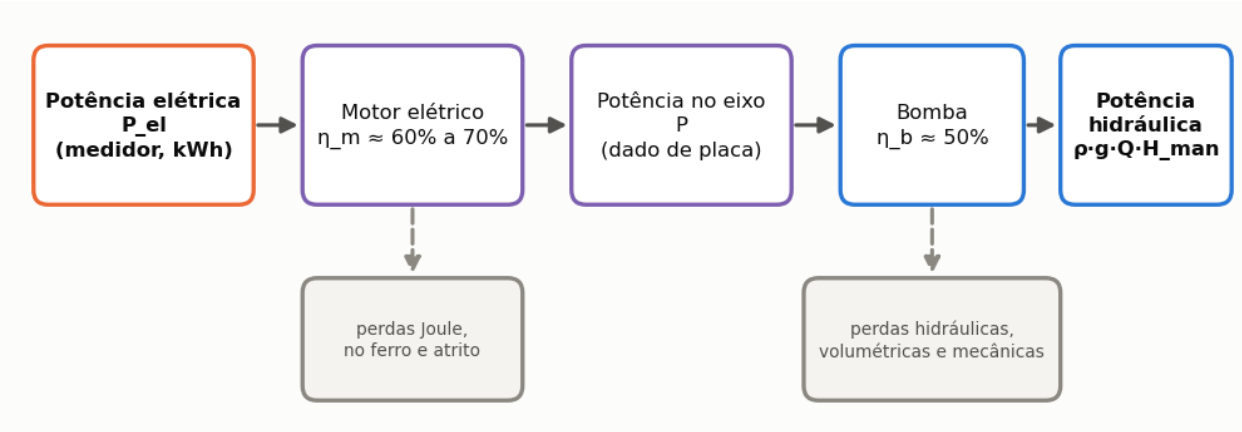


Figure 21 -- Energy chain of pumping. The nameplate power is the power at the shaft, in the middle of the chain: between it and the energy meter is the motor efficiency, eta_m, and between it and the hydraulic power delivered to the water is the pump efficiency, eta_b.

The hydraulic power depends on the manometric head, which adds the head losses of the flow to the geometric difference in level between the water surfaces (Figure 22). Head loss is the energy dissipated per unit weight of fluid and has two parts: the continuous loss, along the pipe, given by the Darcy-Weisbach equation, `h_f = f . (L/D) . v²/2g`, where `f` is the friction factor, `L` the length, `D` the internal diameter and `v` the mean velocity; and the local losses in fittings, valves and bends, `h_loc = K . v²/2g`, where `K` is the coefficient of the singularity. In short building rising mains, the losses typically add 10% to 30% of the geometric difference in level.

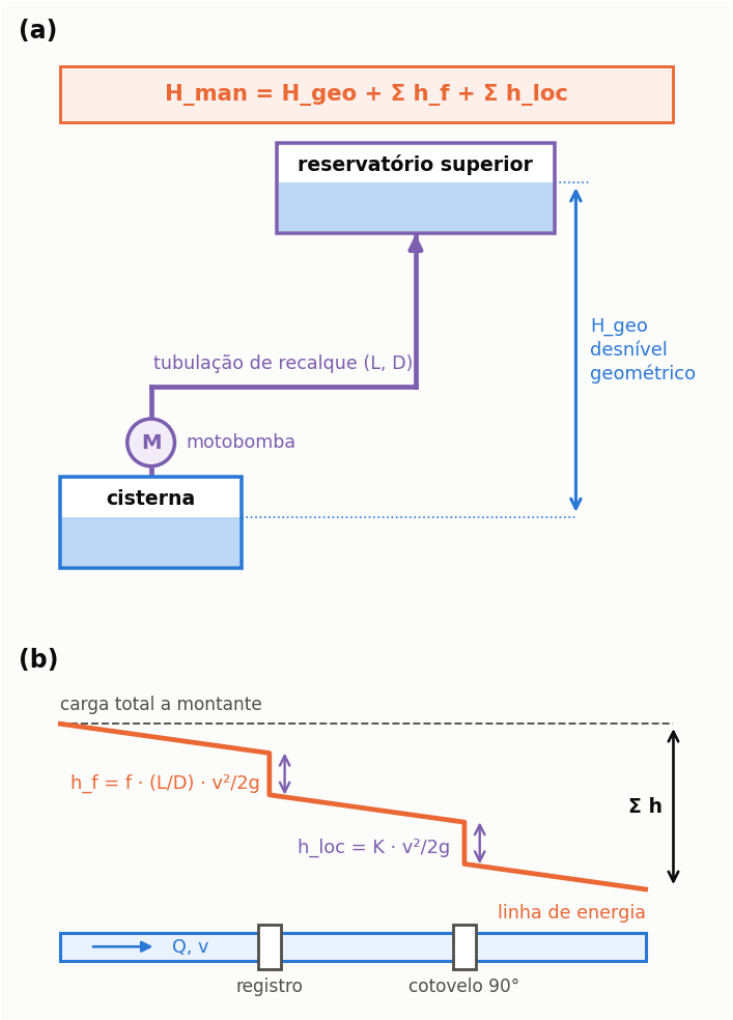


Figure 22 -- (a) Manometric head of a building rising main: geometric difference in level between the water surfaces of the cistern and the upper tank, plus the head losses; (b) energy line along a stretch of pipe, with the continuous friction loss (h_f) and the local losses at a valve and an elbow (h_loc).

2.6.3. The motor efficiency, when not entered

Table 13 of ABNT NBR 17094-2:2016 applies, line IR1, two-pole motors -- which is what residential pump sets are, at 3,450 rpm (Figure 23):

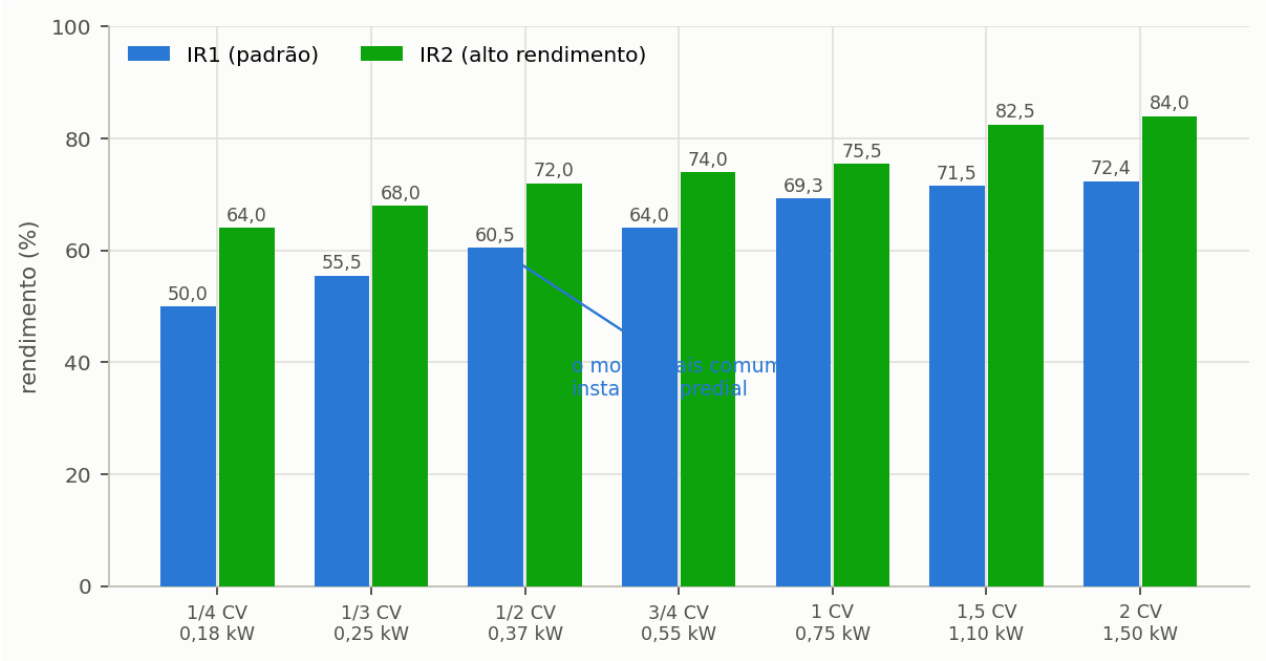


Figure 23 -- Minimum efficiency of two-pole single-phase induction motors, according to Table 13 of ABNT NBR 17094-2:2016. The program adopts class IR1 when the project does not enter the efficiency, as the least demanding and therefore the conservative assumption.

The two values that appear most in building installations are those of half a horsepower, with **60.5%** efficiency in class IR1, and of one horsepower, with **69.3%**. The difference between them is almost nine percentage points, and it carries over entirely to the energy running cost, because it is the efficiency that separates the shaft power from the power actually drawn from the mains.

IR1 is adopted because it is the normative floor and what is found in off-the-shelf pump sets. Anyone with a high-efficiency motor enters the value marked on the nameplate.

For powers not in the table, the standard says to adopt the larger of the two neighbouring efficiencies when the power is at the midpoint or above, and the smaller when it is below.

2.6.4. Why the complete chain matters

The specific consumption of the reference case -- half a horsepower, 12 m of manometric head, 82 m³ pumped and 98 starts per year -- is 0.1080 kWh/m³. Ignoring the motor efficiency, or deducting the pump efficiency twice, changes that number by tens of percentage points, and those are precisely the two most frequent mistakes in sizing spreadsheets.

The first mistake is to treat the nameplate power as the power consumed at the socket; the second, to apply the hydraulic efficiency in a calculation that already starts from the flow actually delivered, which counts it twice. The chain presented in this section is explicit precisely to make both visible.

2.6.5. The pump sizing wizard

When the power, flow rate and manometric head of the pump set are not known, the *Pump Sizing Wizard* button, in step **4. Reservoir & Costs** and in the economic dialog, estimates the three values and selects a commercial model. The wizard follows the usual sizing sequence of a building rising main:

1. **Pumping flow rate.** `Q = V_sup ÷ t_ench`, with the volume of the upper tank and the desired filling time, 45 minutes by default.

2. **Economic diameter.** By Forchheimer's formula, `D = 1,3 . (t_op ÷ 24)^0,25 . raizQ` (1.3 and 0.25 in decimal-point notation), with `D` in metres, `Q` in m³/s and `t_op` the daily operating time, 2 h by default. The diameter adopted is the smallest commercial solvent-weld PVC pipe whose internal diameter meets the calculated value, and never smaller than the outlet of the selected pump set; the suction gets one commercial size above the delivery.
3. **Velocity.** `v = Q ÷ (π . D_i² ÷ 4)`, highlighted when it falls outside the range of 0.6 m/s to 2.5 m/s.
4. **Head loss.** By the Hazen-Williams formula, `J = 10,643 . Q^1,852 . C^-1,852 . D_i^-4,87`, with `C = 140` for PVC, multiplied by the length of the piping and increased by 30% for the local losses.
5. **Manometric head and power.** `H_man = H_geo + h_f` and `P = rho . g . Q . H_man ÷ eta`, with a set efficiency of 55% by default.
6. **Selection.** The first model of the internal catalogue whose flow rate, maximum head and power meet the operating point, with the pump curve plotted against the calculated point.

The calculated manometric head and head loss go back to the project and feed the energy calculation of this section, in place of the fallback value.

2.6.6. The fallback values

Used when the project does not declare them:

Table 55 -- Fallback values adopted when the project does not declare them

Parameter	Value	Origin
manometric head	12 m head	two-storey house with a buried cistern: ~8 m of geometric level difference and ~30% of head loss give 10 to 12 m; the conservative end is adopted
pump efficiency	55%	residential pump sets from half to one horsepower lie between 40% and 60% at the operating point
start-up time	108 s	see the caveat below

> **IN DEPTH . THE MANOMETRIC HEAD IS THE PARAMETER THAT WEIGHS MOST**
> Anyone who knows the project's level difference should enter it. Doubling the head doubles the energy.

> **IN DEPTH . THE 108 S START-UP TIME**
> It represents the priming time of a self-priming pump whose suction line has drained. In an installation with a foot valve in working order, the line stays primed and the following starts are immediate. In the reference case, start-ups account for 19% of the annual consumption; with a 3 s transient, they would account for 0.5%. The value is conservative and is recorded as an open issue of the program.

2.6.7. Direct pressurisation: the energy is time switched on

Everything said so far applies to pumping to a tank, in which the pump fills the upper tank in one go, at its own flow rate, and switches off. The pressure pump works another way. It starts at each flush and stays on while the toilet cistern fills, and it is the cistern's float valve, not the pump, that decides the flow. A larger pump does not fill the cistern faster: it stays on for the same time, drawing more power.

That is why the program does not calculate the energy of the pressure pump by the specific energy `rho.g.H/(eta_b.eta_m)`, which assumes the pump at its best-efficiency point. It calculates it by the time switched on:

$$E = (P \div \eta m) \cdot [V \div \min(Q_{\mathrm{b}}(H), Q_{\mathrm{p}}) + N \cdot t_{\mathrm{r}}] \quad (37)$$

Table 56 -- Variables of Equation 37

Symbol	Meaning	Unit
`E`	annual energy of the pressure pump	kWh/year
`P`	nameplate power, at the shaft	kW
`eta_m`	motor efficiency, from Table 13 of ABNT NBR 17094-2	dimensionless
`V`	rainwater volume consumed	m³/year
`Q_b(H)`	flow rate of the model's curve at the working pressure `H`	m³/h
`Q_p`	flow rate of the point of use	m³/h
`N`	starts per year: residents x 5.05 flushes x 365.25	1/year
`t_r`	switch-off delay of the electronic control	h

It is the form of the «simple method» Ward, Butler and Memon (2012) describe, power times operating duration, with two corrections: the effective flow rate is the smaller of that of the pump and that of the point, and the delay of electronic controls enters at each start. The nameplate power is that of the nominal point, and at low flow the motor draws a little less; the estimate tends to overestimate the energy, but it ranks the models among themselves.

The default point flow rate is 0.15 L/s, that is, 9 L/min: the design flow rate of a toilet with a flushing cistern in Table A.1 of ABNT NBR 5626:1998. The 2020 edition dropped the table and left to the design the task of stating the flow rates at the points of use (item 6.7.1); anyone with the toilet manufacturer's figure should use it.

2.6.7.1. The catalogue

Step **4. Reservoir & Costs** has six residential pressure pumps from five manufacturers, each with the flow x pressure curve from its manual, kept among the documents consulted: Komeco TP 820T and TP 825T, Cardal Potenza, Jacuzzi AcquaHouse, with a curve tested at the Pump Labelling Laboratory of UNIFEI, Syllent Impulse Press and Rowa Max Press 20E. For another model, the «Custom» option asks for the nameplate power, the flow rate read on the manufacturer's curve at the working pressure and the switch-off delay.

The table applies the catalogue to the tutorial's direct pressurisation: the 4,000 L cistern without an upper tank, four residents and the toilet flow rate of 9 L/min.

Table 57 -- Annual energy of six catalogue pressure pumps in the tutorial's direct pressurisation, with the toilet flow limiting the pump flow

Model	Control	Curve flow at 18 m head	Hours per year	kWh per year	kWh/m³	kWh per year at 24 m head
Komeco TP 820T, 1/4 CV	pressure switch	13.3 L/min	110	40.5	0.68	82.0
Cardal Potenza, 1/3 CV	flow switch	18.8 L/min	110	48.6	0.82	does not reach
Komeco TP 825T, 1/2 CV	pressure switch	29.1 L/min	110	66.9	1.13	66.9
Jacuzzi AcquaHouse 5AQH1, 1/2 CV	pressure switch	27.4 L/min	110	66.9	1.13	66.9
Rowa Max Press 20E, 1/2 CV	electronic	12.8 L/min	110	66.9	1.13	does not reach
Syllent Impulse Press, 1/2 CV	electronic	7.0 L/min	162	98.5	1.66	does not reach

At 18 m head, five of the six models deliver more than the toilet's 9 L/min, and all stay on for the same 110 hours a year. What separates them is the power: the 1/4 hp one uses 40 kWh, the 1/2 hp ones, 67. The sixth, with little more than 7 L/min at 18 m, works almost at the end of its curve, stays on longer and also adds ten seconds of delay to each of the 7,373 starts of the year.

At 24 m head the order reverses. The 1/4 hp model drops to 4.4 L/min, and the time switched on doubles; three models do not even reach the pressure. The best becomes a 1/2 hp one. The rule the table teaches is to choose the smallest model that delivers the point's flow rate at the working pressure, and it is what item 6.5.11.12 of ABNT NBR 5626:2020 asks for: that the energy consumption of pumping and pressurisation pumps be minimised by the choice of pump, among the alternatives considered. The screen points to that model and shows the energy of all the others next to it.

> **Two pumps, not one.** Item 6.5.11.2 of the same standard requires at least two pumps, operating independently, in pumping and pressurisation systems. The tutorial follows the standard, and the investment counts the duty pump and the standby. EN 16941-1:2024 asks for a standby pump «if necessary».

2.6.8. Why the energy of pumping to a tank is small, and that of the pressure pump is not

Lifting one cubic metre twelve metres costs about 0.033 kWh of hydraulic energy. With the two efficiencies, it comes to 0.11 kWh per cubic metre delivered, and the tutorial's pumping, with the start-ups, to 0.13 kWh/m³.

Direct pressurisation of the same project uses 0.68 to 1.66 kWh/m³, depending on the model: five to thirteen times more. The numbers reproduce a difference the literature had already measured. In the review by Vieira et al. (2014), the median of the theoretical studies was 0.20 kWh/m³, and that of the empirical ones, 1.40 kWh/m³, because the former ignored the start-ups, standby mode and the real efficiency of motor and pump. The same review reports 1.70 kWh/m³ to feed a toilet cistern directly with a 450 W pump, and a mean of 1.52 kWh/m³ in nineteen Australian homes with direct feed.

The tutorial's pumping stays close to the theoretical median because it works as the theoretical studies assume: few starts per year, at the pump's own flow rate. The pressure pump falls in the empirical range because it works as the measured homes do: one start per use, at the flow rate of the point.

Even so, the energy is small compared with the saving on the bill. The 40 kWh of the smallest pressure pump cost less than thirty reais a year at the tutorial's tariff. The argument that «pumping uses more than it saves» does not hold in small buildings, but the choice of arrangement and model can multiply the energy part by ten, and it is that part that weighs in the carbon of section 2.8.

In a tall building, with thirty metres of pumping, the calculation changes, and it is worth redoing it.

2.7. Economic analysis and tariff regimes

> This section formalises the cash flow and details the six tariff regimes -- the data that most decides the viability of a rainwater system in Brazil.

2.7.1. The saving is the difference between two bills

The program does not calculate the saving by multiplying the volume saved by a mean tariff, and the distinction is not one of refinement: it is one of order of magnitude. For each month of the projection it builds two complete bills, one with the consumption the building would have without the rainwater system and another with the potable water actually bought, and takes the difference between them.

$$E_{\mathrm{mês}} = F(V_{\mathrm{tot}}) - F(V_{\mathrm{pot}}) \tag{38}$$

Table 58 -- Variables of Equation 38

Symbol	Meaning	Unit
`E_mês`	saving obtained in the month	R$
`F(.)`	function that calculates the water and sewage bill for a volume	R$
`V_tot`	monthly volume the building would consume without the rainwater system	m³
`V_pot`	monthly volume of potable water actually bought from the mains	m³

The reason for proceeding this way lies in the progressiveness of Brazilian tariff structures. The cubic metre no longer consumed is always the last one on the bill, which is the most expensive, and never the mean one. In a municipality whose first block costs R$ 3.08/m³ and whose second block costs R$ 14.31/m³, calculating the saving by the mean tariff underestimates it by more than half, and the error grows with the consumption of the building.

2.7.2. The water and sewage bill

The tariff structure of each municipality is represented in the program by a sequence of blocks, and each block is a quadruple made up of the minimum volume, the maximum volume, a value and a code. The code is what gives meaning to the value, because apparently similar structures charge in very different ways.

Before the codes, it is worth seeing what the bill is made of. The monthly water and sewage bill is the sum of parts that behave differently: a fixed part, which does not respond to consumption; one or more consumption blocks, in which the price per volume applies to the volume allocated within each block; and sewage, which in much of Brazil is a percentage of the water charge and therefore follows everything it does. Saving water affects each of these parts in a different way, and it is this asymmetry that Figure 24 shows.

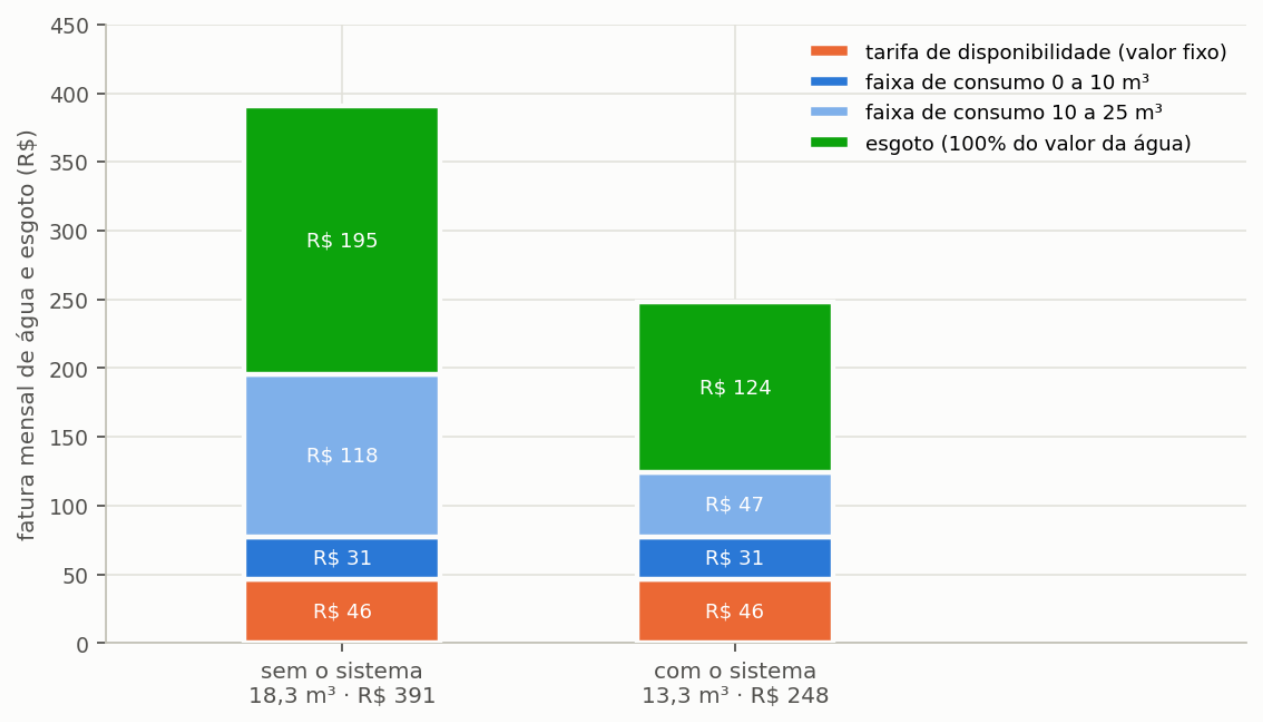


Figure 24 -- Monthly water and sewage bill of the tutorial project, broken down into parts, without and with the system. The availability charge is identical in both cases; the reduction occurs in the upper consumption blocks, which have the highest price per cubic metre, and extends to sewage, charged as a percentage of the water charge.

The code of each block defines how its value is applied (Figure 25). In the consumption blocks, code 1 or 0, the price per volume is charged on the volume allocated within each block; in the availability charge, code -1, the fixed value is added to the consumption part; in the retroactive allowance, code -2, consumption above the allowance makes the whole volume be billed from 0 m³; in the payment floor, code -3, the declared value is the lower limit of the bill; and in the exponential formula, code 2 holds the base price and the code 3 rows hold the table of exponents.



Figure 25 -- The six tariff block codes and the effect of each one on the monthly water and sewage bill. Two structures with the same numerical values result in different bills depending on the code of each block.

Two of these regimes are often confused with each other, although they have opposite effects on the viability of a rainwater system. It is worth examining them with numbers.

- **The availability charge adds.** A consumption part of R$ 55 plus R$ 46 of availability charge gives a bill of R$ 101, and saving R$ 15 of consumption leaves the bill at R$ 86.
- **The floor limits from below.** With a floor of R$ 40 and a billed consumption of R$ 55, saving up to R$ 15 **reduces nothing**.

The difference between the two regimes decides whether each cubic metre saved earns a discount from the first one or whether the saving only starts to count after a threshold is passed, and a viability study that confuses them reaches inverted conclusions. That is why the program codes the regime explicitly instead of treating it as a detail of the table.

2.7.2.1. The exponential formula

In the CORSAN municipalities the water part is not a sum of blocks but a power of the month's consumption:

$$T_{\text{agua}} = \text{PB} \times C^{n} + \text{SB} \tag{39}$$

Table 59 -- Variables of Equation 39

Symbol	Meaning	Unit
`T_agua`	water part of the bill	R$
`PB`	base price of the municipality	R$/m³
`C`	consumption of the month per dwelling unit	m³
`n`	exponent from the Table of Exponents for consumption `C`: 1.00 up to 20 m³, 1.13 between 101 and 150 m³	dimensionless
`SB`	basic service charge	R$

The exponent is read from the **total** consumption of the month and applied to the whole volume. In the residential category there is no cap; only the industrial category, above 1,000 m³, charges the excess at prices of its own, and the program represents that cap with an ordinary block from there on. The monthly saving is the difference between the two bills, `PB x (C■■¹ - C2■²)`, each consumption with its own exponent, and not the product of a unit price by the volume saved; with `n` above 1, the cubic metre saved is worth more than the base price.

2.7.2.2. Individual metering

In multi-family buildings with individual metering, consumption is evaluated at the level of a single dwelling unit and the result is multiplied by the number of units. The treatment matters because of progressiveness: a building of twenty flats treated as a single consumer would fall entirely into the most expensive block of the table, and the resulting saving would come out grossly overestimated, because each cubic metre saved would be valued at a tariff none of the twenty units actually pays.

2.7.2.3. Sewage

$$T_{\text{esg}} = T_{\text{agua}} \times s \tag{40}$$

Table 60 -- Variables of Equation 40

Symbol	Meaning	Unit
`T_esg`	sewage part of the bill	R$
`T_agua`	water part of the bill, already calculated by the progressive blocks	R$
`s`	sewage percentage charged by the utility	dimensionless

The sewage percentage applies to the water part of the bill, not to the total bill, and confusing the two denominators doubles or halves the part calculated. One hundred per cent, the value charged by many Brazilian utilities, means that the sewage part equals the water part, and therefore that sewage accounts for half of the total bill.

There is a third mode, CORSAN's: sewage charged **per cubic metre consumed**, `T_esg = C x p_esg`, without the basic service charge and without the exponent of the water part. Applying the percentage on the water part there would charge sewage on the basic service and on the power, and the bill would come out higher than the utility charges.

2.7.3. The six regimes in the database

The catalogue spreads its 652 municipalities very unevenly across the six regimes. The allowance within the minimum alone accounts for **329** of them, more than half; then come 165 with no minimum at all, 122 with an availability charge and 34 on the CORSAN exponential formula, whereas the retroactive allowance appears in only two and the payment floor in none (Figure 26). The table below shows how each regime is coded and applied in the calculation of the bill.

Table 61 -- Coding and application of each minimum-charge regime in the calculation of the bill

Regime	Coding of the blocks	Calculation of the water part	Effect of reducing consumption
no minimum charge	consumption blocks only (code 1 or 0)	sum, in each block, of the price per m³ applied to the volume allocated within it	every reduction in consumption reduces the bill
availability charge	fixed value (-1) and consumption blocks from 0 m³, with a positive price	fixed value added to the consumption part, billed from the first m³	every reduction in consumption reduces the bill; the fixed value remains
allowance within the minimum	fixed value (-1) and a first block from 0 to N m³ at R$ 0.00/m³	fixed value that includes the N m³ of the allowance; above them, consumption blocks	a reduction below N m³ does not change the bill
retroactive allowance	allowance (-2) from 0 to N m³ with a fixed value, and consumption blocks	up to N m³, fixed value; above, the whole volume billed from 0 m³ by the blocks	reducing consumption to N m³ or less replaces the consumption part by the fixed value
payment floor	floor (-3) and consumption blocks	consumption part by the blocks, limited from below to the value of the floor	a reduction that takes the part below the floor does not reduce the bill
exponential formula	basic service (-1), base price (2) and table of exponents (3) covering all consumption	basic service added to PB x C■, with n from the whole month's consumption	every reduction in consumption reduces the bill, by the difference between the two powers

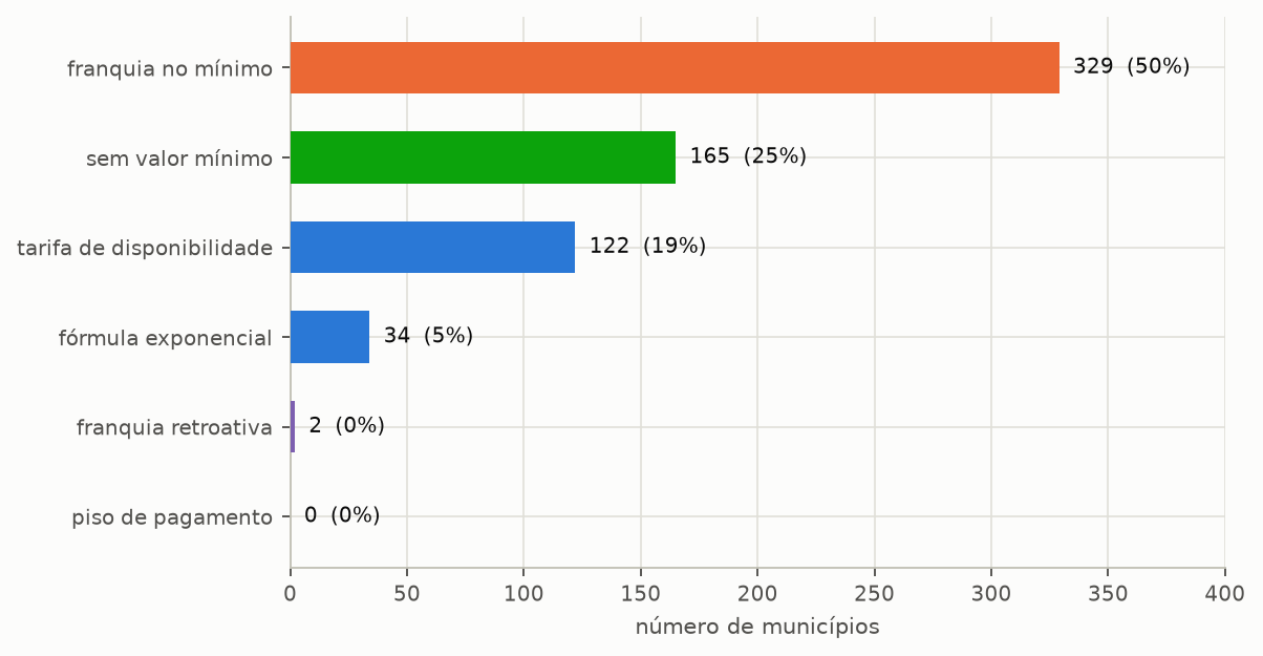


Figure 26 -- Distribution of the 652 municipalities of the catalogue among the six tariff regimes. Half of them apply an allowance within the minimum, a regime in which reducing consumption below the allowance does not reduce the bill.

The payment floor is implemented in the calculation engine and no municipality of the database has so far been classified under it, a situation recorded here for transparency: the absence of occurrences does not mean the regime does not exist in the country, only that the survey did not find it among the 652 municipalities catalogued.

Section 3.3 presents the numerical example comparing Floriano and Florianópolis, from which follows the most important practical lesson of this section: under an allowance regime, saving water below the allowance volume earns no discount at all on the bill.

2.7.4. The investment

$$\mathrm{CAPEX} = C_{\mathrm{cist}} + C_{\mathrm{sup}} + C_{\mathrm{mo}} + C_{\mathrm{tub}} + C_{\mathrm{ace}} + (C_{\mathrm{bomba}} \times n_{\mathrm{b}}) + \mathrm{CAPEX}_{\mathrm{inc}} \tag{41}$$

Table 62 -- Variables of Equation 41

Symbol	Meaning	Unit
`CAPEX`	total initial investment of the system	R$
`C_cist`	cost of the lower tank	R$
`C_sup`	cost of the upper tank, when there is one	R$
`C_mo`	cost of installation labour: in a new project, 30% of the materials, with a floor of R$ 1,400	R$
`C_tub`	cost of the piping: pipes, bends and fittings of the 3D layout	R$
`C_ace`	cost of the accessories: the parts of the 3D layout, one by one, and the control cables	R$
`C_bomba`	unit cost of the pump set	R$
`n_b`	number of pump sets, including the standby when there is one	unit
`CAPEX_inc`	incremental investment of the treatment line, including the first flush when sized	R$

In a new project, `C_tub` and `C_ace` are not allowances: they come from the **bill of quantities of the 3D layout** (section 3.9). Each metre of pipe, each bend, each end fitting, each part -- valves, check and foot valves, solenoid, water meter, level switches, control panel, inlet calming device, floating suction, vents, overflow, air gap, point of use -- and each metre of control cable has a price, and the sum is what enters here. Cistern, upper tank and pump set have terms of their own; the inlet filter and the first flush enter through the treatment line when it sizes them. The designer can type other values in the economic step; the project then uses the typed ones, and the audit spreadsheet records the change. Projects saved before this rule keep the costs they declare.

The labour of a new project is also recalculated at each run: 30% of the sum of cistern, upper tank, pump sets, piping and accessories, with a floor of R$ 1,400 (key `labor_percentage` of `costs_database.json`). It thus grows with the tank: in the volume sweep, each volume has its own cistern, its own usual layout and its own labour. Previously, labour was fixed once, on generic allowances for piping and accessories, and was the same for all volumes. A value typed in the economic step prevails, as in the other items.

The last term follows the incremental rule of section 2.5 and includes, when the physical sizing of the first flush is enabled, the first-flush chamber and the leaf separator; the annual running cost of that subsystem enters the running cost of the treatment line.

Tank prices come from the program's cost database, with the source declared for each material. When the requested volume is not in the table, the program interpolates linearly between the neighbouring points; when the volume exceeds the largest tabulated value, extrapolation is only done if the regression over the known points has a coefficient of determination above 0.90, and the program warns when that criterion is not met, instead of extrapolating silently.

The make-up of the investment surprises those who come to the subject expecting the tank to dominate the bill. In small residential volumes it weighs less than labour and piping, and only comes to dominate when the volume grows enough for the price of the tank to exceed the fixed sum of the installation. The design consequence is direct: doubling the tank does not double the investment, and that non-linearity is part of the reason why the net present value curve has an interior maximum (Figure 27).

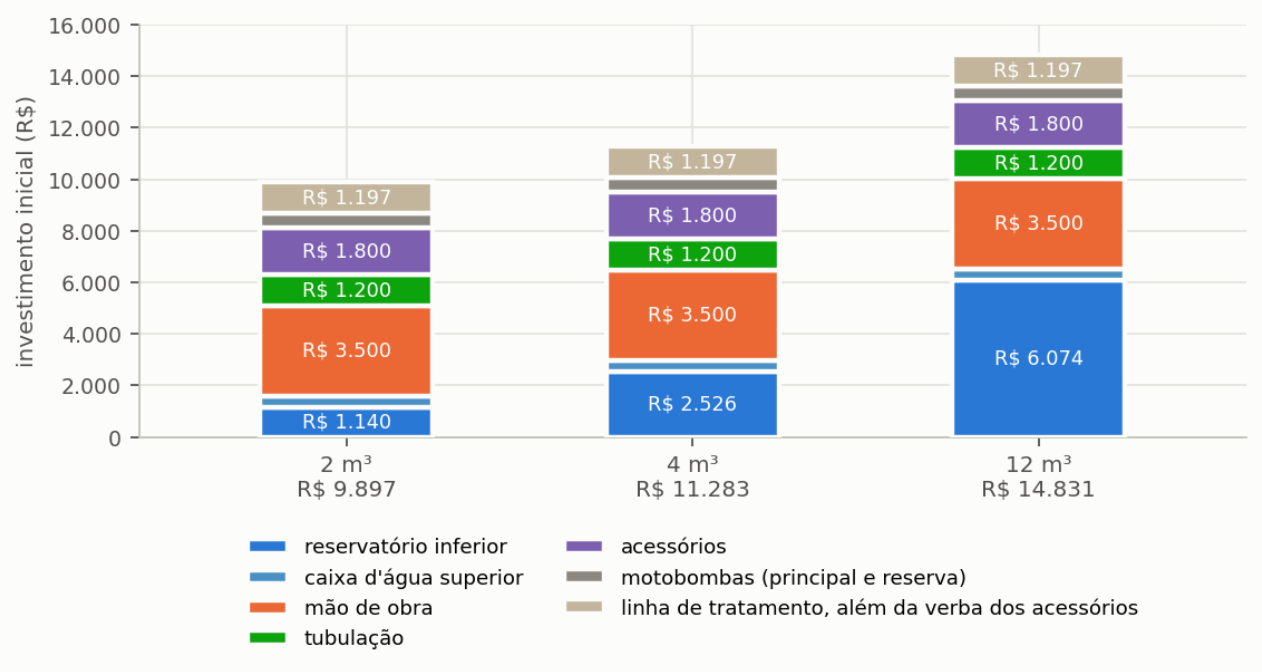


Figure 27 -- Initial investment broken down into the seven items the program adds up, for three volumes of polyethylene tank. With the costs the tutorial declares, only the lower tank grows with the volume; upper tank, labour, piping, accessories, pump sets and treatment line do not change. The treatment line enters for what it adds beyond the filtering allowance already built into the accessories, so as not to count the filter twice. In a new project, piping and accessories come from the 3D layout, which has no such allowance, and labour follows the materials of each volume.

2.7.5. The annual running cost

Table 63 -- Parts of the annual running cost and the origin of each one

Part	Origin
pumping energy	section 2.6, with the tariff flag applied
chlorination	mass of **product**, not of active chlorine
treatment plant	incremental running cost, without counting chlorination twice
first flush	annual maintenance of the first-flush chamber and of the leaf separator, when sized
declared operating costs	the user's free list

The tariff flag (*bandeira tarifária*) adjusts the electricity tariff according to the operating condition of the Brazilian power system, and the program accepts the four current modes: yellow, red level 1, red level 2 and white, the last with a discount in off-peak periods.

2.7.6. The month-by-month projection

The projection is made month by month over the whole analysis period, and applies to each part the inflation and adjustment factors defined below.

The result of this projection is a monthly series of balances, and what matters in it is the moment the cumulative balance stops being negative. That moment is the payback time. Figure 28 aggregates the projection by year: in columns, the saving on the water and sewage bill and the costs, with the investment in year zero; as a line, the discounted cumulative balance, which crosses zero at the payback time.

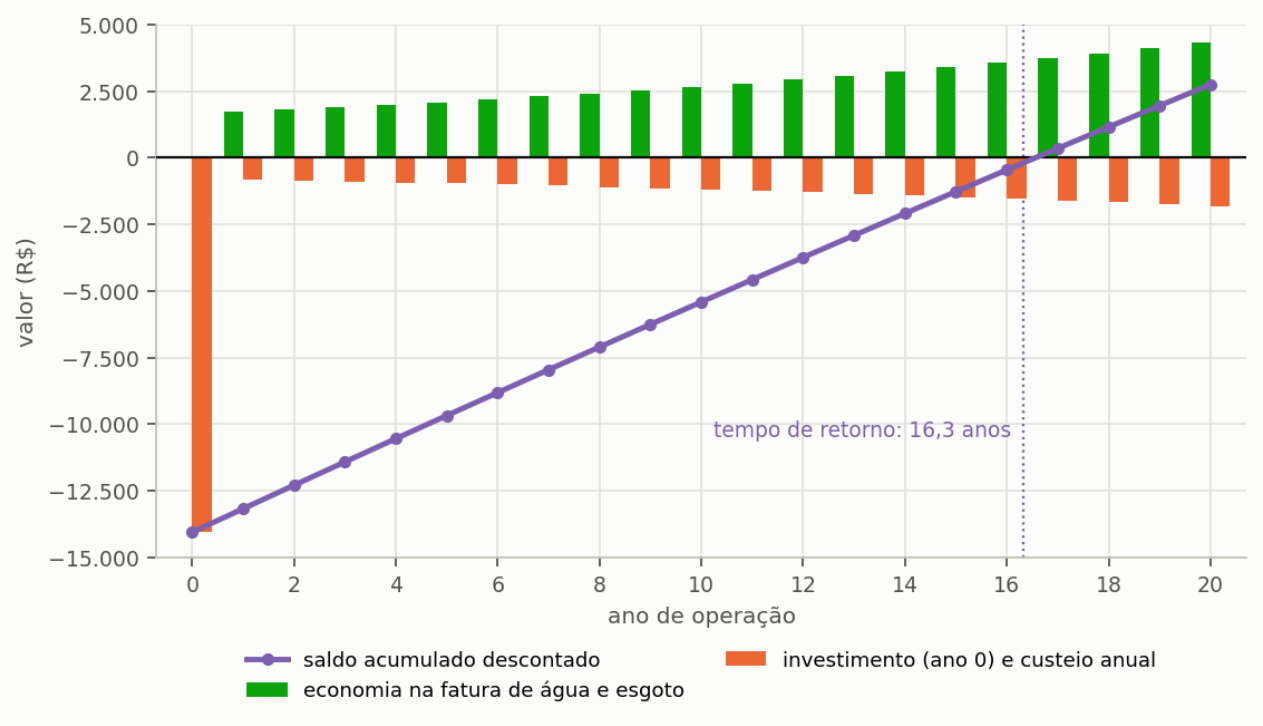


Figure 28 -- Cash flow of the tutorial project over twenty years: annual saving on the water and sewage bill and annual costs, with the initial investment in year zero, in nominal values; and cumulative balance discounted at the minimum attractive rate of return, which crosses zero at the payback time of 16.3 years. 4,000 L concrete cistern with an upper tank, as in section 4.7.

Inflation applies only to the running cost:

$$f_{\mathrm{inf}} = (1 + i)^{m - 1} \tag{42}$$

Table 64 -- Variables of Equation 42

Symbol	Meaning	Unit
`f_inf`	cumulative inflation factor up to month `m`	dimensionless
`i`	declared monthly inflation rate, as a fraction: 0.35% a month enters as 0.0035	dimensionless
`m`	month of the cash flow, counted from 1	month

Tariff adjustments apply only to the bill, in steps:

$$k = \lfloor (m - 1) \div I \rfloor \tag{43}$$

$$f_{\mathrm{j}} = (1 + r_{\mathrm{j}})^{k} \tag{44}$$

Table 65 -- Variables of Equation 44

Symbol	Meaning	Unit
`k`	number of adjustments already applied up to month `m`	dimensionless
`m`	month of the cash flow, counted from 1	month
`I`	interval between adjustments, typically 12	month
■ ■	integer part of the division	--
`f_j`	cumulative adjustment factor of part `j`	dimensionless
`r_j`	adjustment rate of part `j`, which can be water, sewage or energy	dimensionless

Equation 44 is applied separately to each of the three parts, with rates of their own, and the following remark is about that separation.

> **Water and sewage are adjusted by separate indices**, not by a single index on the aggregate bill. The two parts are the subject of separate acts of the utility, and treating them together indexes half the bill by the wrong index.

The green property tax incentive (*IPTU verde*), when the municipality offers it and the user enables it, enters the cash flow as an annual saving divided by twelve, which spreads it evenly over the months of the year. The simplification is deliberate: the discount is granted all at once, when the tax is assessed each year, but spreading it keeps the cash flow from showing a step that has nothing to do with the performance of the system.

2.7.7. The indicators

Net present value:

$$\mathrm{VPL} = -\mathrm{CAPEX} + \sum [\mathrm{FC}_\mathrm{m} \div (1 + \mathrm{TMA})^{m}] \tag{45}$$

Table 66 -- Variables of Equation 45

Symbol	Meaning	Unit
`VPL`	net present value of the investment	R$
`CAPEX`	total initial investment, from Equation 41	R$
`FC_m`	net cash flow of month `m`	R$
`TMA`	monthly minimum attractive rate of return, as a fraction: 0.5% a month enters as 0.005	dimensionless
`m`	month of the cash flow, from 1 to the end of the analysis period	month

Internal rate of return: the rate that makes the NPV zero. When there is no real root -- which happens when the flow never becomes positive -- the program returns null, instead of an invented number.

Discounted payback: the first month in which the discounted cumulative balance crosses zero.

2.7.8. The fragile assumptions

Two of them deserve a sensitivity analysis, and section 3.7 shows how.

The tariff adjustment. Five per cent a year for twenty years multiplies the tariff by 2.65; three per cent, by 1.81; seven per cent, by 3.87. The saving of the last years grows in the same proportion, and that is why the adjustment is among the assumptions that most move the net present value, as the tornado of section 3.7 shows; it is also one of the most uncertain.

The service life. Twenty years is the default. A concrete cistern lasts longer; polyethylene components exposed to the sun last less. The service life enters both the analysis period and the carbon payback of section 2.8.

2.8. Life cycle assessment

> This section presents the seven stages of the inventory and the boundary adopted, and declares what is measured and what is a screening assumption.

Life cycle assessment is the method that accounts for the environmental impacts of a product over its whole existence, from the extraction of raw materials to final disposal. Applied to a rainwater harvesting system, it answers whether the carbon emitted to manufacture the tank, the piping and the pump set is offset by the carbon no longer emitted by not consuming water from the utility.

Netuno's assessment is a screening one, not a full inventory. The distinction matters and the standard recognises it: a screening assessment ranks alternatives and identifies what dominates the result, whereas an inventory conforming to ISO 14040 documents each elementary flow with traceable provenance and undergoes external critical review. The first serves to decide between polyethylene and concrete; the second, to declare an environmental performance publicly.

2.8.1. The four phases of the standard, and where the program stands

The ISO 14040 and ISO 14044 standards, published in Brazil as ABNT NBR ISO 14040 and ABNT NBR ISO 14044, organise every life cycle assessment into four phases that feed back into one another. It is worth going through them, because each one corresponds to a decision the program has already made for you, and that a published study needs to declare.

1. Goal and scope. Defines what the study is for, the **functional unit** and the system boundary. The functional unit is the quantity of service being compared, and it is what makes different alternatives comparable. In the program, the implicit unit is the **harvesting system of a project over the adopted service life**, and the comparison is made against the supply of the same volume of water by the utility. Anyone wanting to compare with the literature can convert it to the most used unit, the cubic metre of water delivered, by dividing the balance by the volume of the service life, as Equation 51 does.

2. Inventory. Surveys the input and output flows of each process: masses of material, energy, consumables. In the program, the inventory is the masses of the components, the pumping energy, the mass of chlorine product and the volume of water replaced.

3. Impact assessment. Converts the flows into category indicators by characterisation factors. The program assesses **one** category, climate change, in kilograms of carbon dioxide equivalent, with the emission factors of appendix C.

4. Interpretation. Identifies what dominates the result, tests the sensitivity to the assumptions and draws conclusions within the limits of the scope. That is what sections 4.8 and 3.7 do with the tank material and with the other assumptions.

2.8.2. The building modules

For buildings, the European standard EN 15978 subdivides the life cycle into modules, and the same nomenclature is used in environmental product declarations according to EN 15804. Declaring which modules the study covers is the shortest and most precise way of declaring the boundary, and it is what a reviewer expects to find.

Table 67 -- EN 15978 life cycle modules and what the program's assessment covers in each one

Module	Stage	Covered by the program
A1 to A3	product: raw material, transport to the factory, manufacture	yes, for tanks, piping, pump sets, first flush and all the parts, fittings and cables of the 3D layout
A4 and A5	construction: transport to the site and installation	no; the excavation for the buried cistern is left out
B1 to B5	use, maintenance, repair, replacement and refurbishment	no; pump and component replacements do not enter
B6	operational energy	yes, the pumping energy
B7	operational water	yes, as the credit for the potable water avoided and the treatment consumables
C1 to C4	end of life: demolition, transport, processing and disposal	no
D	benefits and loads beyond the boundary, such as recycling	no

The reading of the table is that the program does a **cradle-to-gate assessment, plus operation**. It is the most common boundary in studies of rainwater harvesting in buildings, but not the only one, and comparing the program's balance with that of a study that includes end of life or replacements requires adjusting the boundaries before comparing the numbers.

2.8.3. The seven stages

The inventory is organised into seven stages, grouped by the nature of the emission (Figure 29). The carbon balance is the sum of the emissions of the first six minus the credit of the seventh, the only term with a negative sign, and it is this credit that decides whether the system offsets the carbon embodied in its construction.

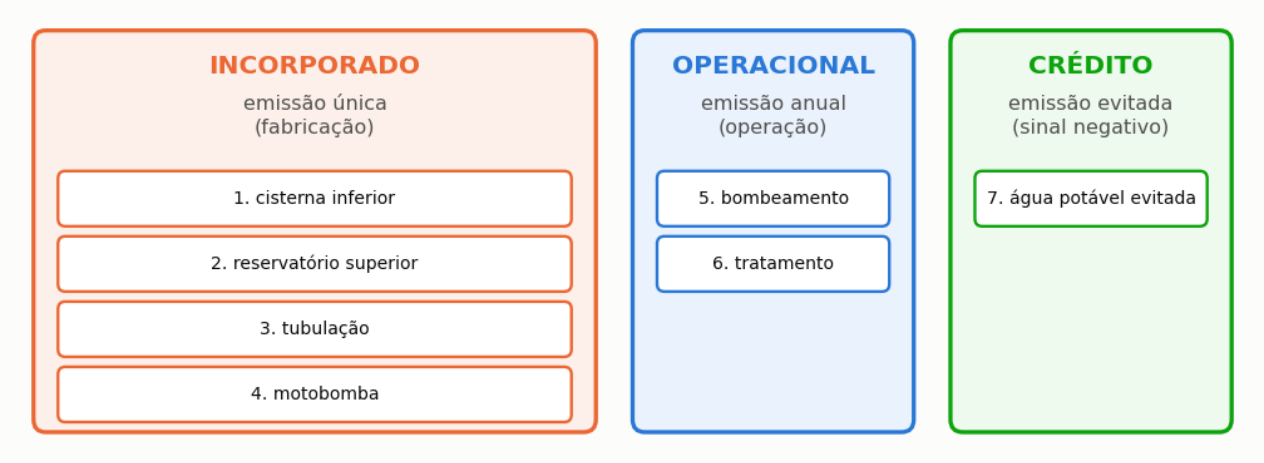


Figure 29 -- The seven stages of the inventory, grouped by the nature of the emission. The first four are embodied and occur only once; the fifth and sixth are operational and repeat every year; the seventh is the credit, and it is the only one that enters with a negative sign.

The first four stages account for the **embodied carbon**, that is, the emissions that occurred in manufacturing the components, before the system goes into operation; when the physical first flush is sized, the first-flush chamber and the leaf separator are added to them. The next two account for the operational emissions, which repeat in each year of use. And the seventh is the **credit**: the emission the utility no longer generated because that water did not have to be abstracted, treated and distributed.

Volume zero without rainwater consumption represents the no-system scenario and, by construction, receives no fixed impact of tank, piping or pump set -- it is the reference against which all the other alternatives are compared.

2.8.4. Embodied carbon

$$E_{\text{comp}} = m_{\text{comp}} \times \text{FE}_{\text{mat}} \tag{46}$$

Table 68 -- Variables of Equation 46

Symbol	Meaning	Unit
`E_comp`	embodied emission of the component	kg CO2e
`m_comp`	mass of the component	kg
`FE_mat`	emission factor of the material, cradle to gate	kg CO2e/kg

The mass of each component follows from its sizing. For the tank, it is obtained from the useful volume multiplied by the characteristic mass density of the material, which varies by more than an order of magnitude between polyethylene and reinforced concrete -- and a good part of the environmental result of the project follows from that variation.

Table 69 -- Mass per useful cubic metre of each tank material

Material	Mass per useful m³	What this means in 5 m³
polyethylene	22 kg/m³	110 kg
fibreglass	15 kg/m³	75 kg
stainless steel	38 kg/m³	190 kg
reinforced concrete	1,200 kg/m³	6,000 kg

> **Mass is carbon.** Reinforced concrete is by far the most massive material on the list, and that is why it dominates the inventory when chosen. Section 4.8 shows the consequence: with concrete, the tutorial system does not pay off its carbon debt within the service life.

2.8.4.1. The inventory of the tutorial project, item by item

The table below breaks down the carbon of the final tutorial project: a 4,000 L polyethylene cistern, a 1,000 L upper tank, two pump sets, the standard piping and the first flush.

Table 70 -- Carbon inventory of the final tutorial project, item by item, with the share of each component in the embodied carbon; negative annual values are credits

Item	Embodied	Share of embodied	Annual
4,000 L cistern, polyethylene	141.9 kg	43.3%	--
1,000 L upper tank, polyethylene	35.5 kg	10.8%	--
PVC piping, DN 100 mm	55.5 kg	17.0%	--
PVC piping, DN 32 mm	3.5 kg	1.1%	--
PVC piping, DN 25 mm	8.9 kg	2.7%	--
two half-hp pump sets	61.0 kg	18.6%	--
first flush: 300 L chamber and leaf separator	21.1 kg	6.5%	--
pumping energy	--	--	0.66 kg/year
tablet chlorination	--	--	0.24 kg/year
potable water avoided	--	--	-20.94 kg/year
total	**327.4 kg**	**100%**	**-20.05 kg/year**

Three readings matter. The first is that, even in polyethylene, the cistern accounts for almost half of the embodied carbon, and the tanks together for more than half. The second is that the piping and the pump sets, which are usually forgotten in simplified calculations, add up to almost two fifths of the total: a study that counted only the tank would underestimate the embodied carbon by almost half. The third is in the annual column: the credit of the water avoided is about 23 times larger than the annual emissions of pumping and chlorination together, and that is why, with the right material, the manufacturing debt pays off.

2.8.5. Operation

Pumping:

$$E_{\text{ener}} = C_{\text{ano}} \times \text{FE}_{\text{rede}} \tag{47}$$

Table 71 -- Variables of Equation 47

Symbol	Meaning	Unit
`E_ener`	annual emission of the pumping energy	kg CO2e/year
`C_ano`	annual energy consumption of the pump set	kWh/year
`FE_rede`	emission factor of the electricity grid, by region and year	kg CO2e/kWh

The annual energy consumption comes from the calculation presented in section 2.6, according to the supply arrangement declared, and the grid emission factor comes from one of the program's databases. The default database, at 0.085 kg CO2e/kWh, is a declared choice among the values published for the Brazilian grid: it is close to the SIDAC and CaDI values, and above the annual MCTI series, which covers only the carbon dioxide of generation. The alternative databases include the values of Vaz, Scolaro and Ghisi (2025) for the 2022 grid, 0.104 kg CO2e/kWh for the fossil indicator and 0.151 for the total indicator, useful for the sensitivity analysis. In the International Edition the factor is typed in.

Energy is small in the program, but it is worth knowing that it tends to be larger in practice. Vieira et al. (2014), in a review of the energy intensity of rainwater harvesting systems, found a median of 0.20 kWh/m³ in theoretical studies and of 1.40 kWh/m³ in empirical studies, and attributed part of the difference to start-ups, standby consumption and real efficiencies worse than catalogue ones. The tutorial, at 0.13 kWh/m³, is in the range of theoretical studies. Ghimire et al. (2014, 2017) reached a converging conclusion by life cycle assessment: in the

systems they analysed, the pump set and the pumping energy were the components that weighed most in most of the impact categories, and the pumpless version outperformed the public supply in almost all of them.

> **When energy decides.** With the Brazilian grid factor, multiplying the tutorial's energy by ten, taking it close to the empirical median of Vieira et al., would add about 6 kg CO2e a year to the emissions and reduce the annual balance from 20.1 to about 14.1 kg CO2e. The annual balance would remain positive, but the carbon payback would go from 16.3 to about 23 years, beyond the twenty years of service life adopted: the system would end its life owing carbon. On grids more fossil-based than the Brazilian one, the effect is even larger.

Treatment: the treatment line **replaces** the chlorination term instead of adding to it. It is the same rule as in section 2.5, and it exists so that the hypochlorite is not counted twice.

2.8.6. What the boundary includes, and what it leaves out

Before discussing any emission factor, it is worth fixing the scope. The program's assessment is a **screening** one: it serves to rank alternatives against each other, not to replace a life cycle inventory conforming to ISO 14040. What enters the calculation and what is left out is declared, because an environmental balance without a declared boundary can be neither checked nor contested (Figure 30). The most limiting exclusion is that of the impact categories beyond global warming: a system with good carbon performance can perform unfavourably in another category, and the assessment does not record it.

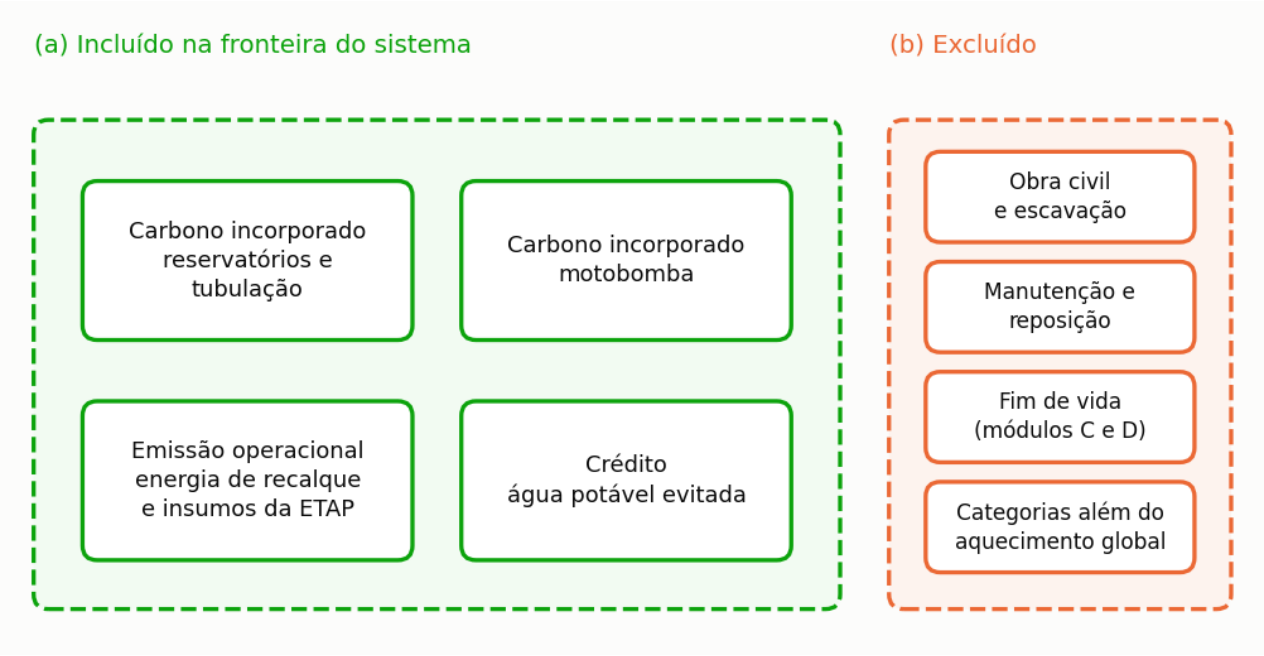


Figure 30 -- Boundary of the program's environmental assessment: (a) processes included -- embodied carbon of the tanks, the piping and the pump set, operational emissions of pumping and of treatment consumables, and the credit for the potable water avoided; (b) processes excluded -- civil works, maintenance and replacement, end of life and impact categories beyond global warming.

2.8.7. The boundary: water delivered, not produced

The emission factor of potable water is per cubic metre **delivered to the building**, not per cubic metre produced at the treatment works.

The difference is the distribution losses, which in Brazil range from **33.9% in the Southeast to 46.9% in the North**. Producing one cubic metre delivers between 0.53 and 0.66.

A published factor at the outlet of the works, applied to the volume delivered, **underestimates the avoided emission by up to a third**. The program makes the conversion and declares the macro-region adopted.

Appendix C lists the 19 water production routes available, with the provenance and boundary of each one.

2.8.8. The indicators

Net annual balance:

$$E_{\text{liq}} = E_{\text{evit}} - E_{\text{ener}} - E_{\text{trat}} \tag{48}$$

Table 72 -- Variables of Equation 48

Symbol	Meaning	Unit
`E_liq`	net annual emission balance	kg CO2e/year
`E_evit`	emission avoided by the utility water no longer consumed	kg CO2e/year
`E_ener`	annual emission of the pumping energy	kg CO2e/year
`E_trat`	annual emission of the treatment consumables	kg CO2e/year

Carbon payback:

$$\text{PB}_{\text{amb}} = E_{\text{inc}} \div E_{\text{liq}} \tag{49}$$

Table 73 -- Variables of Equation 49

Symbol	Meaning	Unit
`PB_amb`	environmental payback time of the system	year
`E_inc`	carbon embodied in the manufacture of the components	kg CO2e
`E_liq`	net annual emission balance, from Equation 48	kg CO2e/year

When the net annual balance is not positive -- a real situation, not a hypothetical one, in systems with a large concrete tank -- there is no environmental payback, because the debt is never paid off. The program returns 999 years in that case, instead of infinity, so that the value can still be ranked in comparisons; the meaning, however, is that the system ends its service life owing carbon.

Balance at the end of life:

$$S_{\text{vida}} = (E_{\text{liq}} \times n_{\text{anos}}) - E_{\text{inc}} \tag{50}$$

Table 74 -- Variables of Equation 50

Symbol	Meaning	Unit
`S_vida`	carbon balance at the end of the adopted service life	kg CO2e
`E_liq`	net annual emission balance, from Equation 48	kg CO2e/year
`n_anos`	service life adopted for the system	year
`E_inc`	carbon embodied in the manufacture of the components	kg CO2e

Negative means the system **emitted more than it avoided** over the adopted life. It is not a theoretical hypothesis: it happens with a concrete cistern and a twenty-year service life, as section 4.8 shows.

Specific avoided emission:

$$E_{\text{esp}} = S_{\text{vida}} \div V_{\text{vida}} \tag{51}$$

Table 75 -- Variables of Equation 51

Symbol	Meaning	Unit
`E_esp`	specific emission of the system over the service life	g CO2e/L
`S_vida`	carbon balance at the end of the service life, from Equation 50	kg CO2e
`V_vida`	total volume of rainwater delivered over the service life	m³

2.8.9. What carries a screening label

Not every factor of the inventory has a primary source. The program marks those that do not, and the report shows the mark.

The masses per useful cubic metre of polyethylene, concrete and stainless steel, which are still geometric scenarios, carry the label, as does any factor typed in without a source. The default emission factors have declared provenance; the last to receive it was that of **hypochlorite**, which now comes from Ecoinvent 3.8: as the database has no calcium hypochlorite, the program uses the sodium one on an active-chlorine basis, 2.65 kg CO2e per kg, which gives 1.72 kg CO2e per kg of 65% tablet. The manufacturing emission of the pump set, which until the September 2026 revision was an arithmetic extrapolation in four of the six power ratings, now combines the catalogue mass with the intensity of an environmental product declaration; it is an order-of-magnitude approximation, and appendix C declares the caveat.

> **In a published study, declare which factors carry a screening label.** An inventory that presents all factors with the same degree of confidence hides where the weakness lies.

2.8.10. What the literature shows

Life cycle assessment of rainwater harvesting systems is an active field without a single method. Vaz et al. (2023), in an integrative review conducted under the PRISMA protocol, concluded that residential buildings are the most studied case, but that there is no consensus on a uniform analytical framework; that several benefits of rainwater, such as runoff reduction, rarely enter the assessment; and that integration with life cycle costing is a promising way of showing the benefits more directly.

For the Brazilian case, Marinoski and Ghisi (2018) assessed a social housing unit in Florianópolis and compared fibreglass, high-density polyethylene and reinforced concrete tanks. The fibreglass ones were the most favourable from the environmental point of view, and the concrete ones, from the point of view of investment. The result converges with that of section 4.8 in the essentials: the tank material weighs on the carbon, and the environmental and economic dimensions can point in different directions.

Ghimire et al. (2017), comparing a commercial system with the municipal supply in eleven impact categories, showed something else this assessment does not reach: the rainwater system outperformed the municipal one in ten of the eleven categories, but not in all, and the result of each category depended on the energy intensity of pumping. It is the strongest argument for declaring that the program's assessment covers a single category.

2.8.11. What the assessment does not cover

- **end of life.** There is no credit for recycling nor burden for disposal. The boundary is cradle to gate plus operation.

- **excavation and civil works.** The earthworks of the buried cistern do not enter.
- **maintenance.** Component replacements over the service life do not enter.
- **impacts beyond carbon.** There is no eutrophication, acidification, land use or toxicity. It is an assessment of **one** impact category.

The last point is the most important. A system with good carbon performance can perform poorly in another category, and this assessment would not see it.

2.9. Certifications

> This section gathers the qualification rules of five sustainability rating systems and distinguishes, in each one, what is a normative requirement from what is a criterion adopted by the program.

Rainwater harvesting systems appear in practically every environmental certification system for buildings, and in each one with a metric of its own. Some measure the saving against the total consumption of the building, others against a reference baseline, and others still require the comparison in more than one dimension at once. The program calculates the qualification in five of these rating systems from the simulation results, and always with the same caveat: the program shows where the project would qualify, and certifies nothing. Certification is decided by the certification body, with the complete documentation of the development, of which water performance is only one piece.

In the International Edition, the program shows LEED, EDGE and GRI only; the two Brazilian rating systems, AQUA-HQE and Selo Casa Azul, are described here because the Brazil Edition evaluates them.

Before going through them one by one, it is worth seeing them side by side. The thresholds of each rating system are on scales that seem incomparable while they stay in separate tables, and become comparable when placed on the same saving ruler (Figure 31). LEED v4 scores the reduction of outdoor use (WEc1: 1 point from 50% and 2 points at 100%) and of indoor use (WEc2: 1 to 6 points between 25% and 50%) separately; LEED v5 gathers the category in credit WEc2, in which the reduction in fixtures and fittings is worth 1 to 3 points between 30% and 40% and the reduction of the whole project, 1 to 8 points between 30% and 65%; AQUA-HQE Residential awards item EAU.1.2 from 20% and EAU.1.3 from 40%; and EDGE requires a minimum reduction of 20% at the same time in water, energy and the energy embodied in materials, without points.

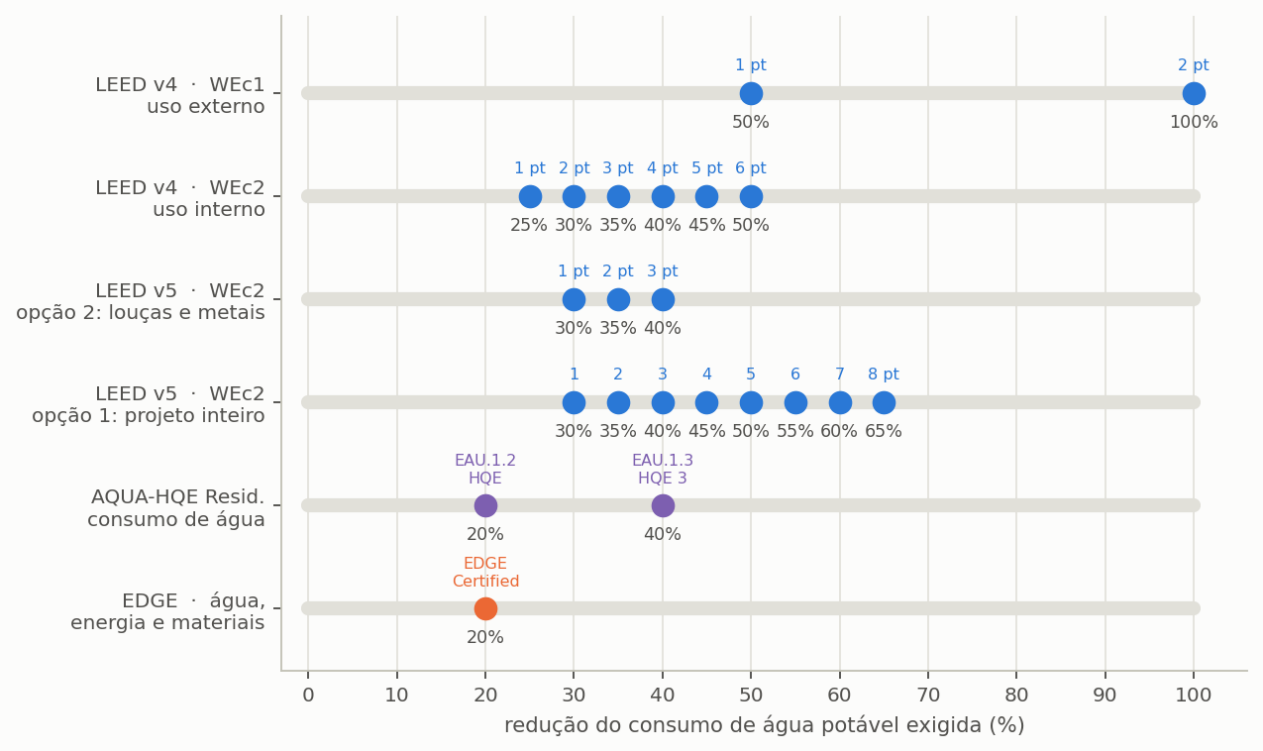


Figure 31 -- The thresholds of reduction in potable water consumption with which the program places the project in LEED v4 (credits WEc1 and WEc2), LEED v5 (credit WEc2, options 1, 2 and 4), AQUA-HQE Residential and EDGE, on the same scale. Selo Casa Azul is not on the ruler, because its criterion scores the existence of the system and the scope of use, not a percentage. The same project crosses different rating systems at different points, which is why declaring the saving obtained is worth more than declaring the certification intended.

2.9.1. LEED v4 and v4.1, Water Efficiency category

LEED is the North American rating system most widespread internationally. Version 5, launched in April 2025 for commercial buildings, coexists with versions 4 and 4.1, which accept registration of new commercial projects until 30 June 2027 and certification until 30 June 2033; the program places the project in both. In versions 4 and 4.1, water efficiency has a category of its own worth eight points spread over two credits. The fundamental distinction it imposes is between indoor and outdoor water use, and the whole score follows from that separation.

Credit WEc1 deals with the reduction of outdoor use, that is, irrigation and the other demands of uncovered areas, and is worth up to two points. The program awards the two points with a 100% reduction in outdoor potable water use and one point from 50%, taking the rainwater allocated to irrigation from the daily balance itself; and a project with no outdoor demand registered does not score, in which case the program records the reason explicitly, so that the user does not read the zero as a performance failure.

Credit WEc2 deals with the reduction of indoor use and is worth up to six points, spread on a scale of five percentage points at a time from a 25% saving. It is preceded by a prerequisite, WEp2, which requires a 20% indoor reduction and is mandatory: without it the development is not eligible for certification, and it awards no points itself.

> The separation between indoor and outdoor use depends entirely on the **breakdown of the demand** described in section 4.2. The program does not invent that separation: it uses the rainwater the daily balance allocated to irrigation, which is supplied after indoor use. Without a broken-down outdoor demand, credit WEc1 appears as not assessed, and the indoor saving is taken, conservatively, as equal to the total saving.

> The score label gives only the points of the water category. The LEED level -- Certified, Silver, Gold or Platinum -- results from the sum of all the categories, and water alone does not define it.

2.9.2. LEED v5, credit WEc2 Enhanced Water Efficiency

In version 5, checked against the *LEED v5 BD+C Rating System* of November 2025, the indoor and outdoor use credits were gathered in credit WEc2, *Enhanced Water Efficiency*, worth up to eight points in new construction and up to seven in *Core and Shell*; the program assesses only new construction. Code WEc1 now designates another credit, *Water Metering and Leak Detection*, of submetering and leak detection, and should not be confused with the outdoor-use WEc1 of LEED v4. Prerequisite WEp2 requires, among other conditions, a 20% reduction in the consumption of fixtures and fittings relative to the rating system's baseline, and prerequisite WEp1 requires the alternative water source, such as the rainwater cistern, to have a meter of its own, separate from the mains one. The project scores by one of two paths: Option 1, which measures the reduction in water use of the whole project, or the sum of Options 2 to 6, which treat each part separately.

Table 76 -- Options of LEED v5 credit WEc2 to which rainwater harvesting contributes

Option	What it measures	Score
1. Whole project	reduction in the project's water use against the baseline	30% 1 point, and 1 point every 5%, up to 65% 8 points; with an alternative water source, each step is worth 1 point more, up to 8
2. Fixtures and fittings	reduction calculated against the WEp2 baseline, which may include an alternative source	30% 1 point, 35% 2 points, 40% 3 points
4. Outdoor use, path 2	reduction of the theoretical irrigation requirement	50% 1 point, 100% 2 points
6. Reuse, path 2	on-site reuse system serving irrigation, flushing, make-up or process	2 points

The program calculates both paths and adopts the one with the higher score: Option 1 with the total saving, and the sum of Options 2, 4 and 6 with the indoor saving, the outdoor saving and the existence of the system, which the program treats as the reuse system of Option 6. The cap of eight points applies to both. Options 3 and 5, for appliances and process water, and path 1 of Option 4, landscaping without irrigation, do not depend on rainwater harvesting and are left out of the calculation. As in LEED v4, the rating system's reductions are measured against a baseline of fixture flow rates and irrigation, not against the modelled demand, so the qualification is indicative.

Rainwater harvesting can also contribute to credit SSc3, *Rainwater Management*. Under Option 1, it awards 1, 2 or 3 points for retaining on site the volume corresponding to the 80th, 85th or 90th percentile of local rainfall (70th, 75th or 80th on a zero-lot-line site), and 1 additional point when part of that volume is reused, without exceeding the cap of 3 points; under Option 2, it awards 3 points to whoever retains the increase in runoff relative to the natural land cover. The program does not assess this credit, because it depends on the runoff of the whole site, not only of the roof; the criteria matrix records it as not assessed.

2.9.3. AQUA-HQE Residential, water consumption theme

AQUA-HQE is the Brazilian adaptation, led by Fundação Vanzolini with CERQUAL Qualitel, of the French HQE rating system. In the current residential buildings scheme, version 00 of 22 November 2024, water is dealt with in the «Water consumption» theme, by items, and each item is classified into a level: prerequisite (PR), HQE, HQE 2 or HQE 3. Three items concern rainwater harvesting:

Table 77 -- Items of the water consumption theme of AQUA-HQE Residential related to rainwater harvesting

Item	Level	Requirement
EAU.1.2	HQE	reduction of at least 20% in potable water consumption relative to the reference consumption
EAU.1.3	HQE 3	saving of 40% of potable water in the development
EAU.1.5	HQE 3	rainwater recovery system, according to ABNT NBR 15527 and 16783

The program compares the simulated potable water saving with the thresholds of 20% and 40%, and records item EAU.1.5 whenever there is a system. The program's percentage is indicative: the rating system measures the saving against a reference scenario of flow rates and uses per device, calculated in the Fundação Vanzolini auxiliary spreadsheet, which adds the effect of water-saving fixtures to that of non-potable water.

> **Correction in this version.** Until the September 2026 revision the program placed the project in «Base», «Good Practice» and «Excellent» profiles, at 10, 25 and 40% saving. The 2024 scheme has no such profiles, and the qualification was redone by items.

2.9.4. Selo Casa Azul + CAIXA, criterion 3.6

Selo Casa Azul is the housing rating system of Caixa Econômica Federal, the Brazilian federal savings bank. In the current guide, version V023 of 29 April 2026, rainwater harvesting is criterion 3.6, of the «Efficient water management» category, and its indicator is not a percentage: it is the **existence** of a harvesting system, with catchment, treatment, storage and points of use shown in the design. The score depends on the scope of use:

Table 78 -- Score of criterion 3.6 of Selo Casa Azul + CAIXA according to the scope of use

Scope of rainwater use	Points
common areas of the development	2
houses in a subdivision, with the required provision	2
dwelling units	4
dwelling units and common areas	5

The criterion is optional for the Cristal/Bronze, Topázio/Prata and Safira/Ouro levels, and mandatory for Diamante. For the single dwelling, Selo Casa Azul Uni applies, version V001 of December 2025, in which criterion 7, «Rainwater harvesting», is worth 1 point and is mandatory for the Ouro seal. The program records the criterion as met when there is a system, and awards the points only when the scope is given.

> **Correction in this version.** Until the September 2026 revision the program graded the project as «Prata», «Ouro» and «Diamante» at 10, 20 and 30% saving. The guide does not link the criterion to a percentage, and the qualification was redone by existence and scope.

> **Declare it in the study.** The qualification is only defensible with the current version of the rating system in hand, and the program does not replace that check.

2.9.5. EDGE, the only one that does not score

EDGE, developed by the World Bank's International Finance Corporation, follows a logic different from all the previous ones. Instead of awarding points to increasing performance, it measures the percentage saving against a local baseline in three independent dimensions -- energy, water and the energy embodied in materials -- and requires **each** of them to cross the established threshold. There is no compensation between dimensions: an exceptional water performance does not make up for a deficiency in energy.

EDGE has two further levels above the first. **EDGE Advanced** requires, in addition to the 20% in the three dimensions, an on-site energy saving of at least 40%; **Zero Carbon** requires the previous one plus 100% renewable energy or full offsetting of the remaining emissions.

It is worth noticing which bar the advanced level raises. EDGE Advanced raises the energy requirement, not the water one. The practical consequence for those designing rainwater systems is that the system secures the water bar with relative ease, whereas the certification level achieved is decided entirely outside the scope of the water design. The program assesses only the water bar, by measure WEM14, whose input in the IFC application is the catchment area in square metres.

> **The honest limit is in the denominator.** The EDGE baseline is local, varies with the country and the building type, and is set by the IFC application itself. The program accepts the baseline declared by the user; in its absence, it adopts the modelled demand of the project and warns that the resulting percentage is a **floor**, not the value certification will recognise.
>
> The energy and embodied energy bars are beyond the program's reach, and it declares that limitation in the report. Claiming that a project is EDGE Certified with a third of the necessary evidence would be irresponsible.

The thresholds above were taken from the IFC document *EDGE Certification Process*, and the codes of the water measures were checked against an EDGE v3.0.0 assessment report of a project already certified.

2.9.6. GRI 303

The Global Reporting Initiative is not a building certification system but a standard for corporate disclosure of social and environmental performance. Its 303 series deals with water and effluents, and two of its indicators are fed directly by the program's results: GRI 303-3, which reports the abstraction avoided in cubic metres per year and accumulated over the period, and GRI 303-5, which expresses the water circularity rate of the development.

2.9.7. What to declare

Any study reporting qualification in certification systems needs to make four points explicit, and omitting any of them compromises the verifiability of the result. The first is that the water quality classes 2 to 4 adopted by the program are internal adaptations, not classes of ABNT NBR 15527. The second concerns EDGE, and consists in declaring that only the water bar was assessed, identifying the baseline used. The third applies to LEED, and consists in declaring the version of the rating system adopted and stating whether the split between indoor and outdoor use was broken down from project data or whether, in its absence, the indoor saving was taken as equal to the total saving. The fourth, finally, is the general caveat: the qualification the program produces is indicative, and certification is decided by the certification body in the light of the complete documentation of the development.

2.10. Multi-criteria decision

> This section formalises the two ranking methods the program offers and declares three departures from the canonical formulation that a published study needs to cite.

Multi-criteria analysis is the branch of operations research that deals with choosing between alternatives when the evaluation criteria disagree with one another and cannot be reduced to a single number without someone declaring what is worth more. That is exactly the situation of a rainwater harvesting project: the volume that delivers most water is not the one that yields most money, and neither is usually the one with the lowest carbon impact.

What the methods of this section do is not discover the right answer. They make explicit the weighting that would otherwise remain implicit in the choice of whoever decides. That is the whole difference between a study that declares its values and one that hides them behind an apparently objective result.

2.10.1. The seven criteria

The program evaluates each volume of the sweep by seven criteria, grouped into three dimensions. Each criterion has a direction of preference: in some, more is better; in others, less.

Table 79 -- The seven decision criteria, the direction of preference and the group of each one

#	Criterion	Direction	Group
1	saving potential	maximise	technical
2	overflow	minimise	technical
3	net present value	maximise	economic
4	discounted payback	minimise	economic
5	internal rate of return	maximise	economic
6	CO2e at end of life	minimise	environmental
7	potable water consumption	minimise	environmental

The alternatives submitted to the ranking are the volumes simulated in the sweep, and each of them appears with the seven values above, calculated in sections 2.1, 2.7 and 2.8. No criterion is invented at this stage: the method only ranks what the previous dimensions have already produced.

2.10.2. TOPSIS

TOPSIS -- *Technique for Order of Preference by Similarity to Ideal Solution*, proposed by Hwang and Yoon in 1981 -- starts from a simple geometric idea. If each alternative is a point in a seven-dimensional space, one dimension per criterion, then there are two reference points: the **ideal solution**, made up of the best value of each criterion, and the **anti-ideal solution**, made up of the worst. No real alternative occupies either of them, because none wins at everything or loses at everything, and the best alternative is the one that is at the same time closest to the ideal and furthest from the worst case.

The method carries out five steps. It builds the decision matrix, with the alternatives in the rows and the criteria in the columns; normalises that matrix, so that criteria measured in reais, in percentage points and in kilograms come to occupy comparable scales; applies the declared weights; identifies the ideal and anti-ideal solutions; and ranks the alternatives by relative closeness to the ideal.

$$C_{\mathrm{i}} = d_{\mathrm{i}}¤ \div \left(d_{\mathrm{i}}¤ + d_{\mathrm{i}}¤\right) \tag{52}$$

Table 80 -- Variables of Equation 52

Symbol	Meaning	Unit
`C_i`	closeness coefficient of alternative `i`, between 0 and 1	dimensionless
`d_i■`	Euclidean distance from alternative `i` to the ideal solution	dimensionless
`d_i-`	Euclidean distance from alternative `i` to the anti-ideal solution	dimensionless

The coefficient of Equation 52 would be 1 for an alternative that coincided with the ideal solution on all criteria, and 0 for one that coincided with the anti-ideal. As no real alternative occupies these extremes, what matters is not the absolute value of the coefficient but the ranking it produces.

Applied to the full sweep of the tutorial project, from 1 to 20 m³ in steps of 1 m³ with equal weights, the score rises to an interior maximum and falls at the larger volumes, where the investment and the embodied carbon come to weigh more than the gain in water (Figure 32). The maximum, at 7 m³, differs from the 5 m³ obtained in section 4.9 with six candidates, because the ideal and the anti-ideal are built from the set evaluated: adding or removing alternatives shifts the reference points and can change the ranking, a phenomenon known as rank reversal.

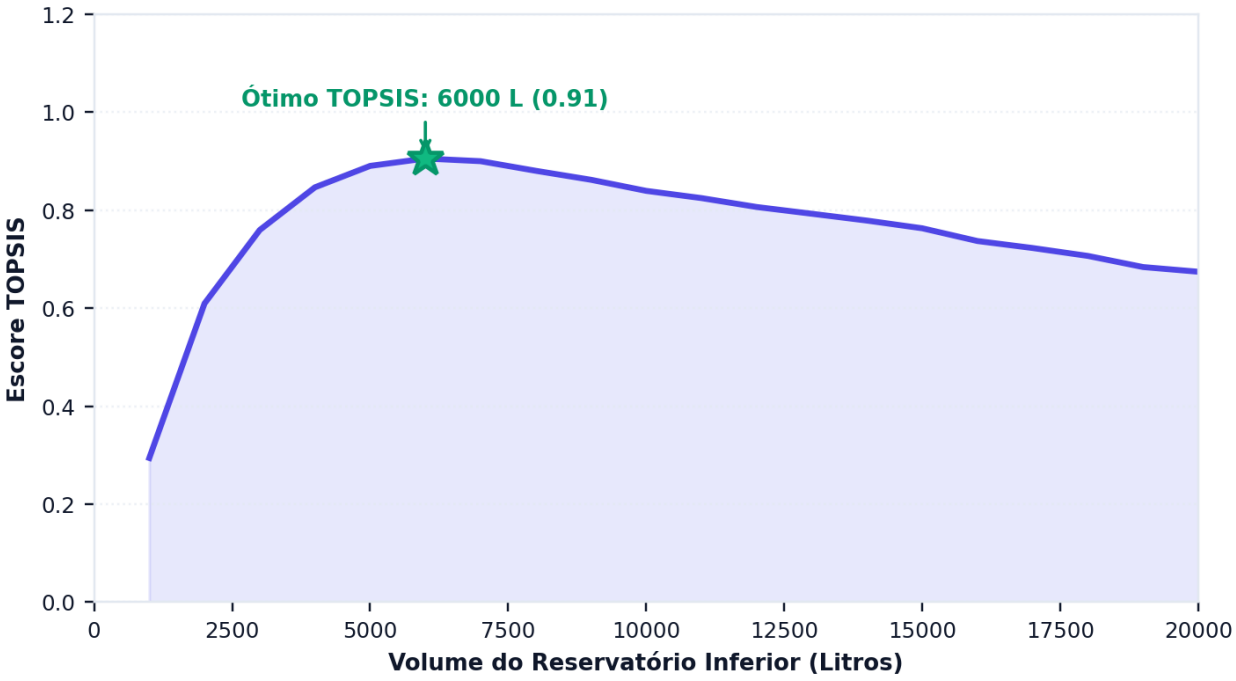


Figure 32 -- TOPSIS score of each volume of the sweep of the tutorial project, from 1 to 20 m³, with a polyethylene cistern and equal weights among the seven criteria; the star marks the volume with the highest score. Figure generated by the program itself.

2.10.3. Three departures to declare

The program's implementation departs from the canonical formulation in three points, all deliberate, and a published study needs to declare them so as not to attribute to the original methods a behaviour that belongs to the program.

The normalisation is a variant. Hwang and Yoon's canonical formulation uses vector normalisation, which assumes non-negative values on all criteria. The net present value, however, is often negative at small volumes, where the investment does not pay off. The program works around the restriction by translating each criterion by `x - min(x)` before normalising. The choice is necessary and deliberate, but presenting the result as canonical TOPSIS would be incorrect: declare the variant.

Opricovic's conditions are not implemented. The VIKOR method, in Opricovic's formulation, requires two checks on the compromise solution found -- acceptable advantage and acceptable stability -- and, when either fails, returns a **set** of equivalent solutions instead of a single winner. The program always returns the best alternative, without carrying out the test. In practice, this means that the VIKOR compromise solution should be read alongside the weight sensitivity analysis, and never alone.

The attribution of the code. The implementation of VIKOR and SMAA-VIKOR was written independently and checked against the `crispyn` library, by Aleksandra B■czkiewicz. TOPSIS was checked against Hwang and Yoon's original formulation in a case with all values positive, in which the translation described above is harmless and the two formulations must coincide.

2.10.4. SMAA-VIKOR

SMAA -- *Stochastic Multicriteria Acceptability Analysis* -- exists to answer the question TOPSIS does not: what if the weights were different? Instead of requiring a set of weights defined beforehand, the method draws thousands of weight combinations and returns, for each alternative, the probability of it occupying each position of the ranking.

Reading the result is direct and informative. A volume that wins in 80% of the draws is a robust recommendation, which does not depend on a specific weighting to hold. A volume that wins in 30% is a fragile recommendation, and the text of the study needs to say so instead of presenting the winner as if it were consensual.

> It is the right tool when there is no consensus on the weights -- which is the normal case in real projects, where the client, the designer and the environmental agency usually value different dimensions.

The most useful reading of the stochastic analysis is not the winner but the margin by which it wins. A volume that comes first in three quarters of the weight draws is a stable choice, which survives a change in the decision-maker's priorities; two technically tied volumes indicate the opposite, and in that case declaring the weight adopted stops being a formality and becomes the only information supporting the recommendation (Figure 33).

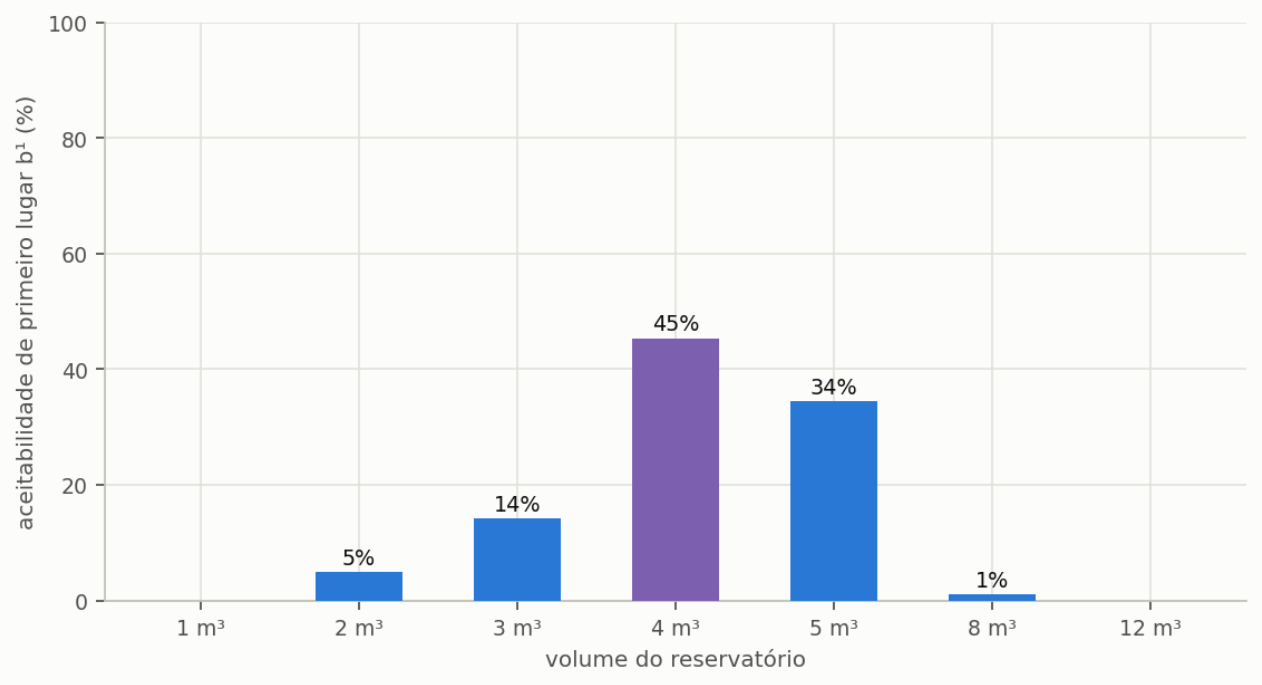


Figure 33 -- First-rank acceptability (b^1) of each candidate volume, obtained with five thousand weight vectors drawn on the simplex. The highlighted volume is the one that most often comes first, and the others share the rest.

2.10.4.1. The tutorial, by both methods

The table sets side by side the TOPSIS score, with equal weights among the three dimensions, and the SMAA-VIKOR first-rank acceptability, with five thousand weight vectors drawn, for the seven volumes of the tutorial in polyethylene with an upper tank.

Table 81 -- TOPSIS closeness score with equal weights across the dimensions and first-rank acceptability in SMAA-VIKOR, with 5,000 random weight vectors, for the tutorial volumes

Volume	TOPSIS score	TOPSIS rank	First rank in SMAA-VIKOR, b[1]
1,000 L	0.401	7th	0.0%
2,000 L	0.688	5th	5.0%
3,000 L	0.800	3rd	14.2%
4,000 L	0.854	1st	45.4%
5,000 L	0.847	2nd	34.5%
8,000 L	0.693	4th	1.0%
12,000 L	0.542	6th	0.0%

The two methods agree on the winner, 4,000 L, and disagree on the margin. TOPSIS separates first from second place by seven thousandths of score, a difference no input data supports. SMAA measures that fragility: 4,000 L wins in 45% of the draws and 5,000 L in 35%, and the two together account for four out of every five draws. By the yardstick of section 3.7, the recommendation is reasonable, and the study should declare 5,000 L as the second option. The extremes, 1,000 L and 12,000 L, win in no draw at all: no plausible weighting recommends them, and that is also a result.

2.10.5. The weights

The program accepts one weight per dimension, technical, economic and environmental, or one weight per criterion. With equal weights among the dimensions, each is worth a third, shared equally among its criteria: the two technical and the two environmental ones are worth a sixth each, and the three economic ones, a ninth each. The choice, however, is not technical: it is a statement of value, and therefore needs to appear in the text of the study. The statement «optimal volume by TOPSIS», without the weights, is impossible to reproduce, and two studies with different weights reach different volumes without either of them being wrong.

Three courses are defensible, and the choice between them depends on how much is known about the priorities of whoever decides. Equal weights, explicitly declared as an absence of preference, are the neutral position. Justified weights, with the reason for each one, are preferable when the priorities are known. And SMAA, reporting the range of results instead of a single winner, is the most complete course when there is no consensus.

2.10.6. The best by each criterion alone

Besides the weighted ranking, the program identifies the winner of each group of criteria -- technical, financial and environmental -- and reports the three separately. They are numbers worth publishing together, because each one answers a legitimate and different question.

When the three coincide, the recommendation is strong and the decision needs no weighting. When they diverge, which is the common case and that of the tutorial, the divergence **is** the result: hiding it behind a single weighted number impoverishes the study and hands the reader a conclusion they have no way of evaluating.

2.10.7. What the method does not do

It is worth ending with the limits, because multi-criteria methods are often presented with an authority they do not have.

They do not choose the weights. The weighting belongs to the analyst, and it is a decision of value, not of technique. No statistical procedure replaces it.

They do not validate the criteria. If the net present value is wrong because the municipality's tariff regime was entered incorrectly, TOPSIS will rank the error with impeccable precision. The method ranks what it receives.

They do not incorporate constraints. Space available on the site, maximum budget, a regulatory requirement: all of these are filters prior to the ranking, applied by the designer when deciding which volumes even enter the comparison.

3. Guides available in the program

3.1. Comparing cities

> **Question this guide answers:** would the same project yield the same in another city?

It would not, and the difference does not go in the direction intuition suggests. The common expectation is that more rain means more yield, and that expectation is wrong often enough to deserve a whole section.

> **Edition.** The Brazilian cities of this guide come from the national climate database of the Brazil Edition. In the International Edition the same comparison is made with rainfall files of your own, one per city; everything else in the guide applies as written.

3.1.1. How to make the comparison

The procedure is simple: keep all the parameters of the project and change only the precipitation series. In step **1. Project & Rainfall**, choose another city from the national climate database or load a file of your own, and run the simulation again. To compare several cities without redoing the project for each one, save one file per city, as section 3.6 recommends.

3.1.2. The result

The tutorial project -- 150 m² of catchment, 600 L/day of demand, a 28% suppliable fraction and a 4,000 L cistern -- taken to four Brazilian state capitals produces the table below.

Table 82 -- The same tutorial project simulated in four Brazilian state capitals

City	Annual rain	Days with rain	Ratio between the wettest and the driest month	Potential with 4,000 L	Ideal volume
Florianópolis	1,795 mm	46%	3.2x	27.12%	4,000 L
São Paulo	1,624 mm	42%	4.0x	25.74%	5,000 L
Brasília	1,376 mm	35%	1,191.7x	15.45%	4,000 L
Manaus	2,155 mm	51%	17.3x	20.96%	5,000 L

Figure 34 separates the two quantities, each one on its own scale.

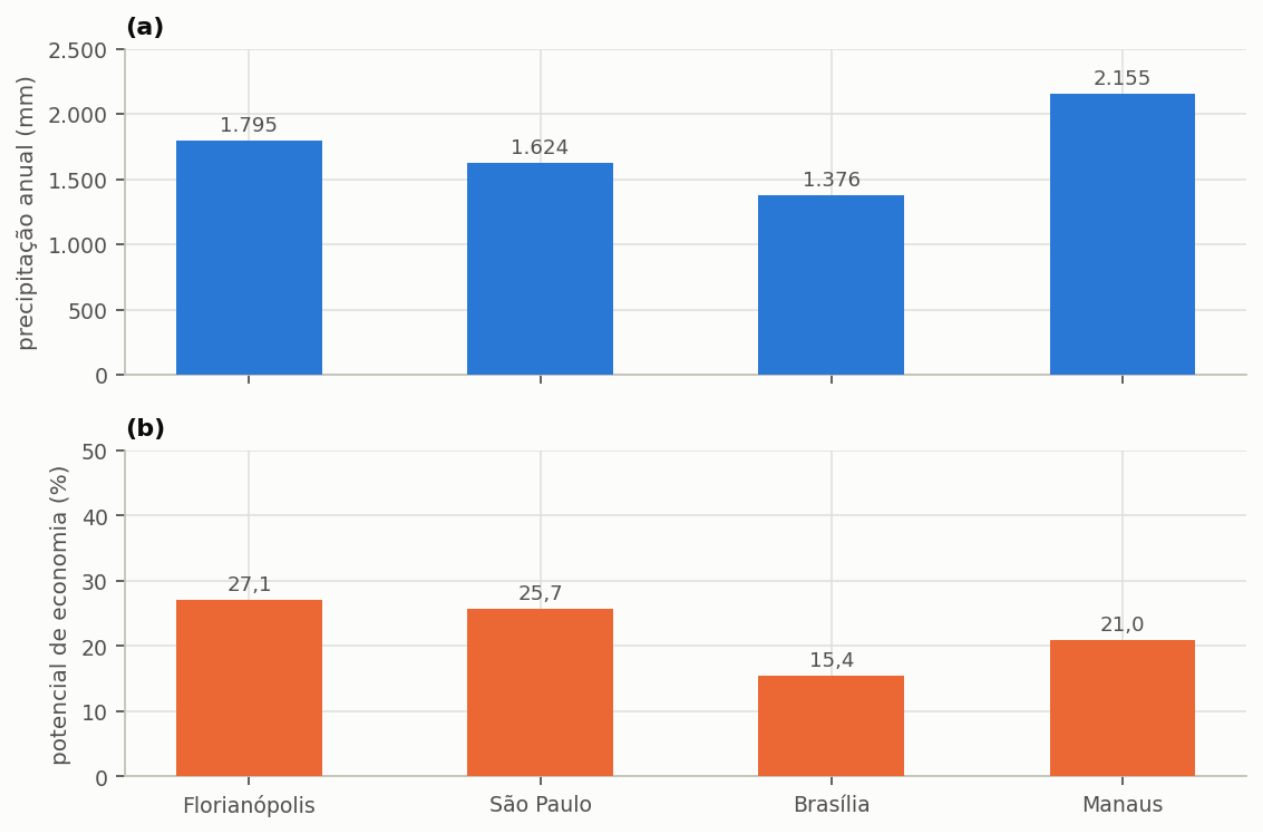


Figure 34 -- (a) Mean annual precipitation and (b) potable water saving potential of the tutorial project, with 150 m² of catchment, 600 L/day of demand, a 28% suppliable fraction and a 4,000 L cistern, in four state capitals. Manaus has the highest annual precipitation and does not have the highest saving potential.

3.1.3. The regime matters more than the depth

Read side by side, the annual rain and saving potential columns do not move together, and the two extreme cases of the table show why. Manaus receives 20% more rain than Florianópolis -- 2,155 mm a year against 1,795 mm -- and still yields 23% less: 20.96% against 27.12% potential. Brasília, at the opposite extreme, receives 23% less rain than Florianópolis and yields little more than half, with 15.45%.

The explanation lies in the ratio between the wettest and the driest month of each city, which measures how much the rain concentrates over the year.

Table 83 -- Ratio between the wettest and the driest month in the four capitals

City	Ratio	What it means
Florianópolis	3.2x	it rains all year round, with a wetter season
São Paulo	4.0x	likewise, slightly more marked
Manaus	17.3x	a short but real Amazonian dry season
Brasília	1,192x	**the driest month is truly dry**

In Brasília, June, July and August record practically no rain. A tank sized to get through ninety days of dry spell would have to store ninety days of rainwater demand, which in this project would be 15,120 litres -- a volume no residential cistern can hold. The consequence is that the system stays idle for a whole quarter every year, and the annual potential reflects that idleness.

> **The practical conclusion.** Rainwater harvesting yields more where rain is frequent than where it is abundant. Cities with a humid subtropical climate and well-distributed rain are the best Brazilian case; cities with a tropical climate and a marked dry season are the worst, however intense the rain of the wet season.

Figure 35 presents the monthly regime of the four capitals on the same scale.

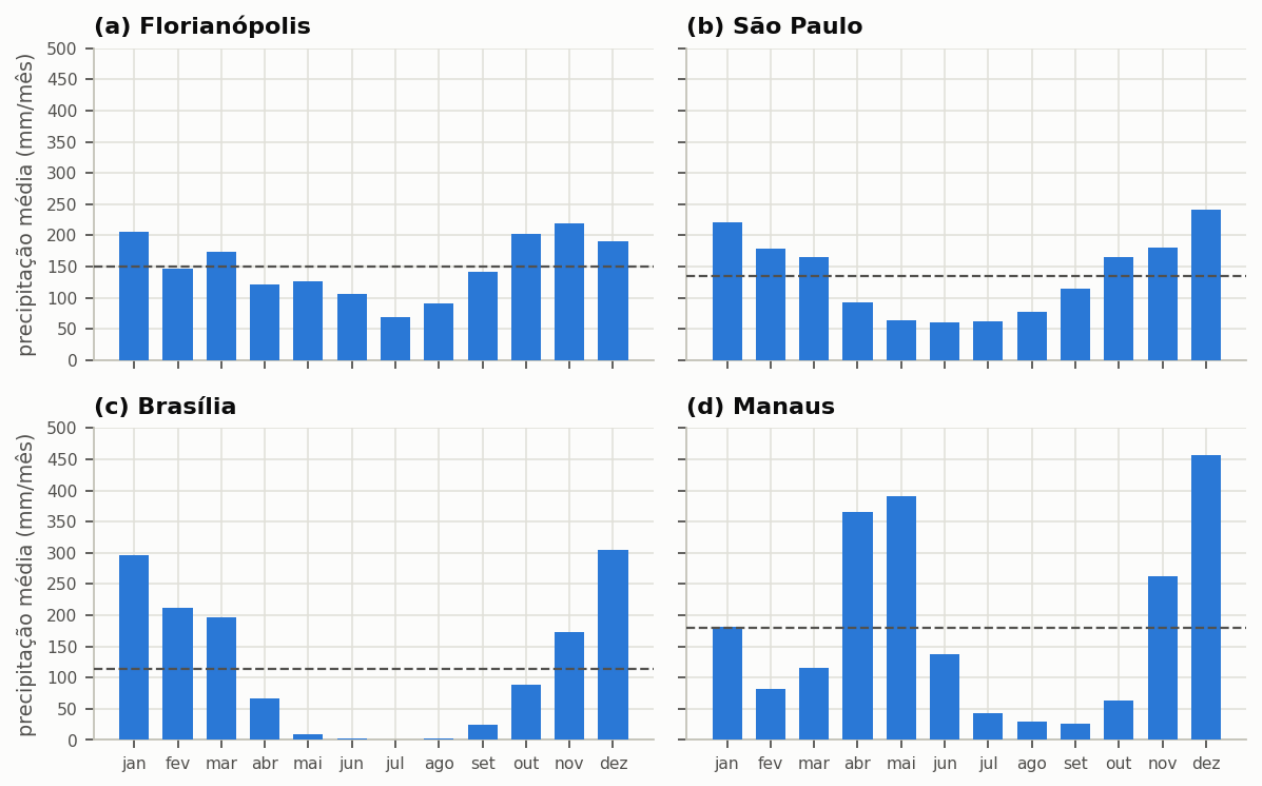


Figure 35 -- Mean monthly precipitation in (a) Florianópolis, (b) São Paulo, (c) Brasília and (d) Manaus, on the same scale; the dashed line is the monthly mean of each series. Florianópolis and São Paulo spread the rain over the whole year; Manaus has a short dry season, from July to October; and Brasília receives practically no rain from May to September. It is this distribution, not the annual total, that decides the performance of the tank.

3.1.4. What the literature had already found

The result of the four capitals is not a peculiarity of this tutorial. Ghisi and Souza (2012) simulated houses in thirteen cities in different countries, with Netuno itself, and reached the same conclusion by another route: the largest saving potential appears where rain is constant, which does not always coincide with the largest annual depth, and cities with a well-defined dry season need larger tanks for the same performance.

In Brazil, two surveys show the range of variation within a single state or a single city. Ghisi, Lopes and Rupp (2016) ran 2,700 simulations for houses in 60 cities of Santa Catarina, with long daily series, and obtained ideal tanks of 1,000 to 16,000 L and mean savings of 75 to 461 litres per day per house. Ghisi and Custódio (2019), with 33,720 scenarios for the residential sector of Joinville, found a mean potential of 18.5% with rainwater only in toilets, and of 40.8% when the washing machine also receives it -- the same choice of use that section 4.2 discusses.

The lesson for anyone comparing cities is the same in both studies: the local series decides, and a volume or a potential brought from another city, even a neighbouring one, does not replace the simulation.

3.1.4.1. Many cities at once

To compare more than three or four cities, changing the series by hand stops being practical. The national climate database has series for 652 municipalities, and the scenario spreadsheet of section 3.6 runs the same project against several of them in a single run, saving one result per row. Keep the same building and tank parameters on all rows of the spreadsheet, and vary only the city: that is what isolates the effect of the climate.

3.1.5. What else changes between cities

Precipitation is not the only quantity that varies between municipalities, and the other two can reverse the economic conclusion of a comparative study.

The first is the **tariff**. The price of a cubic metre varies considerably across the country, and the minimum-charge regime varies even more. A 15% potential in a municipality with a high tariff and an availability-charge regime can yield more money than a 27% potential in a municipality with an allowance on the minimum consumption, and section 3.3 deals with that mechanism in detail.

The second is the **electricity grid emission factor**, which changes at the same time the emission avoided per cubic metre of water replaced and the environmental cost of pumping. The Brazil Edition has the factor broken down by region; in the International Edition it is entered for each project.

> **When comparing cities in a study, make clear what varied.** Comparing only the precipitation, keeping tariff and emission factor fixed, isolates the climate effect and is a legitimate comparison, as long as it is declared as such. Varying all the quantities at once produces a number that cannot be explained, because it adds effects of possibly opposite signs.

3.1.6. The ideal volume also changes, but little

The last column of the table holds a surprise: the ideal volume varies from 4,000 to 5,000 L across the four capitals, a modest range compared with the difference in regime that separates them.

The explanation lies in the nature of the criterion. The ideal volume by marginal gain locates the knee of the yield curve, and the position of that knee is governed by the **typical size of the rain events**, which is reasonably similar across the four cities. What varies strongly is the **plateau** the curve reaches after the knee, not the point at which it bends.

Put another way: in Brasília you get the same volume right as in Florianópolis and harvest considerably less water with it.

3.2. Using a future climate scenario

> **Question this guide answers:** will the system I size today still serve in 2060?

> **Brazil Edition.** The climate projection database described here ships only with the Brazil Edition. In the International Edition, the method of this guide applies as it stands to projected series of your own -- for example, bias-corrected series from a national climate service -- loaded as rainfall files.

A rainwater tank lasts thirty years or more, and the rain that will fall on the building over that period will not be the same as the rain that fell in the recent past. Sizing a long-lived system only on the historical series implicitly assumes a stationary climate -- an assumption the available projections do not support. Netuno brings a database of climate projections precisely so that the sizing can take this question into account explicitly.

3.2.1. What the database has

Table 84 -- Content of the national climate database distributed with the program

What the database holds	Quantity
Brazilian cities	652
CMIP6 climate models	19
emission scenarios	2, SSP2-4.5 and SSP5-8.5
historical simulation	1980 to 2013
future simulation	2015 to 2100

CMIP6 is the sixth phase of the *Coupled Model Intercomparison Project*, the internationally coordinated exercise that feeds the IPCC reports. Each model is a global climate model from a different research centre.

3.2.2. Where the series come from



Figure -- Official logo of CLIMBra (Climate Change Dataset for Brazil), developed for hydroclimatic studies and climate change impacts in Brazil.

The program's database is extracted from **CLIMBra** (*Climate Change Dataset for Brazil*), a reference dataset published by Ballarin et al. (2023) in the journal *Scientific Data* (Nature) (doi:10.1038/s41597-023-01956-z), with public data available in the *Science Data Bank* (doi:10.57760/sciencedb.02316) under the open CC0 licence. Three features define what the series represent, and it is worth knowing them before using them:

They are bias-corrected projections. Global models have a resolution of tens to hundreds of kilometres and reproduce local rain poorly: they tend to get the frequency of rainy days, the intensity and the extremes wrong. CLIMBra corrected the daily series of the 19 models by the *Quantile Delta Mapping* (QDM) method, which adjusts the simulated distribution to the observed one and, at the same time, preserves the relative changes of the quantiles between the historical and the future period. The observed reference was the daily grid of Xavier, King and Scanlon (2016), built from thousands of rain gauges and stations in Brazil, with a resolution of 0.25°, that is, about 28 km.

They are grid cells, not stations. The series of a city is that of the 0.25° cell that contains it. It represents the mean rain over an area of the order of 700 km², not that of a rain gauge. In rugged terrain or on the coast, the difference between the rain of the cell and that of the plot can be large.

The database's history is simulated. The 1980 to 2013 series is not the observed rain, but each model's simulation for that period, corrected to have the same **statistics** as the observation. The chronology does not coincide: a rainy day in the model series does not correspond to a real rainy day. That is why the right comparison for a future scenario is with the **history of the same model**, not with the observed series that comes with the program. Mixing the two would attribute to the future climate the difference between observation and model.

Maraun (2016), in a critical review of bias correction, warns of the limit of the procedure: it presupposes that the model represents the processes and the change of the climate plausibly, and it cannot correct an implausible trend. The correction improves the statistics; it does not turn a bad model into a good one.

> **Database compressed and embedded in the program:** the 37,164 text files of CLIMBra (which originally take up about 6.5 GB as uncompressed CSV) were consolidated and compressed by Netuno 5 into a high-performance binary SQLite database (`climbra.db`). That database is built into the executable and the installer, allowing instant queries (< 1 ms per series) and immediate runs for any of the 652 cities without large external downloads. The original CSV files are not distributed and do not need to be installed.

3.2.3. Choosing model, scenario and window

The **scenario** represents the future path of greenhouse gas emissions, and the two available follow the *Shared Socioeconomic Pathways* adopted by the IPCC. They are not forecasts, but coherent hypotheses about global socioeconomic behaviour over the century (Figure 36). The name combines the socioeconomic pathway, from 1 to 5, with the approximate radiative forcing in 2100, in watts per square metre: SSP2-4.5 is the middle path, with a forcing of 4.5 W/m², and SSP5-8.5 is the fossil-fuel-intensive development path, with 8.5 W/m². Strong mitigation scenarios, such as SSP1-2.6, are not part of CLIMBra.

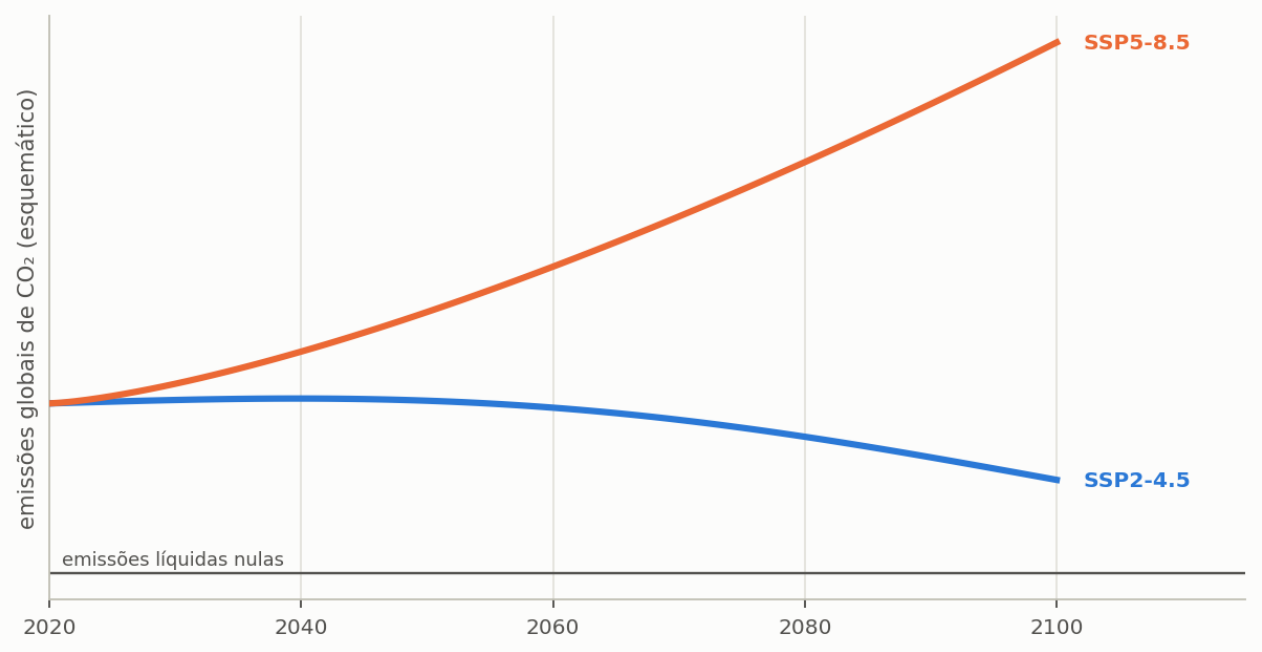


Figure 36 -- The shape of the two emission pathways available in the database. SSP2-4.5 is the intermediate pathway, with emissions that stabilise and then fall; SSP5-8.5 assumes high and growing emissions, without effective mitigation. The figure shows the shape of each one, not projected values.

The **time window** is the slice of the future series, from 2015 to 2100, on which the study is made. The program lets you trim any period, with the start and end dates of step **1. Project & Rainfall**. The most common convention is to take thirty-year windows, the same length as climatological normals, and the **Period shortcut** field fills the dates with three of them: 2021-2050, 2041-2070 and 2071-2100, besides the full series. The shortcut only fills the dates, which remain editable, and the step warns when the slice is shorter than ten years. The choice of window should relate to the intended service life of the system: assessing a twenty-year tank, installed now, against the end of the century assesses a period in which it will already have been replaced.

The choice of the **climate model** is the most delicate of the three decisions, and a subsection of its own is devoted to it further on.

3.2.4. How to conduct the study

The right procedure is to run the same project in each combination of interest and compare the results with one another, instead of sizing directly on the projected series. Four steps summarise it.

1. Close the sizing with the **historical** series. It is the reference.
2. Run the same project on the **simulated history** of each model you will use, from 1980 to 2013. It is against it, not against the observed series, that the future is compared.
3. Keep all the other parameters and change only the series, for each scenario and window.
4. Compare the **saving potential** and the **days of full supply**, always as a difference between the future and the history of the same model. The second usually degrades more than the first, because what changes first is the distribution, not the total.

> **An example of reading.** Suppose that, with one model, the potential is 36% in the simulated history and 33% in SSP5-8.5 for 2041 to 2070, and that the observed series gives 37%. The change attributable to the climate is three points, not four: the remaining point is the difference between model and observation, which also exists in the past.

> **What not to do.** Size directly on the future series and present the result as a design. The future series is a projection with large uncertainty; it serves to test the **robustness** of a sizing done on the historical series, not to replace it.

3.2.5. The choice of model, and why it matters so much

Nineteen climate models project nineteen different futures for the same city under the same emission scenario, and that spread is not noise to be removed: it is the honest expression of the physical uncertainty of the problem. For rain, the spread between models usually exceeds the difference between emission scenarios, which means that the choice of model weighs more on the result than the choice of socioeconomic future.

Hawkins and Sutton (2011) broke down the uncertainty of precipitation projections for regions of the order of 2,500 km, with the models of the previous phase of the exercise, CMIP3, into three sources: the internal variability of the climate, the difference between models and the difference between scenarios. For precipitation, unlike temperature, internal variability accounts for 50% to 90% of the uncertainty in the first decade and remains dominant, in many regions, until the third; the uncertainty between models comes to dominate at longer horizons; and the scenario uncertainty is small or negligible in all regions and horizons, except near the poles at the end of the century. The authors also show that the signal-to-noise ratio of the change is below one over almost the whole planet, and conclude that adaptation decisions will have to be taken with high uncertainty about regional rain. The consequence for the design is direct: running both scenarios with a single model informs less than running one scenario with several models.

Three courses of action are defensible in the face of this spread, presented here in decreasing order of rigour.

Ensemble. Run all the available models and report the range of results, with the median. It is the most rigorous and the most expensive in computing time. Section 3.6 shows how to do it in batch.

Justified subset. Choose three to five models with documented good performance for South America, and declare the selection criterion and the source.

Single model, declared as such. Acceptable for an exploratory analysis, as long as the text does not present the result as a robust projection.

> **What never to do:** run several models, pick the one that gives the most convenient result, and report only that one.

3.2.6. What the literature shows about the tank in a future climate

Studies that simulate rainwater harvesting systems under climate projections converge on one point: the change in the annual total of rain matters less than the change in its **distribution**. Kisakye et al. (2018), with six models for a district of Uganda, found conflicting projections between models, but agreement about the season: saving and water security would fall in the seasons that were already dry and rise in the wet ones, with a reduction of more than 50% in one of the seasons at the end of the century. Lash et al. (2014), sizing a tank for a university building in the United Kingdom with 3,000 equiprobable rainfall series from the British climate projections, concluded that, for it to be likely to meet in 2080 the same demand as today, the tank would need to be 112% larger, and 225% larger for that to be very likely.

The two studies illustrate what section 3.1 showed for Brazilian cities today: cities with a marked dry season need much larger tanks for the same performance. A future climate with longer dry spells behaves, for the tank, like a change of city.

3.2.7. What to declare

A study with a climate scenario must state, in the method:

- the **model or models**, by name, and the selection criterion;
- the SSP **scenario**;
- the **time window**, with the years;
- the **historical series** used as the reference;
- whether there was downscaling or bias correction, and which; in the program's database, that of CLIMBra, by *Quantile Delta Mapping* against the grid of Xavier et al. (2016);
- the **source of the database**, with the version: Ballarin et al. (2023);
- that the future was compared with the **simulated history of the same model**.

Without this, the result is not reproducible. Appendix E has the source of the program's database.

3.2.8. A caveat of method

Climate projections carry two stacked uncertainties of different kinds: that of the scenario, which is socioeconomic and concerns future human choices, and that of the model, which is physical and concerns the representation of the climate system. Neither is reduced by running more simulations with the same material, because it is not sampling uncertainty.

A study conducted honestly presents the result as a range, not as a single number, and avoids conclusions that depend on the third decimal place of a projection for 2060.

3.3. Adjusting the tariff of your municipality

> **Question this guide answers:** is my municipality's tariff in the database, and is it right?

> **Edition.** The catalogue of 652 municipalities and step **12. Tariff Manager** come only with the Brazil Edition. In the International Edition the tariff is typed in, in the economic step, with the same block table, the same codes and the same six regimes described here -- so this guide is also the reference for entering any tariff structure.

This is the guide that changes results the most. The tariff structure decides the economic feasibility of a rainwater system more than any design parameter, and it is also the input that most often appears wrong in published studies on the subject.

3.3.1. How the water and sewage bill is made up in Brazil

Before checking the municipality's tariff, it is worth understanding what is being checked. The water bill in Brazil is not the product of consumption by a single price, and five features set it apart from the electricity bill, which is the reference most familiar to most readers.

Provision is local, and so is the price. Each municipality is served by a state company, a municipal authority or a private concessionaire, and each provider has its own table, approved by a regulatory agency -- state, municipal or intermunicipal. There is no national water tariff, and the difference between two neighbouring municipalities can be more than double.

The tariff is progressive by consumption block. A cubic metre does not cost the same at any quantity: consumption is divided into blocks, and each block has its own, increasing price. In the CASAN table in force in Florianópolis, the first ten cubic metres cost R$ 3.08 each and the next fifteen cost R$ 14.31 each. Progressivity is what makes saving water more valuable than it seems: the cubic metre no longer consumed is always that of the highest block on the bill, the most expensive one.

There is a part that does not depend on consumption. Almost every utility charges something even to those who consume nothing, and the name and nature of that part vary from place to place. It is the subject of the subsection on the six regimes, below, and it is the point that most decides whether a rainwater system pays for itself.

Sewage is charged as a percentage of water. Most providers bill sewage by applying a percentage to the water part, commonly 80% or 100%, on the argument that the volume discharged is proportional to the volume consumed. When that percentage is 100%, the sewage part equals the water part, and sewage accounts for half of the bill -- which means each cubic metre saved is worth twice the price of water.

There are different categories of consumer. Every utility publishes separate tables for the residential, commercial, industrial and public categories, with prices that are often double the residential one or more. In the CASAN table, the first block costs R$ 3.08/m³ for residential and R$ 6.81/m³ for commercial.

> **What the program brings, and what you need to enter.** The program's catalogue of 652 municipalities records the **residential category** of each one. The commercial, public and industrial categories are added municipality by municipality, transcribed from the provider's official table, and can only be chosen where they have been collected -- in the 34 CORSAN municipalities, for example; they are never estimated from the residential one. The other categories exist, are public and are in the resolutions of the regulatory agencies -- what does not exist is a national catalogue of them, which would be a survey in its own right. To study a school, a shop or an industry, enter the structure of the right category in the custom mode of this same screen: the program accepts any block table, with any minimum-charge regime, and section 3.5 shows the procedure.

3.3.2. Start by checking what is already in the database

The program comes with a catalogue of 652 Brazilian municipalities, and each record has the complete tariff structure, the sewage percentage charged, the minimum-charge regime adopted, the effective date of the table and the documentary source from which the data were taken. The catalogue covers the residential category, as the previous subsection explains.

In step **12. Tariff Manager**, choose the state and the municipality. Besides the structure, the program shows the **confidence grade** given to the record, which sums up the quality of the source and the date of the survey. Three items deserve checking against your utility's current resolution, and the order matters because the first invalidates the others.

1. **the effective date** -- tariffs are adjusted every year;
2. **the block values**;
3. **the minimum-charge regime**, which is item 3.3.4.

If any of the three differs, edit the record. The edit applies to your project, does not change the catalogue of other users, and is saved inside the `.netuno` file -- which means whoever receives your project will reproduce the same calculation, with the same table.

3.3.3. How the table is built

Each row of the tariff structure is represented in the program by a quadruple of values: the minimum volume of the block, the maximum volume, a monetary value and a code. The code is the decisive element, because it is what says what that value means -- and structures that look alike charge in very different ways depending on the code of each row.

Table 85 -- The six tariff block codes and the effect of each one on the bill

Code	What it is	How it enters the bill
1 or 0	consumption block	the price is charged on the volume that falls **within** each consumption block
-1	availability charge	fixed value **added** to the consumption part
-2	retroactive allowance	up to the allowance volume, the fixed value is paid; beyond it, **everything from zero** is charged
-3	payment floor	the bill does **not** go below this value
2	exponential base price	`PB` of the formula `PB x C`■, valid up to the row's maximum
3	exponent of the exponential table	the `n` applied to the consumption **of the whole month** that falls in the row

The Florianópolis structure, taken as an example, has five rows: one availability charge and four progressive blocks.

```
(0.00 ; 0.00 ; 46.34 ; -1)     availability charge, always added
(0.00 ; 10.00 ;  3.08 ;  1)    first block
(10.00 ; 25.00 ; 14.31 ; 1)    second block
(25.00 ; 50.00 ; 19.14 ; 1)    third
(50.00 ; 999999 ; 24.08 ; 1)   last
```

3.3.4. The six regimes, and why they decide projects

The «minimum consumption» appears on the bill of almost every Brazilian municipality, and it means **different** things in each one. The program distinguishes six regimes:

Table 86 -- The six tariff regimes and what happens when saving in each one

Regime	What happens when saving	Municipalities in the database
no minimum	each m³ saved is deducted	165
availability charge	likewise; the fixed fee remains, but it does not prevent the saving	122
exponential formula	likewise; the m³ saved is worth the base price raised to the exponent	34
allowance in the minimum	below the allowance, saving **deducts nothing**	329
retroactive allowance	beyond the allowance, everything from zero is charged	2
payment floor	the bill does not go below the floor	0

The case that brings down studies. Compare two real structures of the database, side by side, for the same consumption:

Table 87 -- Effect of the water saving on the bill in two opposite tariff regimes

Consumption	Floriano (PI), allowance in the minimum	Florianópolis (SC), availability charge
14 m³	R$ 92.15	R$ 134.38
12 m³	R$ 72.52	R$ 105.76
10 m³	R$ 52.89	R$ 77.14
8 m³	**R$ 52.89**	R$ 70.98
5 m³	**R$ 52.89**	R$ 61.74

In Floriano, the bill **stops falling at 10 m³**. A house that consumes 12 m³ and installs a system that saves 4 goes down to 8, and saves R$ 19.63 -- not the R$ 24.17 the average tariff of R$ 6.04/m³ would suggest. If it saved 7, it would go down to 5 and save the **same** R$ 19.63: the three additional cubic metres are worth nothing.

In Florianópolis, each cubic metre saved reduces the bill, down to the first one.

> Running the Floriano case with the Florianópolis structure produces a payback that does not exist. Three hundred and twenty-nine of the 652 municipalities of the database are in the allowance regime. It is the majority, and it is the most common scenario in the country.

Figure 37 shows the three responses side by side. What distinguishes the regimes is not the price charged, but the shape of the curve: where the curve is horizontal, saving water does not reduce the bill at all.

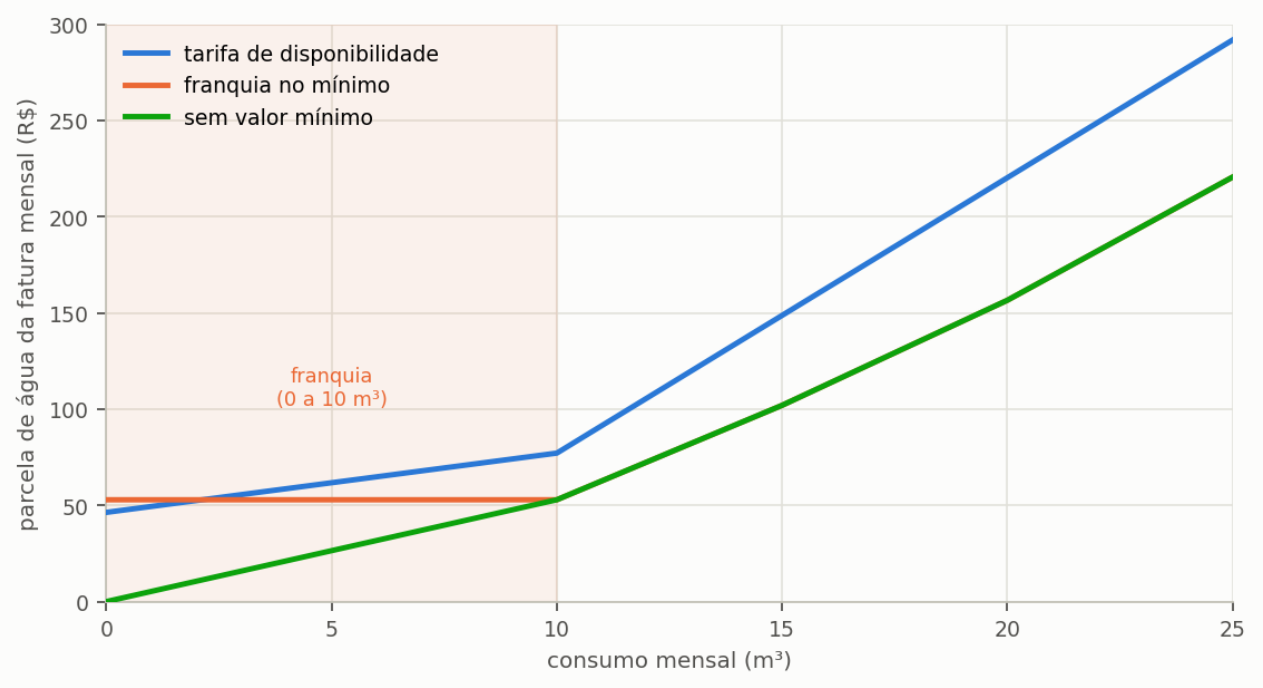


Figure 37 -- Water part of the monthly bill as a function of consumption, in three minimum-charge regimes. Sewage is not in the chart because it is charged as a percentage of this part and only multiplies the curve, without changing its shape. With the availability charge and with no minimum, each cubic metre saved reduces the bill from the first one; with the allowance in the minimum, the horizontal stretch marks the consumption range in which saving water does not reduce the bill.

3.3.5. The CORSAN exponential formula

The 34 municipalities of Rio Grande do Sul in the database served by CORSAN do not charge water by blocks. The water part of the bill is given by a formula:

```
Water = PB x C■ + SB
```

where `PB` is the base price of the municipality, in R$/m³; `C` is the month's consumption, in m³; `n` is the exponent the utility's **Exponent Table** gives to that consumption; and `SB` is the basic service, charged even without consumption. In the residential category, `n` is 1.00 up to 20 m³ and rises with each cubic metre from there on:

Table 88 -- Exponent n of the CORSAN Exponent Table, residential category

Month's consumption	n	Month's consumption	n
up to 20 m³	1.00	36 to 45 m³	1.10
21 m³	1.01	46 to 100 m³	1.11
22 to 27 m³	1.02 to 1.07	101 to 150 m³	1.13
28 and 29 m³	1.08	151 to 1,000 m³	1.1287 to 1.12
30 to 35 m³	1.09	above 1,000 m³	1.11 to 1.0858

The table continues beyond 1,000 m³: only the industrial category has a cap at that volume, above which the excess is charged at its own prices. **Sewage** is not in the formula: it is charged per cubic metre consumed, without the basic service and without the exponent -- R$ 6.40/m³ treated in the municipalities of ANNEX 5, 70% of the base price truncated to the centavo.

Two differences from the blocks decide the calculation:

- **The exponent applies to the whole consumption.** With blocks, each cubic metre is charged at the price of the block it falls in; with the formula, the `n` of the row in which the total consumption falls applies to the whole

volume. Going from 20 to 21 m³ costs R$ 15.09 of water, not the R$ 9.15 of the base price, because the whole 21 m³ are now raised to 1.01.

- **There is no price step, there is a curve.** The mean price of a cubic metre of water is PB x C■-¹: constant up to 20 m³ and growing after that.

Table 89 -- Monthly bill in Alegrete (RS) by the exponential formula, effective 01/07/2026

Consumption	n	Water	Water and sewage	Mean water price
10 m³	1.00	R$ 134.89	R$ 198.89	R$ 13.49/m³
20 m³	1.00	R$ 226.39	R$ 354.39	R$ 11.32/m³
25 m³	1.05	R$ 312.08	R$ 472.08	R$ 12.48/m³
30 m³	1.09	R$ 416.20	R$ 608.20	R$ 13.87/m³

The R$ 134.89 for 10 m³ are exactly the «minimum tariff without a water meter» of the CORSAN table, 43.39 + 9.15 x 10, which confirms the transcription. For rainwater harvesting, the consequence is that **every saving is deducted from the first cubic metre**, and deducted more where the exponent is larger: a house that goes down from 30 to 20 m³ saves R$ 253.81 per month. The straight line used up to 5.0.0, with 70% sewage on the whole bill, gave R$ 155.55 for the same reduction.

In the program, the formula enters the same table of quadruples, with two codes of its own. The Alegrete structure starts like this:

```
(0.00 ; 0.00   ; 43.39 ; -1)   basic service, always added
(0.00 ; 999999 ;  9.15 ;  2)   base price PB
(0.00 ; 20.00  ;  1.00 ;  3)   n for consumption in (0, 20]
(20.00 ; 21.00 ;  1.01 ;  3)   n for consumption in (20, 21]
...                            one row per interval of the Exponent Table
```

In the **Tariff Manager**, the structure appears with the label «Exponential formula», the «Exponent n» rows can be edited, the simulator shows the PB x C■ calculation for the chosen consumption, and the upper chart shows the mean price PB x C■-¹ instead of the steps. The sewage field becomes «Sewage (R$/m³ consumed)», and the economic step offers the same mode, «Per m³ consumed», next to the fixed value and the percentage of water.

3.3.6. When the minimum is diluted into the tariff

A frequent error in public databases: the utility states «minimum tariff of R$ 52.89 for 10 m³», and whoever registers it records it as R$ 5.289 per m³ up to 10 m³.

The difference is enormous. In the first reading, someone who consumes 4 m³ pays R$ 52.89; in the second, R$ 21.16. And, worse for a rainwater study, in the second reading saving below 10 m³ **is deducted**, when in reality it is not.

The program audits the database for this pattern and recodes the cases found, recording the correction. If you register a new municipality, **check this**.

3.3.7. Sewage

The sewage percentage applies to the **water part** of the bill, not to the total bill.

One hundred per cent, as in Florianópolis, means that the sewage part equals the water part -- and so sewage is **half the bill**. Saving water saves both parts, and that is why the effective value of the cubic metre saved almost doubles.

Some municipalities charge sewage by blocks of their own, not by a single percentage. In those cases the single percentage is an approximation, and the program records the limitation.

3.3.8. Where to find the data

In order of reliability:

1. **Tariff resolution** of the state or municipal regulatory agency. It is the official document, and it has the effective date.
2. **Table published by the utility**, usually under «tariffs» on its website.
3. **Bill simulator** of the utility, when there is one. Good for checking: simulate two consumptions and see whether your table reproduces both values.
4. **Real water and sewage bill**, of the property itself. It is the best evidence, and it allows the whole structure to be checked.

> **Always record the effective date and the source.** A study that does not say when the tariff dates from cannot be reproduced two years from now.

3.3.9. Check before you trust

After building the table, do the test that item 3.3.7 suggests: simulate two consumptions, one inside and one outside the minimum block, and compare with the real bill or with the utility's simulator.

If both match, the structure is right. If only one matches, the minimum-charge regime is wrong -- and that is exactly the error that most compromises a rainwater study.

3.4. Modulating the demand by calendar

> **Question this guide answers:** and when consumption is not the same every day?

Schools close in January and July, beach hotels fill up in summer and offices empty at weekends. Treating any of these buildings as if it had a uniform consumption errs in both directions at once: it overestimates the demand in the empty periods and underestimates it at the peaks. The error does not cancel out over the year, because what matters to the rainwater system is not the total consumed but the match between consumption and the rain available on that day.

3.4.1. The four modes

The program offers four modes of demand modulation, which differ in the number of values the user enters and in the situation in which each one is justified (Figure 38).

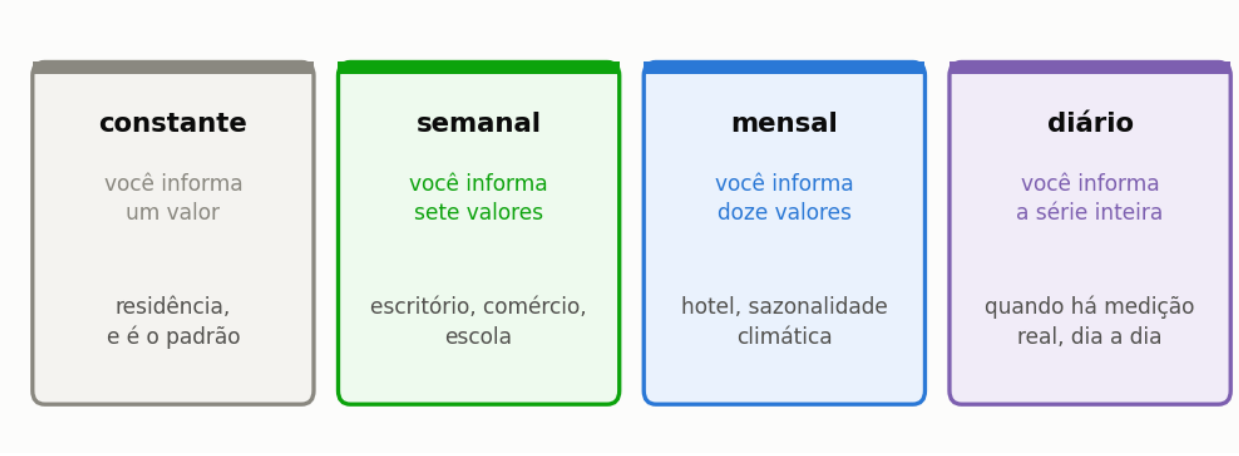


Figure 38 -- The four modulation modes, what each one asks of the user and the situation in which each one is justified. The daily mode only makes sense when there is real measurement, because inventing a daily series adds apparent precision without adding information.

In any of them you can modulate three things independently: the **per capita demand**, the **number of occupants** and the **fraction suppliable by rainwater**.

Modulating occupancy is usually more faithful than modulating demand: in a school on holiday it is not that the pupil drinks less water, it is that there is no pupil.

3.4.2. How to fill it in

In step **2. Building & Demand**, choose the mode. The table of the period opens.

- An empty or zero cell **falls back to the constant value**. This is deliberate: it lets you modulate only the months of interest and leave the rest at the default.
- The values can be typed or loaded from CSV.
- In the daily mode, the first day of the demand series lines up with the first day of the **rainfall series**. If the two have different lengths, the demand series is recycled cyclically.

3.4.3. Why this changes the result

Modulating the demand does not change only the annual total consumed; it changes the timing match between supply and demand, which is precisely the problem the tank exists to solve. Two opposite cases illustrate the reach of this effect, and both are common in Brazil.

For. A school that closes in July, in the south of the country, closes precisely when it rains least. The cistern gets through the dry spell without being drawn on and arrives full in August. The system yields **more** than the annual mean would suggest.

Against. A beach hotel fills up in January and February, when it rains most on the southern coast. It looks good, but the concentrated demand empties the cistern in days, and for the rest of the year it stays idle. The system yields **less**.

The two cases look opposite and are not, and it is worth stating the rule carefully, because the intuitive wording -- «demand in phase with the rain yields more» -- does not survive the second example. The hotel has a demand **in phase** with the rain, since both peak in summer, and still yields less. What separates the two cases is not the phase, it is the **concentration**.

A demand spread over the year makes better use of the same tank; a demand concentrated in a few months empties it in days and leaves it idle for the rest of the year, even if those months are the wettest. The school benefits because its idle period coincides with the dry spell, not because it is in phase with the rain.

Run with the Florianópolis series, the same annual consumption and the same 4,000 L tank, the three hypotheses separate: a constant demand yields 27.1%, the school profile yields 27.3% and the beach hotel profile yields 25.7%. A difference of almost one and a half percentage points, against a ceiling of 28%, is large for a change that does not alter a single litre of the annual consumption, and it is entirely invisible to any calculation made on means -- which is exactly why Netuno simulates day by day. The three profiles are in Figure 39, and the yield of each one in Figure 40.

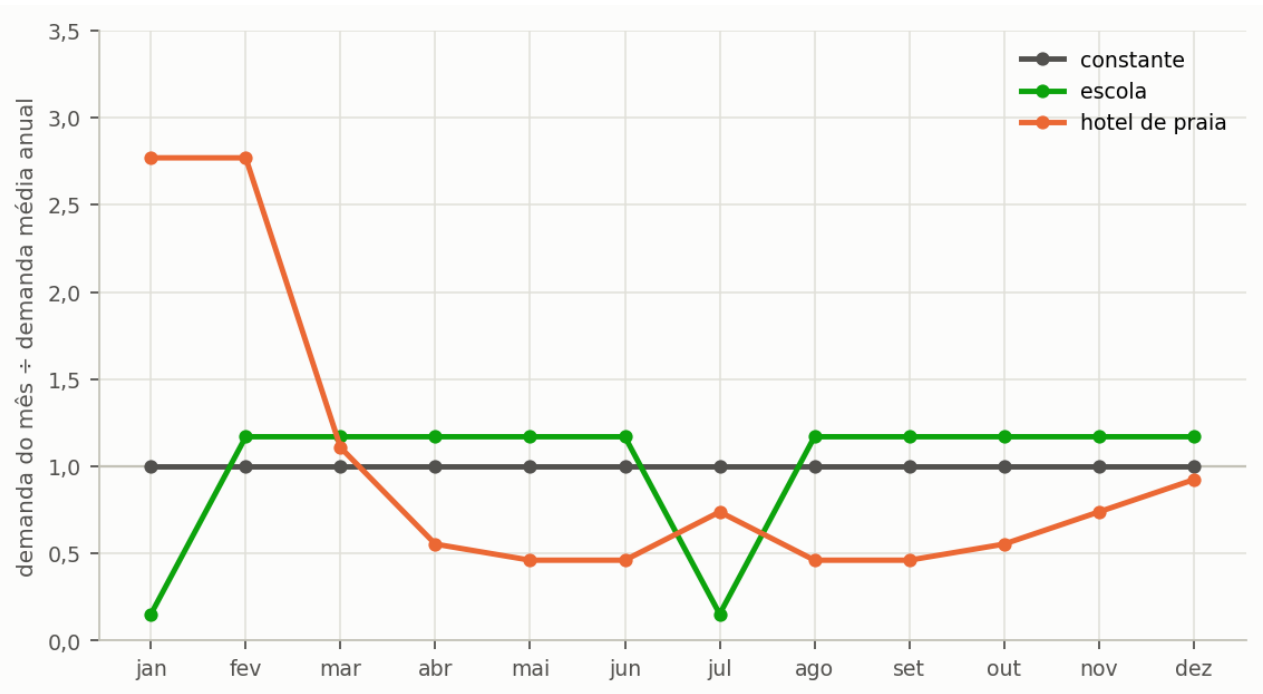


Figure 39 -- The three demand profiles used in the comparison, expressed as the ratio between the month's demand and the mean demand. All three have exactly the same annual consumption; what changes is how it is spread over the twelve months.

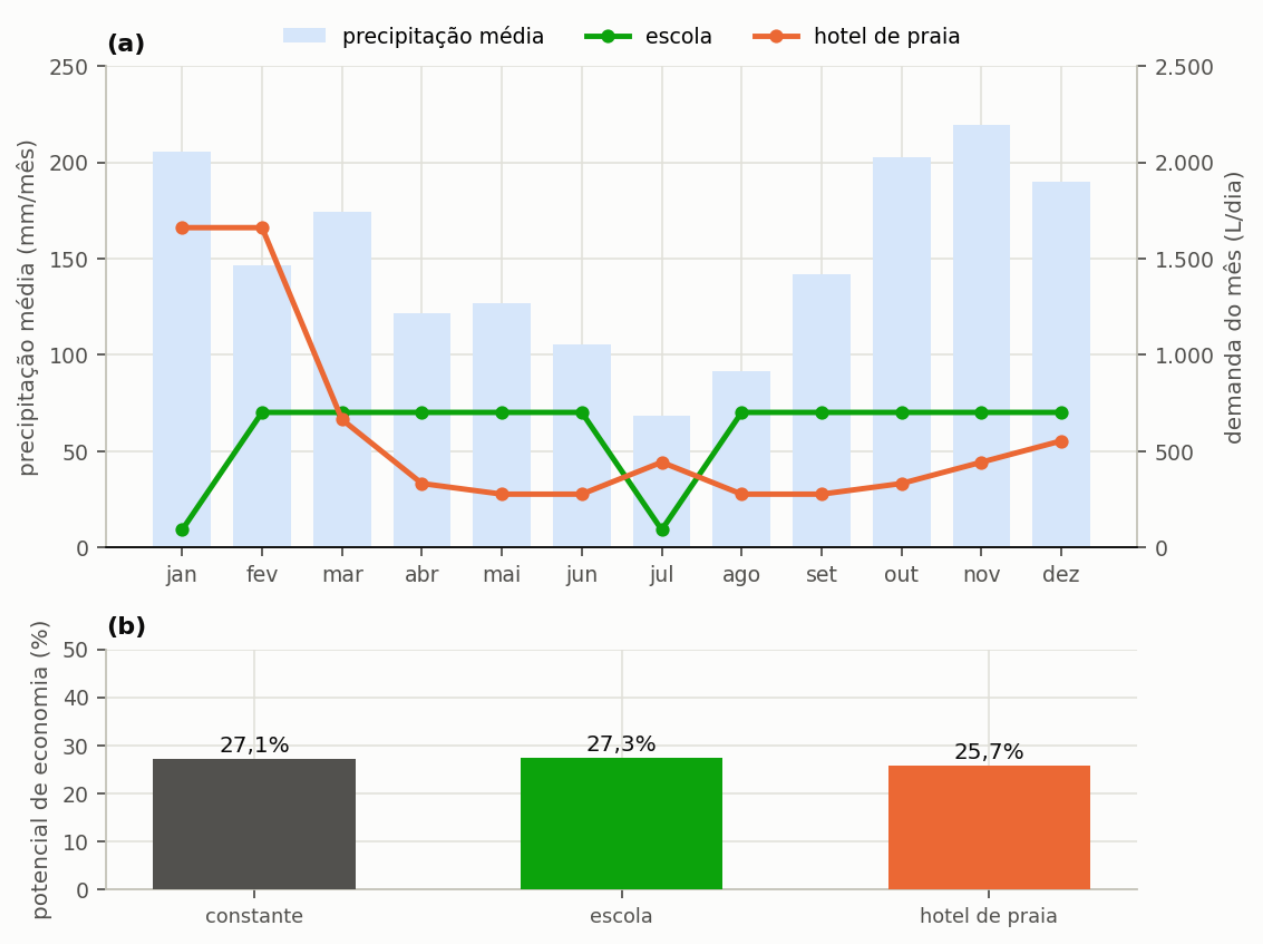


Figure 40 -- (a) Monthly demand profiles of the school and the beach hotel over the mean monthly precipitation of Florianópolis; (b) saving potential of each profile with the same 4,000 L tank. The hotel profile concentrates almost a third of the annual consumption in two months, and it is this concentration, not the phase, that costs it almost one and a half percentage points.

3.4.4. The suppliable fraction can also vary

Less obvious and sometimes more important. In a school, the percentage of non-potable use changes between term time and holidays: during the holidays, floor washing and garden watering remain, and they are uses **entirely** suppliable by rain.

Modulating the fraction captures this. Keeping the fraction constant underestimates the yield in the period of low occupancy.

In the table, the «Indoor Non-Potable (%)» column is a percentage, like every field you type in: 28 means 28%. In the project file, the same column is saved as a fraction, 0.28, and section 4.2 explains why. A cell with a value between 0 and 1 turns orange.

3.4.5. Precautions

Do not modulate what you have not measured. An invented modulation adds apparent precision without adding any information, and passes on to the reader the impression of a rigour the data do not have. When in doubt, an explicitly declared constant demand is more honest than an arbitrary monthly profile.

Declare the origin of the profile. It may come from your own measurement, from twelve monthly water and sewage bills, from a bibliographic reference or from an estimate, and those four origins do not carry the same weight of evidence in a published study.

Check the annual total after modulating. The program shows the mean daily demand resulting from the modulation, and a difference between that value and what was expected is the quickest indicator of an error in filling the table.

3.4.6. The simplest way, when there is one

If you have **twelve water and sewage bills** of the building, you have the measured monthly profile, which is the best possible source. Divide each month's consumption by the number of days and by the number of occupants, and use the monthly mode.

Twelve bills are worth more than any reference from the literature.

3.5. Simulating schools, hospitals and shops

> **Question this guide answers:** does the program work for non-residential buildings?

It does, and it works well for the water side. There is, however, a weighty caveat about the economic side, and it must be known before the study starts, not after it is finished.

3.5.1. The caveat: the tariff category

Brazilian utilities charge different tariffs depending on the consumer category -- residential, commercial, industrial and public --, and the non-residential categories usually cost twice the residential one or more. In the CASAN table in force in April 2026, for example, the first block costs R$ 3.08/m³ for a house and R$ 6.81/m³ for a commercial establishment.

These tables exist and are public: they are in the resolutions of the regulatory agencies and on the utilities' own websites. What the program's catalogue records, for the 652 municipalities, is the residential category of each one, because surveying the four categories of every provider would be a job of another order of magnitude. The non-residential tariff is not missing from the program: it is missing from the catalogue, and you can enter it.

When you choose a non-residential building type, the Brazil Edition **warns** you and says in which direction the error goes:

> The building type chosen is «School», which the utility bills in the Public category. The program's catalogue has only the Residential category, so the water and sewage bill is being calculated with the wrong tariff. The error has a known direction: non-residential categories are more expensive, and therefore the saving and the return are **underestimated**.

> **What to do, in five minutes.** Look up your utility's current tariff resolution, which is usually on its website under «tariffs» or «tariff structure». Find the table of the category of interest. Enter the blocks in the custom mode of step **12. Tariff Manager**, with the corresponding minimum-charge regime. Section 3.3 describes each field. Once that is done, the economic result becomes that of your case, and not a residential approximation.

In the International Edition there is no catalogue: the tariff is always typed in, and so the program does not show this warning. Enter directly the tariff of the category the building is billed in.

There is a consolation in this limitation, and it is methodological: the error has a known direction and it is conservative. Non-residential categories are more expensive than the residential one, so the real project returns more than the program indicates, and never less. A study that declares the limitation therefore presents a feasibility floor -- which is a defensible position, unlike the reverse.

3.5.2. The available building types

The program's eight building types are grouped by the tariff category the utility bills them in, and each one has its own unit of consumption (Figure 41). The unit of consumption changes the meaning of the per capita value entered, and the category decides the block structure of the bill: simulating a school with a residential tariff is one of the mistakes described in section 5.1.

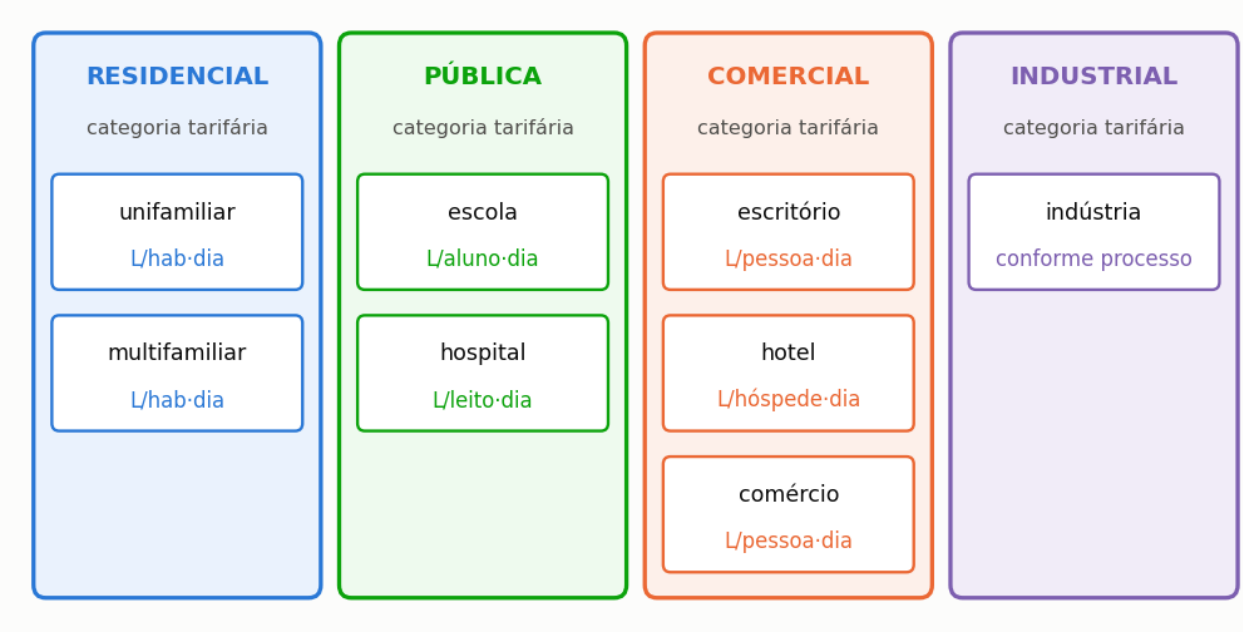


Figure 41 -- The eight available building types, grouped by the tariff category the utility charges. The unit of consumption changes with the building type, and it is what gives meaning to the per capita value entered.

3.5.3. What changes with each building type

School. It is one of the most favourable cases for rainwater harvesting in Brazil, for two reasons that add up. The non-potable share of consumption is high, because the toilet dominates use and many units have neither showers nor a kitchen. And the roof area per pupil is usually large, so supply is rarely the limiting factor.

Modulation by calendar, however, has an effect that depends on the region, and it is better to simulate it than to assume the result. In the Brazilian Southeast and Centre-West, the rains concentrate from October to March and the dry season runs from June to August. The July break coincides with the dry season, and the cistern gets through the dry period without being drawn on; the long holidays from December to February, on the contrary, coincide with the peak of the rains, and during that period the cistern fills and overflows unused. In the South, where rain is better distributed over the year, the school calendar weighs less. Section 3.4 shows how to represent the school days, and section 3.1 shows how the distribution of rain changes from one state capital to another.

Hospital. The most delicate case. High, continuous consumption, without relevant seasonality, which is good for harvesting. But the **quality requirement** is different: there are uses that do not admit rainwater under any circumstances, and the cooling tower, which would be an ideal use, requires microbiological control against Legionella. Treat the quality class with care, and do not use the adapted classes 2 to 4 as if they were a normative requirement.

Office and shop. Demand concentrated on working days, five days out of seven. Modulate by week. The non-potable share is high because almost all consumption is toilets, urinals and cleaning: Proença and Ghisi (2010), who estimated the end uses of ten office buildings in Florianópolis through interviews with the occupants and flow measurements, found 56% to 86% of consumption in non-potable uses, and concluded that the toilet should be the first target of any saving programme in these buildings. The width of the range is itself a warning: two offices in the same city can differ by thirty percentage points, and the suppliable fraction of a study should come from the building studied whenever possible.

Hotel. Strong seasonality, and the case in which monthly modulation matters most. See the warning of section 3.4 about concentrated demand: a coastal hotel fills up precisely in the wettest months and still yields less than a constant demand, because it empties the tank in a few days and leaves it idle for the rest of the year.

3.5.4. Where to get the non-potable fraction from

The suppliable fraction is the ceiling of the saving potential, as section 2.3 shows, and it is therefore the parameter that weighs most on the result of a non-residential building. There are three ways of obtaining it, in decreasing order of reliability.

Measuring the end uses. Sub-meters or flow loggers on the branches, for a few representative weeks. It is the only way that produces a number for the building, and not an average.

Survey by interview and flow. The method of Proença and Ghisi (2010): ask the occupants the frequency and duration of each use and multiply by the measured flows of the fixtures. It costs little and captures the real behaviour of the building, but it depends on the memory of those interviewed.

Reference value from the literature. Acceptable in a preliminary study, as long as the reference is of the same building type, from the same country and, ideally, from the same region. Avoid transposing residential fractions to other building types, and even among dwellings the variation is large: in a multi-family residential building in Florianópolis, Maykot and Ghisi (2020) found the toilet with only 9.2% of consumption and the washing machine with 21.3%, far below the fractions of offices and schools.

> **Declare the route.** «Suppliable fraction of 60%, measured» and «suppliable fraction of 60%, from a reference» are different studies, and the reader needs to know which of the two they are reading.

3.5.5. Individual metering and the number of dwelling units

In a multi-family building with individual metering, enter the **number of dwelling units**. The program evaluates consumption at the level of **one** unit and multiplies by the number of units.

This matters because of the progressive blocks: a building of twenty apartments with individual metering pays twenty bills in the first block, not one bill that adds up all the consumption and falls in the most expensive block. Treating the building as a single unit would grossly overestimate the saving.

3.5.6. What to declare

Besides what section 4.10 already asks for:

- the **building type** and the unit of consumption adopted;
- the **origin** of the per capita consumption -- measurement, reference or standard;
- whether the **tariff category** was corrected or remained the residential one of the catalogue, and in the latter case that the economic result is underestimated;
- the **modulation profile** and its origin;
- for a hospital, the quality class adopted and what it covers.

3.6. Comparing many cases at once

> **Question this guide answers:** how do I compare dozens of alternatives without redoing the whole project for each one?

The program was built to compare alternatives, and it offers three different ways of doing it. The choice between them is not a matter of preference: each way answers a different question, and using the wrong one costs time without producing the information wanted.

3.6.1. The three ways

Table 90 -- The three ways of comparing alternatives depending on what varies between them

If what changes is	Use	Where
only the cistern volume	the sweep	step 4. Reservoir & Costs
the volume, and you want to decide	the multi-criteria table	step 8. Multi-Criteria Decision
any other assumption	saved projects, one per case	`.netuno` file

The first is automatic, the second too, and the third needs organisation.

3.6.2. The sweep: dozens of volumes in one pass

In step **4. Reservoir & Costs**, choose the **sweep** mode and enter the maximum volume and the step. The program simulates all the volumes of the grid and draws the curve.

The sweep is the normal way of working, not an advanced function reserved for special cases: it is almost always the right answer, because «which volume should I adopt?» is the central question of every rainwater harvesting project. Section 4.4 goes through a complete sweep step by step, and section 2.2 formalises how the grid of volumes is built.

The computational cost is modest and does not justify hesitation: a sweep from 0 to 20,000 L with a 1,000 L step means twenty-one simulations of thirty years each, and it finishes in a few seconds on an ordinary computer.

3.6.3. The multi-criteria table: all volumes, all dimensions

Step **8. Multi-Criteria Decision** shows, in a single table, each volume of the sweep with:

- the water saving and the days of supply;
- the net present value, the discounted payback and the internal rate of return;
- the net carbon at the end of the service life and the environmental payback;
- the TOPSIS score and the SMAA-VIKOR acceptability.

It is the densest way of comparing alternatives the program offers: ten answer columns for each volume row. The **decision profile** changes the weights of the dimensions and re-ranks the table on the spot. Section 2.10 explains the methods behind it.

3.6.4. When something other than the volume changes

City, tariff, demand, tank material and building type are parameters the sweep does not go through, and to vary any of them the way is to keep one saved project per case. The procedure below looks bureaucratic at first sight and saves hours of rework, above all when the study is picked up again weeks later.

1. Build the complete project once, in full detail.
2. Save it with a name that describes the case: `escola_florianopolis_5000.netuno`.
3. Change **one** thing, and save it under another name.
4. Run each one, and note the result in a spreadsheet of your own.

The example files of the manual were made this way: `05_manaus.netuno`, `06_brasilia.netuno` and `07_saopaulo.netuno` are the same project, with the rainfall series swapped. That is what section 3.1 compares.

> **Change one thing at a time.** If you change the city and the tariff together, the result changes and there is no way of knowing which of the two moved it. It seems obvious, and it is the most common error in comparative studies.

3.6.5. How to organise the files

A naming convention saves more time than it seems:

```
school_study/
   base.netuno
   city_florianopolis.netuno
   city_manaus.netuno
   tariff_commercial.netuno
   demand_measured.netuno
   results.xlsx
```

And three habits that avoid rework:

Keep the project, not just the result. The `.netuno` file has the rain inside it and reproduces the whole study; the results spreadsheet does not.

Note what changes in the file name. Six months later, `test3.netuno` means nothing.

Export to a spreadsheet at the end of each round. Step **10. Export** saves the tables in `.xlsx`, and that is where the comparison between cases is put together.

3.6.6. How many cases fit

The program imposes no limit on the number of cases compared, and the real limit is the analyst's patience: each thirty-year simulation takes fractions of a second, but building and checking each project takes minutes, and it is the checking that takes the time.

In practice, a well-run comparative study gathers between five and twenty cases. When the number grows far beyond that, it is usually a sign that the research question was not narrowed down enough -- or that the work is a large-scale study, which needs other tools.

For large-scale studies -- factorial matrices of city, area, demand and volume, or uncertainty propagation -- the program offers a command-line interface, described next.

3.6.7. The command line

Installing the program as a Python package registers the `netuno` command, which runs the same calculation routines as the graphical interface without opening the window; the Windows installer with the graphical interface does not include it. The command line reads `.netuno` files, projects in structured JSON and the Netuno 4 CSV format, and writes the results in text, JSON, CSV, HTML or XLSX. It is the interface suited to reproducible studies, because the command itself documents the assumptions changed.

Table 91 -- Subcommands of the command line and the purpose of each one

Subcommand	Purpose
`validate`	checks the structure of the project, the rainfall series and the essential assumptions, without simulating
`run`	simulates the project; by default, only the technical analysis
`analyze`	runs the complete chain: technical, economic, environmental and TOPSIS
`econ`	runs the economic analysis
`init`	creates a template `.netuno` project, editable, with the rainfall series referenced or embedded
`init-catalog`	creates a template JSON catalogue of building or climate assumptions
`matrix`	combines a catalogue of assumptions with a catalogue of climates and simulates every combination
`matrix-projects`	runs a list of projects, a folder of rainfall series or the factorial matrix declared in the project
`batch`	runs the scenarios of a CSV or XLSX spreadsheet, from a base project
`uncertainty`	propagates uncertainties by Monte Carlo for one tank volume
`series-length`	answers how many years of rain the project needs, by the method of Geraldi and Ghisi
`stochastic-demand`	draws the daily consumption at random and returns the performance as a range, not as a single number

The subcommands `run`, `analyze` and `econ` accept the same options: `--analysis` chooses the analysis (`technical`, `economic`, `environmental`, `topsis` or `all`); `--volume` fixes the lower tank volume, in litres; `--rainfall` replaces the project's rainfall series; `--set FIELD=VALUE` changes a project field and can be repeated; `--weights` gives the technical, economic and environmental TOPSIS weights; `--weight-sensitivity N` draws N weight vectors and reports how often each volume wins; `--format` and `--output` set the output format and file; and `--seed` fixes the seed of the stochastic analyses, so that the result is reproducible.

Applied to the tutorial project, `validate` and `run` produce (the command line prints its messages in Portuguese -- the first one reads «valid project: structure, rain and essential assumptions checked»):

```
netuno validate 01_tutorial_florianopolis.netuno
Projeto válido: estrutura, chuva e premissas essenciais verificadas.

netuno run 01_tutorial_florianopolis.netuno --volume 4000 --format json --output r.json
```

The file `r.json` records the provenance -- source project, number of days of the series and start date -- and, for the 4,000 L volume, the saving potential of 27.30%, the rainwater volume consumed of 59.8 m³/year, the overflow of 127.5 m³/year and the classification of the days into 97.3% full, 0.5% partial and 2.2% none.

The `uncertainty` subcommand draws, in each sample, the precipitation, the catchment area, the demand and the investment with a distribution of declared coefficient of variation -- by default, 10% for precipitation, 0% for area, 10% for demand and 15% for investment, in 200 samples -- and reports the 5%, 50% and 95% percentiles of the saving potential and of the net present value, with the probability of the net present value being positive. In the tutorial, with 50 samples and 4,000 L, the saving potential lies between 26.8% and 27.6%, with a median of 27.3%. The range is narrow because the system already works close to the 28% ceiling: a 10% variation in rain or demand

changes little in a tank that almost never runs dry.

The `series-length` subcommand does, with no screen time limit, the analysis the sweep step offers in the «is the rainfall series long enough?» button, described in section 2.2. Without `--windows`, it goes through **all** the possible windows, which is the exhaustive mode of the papers; with `--windows N`, it draws N windows per length, with the declared seed. The options `--max-volume`, `--step` and `--threshold` define the grid and the marginal-gain threshold, and `--confidence` and `--tolerance` define the similarity criterion.

The `stochastic-demand` subcommand draws the consumption **day by day** and runs the simulation many times, returning the saving potential, the days of full supply and the overflow as a range between the 5% and 95% percentiles. There are two variation parameters, and the difference between them is what the subcommand has to teach: `--daily-cv` is the variation from one day to the next, and `--level-uncertainty` is the uncertainty about how much the building consumes on average. The first hardly moves the result -- drawn independently every day, it cancels out over a thirty-year series --, and the second does. With `--absence` you declare the probability of a day with no consumption at all, such as a trip, and with `--ideal-volume` the program also works out the ideal volume of each realisation, which is what measures whether the design decision withstands the uncertainty of consumption.

This subcommand has no equivalent in the graphical interface, and that is a design decision: there are many parameters, reading them requires care, and on the screen they would confuse more than they would help.

> The subcommands `compare-mass` and `run-vaz-study` run research studies of the group on climate databases of tens of thousands of series, and require the files of those databases; they are not part of the everyday use of the program.

3.7. Sensitivity analysis

> **Question this guide answers:** which assumption moves my result the most, and how much can it take?

Every rainwater harvesting study rests on uncertain assumptions, and the three most influential are usually precisely the least verifiable: will the tariff really rise 5% a year for twenty years? Is the demand really 150 litres per person per day? Will the cistern cost what the quotation says?

Sensitivity analysis measures how much of these doubts reaches the final result. It does not remove the uncertainty, and does not try to; it measures the size of the uncertainty, which is exactly the information the reader needs to decide how far to trust the number presented.

3.7.1. The procedure, in three steps

The procedure needs no specialised tool, and it is the same as in any engineering analysis faced with a poorly known parameter: vary one thing at a time and observe the effect.

1. **Choose the assumption** you suspect is the most influential.
2. **Run the project three times**: with the likely value, a low value and a high one.
3. **Note the result of the three** and compare.

Repeat the procedure for each doubtful assumption. In the end you will have identified which of them actually move the result and which are irrelevant -- and that separation is worth more than any refinement of the central value, because it shows where to invest data-collection effort. Section 3.6 shows how to organise the files so that the set of runs does not turn into confusion.

3.7.2. Which range to adopt for each assumption

The range of variation to adopt for each assumption is not a matter of guesswork: it follows from what is known about the origin of the data, and a measured value admits a much narrower range than a value taken from the literature (Figure 42). The tariff escalation is expressed in percentage points per year; the other ranges, as a percentage of the value adopted.

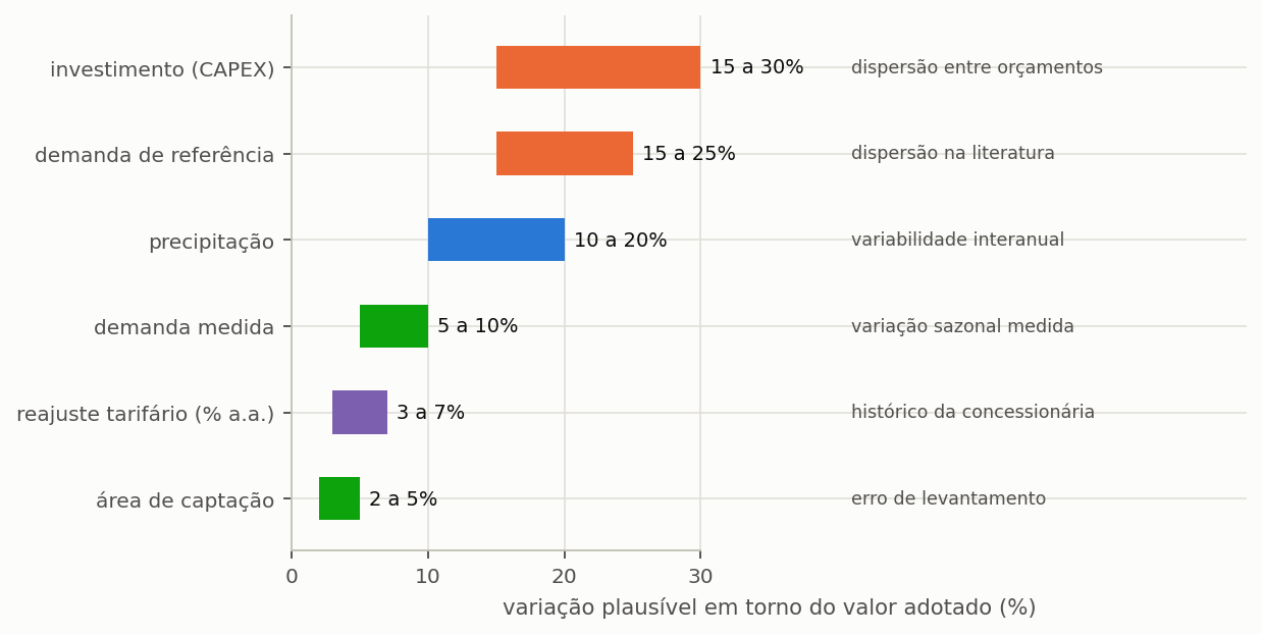


Figure 42 -- Plausible range of variation of each assumption in a sensitivity analysis, with the origin of the uncertainty. The widest ranges correspond to assumptions taken from a reference, not measured in the project.

> **Do not perturb what you know.** If the area was measured in a survey, the range is narrow; widening it produces an artificial uncertainty that dilutes the result and hides the assumption that really matters.

3.7.3. The assumptions that dominate the economic result

In economic studies of rainwater harvesting, the assumptions that move the result most are those that act over the whole analysis horizon: the analysis period itself, the volume of rainwater actually consumed, the minimum attractive rate of return and the **tariff escalation**. The escalation deserves particular attention because it is the least verifiable and it compounds: five per cent a year for twenty years multiplies the tariff by 2.65; three per cent, by 1.81; and seven per cent, by 3.87. The saving of the last years follows that factor, and the choice of escalation rate can decide a good part of the study's economic conclusion.

Try it on your case: in step **6. Economic Results**, change the annual escalation to 3%, note the NPV, repeat with 5% and with 7%. Three runs, three minutes, and a range instead of a number.

> **In a paper, show that range.** A single net present value, calculated with a 5% annual escalation and presented without sensitivity, suggests a precision the assumption does not have.

3.7.3.1. The tutorial, assumption by assumption

The table below applies the procedure to the final tutorial project -- 4,000 L in polyethylene, pumping to a 1,000 L tank, a likely NPV of R$ 5,524 -- and varies five assumptions, one at a time, between a low and a high value.

Table 92 -- Potential, net present value and payback of the final tutorial project with each assumption varied alone, from the low to the high value

Assumption	Low	Likely	High	Potential	NPV over 20 years	Discounted payback
water and sewage tariff escalation, % per year	3.0	5.0	7.0	27.30% to 27.30%	R$ 860 to R$ 11,360	16.8 years to 10.8 years
minimum attractive rate of return, % per month	0.3	0.5	0.7	27.30% to 27.30%	R$ 10,014 to R$ 2,241	10.9 years to 15.5 years
per capita consumption, L/person.day	120.0	150.0	180.0	27.73% to 26.72%	R$ -51 to R$ 10,723	19.6 years to 9.6 years
rainwater-suppliable fraction, %	20.0	28.0	40.0	19.91% to 36.96%	R$ -2,532 to R$ 16,058	no payback to 7.7 years
cistern price, % of the quotation	70.0	100.0	130.0	27.30% to 27.30%	R$ 6,282 to R$ 4,766	11.8 years to 13.7 years

The reading separates the assumptions into three groups.

The one that decides the project is the suppliable fraction. From 20% to 40%, the NPV goes from negative to almost three times the likely value, and the payback from «no payback» to less than eight years. It is the assumption section 4.2 fixed and that depends on the decision about uses, not on measurement: including or not the washing machine, with the quality criterion item 4.1.8 of NBR 15527:2019 requires, moves the result more than any economic parameter.

Those decided by the market weigh next. The tariff escalation and the minimum attractive rate of return shift the NPV by thousands of reais each, but do not touch the potential. Per capita consumption moves both: with less consumption, the suppliable fraction applies to a smaller demand, the cistern almost never runs short, and the potential rises a little while the saving in money falls.

The one that seemed important hardly weighs. Thirty per cent more or less on the price of the cistern changes the NPV by R$ 760 and the payback by one year. The polyethylene cistern is a small share of the investment, and labour, piping and accessories, which do not vary with the volume, dominate the account.

The result reverses the intuition of anyone who prices before simulating: spending time quoting the cistern is worth less than spending time defining the uses.

3.7.4. The sensitivity the program does by itself

There is one case in which the sensitivity analysis comes ready and needs no additional runs, and it concerns the final decision of the study: the choice of volume as a function of the weights given to the criteria.

In step **8. Multi-Criteria Decision**, the **SMAA-VIKOR acceptability** column is already a ready-made sensitivity analysis. It answers: if the weights of the dimensions were drawn at random instead of fixed, how often would each volume come first?

Read it like this:

Table 93 -- Reading the SMAA-VIKOR acceptability of the winning volume

Acceptability of the winning volume	What it means
above 70%	robust recommendation; the weights hardly matter
between 40 and 70%	reasonable recommendation, and it is worth declaring the second option
below 40%	the choice depends on the weights, and the text must say so

The distinction this column allows is epistemological, not only statistical: a volume that wins in 80% of the weight draws is a recommendation, whereas a volume that wins in 30% is a preference of the analyst, legitimate as long as it is presented as such. Section 2.10 sets out the method that produces this number.

The tornado chart (Figure 43) is the established way of presenting this set: one bar per assumption, ordered by range, with the two variations in different colours. Read from top to bottom, it answers the question the reviewer will ask first, which is on which assumption the result depends.

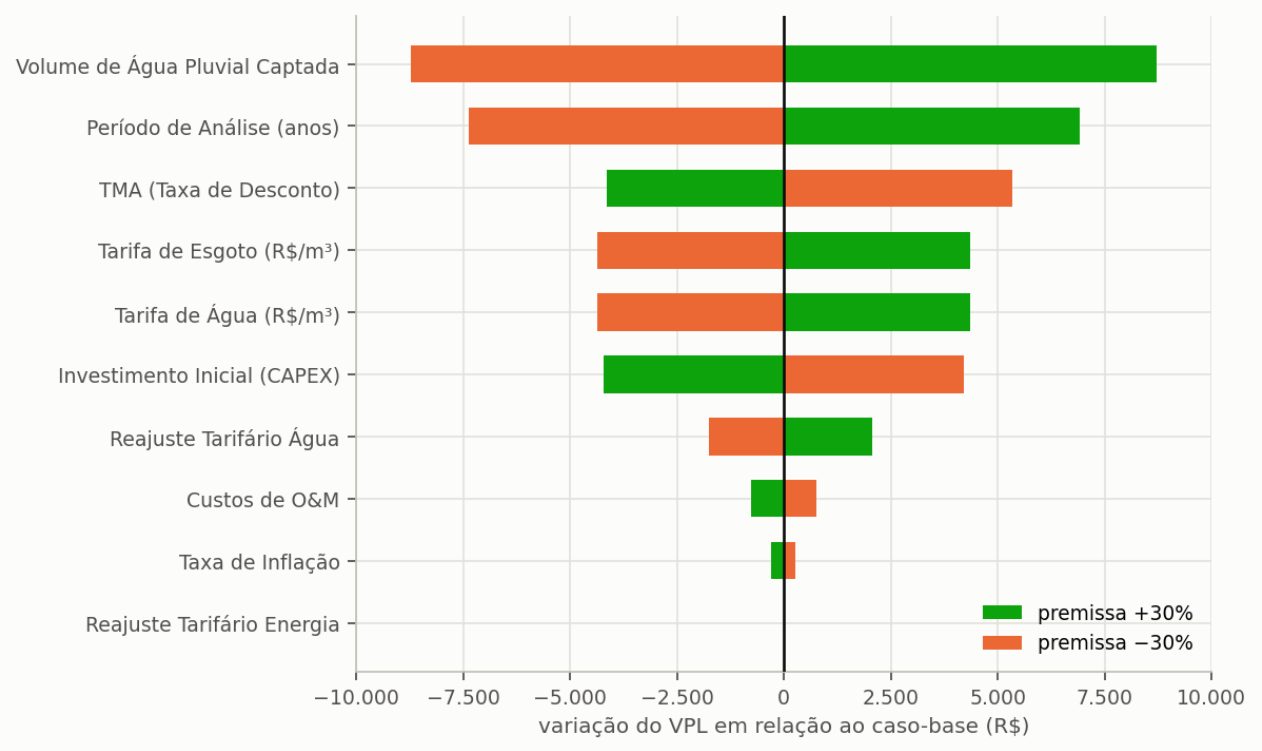


Figure 43 -- The variation of the net present value when each assumption is changed by 30% up and down, keeping the others. The colour identifies the variation applied, not the sign of the effect: that is what makes it possible to see that raising the minimum attractive rate lowers the present value, whereas raising the tariff increases it. The base case of this figure comes from the program's simplified economic model, which discounts an annual flow with a single marginal tariff and the declared escalation; that is why the base-case present value does not match that of the tutorial's monthly cash flow. What the tornado measures is the relative sensitivity, not the absolute value.

3.7.5. How to present it

Three ways, from the simplest to the most complete:

Scenario table. Pessimistic, likely and optimistic, with the assumptions of each one declared. Simple, honest and enough for most work.

Tornado chart. One bar per assumption, showing how much the result varies when it varies alone. It ranks the assumptions by importance and is the most informative presentation per area of page.

Distribution. The full range of the result, when all the assumptions vary together. It requires declaring the distributions adopted, not just the ranges.

> **IN DEPTH . PROPAGATION BY MONTE CARLO**
> The third way is done by drawing thousands of combinations of assumptions and accumulating the distribution of the result. The program does this propagation with the `uncertainty` subcommand of the command line, described in section 3.6. For a course or design study, the scenario table of the first item answers the same question with a fraction of the effort -- and it is easier to defend, because each scenario has assumptions that can be read.

3.7.6. What to declare

- the **ranges** adopted and the origin of each one;
- how many runs were made, and with which values;
- for the choice of volume, the **acceptability** of the winner;
- **which assumptions were not perturbed**, and why.

The last is the one most often forgotten, and the one a reviewer most often asks for.

3.8. Importing a Netuno 4 project

> **Question this guide answers:** do my old studies open in Netuno 5?

They do, and most of the data cross the conversion without loss. The care this section asks for comes from two facts that sit badly together: some fields do not cross, and the import does not say which. A study reused without checking can therefore keep looking like the original study while simulating something else.

3.8.1. How to import

Just open the old file from the open-project menu. The program recognises the legacy format, which is semicolon-separated text, and converts it into the internal structure of the current version. Once converted, save the project: it is then saved in Netuno 5's JSON format, with the fields described in appendix A, and the original file stays intact on disk, which is worth keeping as evidence of the previous study.

3.8.2. What comes across

Practically the whole input core of Netuno 4 has a direct counterpart in the current version, and is converted without user intervention. The table below lists what the import recovers.

Table 94 -- Netuno 4 data recovered by the import

Item	Comes across?
rainfall file and start date	yes
catchment area, runoff coefficient	yes
first flush and mode	yes
total demand and number of residents	yes
percentage suppliable by rainwater	yes
upper tank, pumping trigger level	yes
lower tank, range and step	yes
difference between potentials	yes
water and sewage tariffs, taxes	yes
initial costs: labour, piping, accessories	yes
pump: power, efficiency, flow rate, start-up time, cost	yes
electricity tariff and taxes	yes
inflation, escalation, analysis period, minimum attractive rate	yes
operating costs	yes

3.8.3. What does not come across

> **Warning, and always check.** The reader of old files **does not recognise any variable demand key**. A Netuno 4 project that used a monthly or daily demand is imported as a **constant** demand, without any warning.

The consequence of this limitation deserves to be spelled out, because it changes results without changing the appearance of the project. If the original study modulated the demand by calendar -- a school on holiday, a seasonal

hotel -- the import degrades it to a uniform consumption, and the mismatch between supply and demand that the modulation represented disappears from the calculation. Check the demand mode right after importing, and re-enter the profile by hand if it is on «constant» when it should not be. Section 3.4 shows how.

There is also a set of parameters that do not come across for a different reason: they simply did not exist in Netuno 4. They are irrigation and the suppression of watering by rain, the water quality class and the treatment line, the tank materials and the pipe lengths per diameter, the life cycle assessment parameters, the green property-tax incentives (IPTU Verde), the building type and the number of dwelling units. All take the default value of version 5, and all deserve review before the imported project is considered ready.

3.8.4. Why the result changes

Even if the import recovers all the input data perfectly, the simulation result will not be identical to that of the previous version. The difference is deliberate, has a known cause and is quantified.

The main cause is the **treatment of the first flush**. Netuno 4 treats the first-flush depth as an event threshold: when the day's rain does not reach the depth, nothing is collected; when it exceeds it, the whole rain is collected, without subtraction. Netuno 5 subtracts the depth from the day's precipitation, which is physically more faithful, because the water diverted to the self-cleaning device really does not reach the cistern.

Over 4,050 reference cases, Netuno 5 estimates on average **2.3 to 2.7 percentage points less** potable water saving than Netuno 4. Half of that divergence comes from the treatment of the first flush: 1.36 percentage points of the 2.71 measured in the Manaus and São Paulo set. The other causes are broken down in section 6.1.

3.8.5. Comparing the two versions correctly

When the work explicitly compares the two versions, there is a simple procedure that isolates the behaviour of the water balance from the deliberate methodological divergence, and it needs no feature of the program: **run both versions with a first flush of zero**. Without a first flush, the two rules coincide by construction, and any remaining difference is attributable to the implementation, not to the method.

Once that comparison is made, repeat it with the project's real first flush to measure the total difference. Taken together, the two numbers separate what is a deliberate change of method from what would be an eventual divergence of implementation, and that separation is exactly what a reviewer asks for.

> The internal verification protocol mentions an option to emulate the old first-flush rule automatically. **That option is not implemented.** Use the zero-first-flush route, which produces the same result with a little more work and no surprises.

3.8.6. After importing

Seven checks are enough to turn an imported project into a reliable one, and they should be done in the order below, because the first ones affect the result of the following ones.

1. **Demand mode** -- was it degraded to constant?
2. **Building type** -- is it the right one?
3. **Tank materials** -- the default adopted is polyethylene.
4. **Water quality class** and disinfection technology.
5. **Tariff regime** of the municipality -- Netuno 4 did not distinguish the regimes, and section 3.3 shows what that changes.

6. **Motor efficiency** of the pump, which Netuno 4 did not have.
7. **Run the project and check the potential** against the old result before trusting the imported file.

And keep the original file: it is the documentary evidence of the previous study, and no conversion replaces it.

3.9. Exporting to CAD, BIM and spreadsheets

> **Question this guide answers:** how do I take the result to the construction design and to my own analysis?

3.9.1. What each format carries

Table 95 -- Export formats and the purpose of each one

Format	Extension	What it carries	What for
report	HTML, PDF	complete result and assumptions	delivery
spreadsheet	XLSX	daily series and sweep	re-analysis
audit	XLSX	every field of the assumptions, the tariff, the treatment, the LCA, the economics and the arrangement	checking
drawing	DXF	sized hydraulic diagram	construction design
BIM manifest	JSON	components with type, position, level and parameters	import into Revit
project	NETUNO	all the steps and the rainfall series	reopening and auditing

3.9.2. The project file

The `.netuno` format is the only one that keeps the whole study: the nine steps, the multi-criteria weights, the LCA routes and the **daily rainfall series**, embedded in the file itself. A `.netuno` file reopens on another computer without the source CSV and reproduces the same number; it is with it that the cases of appendix J are distributed. The Netuno 4 CSV is still available in the export, marked as legacy, because it does not keep the steps added since then.

3.9.3. Spreadsheet

Exporting to a spreadsheet is the way for anyone who intends to do their own analysis of the results, whether to build a chart the program does not offer, or to combine Netuno's numbers with data from another source. Two data sets are available, and they answer different questions.

The **daily series** has one row per simulated day, with the recorded rain, the effective rain after the first flush, the collected volume, the day's demand, the rainwater consumption, the mains consumption, the overflow, the volume stored in both tanks, the volume pumped and the supply class. It is the basis of any chart or statistic the program does not offer ready-made, and it is what is used when the intended analysis differs from the one the program does.

The **sweep** has one row per simulated volume, with all the indicators calculated for each. It is the basis of the diminishing-returns curve presented in section 4.4, and the right format for anyone who wants to redraw it with another presentation.

Step **10. Export** builds the spreadsheet from six sets, ticked one by one: the executive summary with the project metadata, the daily water balance, the sweep curve with reliability, the year-by-year cash flow, the life cycle inventory and the multi-criteria decision matrix. Three choices complete the export:

Table 96 -- Spreadsheet export options and when to use each one

Option	Choices	When to use
format	XLSX, CSV or JSON	XLSX to open in Excel; CSV for R, Python or another program; JSON for integration
CSV separator	semicolon or comma	semicolon in Excel set to a decimal-comma locale; comma in R and most international tools
time filter	first typical year or full series	the year to illustrate; the full series for any statistic
economic horizon	10, 20, 25 or 30 years	20 years, the minimum design life of the plumbing system (ABNT NBR 15575-1:2025, Table 25), unless justified

3.9.3.1. Exporting everything, for audit

The **Export EVERYTHING (audit)** button, next to the spreadsheet export, saves a single Excel file with the six sheets above and, besides them, sheets nobody chooses: they go through **every** field of what the program used. They are the project assumptions, one row per parameter, with the value and the description taken from the code itself; the tariff applied, with the meaning of each block code, the regime, the sewage mode and the bill for consumptions from 0 to 50 m³; the treatment line sized in the quality step, with the stages, the doses and the carbon the LCA adds; the life cycle inventory item by item; the results of the simulation, the economics and the multi-criteria decision; and the 3D arrangement, with each part, each run and the take-off by diameter. The «0_Leia_me» (read-me) sheet says what is in each one.

It is the file for anyone who needs to check a calculation from start to finish, or to answer an assessor who asks what went into the carbon or the saving. A new parameter in the program appears in it without anyone having to remember to add it.

> **A warning about size.** Thirty years of daily series are 10,950 rows per simulation. A sweep of twelve volumes with a detailed series generates a hundred and thirty thousand rows. Export the detailed series only for the volume of interest.

3.9.4. Drawing in DXF

The program generates the hydraulic diagram of the sized system: cistern, filter, first flush, pump when there is one, upper tank, and the piping with the diameters adopted.

What the program produces is a **diagram**, not a construction design. It represents the components of the system and the logical connections between them, with the diameters adopted, and serves as a basis for the designer to draw the installation in plan, with the levels, the routing and the real clashes of the building. A diagram does not replace a plumbing design, and delivering it as one would be an error of professional responsibility.

Nor does the diagram carry -- and the construction design must add -- the building installation requirements of ABNT NBR 15527:2019 for non-potable water:

- rainwater piping painted or identified in **purple** (Munsell 2.5RP 3/10 or similar), with the inscription «ÁGUA NÃO POTÁVEL» (non-potable water) every 3 m on exposed pipes (item 4.5.5);
- a distribution system independent of the potable one, without cross-connection (4.5.8), with an air gap on the backup supply from the mains, according to ABNT NBR 5626:2020 (6.15.2);
- restricted-use points, such as the garden tap, with a «ÁGUA NÃO POTÁVEL» sign (4.5.9);
- separate potable and non-potable tanks, at least 60 cm apart between outer faces when side by side, and the non-potable one below when stacked (4.5.10 and 4.5.11);
- pumping according to ABNT NBR 5626 (4.5.2), which includes the two pumps of item 6.5.11.2 of that standard.

In a European project, the identification follows the national rule called for by EN 16941-1:2024 instead, and the repository document `docs/normas_nbr_x_en16941.md` lists the differences.

What the program does **not** size:

- routing and slope of the piping;
- real head loss of the installation;
- hydraulic arrangement of the first-flush device, of which the program calculates only the volume, `A x d_f`, and the cost;
- overflow and connection to the drainage;
- foundation and waterproofing of the cistern.

3.9.5. Block library and BIM manifest

In step **10. Export**, the *Block Library for Plans + BIM JSON* mode saves the DXF blocks of the components at 1:1 scale, in millimetres, and, next to them, the `*_netuno-bim.json` file. The program does not export IFC. The JSON manifest is the contract between the sizing and the model: each instance declares the component type, the level, the local position in millimetres, the rotation and the sized parameters, such as capacity and diameter, and the structure is validated before saving.

The integration with Autodesk Revit is done by an importer distributed with the program, run in the CPython 3 engine of pyRevit, with a document open in Revit 2022 or later, and it inserts the instances of the manifest using the families indicated in `family_mapping.json`, which associates each program type -- cistern, upper tank, filter, first flush, pump set, valves, non-return valve, air gap, overflow, non-potable point of use, calmed inlet, floating suction, level switch, solenoid valve, water meter, foot valve with strainer, vent and control panel -- with a family and a type of the office.

With no saved organisation, the program pre-fills the usual arrangement of these installations, from the study: roof collection, filter and first flush; cistern with a calmed inlet, abstraction by floating suction with a foot valve and strainer, a level switch against dry running, a screened vent and an overflow with a trapped outlet and anti-backflow valve to the storm drain; isolation valves on both sides of the pump set, a non-return valve on the rising main and a control panel; an upper tank with a level switch, screened vent, overflow with visible discharge, cleaning outlet with a valve and distribution with a valve to an identified point of use; and the potable backup supply -- water meter, valve and solenoid valve -- which only opens when rainwater runs out and discharges into the upper tank through an air gap, as ABNT NBR 5626:2020 requires. The control panel decides: with the upper tank at the start level and water in the cistern, it starts the pump; with the cistern at the minimum, it blocks the pump and opens the solenoid, which fills the tank only up to the backup float, lower, so as to leave room for the next rain. Pump and solenoid never run together. In the volumetry, the control cables -- from the level switches to the panel and from it to the pump and the solenoid -- appear dashed, and the complete logic is on the panel's data sheet. The arrangement is a starting point: positions, routes and parts belong to the designer.

The arrangement follows two choices of the project:

- **Number of pump sets.** With two, the standby is drawn in parallel with the duty pump, with its own isolation valve on the suction, non-return valve and isolation valve on the delivery. It connects to the duty pump's lines by tees.
- **Supply arrangement.** It changes what exists in the installation:
  - **Pumping to an upper tank:** it is the arrangement described above.
  - **Direct pressurisation:** there is no upper tank. The pressure pump takes the water from the cistern straight to the point of use. The potable backup supply discharges into the cistern itself, also through an air gap, and only up to a backup float installed in it.

- **Gravity:** there is no pump. The cistern sits on a base, above the point of use, and the water leaves by a low side outlet with a valve. The potable backup supply discharges into the cistern, as with pressurisation.

While the runs have not been organised, the arrangement is rebuilt at each change of the volume, the supply or the number of pumps.

With these data the program also builds the sheet of the 3D diagram, not to scale and on an A3 landscape sheet. Only the levels that organise the installation -- cistern, upper tank, pump set, filter and first flush -- get a level plane; the small parts appear on the drawing by their code, and the type and level of each one are in the «Parts and accessories» table, next to the assembly sequence and the take-off by diameter.

The importer does not create families or replace components by approximation: an instance without a family or without a corresponding level is listed as pending at the end of the import. The coordinates are local, and the tie to the project base point and north is done in Revit, by the designer. The manifest serves coordination and quantity take-off, and does not replace the modelling of the plumbing installation.

3.9.5.1. From the arrangement to the CAPEX and the LCA

The same arrangement feeds the cost and the carbon. The program takes off, part by part:

- the metres of pipe per diameter;
- the bends, one per change of direction of the route;
- the fittings, one at each end of a route connected to a part;
- each part of the manifest, by type;
- the metres of control cable.

Each item receives a price and a mass. The price goes to the CAPEX: pipes and fittings under *Piping*, parts and cables under *Accessories*. The mass, times the material factor, goes to the LCA in stage A1-A3. Cistern, upper tank and pump set already have their own line in both analyses and are not counted again. The filter and the first flush enter the CAPEX through the treatment line, when it is active, and the LCA through the first-flush subsystem, when it is active.

An automatic test checks that no part, no metre of pipe, no bend, no tee and no cable of the arrangement is left without cost or without carbon. The test covers the three supply arrangements, with one and with two pump sets. The `Auditoria_Quantitativo` sheet of *Export everything* lists each item with its destination in the CAPEX and the LCA, and closes the sum against the values the analyses used. The same sheet also checks whether the arrangement is consistent with the project: an arrangement organised by hand can fall behind when the project changes the supply or the number of pumps, and the divergence shows up there.

The prices and masses of the parts are a reference scenario, not a quotation. They are in `costs_database.json`, under the `pecas_do_arranjo` key, and can be changed there. In the economic step, the designer can also type other piping and accessory values: from then on the program uses the typed values. A new project starts from the arrangement. An old file keeps the costs it already declared. The reference prices are Brazilian, in reais; in the International Edition, and whenever the currency of the project is not the real, the program does not apply them, and the costs are typed in.

3.9.6. Reports

The three of section 4.10: executive, environmental with ESG, and research. In HTML for consultation and in PDF for delivery.

> The program has two PDF generators and uses the most complete one the computer allows. When it falls back to the alternative, the document comes out simpler, with the same content and the same numbers.

3.9.7. A filing tip

When closing a study, keep together, in the same folder:

1. the `.netuno` file;
2. the **rainfall series** used, or its exact identification in the database;
3. the research report in PDF;
4. the sweep spreadsheet;
5. if there was a batch, the **script** with the commands;
6. a text file with the **program version**.

Six items that make the study reproducible five years from now. Without them, the PDF report is just a claim.

4. Tutorial: one example from start to finish

4.1. The project and the rain

> By the end of this section you will have a project created, a precipitation series loaded and the catchment defined. And you will know why each of these three numbers matters more than it seems.

The ten sections of this chapter build **a single project**, from start to finish. With each section the study gains a layer, and the result changes. Keep the file: it is the same up to section 4.10.

4.1.1. The project

A single-family house in Florianópolis, in southern Brazil. Four residents, a 150 m² roof in ceramic tiles, a cistern buried in the back yard. It is the most common case in Brazil, and the most studied.

The choice of Florianópolis is not sentimental. The available precipitation series is long; the tariff structure of CASAN, the water and sewage utility of the state of Santa Catarina, is public and auditable; and the local rainfall regime is well distributed over the year, which -- as section 4.4 will show -- changes the choice of volume profoundly.

> **International Edition.** The tutorial uses Brazilian data because it is the case the program's results were checked against. In the International Edition you can follow it with the Florianópolis series that comes with the program and type in the tariff of section 4.7; the hydraulic results are the same in both editions.

4.1.2. The precipitation series

Open step **1. Project & Rainfall**. There are two ways to provide the rain.

Your own file. A CSV with one daily reading per line, in millimetres, in a single column. It is the way for anyone with data from a specific station -- in Brazil, from INMET or from the National Water Agency; elsewhere, from the national meteorological service.

National database (Brazil Edition). 652 Brazilian cities, 19 CMIP6 climate models and two emission scenarios, with the historical simulation of each model. It is the way for anyone studying other cities or future scenarios, and section 3.2 shows how to use it.

For our project, use the Florianópolis series that comes with the program and trim the **first 30 years**.

Table 97 -- Characteristics of the Florianópolis precipitation series used in the tutorial

The Florianópolis series	Value
days in the series	10,950
years	30.0
mean annual precipitation	1,795 mm
days with rain	46%
mean rain on rainy days	10.6 mm
largest daily rain	128.1 mm
days below the first-flush depth	7%

Three numbers in this table decide the project, and it is worth looking at them before any simulation.

1,795 mm per year is a lot of rain. Florianópolis is among the rainiest state capitals in Brazil, and that means the limiting factor of the system will not be the supply of water, but the demand and the size of the tank.

It rains on 46% of the days. Frequent, well-distributed rain favours small tanks: the cistern fills many times a year instead of having to get through a long dry spell. Compare with Brasília, in section 3.1, where the rain concentrates from October to April and the same tank yields little more than half.

Seven per cent of the days have rain, but below 2 mm. Those days will be diverted entirely, and the item on the first flush, further on, explains why.

> **How many years of series to use.** Thirty is comfortable. Below ten, the result starts to depend on the stretch chosen; below five, the bias of the empty tank at the start of the simulation weighs noticeably. Section 5.1 deals with this mistake in detail.

It is worth looking at a whole series before using it. Thirty years of Florianópolis are not thirty similar years: the driest year is almost a third below the wettest, and it is this variation between years that an annual mean erases. The daily distribution tells the rest of the story -- most days have no rain at all, and the annual volume is delivered by a small number of intense days (Figure 44).

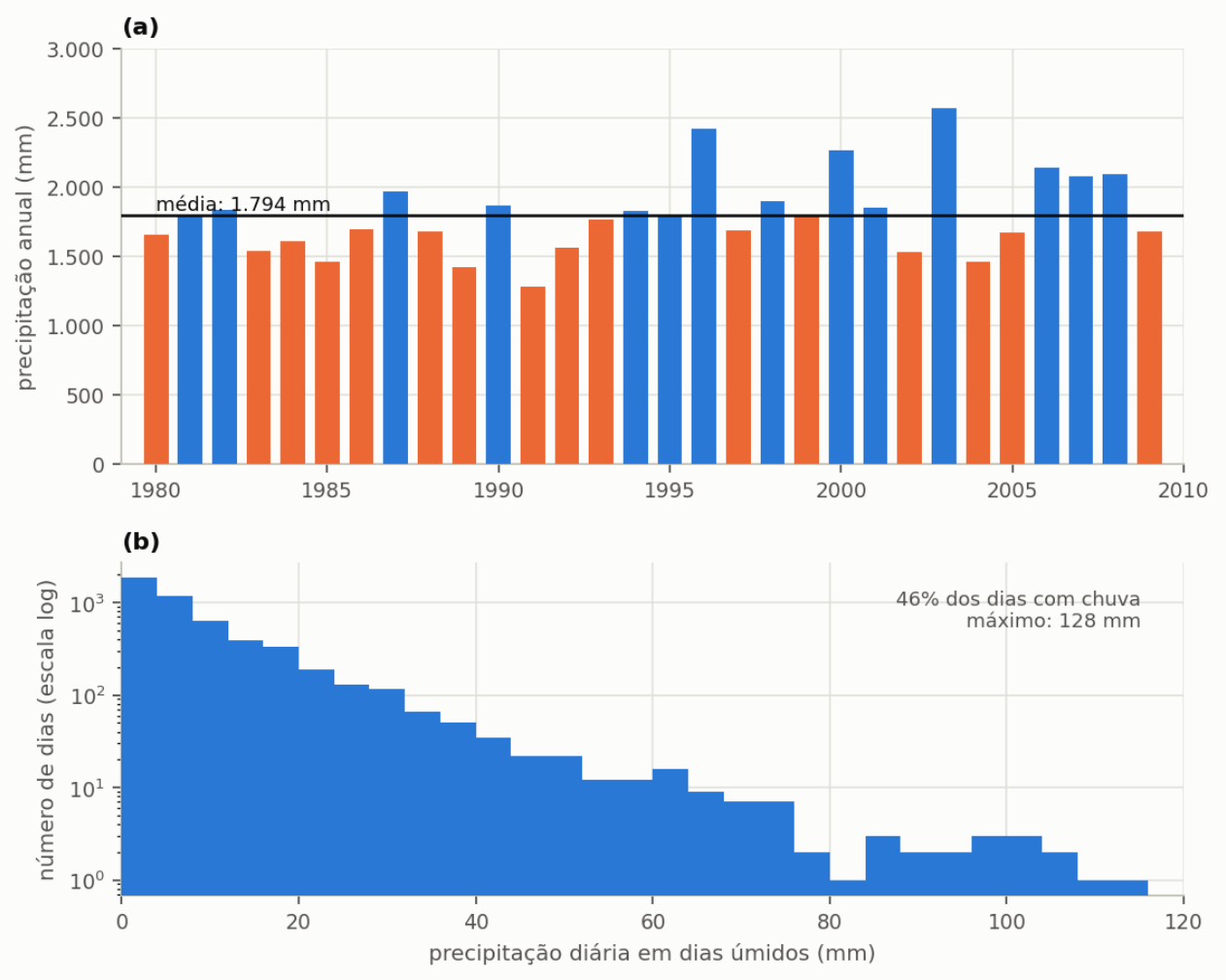


Figure 44 -- Florianópolis precipitation series used throughout the tutorial: (a) total precipitation of each year, with the thirty-year mean as a solid line and the years above the mean in blue and below it in orange; (b) frequency distribution of daily precipitation on wet days, on a logarithmic scale, which shows how few days account for the largest volumes.

4.1.3. The catchment area

The tutorial project adopts a catchment area of 150 m², and this is the first field where people often go wrong.

The catchment area is the **horizontal projection** of the roof, not the sloping area of the roof. A roof with a 30% slope over a 150 m² plan has about 157 m² of tiled surface, and still collects the same 150 m² of rain: precipitation falls vertically, and what it meets is the shadow of the roof projected on the ground. The definition is that of item 3.6 of ABNT NBR 15527:2019: the catchment area is the «horizontally projected area of the impermeable roof surface where the water is collected», and it is this area `A` that enters the availability equation of item 4.1.6. The European standard EN 16941-1:2024 uses the same horizontal projection.

> **Catchment area is not contributing area.** ABNT NBR 10844:1989 uses another area, for another purpose. To size gutters and downpipes by the peak flow, its item 5.2.1 adds to the projection the increments due to the roof slope, `(a + h/2).b` in its Figure 2, and the walls that intercept wind-driven rain. That area serves the design flow of a five-minute storm (item 5.3.1), and it overestimates the volume the roof collects over the year. In Netuno, enter the horizontal projection; the contributing area belongs to the gutter design, which the program does not do.

> **Warning.** Using the sloping area overestimates the collection by the ratio `1/cos(slope)`. On a 30% roof it gives 4.4% more water that does not exist. On steeper roofs the error grows quickly.

Once the series is loaded, the program already shows the regime it describes. It is worth checking this drawing before going on: a series from the wrong city, or with the wrong unit, usually shows up in the seasonal cycle of Figure 45 before it shows up in any result.

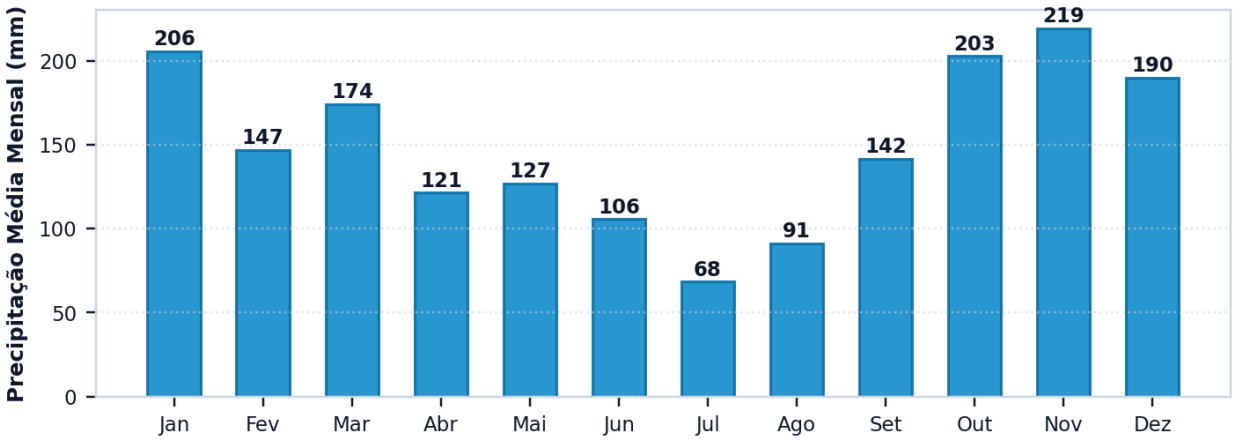


Figure 45 -- The mean seasonal cycle of the series, calculated and drawn by the program as soon as the precipitation is loaded. In Florianópolis, July receives about a third of what November receives, and it is this difference that the tank has to get through. Figure generated by the program itself.

4.1.4. The runoff coefficient

The tutorial types 0.85 in the coefficient field. The program, in turn, opens with ceramic tiles selected and applies 0.75, the value the list of materials gives them, which is also the value it assumes when no material is entered. Take care when reproducing the tutorial: to get the numbers of this manual, choose «Other / Custom» and type 0.85.

The origin of 0.85 deserves precision, because it is often quoted as a runoff coefficient and it is not one. ABNT NBR 15527:2019, in item 4.1.6, writes the available volume as the product of four terms -- precipitation, area, surface runoff coefficient C and efficiency of the collection system eta -- and recommends 0.85 **for eta**, in the absence of manufacturer data. That eta already accounts for the solids diversion device and for the diversion of the initial runoff. Netuno has a single field for the fraction of the rain that reaches the cistern and models the first flush separately, in millimetres. Typing the standard's eta there and also applying the 2 mm first flush counts the loss of the initial runoff twice, so the tutorial is conservative. In a study, declare the value adopted in the field, its origin and the first flush, and do not attribute to the standard a runoff coefficient it does not set.

> **International Edition.** EN 16941-1:2024 writes the yield as the product of a surface yield coefficient `e`, given by roof type in its Table 2, and a filter efficiency `eta`, 0.9 by default, with no first-flush depth. Netuno keeps its own list of coefficients and the first flush in both editions; to approach the European standard, type `e x eta` as a custom value and set the first-flush depth to zero. The repository document `docs/normas_nbr_x_en16941.md` lists the differences.

The surface runoff coefficient represents the fraction of the incident rain that actually reaches the cistern, and the difference to one corresponds to what is lost on the way. The losses have four origins: absorption by the roof material, evaporation on the heated surface, splashing out of the gutter and the initial wetting of the tiles at each new rain event. Figure 46 shows the coefficient the program gives to each roof material.

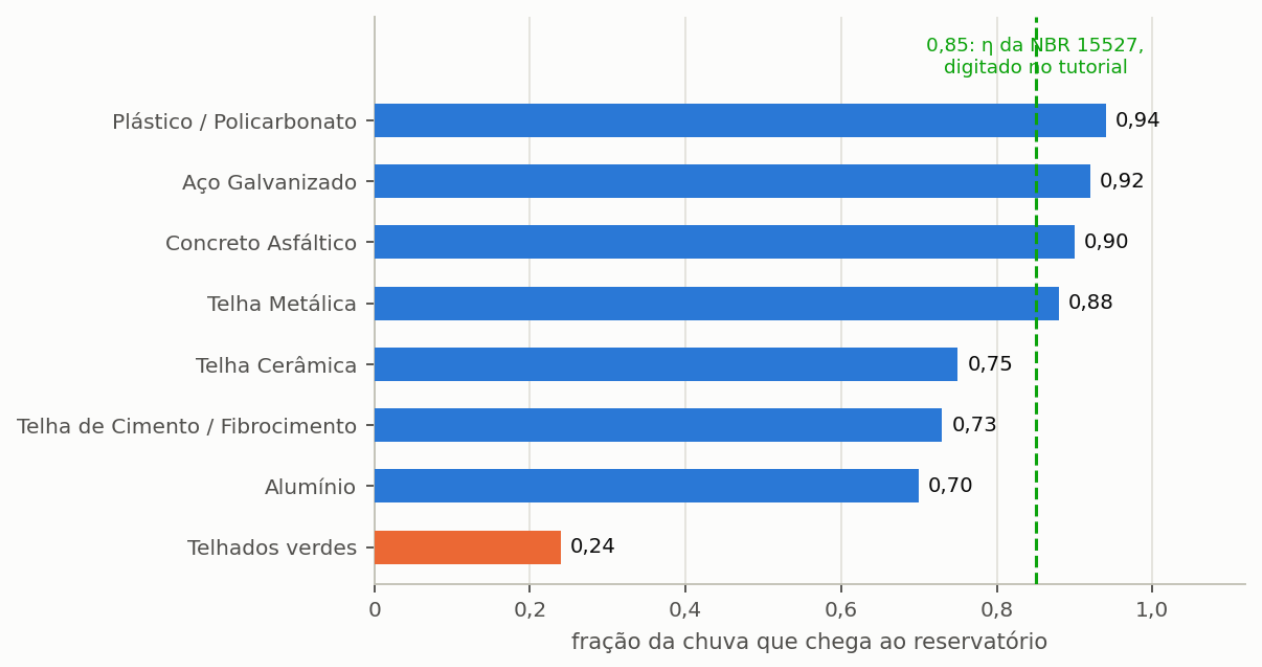


Figure 46 -- The runoff coefficient the program gives to each roof material, with the values of the study of materials published in the journal MIX Sustentável (2020). The dashed line marks 0.85, the collection factor of ABNT NBR 15527 adopted in the tutorial. The green roof is the only one that departs from the group, because it retains part of the rain in the substrate.

The program accepts any value between zero and one, and refuses values outside that interval naming the field -- a coefficient larger than one would mean that the roof delivers more water than the rain brings it. Step **11. Documentation** has the same table, with the source of the values. For a mixed roof, the coefficient to type is the mean of the coefficients of each material weighted by their areas.

4.1.5. The first flush

The tutorial diverts 2.0 mm on every rain event.

The first water that washes the roof carries dust, leaves, bird droppings and everything else that has settled since the previous rain. Diverting it, instead of leading it to the cistern, is the cheapest and most effective quality measure a rainwater system has, and that is why it is practically universal in well-run projects. ABNT NBR 15527 recommends a 2 mm first flush when there is no local data to justify another value, and that is the value adopted here.

The program offers three ways of applying the first flush, which correspond to three types of self-cleaning device.

Table 98 -- The three ways of applying the first flush

Mode	When it diverts	When to use
1	on every rain event	the default, and the most conservative
2	only on the first event after a dry spell	a device that does not empty by itself between close rains
3	every X days	a device with scheduled emptying

> **Warning, and this is the point that surprises the most.** When the first flush applies and the day's rain **does not exceed** the depth, the **whole** day is lost, not the difference. A day of 1.6 mm with a 2.0 mm first flush collects **zero**. In the Florianópolis series this affects 7% of the days. In climates with frequent drizzle, such as the southern Brazilian coast in winter, it can exceed 20%.

The reason for this behaviour is physical, not conventional: the self-cleaning device has to fill completely before any water goes on to the cistern. When the day's rain is not enough to fill it, nothing passes, and the retained volume is lost to the drainage together with the dirt it washed.

> **The depth is not the device.** `d_f` is a depth in millimetres. The hydraulic volume of the self-cleaning chamber is `A x d_f`, which in our project gives 150 x 2 = **300 litres**. Step **3. Quality & Treatment** sizes and prices that chamber (section 2.5), and section 4.7 shows its cost in the investment.

4.1.6. What we have so far

Table 99 -- State of the tutorial project at the end of defining the rain and the catchment

Item	Value
series	Florianópolis, 30 years
catchment area	150 m²
runoff coefficient	0.85
first-flush depth	2.0 mm, every event

With this the program already knows **how much water comes in**. It remains to say how much goes out, which is section 4.2.

A back-of-the-envelope calculation, to calibrate the expectation: 1,795 mm per year on 150 m² at a coefficient of 0.85 gives about 229 m³ per year of raw water, before the first flush. Section 4.3 shows how much of that volume turns into an actual saving -- and the difference between the two numbers is the most important lesson of the tutorial.

4.2. The building and the demand

> By the end of this section the program will know **how much water goes out** and, more importantly, **which part of it rain can supply**. That fraction is the ceiling of everything that follows.

Section 4.1 defined the supply. This one defines the demand. And there is an asymmetry worth understanding from the start: **rain cannot supply all the demand**, however much water there is. There is a ceiling, it is defined here, and no tank goes beyond it.

4.2.1. The building type

In step **2. Building & Demand**, choose *Single-family residential*. The program has eight building types, each with its reference per capita consumption (Figure 47) and the usual split of end uses. Industry has no reference range, because consumption depends on the production process, not on occupancy: in that case, the value must come from measurement.

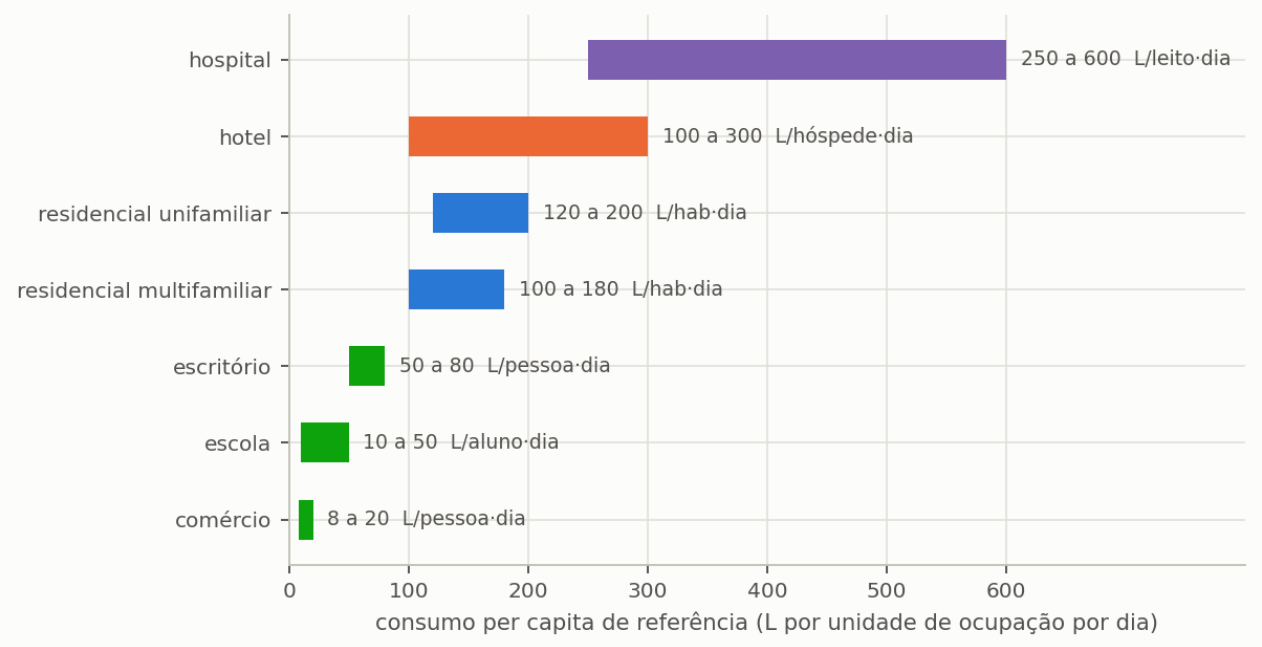


Figure 47 -- The usual ranges of per capita consumption by building type. The width of each range is large on purpose: they are reference values for those who have not yet measured, and measurement, when it exists, always prevails.

The building type is a starting point, not an imposition: the value remains editable. Section 3.5 deals with non-residential cases, which have a trap of their own in the tariff.

4.2.2. The residents and the consumption

Four residents, 150 L/person.day. A total of **600 L/day**.

The value of 150 L/person.day is the consumption of a middle-class Brazilian house with a regular supply. If you have the building's water and sewage bill, use it: divide the monthly consumption in litres by thirty and by the number of residents. The measured number is worth more than any reference.

The program accepts a **constant** demand or one modulated by week, month or day. The tutorial uses a constant one; section 3.4 shows the modulation, which matters in schools, hotels and any building with seasonality.

4.2.3. The fraction rain can supply

28%. It is the most important number in this section.

Rainwater does not go to the shower, the kitchen sink or for drinking. In a house, it goes to the toilet, floor washing, garden watering and vehicle washing. The share of these uses in the total consumption is the **ceiling of the saving potential**.

The split adopted in the tutorial is shown in Figure 48, with the uses rain supplies marked. The toilet accounts for 20% of consumption and cleaning and watering for 8%; shower, kitchen sink, washbasin and washing machine, which account for the rest, stay on potable water.

Adding the two that rain supplies gives the 28% of the tutorial. It is a conservative value for a single-family house without a large garden, and it is conservative on purpose: it keeps to the uses the standard admits.

> **The washing machine is not on the standard's list.** ABNT NBR 15527:2019, in item 4.1.7, admits seven non-potable uses for rainwater: cooling systems, flushing of toilets and urinals, vehicle washing, floor washing, fire-fighting reserve, ornamental use and landscape irrigation. ABNT NBR 16783:2019, which deals with alternative sources in general, also leaves laundry out of item 4.2.1. For any other use, item 4.1.8 of NBR 15527 requires the responsible professional to study the specific quality parameters and treatment. Use in the washing machine is common in the literature and in practice, and that is why many studies work with suppliable fractions larger than this tutorial's. In Joinville, Ghisi and Custódio (2019) obtained a mean potential of 18.5% with rainwater only in toilets, and of 40.8% with toilets and the washing machine. The tutorial does **not** adopt it: the machine stays on potable water, and the suppliable fraction falls to the 28% of the toilet and the outdoor uses. A project that includes laundry must declare that it is outside the normative list and which quality criterion was adopted for it. The European standard EN 16941-1:2024, by contrast, lists laundry among the basic uses, together with toilet flushing and garden watering.

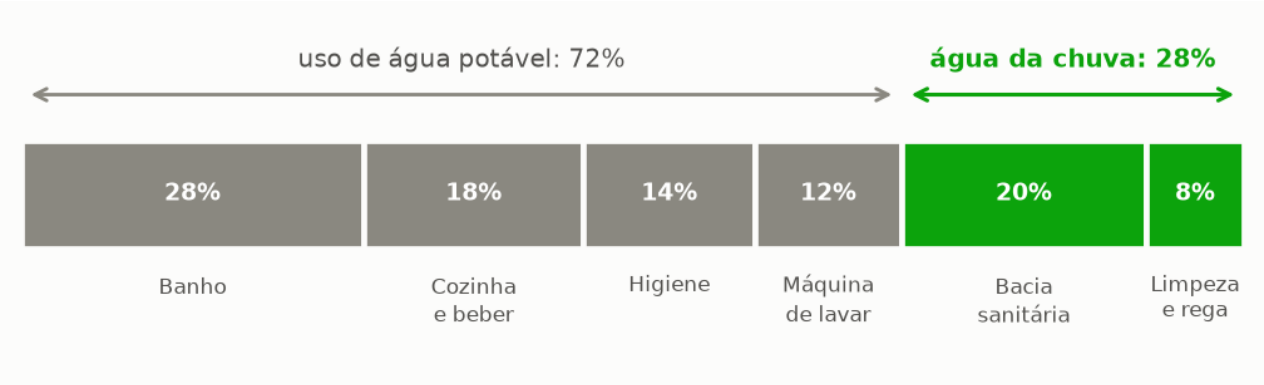


Figure 48 -- Typical split of the consumption of a house by end use. In grey, the uses that require potable water; in green, those where rainwater can be used, whose sum is the suppliable fraction and the ceiling of the saving potential: no tank, however large, saves more than this fraction.

> **This number is the ceiling.** With a 28% suppliable fraction, the saving potential **never** exceeds 28%, however large the tank and however much it rains. The table of section 4.4 shows the curve approaching exactly this limit and never crossing it.
>
> Anyone who wants to go beyond 28% must change the **use**, not the tank: add a use, such as laundry, with the declared quality criterion that item 4.1.8 of the standard requires, or reduce potable consumption.

4.2.3.1. Percentage on screen, fraction in the file

The same quantity appears in two forms, and each one has its place. In every field where you type -- the fractions of the building screen, the weekly and monthly tables, the scenario tables, the batch spreadsheet and the `--share` argument of the command line -- the fraction is a **percentage, from 0 to 100**: 28 means 28%. In the project file, in the parameter file and in `rainwater_share` and `non_potable_indoor_share`, it is a **fraction, from 0 to 1**: 0.28.

The split is not arbitrary. The percentage is the form in which the quantity circulates among designers, standards and water bills, and it is the form least prone to typing errors. The fraction is the form in which it enters the balance equation, multiplying the indoor demand (section 2.1), and it is the form that needs no conversion for anyone reading the file with another program.

What is not allowed is mixing them. Up to this version, twelve places in the program accepted both forms and decided by the rule «above 1 is a percentage». The rule fails precisely on small values, where the error cannot be seen: 1 became 100%, not 1%; 0.5 typed in a percentage column became 50%. Now the unit is fixed by the place, and never guessed from the value. A percentage above 100 is refused; a fraction above 1 is refused with the remark that it looks like a percentage. Figure 49 shows the three states of the field in the interface.

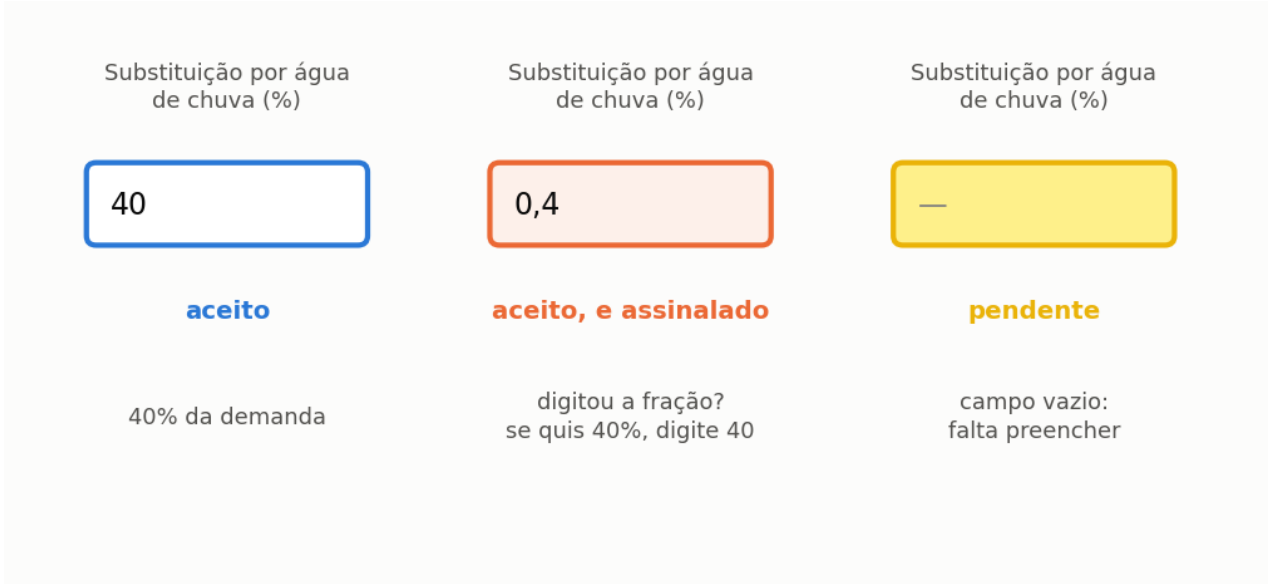


Figure 49 -- States of the suppliable-fraction field in the interface. The field receives a percentage from 0 to 100; in the project file and in the calculation parameters the quantity is stored as a fraction from 0 to 1. A value between 0 and 1 typed in the field is flagged in orange, without blocking the calculation, because 0.4% is possible, although rare.

> **The orange marker.** A value between 0 and 1 in a percentage field is accepted, because less than 1% replacement is possible, but the field turns orange and the tip explains what was understood. In the scenario tables and the batch spreadsheet, the program asks before simulating.

4.2.4. Breaking down the uses, or not

The program accepts two levels of detail in the description of the demand, and the choice between them depends on what the study intends to answer.

In the **simplified mode**, two things are entered: the non-potable fraction of indoor consumption -- toilets and laundry -- and, if there is one, the outdoor demand in litres per day. The **total** suppliable fraction comes from the two,

```
f = (indoor fraction x indoor demand + outdoor demand) ÷ (indoor demand + outdoor demand)
```

and appears in a field that cannot be edited, because it is a result and not an input. Without outdoor demand, the total is the indoor fraction itself: the 28% of this tutorial. It is the mode adopted here and it is enough for most studies.

> **Why the total is not typed.** Up to version 5.0.0 the total-fraction field was editable, but the simulation used the indoor fraction; and, when a project was opened, the screen reset the indoor fraction to 35% and the outdoor one to zero, whatever had been saved. A project that declared 28% was simulated with 35%. Today the screen restores both, the total is calculated, and an old file whose fractions disagree is opened with the total it declares, with a warning.

In the **detailed mode**, the demand is split into three parts: the indoor potable, which rain never supplies; the indoor non-potable, which is the system's priority; and irrigation, supplied last. The split changes two things. It changes the order of supply on days of scarcity, when indoor use takes precedence over the garden; and it makes it possible to distinguish indoor from outdoor use in certification credits, a distinction LEED requires in separate credits, as section 2.9 describes.

It is worth breaking down the uses when there is a garden of significant area, when certification is on the project's horizon, or when irrigation can be suppressed on rainy days -- a situation the next item deals with.

4.2.5. Irrigation and suppression by rain

Once an irrigation volume is declared, the program offers a rain suppression threshold: on days when the precipitation exceeds that threshold, the garden is not watered. The behaviour reproduces that of any irrigation controller with a rain sensor, a common and cheap piece of equipment, and ignoring it would overestimate the outdoor demand over the whole year. The default threshold is 2 mm per day.

> **The rainfall series is read twice.** One reading is converted into collected volume, after the first flush. The other decides whether the garden is watered. The two are independent: a day can suppress irrigation **and** yield collection.

One choice remains that deserves particular attention, because it changes the economic result and is not neutral: the **backup supply** of irrigation. With it on, the mains waters the garden on the dry days when the cistern cannot cope, and that volume enters the water and sewage bill as potable consumption. With it off, the garden is simply not watered on that day, the building does not buy that water, and the corresponding volume appears in the balance as *unmet demand* -- the third part dealt with in section 2.1.

The distinction between the two settings is conceptual before it is numerical. A garden that must be watered every day represents a real demand, which someone will pay for; a garden that accepts not being watered on a dry day represents a convenience, and treating it as demand inflates both the consumption and the apparent saving of the system.

4.2.6. What we have so far

Table 100 -- State of the tutorial project at the end of defining the demand

Item	Value
building type	single-family residential
residents	4
per capita consumption	150 L/person.day
total demand	600 L/day
fraction suppliable by rain	28%
rainwater demand	**168 L/day**

The program now knows the supply and the demand. Before simulating, a comparison worth keeping:

Table 101 -- Comparison between the annual rainwater supply and the suppliable demand

	m³/year
precipitation falling on the catchment area	229
volume left after the first flush	187
rainwater demand of the year	**61**

Supply exceeds demand: the available volume is **three times** what is intended to be used. If supply were the only limit, the system would meet 100% of the rainwater demand and reach the 28% ceiling.

It does not, and section 4.3 shows why.

4.3. The first simulation

> By the end of this section you will have the first result on screen and will know how to **read** the four numbers that matter. Reading is the hard part: the program delivers many numbers, and three of them answer different questions.

One input is missing before simulating: the volume of the tank. Let us guess.

3,000 litres. A 3 m³ concrete cistern is a common back-yard size, it fits the budget and it is what a designer would propose without much thought. Section 4.4 will ask whether it is the right volume. Here we only want to see the program run.

In step **4. Reservoir & Costs**, choose a fixed volume, enter 3,000 L, leave the upper tank unticked, and run.

4.3.1. The result

Table 102 -- Result of the first tutorial simulation, with a 3,000 L tank

Result with 3,000 L	Value
potable water saving potential	**26.48%**
rainwater consumed	158.9 L/day
mains water consumed	441.1 L/day
water overflowed	354.1 L/day
days of full supply	93.9%
days of partial supply	1.3%
days of no supply	4.9%
intended rainwater demand	168 L/day
annual volume saved	58.0 m³/year

The screen returns several dozen values, and four of them are enough to judge the project at this stage. Each one answers a different question, and it is precisely because they answer different questions that they do not replace one another: a project can be good by the first and bad by the third, and reading them together is what avoids a hasty conclusion.

4.3.2. The saving potential: 26.48%

This indicator answers the question that motivates most projects: how much of the water and sewage bill stops being paid. It is the central value the program produces and the one that appears in the summary of any study, and it is the ratio between the rainwater consumed and the total demand of the building over the whole series, as in Equation 14.

The denominator deserves attention, because everything else follows from it: it is the **total** demand, not the rainwater demand. It is this choice that sets the ceiling of the indicator at 28%, which is the suppliable fraction defined in section 4.2, and not at one hundred per cent.

The system reached 26.48% against a ceiling of 28%, which means it delivers 95% of what it set out to deliver. The remaining 5% correspond to the days when the cistern did not have enough water to meet that day's rainwater demand.

> **Do not confuse the two percentages.** «A saving of 26%» and «95% of the rainwater demand met» describe the same system and are different numbers. Published studies sometimes swap one for the other. Declare which one you are using.

4.3.3. The overflow: 354 L/day

It answers: is the tank big enough?

Three hundred and fifty-four litres per day, on average over the thirty years, overflow from the cistern and go to the drainage system. The overflow is more than twice the 168 L/day the system intends to consume, and the tank lets through more water than it uses.

It looks like the definitive argument for enlarging the tank, and it is not. The roof offers three times what the house consumes (section 4.2), and a good part of that surplus would overflow from any cistern: section 4.4 shows that not even 20,000 L brings it below 340 L/day. The overflow only says something about the volume when compared with that of other volumes, and that comparison is what section 4.4 does.

> A high overflow is not a defect of the program or of the project. It is the signature of the mismatch between the rain, which arrives concentrated in a few days, and the demand, which is constant. The tank exists to buffer this, and every finite tank fails in some event.

4.3.4. The days of supply: 93.9% full

It answers: how often can I count on the system?

This is the **time** reliability of the system, and it answers a question the saving potential does not reach: how often rainwater can be counted on. Each day of the simulation receives one of three classifications, according to the degree of supply achieved.

Table 103 -- Classification of each day of the simulation by the degree of supply

Class	What it means
full	rain met **all** the rainwater demand of the day
partial	it met part of it, and the mains made up the rest
none	the cistern was empty; everything came from the mains

On 93.9% of the days the system met the rainwater demand on its own, and on the remaining 6.1% the mains came in, fully or partly. The distinction has a direct consequence on the plumbing design: a system with many days of no supply needs a well-sized automatic backup supply, and the user notices the switching between the two sources over the year. A system with 99% full days is, in practice, independent of the mains for the uses it supplies.

> **Potential and reliability do not go together.** It is possible to have a high potential with low reliability: a tank that fills in the rainy season and dries out in the dry season delivers a lot of volume over the year and fails half the days. The table of section 4.4 shows the two columns diverging.

4.3.5. The annual volume saved

It answers: how much is this worth in money and in carbon?

This is the value that carries over to sections 4.7 and 4.8, because both the saving on the water and sewage bill and the avoided carbon emission are calculated from it. It is 58.0 m³/year, less than a third of the 187 m³/year the roof makes available after the first flush, and the whole difference corresponds to what overflowed.

Before discussing why the potential does not reach 28%, it is worth seeing the tank at work. The results tab has the daily series of the cistern level, and it is there that the two regimes limiting the yield can be seen at once: overflow and shortage (Figure 50).

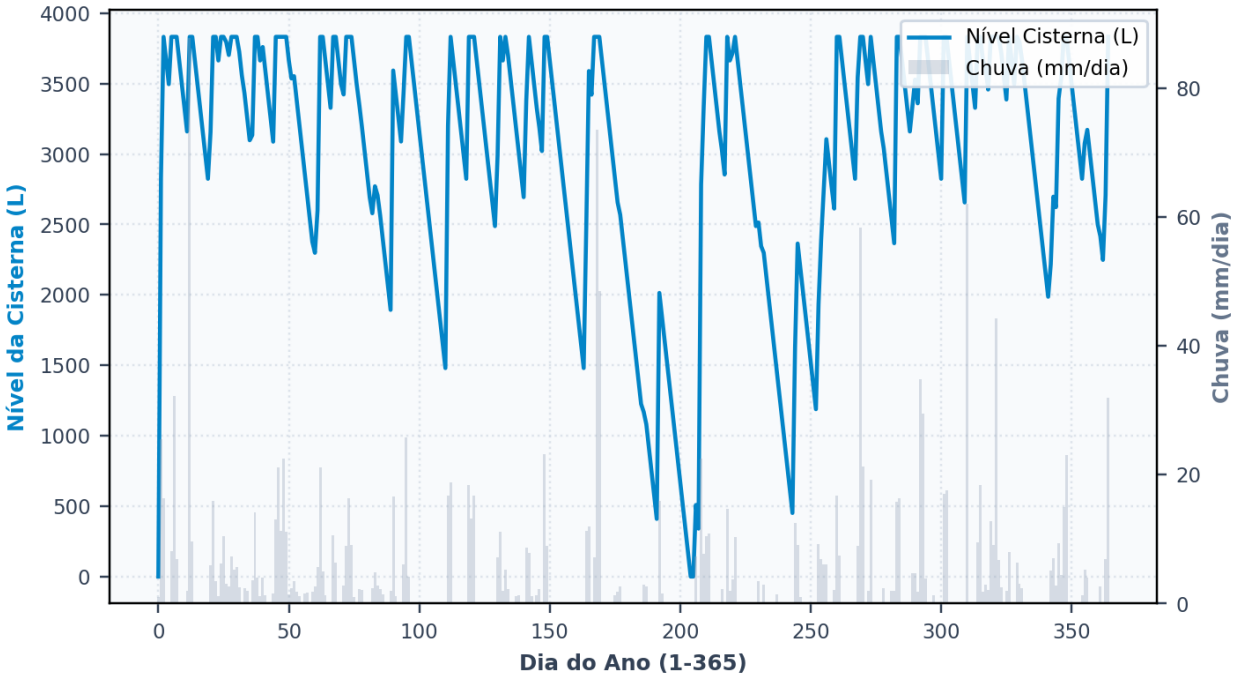


Figure 50 -- The level of the cistern over the first year of the series, with the daily rain in the background. The plateaus at the top are the days when the tank is full and overflows; the valleys at zero are the days when it ran dry and the mains supplied everything. Figure generated by the program itself.

4.3.6. Why we do not reach 28%

Section 4.2 ended with an apparently favourable comparison: 187 m³ per year of supply against 61 m³ of rainwater demand, that is, three times what is needed. Faced with this surplus, it would be reasonable to expect full supply, and yet the system delivers 95% of what was intended.

The explanation lies entirely in **time**. The surplus does not occur when the water is needed: precipitation arrives concentrated, fills the 3,000 L cistern in a single day of intense rain, overflows on the same day, and is missing two weeks later. The annual surplus and the daily supply are quantities that do not communicate, and it is precisely this lack of communication that makes the daily simulation indispensable.

The 3,000 L tank stores the equivalent of eighteen days of rainwater demand. Any dry spell longer than that empties it, and from then on the system depends entirely on the mains until the next rain.

That is why enlarging the tank works -- and that is why enlarging has a limit. Section 4.4 measures exactly where that limit lies.

4.3.7. What we have so far

Table 104 -- State of the tutorial project after the first simulation

Result with 3,000 L	
saving potential	26.48%
out of a ceiling of	28%
overflow	354 L/day
days of full supply	93.9%

A defensible result, and already close to the ceiling. But we chose 3,000 L by guesswork: it may be too little, and it may be too much.

What would be right?

4.4. Choosing the volume

> By the end of this section you will have the tank's return curve, will know where it stops paying off, and will understand why there is **no** absolute optimal volume -- there is an optimal volume for a declared criterion.

Section 4.3 took 3,000 L as a first guess and found 354 L/day overflowing, more than two thirds of everything the roof collects, and a supply already close to the ceiling. The two readings pull in opposite directions, and the question that arises is how much tank is worth having. It cannot be answered by an isolated attempt: it is answered by simulating the same project for many volumes and watching how the result evolves. That is what the program calls a **sweep**.

4.4.1. Running the sweep

In step **4. Reservoir & Costs**, choose the sweep mode and enter the maximum volume and the step. For our project: a maximum of 20,000 L, a step of 1,000 L.

Two particularities of how the grid is built are worth knowing before interpreting the result. The first is that the step entered is rounded up to a multiple of 250 L, the commercial quantum of tanks -- nobody buys a 1,300 L cistern, and an optimum pointing at that volume would be impracticable. When the step asked for is not a multiple of 250, the program adjusts it and records the adjustment, so that the report can declare the step actually used.

The second is that the grid is limited to forty positive volumes. A sweep from 0 to 50,000 L with a 100 L step would need five hundred simulations of thirty years each, and the gain in information does not make up for the computing time, because the curve is smooth and forty points describe it amply. The maximum volume entered is always kept as the last point of the grid, so only the last interval can be smaller than the effective step.

Zero volume is part of the grid as the reference alternative: it represents the no-system scenario, against which all the others are compared.

4.4.2. The curve

Table 105 -- Volume sweep of the tutorial project, with the marginal gain of each added cubic metre

Volume	Potential	Gain over the previous one	Gain per added m³	Overflowed	Full days
L	%	percentage points	%/m³	L/day	%
0	0.00	--	--	513	0.0
1,000	21.03	21.03	21.03	387	70.3
2,000	25.03	3.99	3.99	363	87.9
3,000	26.48	1.45	1.45	354	93.9
4,000	27.12	0.64	0.64	350	96.6
5,000	27.47	0.35	0.35	348	97.9
6,000	27.70	0.23	0.23	346	98.8
8,000	27.92	0.23	0.11	345	99.7
10,000	27.99	0.06	0.03	344	99.9
12,000	28.00	0.01	0.01	344	100.0
16,000	28.00	0.00	0.00	344	100.0
20,000	28.00	0.00	0.00	343	100.0

Figure 51 plots the results of the table: the saving potential converges to the suppliable fraction, and the marginal gain falls by three orders of magnitude across the sweep.

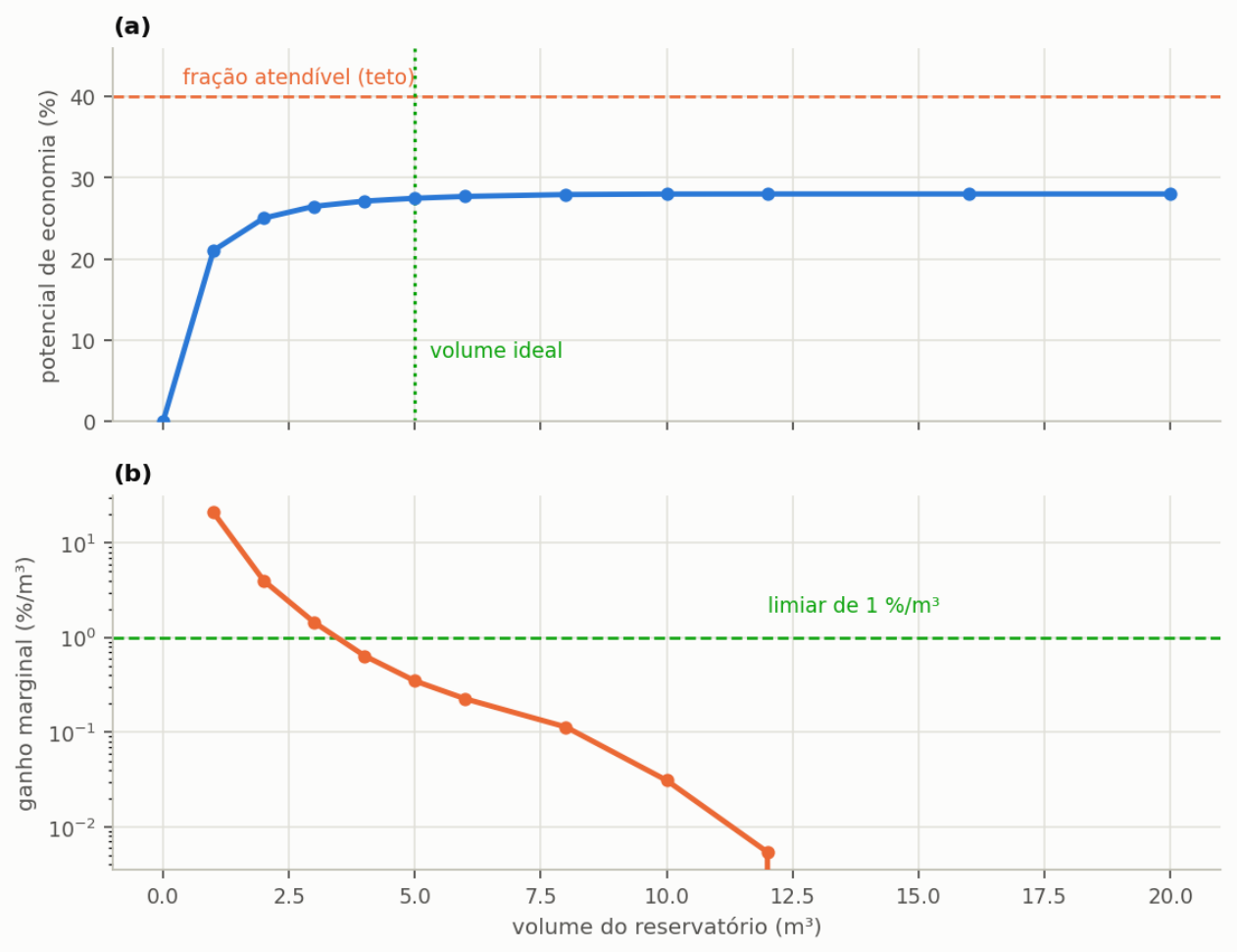


Figure 51 -- (a) Saving potential as a function of the tank volume, approaching the 28% ceiling without crossing it, with the ideal volume marked; (b) marginal gain, on a logarithmic scale, with the 1 %/m³ threshold adopted in the tutorial.

The decisive column of the table is the **gain per added cubic metre**, which is the discrete derivative of the potential curve and expresses how much saving each extra cubic metre of tank adds. The table below highlights a few intervals, and the pattern it reveals is the central argument of this section.

Table 106 -- Marginal saving gain for each cubic metre added to the tank

From -> to	Gain per m³
0 -> 1,000 L	21.03 %/m³
1,000 -> 2,000 L	3.99 %/m³
2,000 -> 3,000 L	1.45 %/m³
3,000 -> 4,000 L	0.64 %/m³
4,000 -> 5,000 L	0.35 %/m³
10,000 -> 12,000 L	0.005 %/m³
16,000 -> 20,000 L	0

The first cubic metre of tank delivers 21 percentage points of saving; those from 10 to 12 m³ deliver five thousandths of a point each, and those from 12 to 20 m³, nothing. The difference is four orders of magnitude, and the cubic metres cost the same concrete, the same excavation and the same labour. From 12 m³ on, the system supplies every day of the series, and no further volume has anything to add.

This behaviour is not a peculiarity of the Florianópolis rainfall regime. It is the characteristic shape of every rainwater harvesting curve, and it follows from two facts acting together. The first litres of storage capture the

small, frequent events, which make up most of the rains in any regime; and the ceiling imposed by the suppliable fraction, 28% in this project, pulls the curve towards the horizontal as the system approaches it, so that the room left for gain runs out.

4.4.2.1. The four charts the screen shows

The sweep does not return only the potential curve. The results tab has four charts that answer different questions about the same set of simulations, reproduced below with the data of this tutorial: the saving potential (Figure 52), the marginal gain (Figure 53), the overflowed and consumed volumes (Figure 54) and the classification of the days (Figure 55).

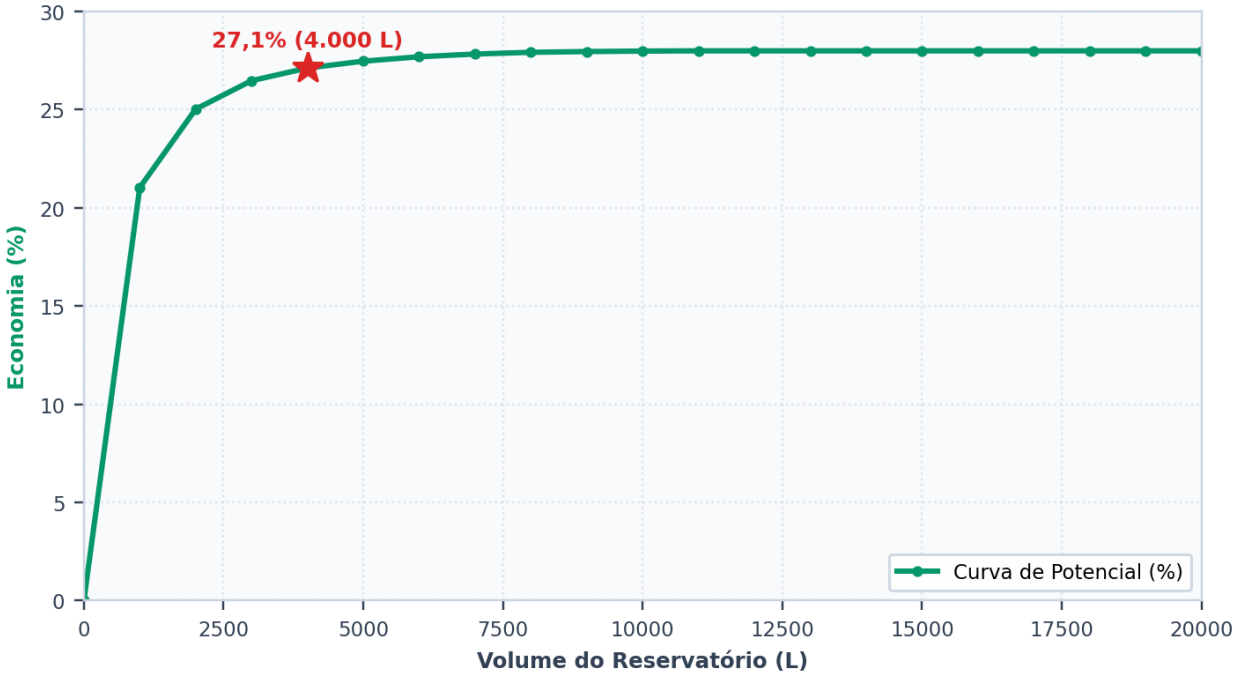


Figure 52 -- The first chart of the results tab: the potable water saving potential as a function of the lower tank volume. The star marks the selected volume. Figure generated by the program itself, with the tutorial data.

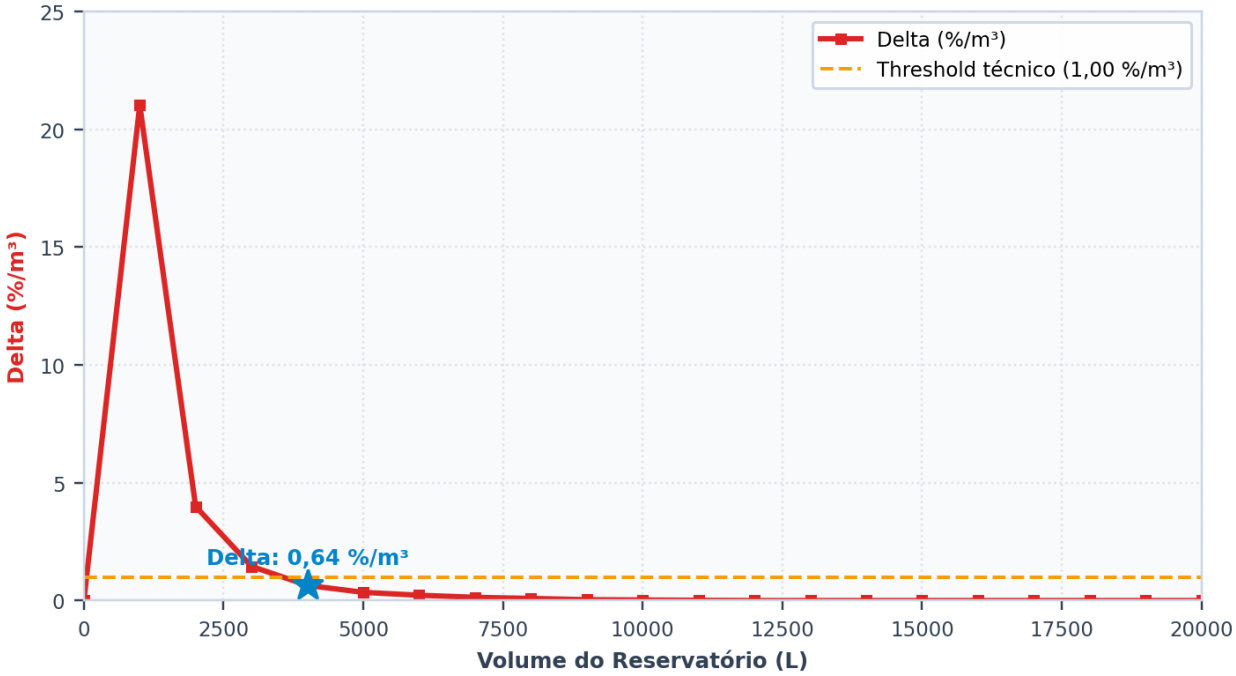


Figure 53 -- The second chart: the marginal gain, in percentage points per added cubic metre. It is the derivative of the previous curve, and it is on it that the ideal-volume criterion operates. Figure generated by the program itself.

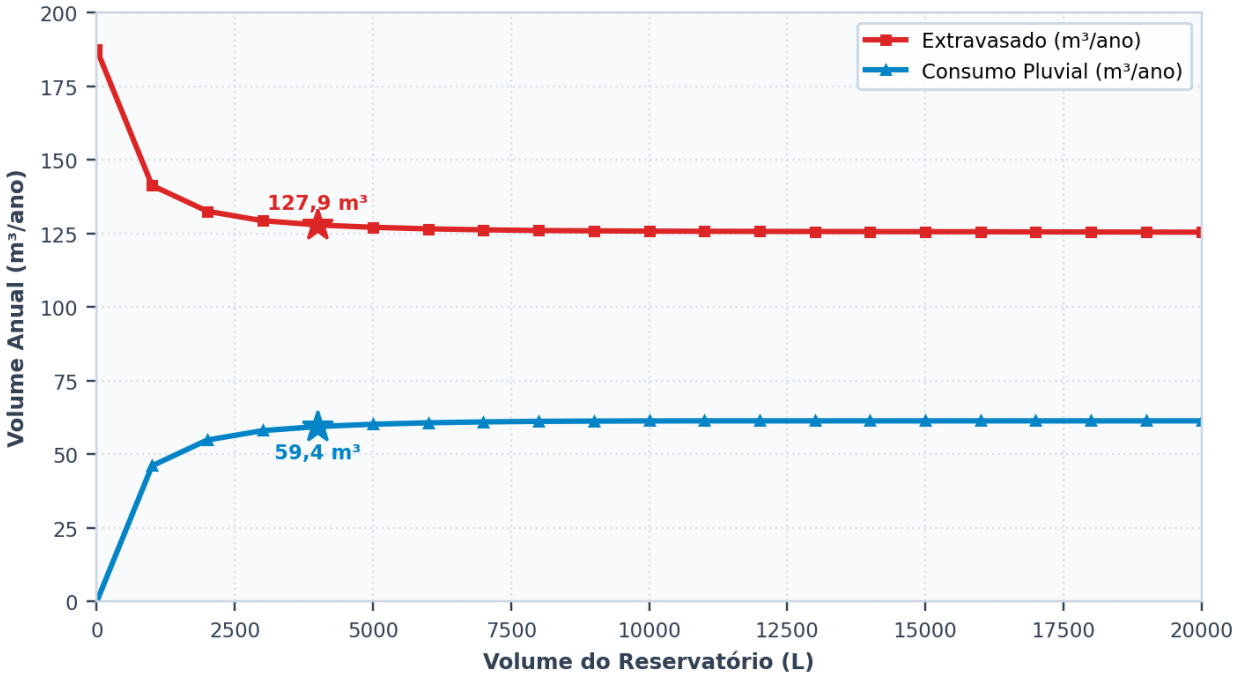


Figure 54 -- The third chart: the overflowed volume and the rainwater volume consumed, both annual, on the same scale. The two curves explain each other, because all the water that stops overflowing becomes consumed. Figure generated by the program itself.

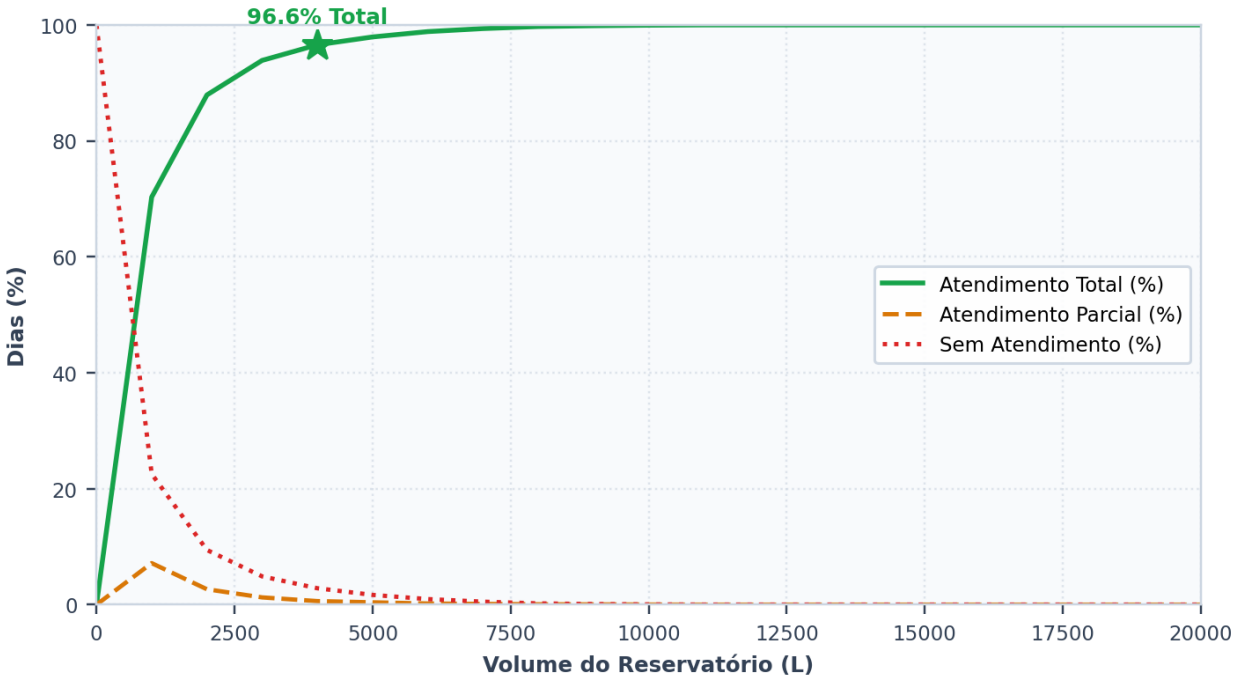


Figure 55 -- The fourth chart: the split of the days of the period into fully supplied, partly supplied and not supplied. It is the time reading of the same sweep, and the only one that answers how often the system can be counted on. Figure generated by the program itself.

4.4.3. The ideal volume

Faced with a curve whose yield falls off in this way, the question «what is the ideal volume?» only admits an answer once it is established what no longer paying off means. Netuno uses the marginal-gain criterion for this: the ideal volume is the smallest volume whose gain per added cubic metre has already fallen below a threshold declared by the user, as formulated in section 2.2.

The program opens with a threshold of 5 %/m³, and the tutorial adopts a stricter one, of 1 %/m³. The difference is not a detail: with the default, the ideal volume of this project would be 2,000 L; with the tutorial's threshold, the first volume to fall below it is 4,000 L, with a marginal gain of 0.64 %/m³. Anyone redoing the tutorial must therefore change the threshold in the program before running the sweep.

Table 107 -- Indicators of the ideal volume obtained with the 1 %/m³ threshold adopted in the tutorial

Indicator	Value
ideal volume, 1 %/m³ threshold	4,000 L
potential at that volume	27.12%
out of a ceiling of	28%
days of full supply	96.6%

The threshold, however, is the analyst's choice, and it decides the result as much as the input data. The table below shows the ideal volume each plausible threshold would produce for this same project.

Table 108 -- Ideal volume obtained with each marginal-gain threshold, for the same project

Declared threshold	Ideal volume	Potential
3.0 %/m³	3,000 L	26.48%
1.0 %/m³	4,000 L	27.12%
0.5 %/m³	5,000 L	27.47%

Table 109 -- Cost and performance compared between the ideal volume and the largest volume swept

	4,000 L	20,000 L
saving potential	27.12%	28.00%
price of the concrete cistern	R$ 5,304	R$ 20,607
mass of concrete	4,800 kg	24,000 kg
embodied carbon of the system	858 kg CO2e	3,546 kg CO2e
price ratio		**3.9x**
potential gain		**+0.88 points**

The program gives no opinion on which threshold would be right, and it could not: the choice depends on the cost of the tank, the value of water in the municipality and the space available on the plot, none of which is hydrological data. What the program does is apply the declared threshold faithfully and record it in the report.

> **In a published study, declare the threshold.** Two papers with an «ideal volume» from different thresholds are not comparable. Between the thresholds of the table, the difference is 2,000 L, and the 5,000 L volume costs a third more than the 3,000 L one to deliver one percentage point.

4.4.4. Why the marginal-gain criterion is not enough

The marginal-gain criterion is purely hydraulic: it looks at the yield curve and nothing else. It does not know how much the tank costs, how much a cubic metre of water is worth in the municipality in question or how much carbon the chosen material carries with it. Three consequences follow from this blindness, and each one can move the recommendation in a different direction.

- If water is very expensive, the gain of 0.35 %/m³ of the fifth cubic metre may pay for itself well, and the economic volume will be **larger** than the hydraulic one.
- If the cistern is expensive or space is scarce, the opposite.
- A 20 m³ concrete cistern weighs twenty-four tonnes and carries, with the rest of the system, about **3.5 tonnes of embodied CO2e**, against less than one tonne for the 4 m³ one. The last cubic metre adds no saving at all, and has no way of offsetting the difference.

Three criteria and three different answers to the same question, and none of them is more legitimate than the others in the abstract. That is why the tutorial does not end here: sections 4.7 and 4.8 add money and carbon to the analysis, and section 4.9 puts the three dimensions on the same table, with declared weights, to reach a compromise volume that can be defended.

> **The most common mistake, and it is in section 5.1.** Publishing the volume with the largest water saving as if it were the recommended volume. According to the table, 20,000 L gives the largest potential, 28.00%, the same as 12,000 L already gave. It also costs **3.9 times more** than 4,000 L, and carries **4.1 times more embodied carbon**, to deliver 0.88 percentage points more.

The sweep has one more reading that the table does not show: how much of the capacity the consumption actually uses. Measuring the tank by its level is misleading, because a large cistern always looks full precisely because it stores water the demand never gets to withdraw. Figure 56 measures the capacity by consumption, in two ways. The first is the **stored autonomy**, the volume at the end of each day divided by the daily rainwater demand of 168 L, shown as a duration curve. The second is the number of **turnovers per year**, the ratio between the annual rainwater consumption and the capacity, which says how many times the volume of the cistern is actually consumed and replenished.

The 2 m³ cistern turns its volume over 27 times a year and ends the day empty on 13% of the days: the capacity is fully used, and water runs short. The 4 m³ one turns over 15 times and empties on 4% of the days. The 20 m³ one turns over only 3 times a year and never empties, but on half of the days it holds more than 110 days of demand, far beyond the dry spell the series records. That reserve the consumption never mobilises is what the additional investment buys, and it is the physical reading of the diminishing returns of the potential curve.

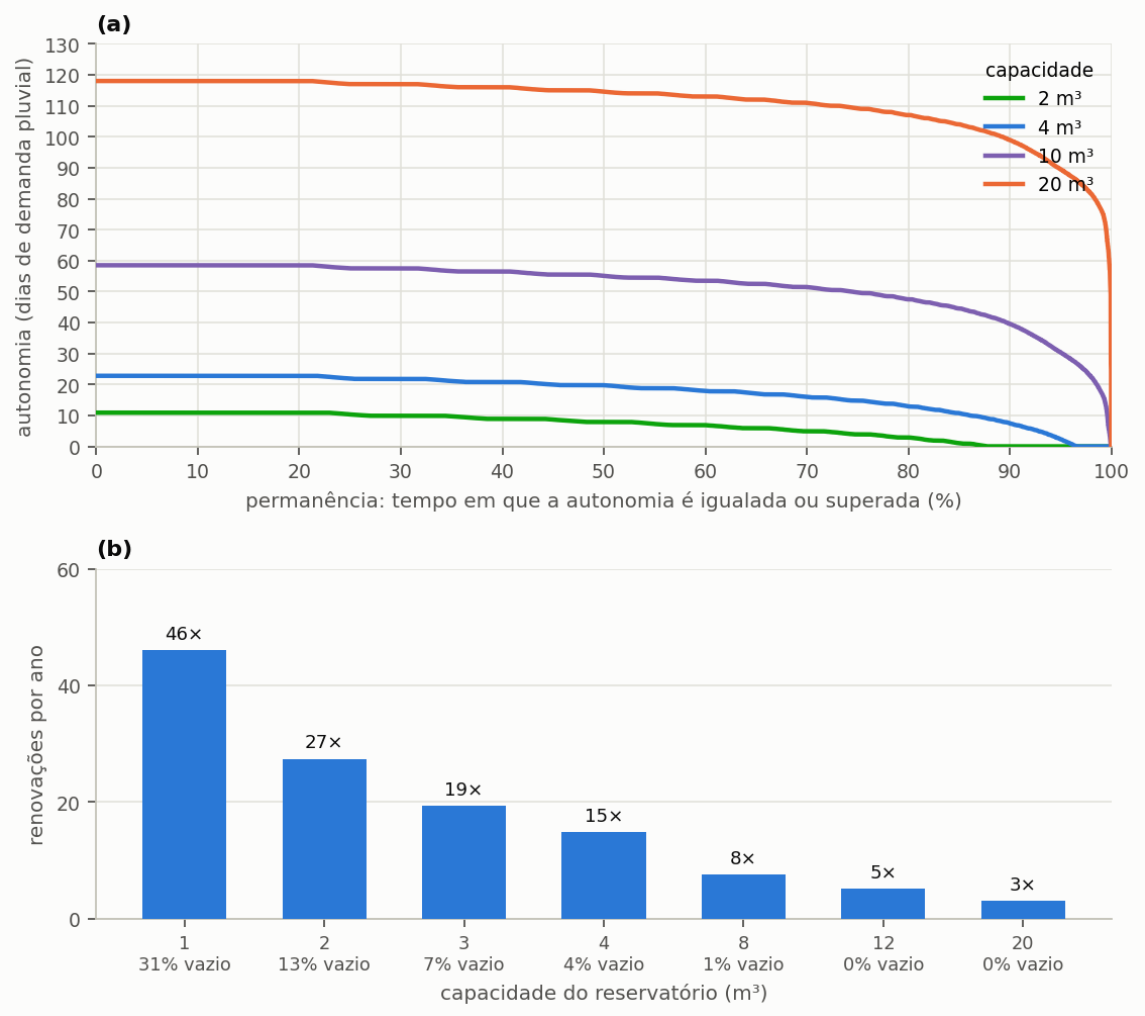


Figure 56 -- Use of the lower tank capacity in the tutorial project, with a rainwater demand of 168 L/day: (a) duration curves of the stored autonomy at the end of each day, in days of rainwater demand, for 2, 4, 10 and 20 m³; (b) annual turnovers of the volume, given by the ratio between the annual rainwater consumption and the capacity, with the fraction of days on which the tank ends empty (up to 5% of the capacity) shown under each volume.

4.4.5. Two more readings of the table

The overflow never reaches zero. It starts at 513 L/day without a tank, falls to 343 L/day with twenty cubic metres and levels off there. Two thirds of the water the roof collects will keep going to the drainage, for a reason no enlargement of the tank solves: the demand simply does not consume it. Rainwater harvesting systems do not hold water for the sake of holding it, and anyone seeking flood attenuation needs a detention tank, which is another project, with another sizing criterion.

Reliability grows faster than the potential. Going from 3,000 to 4,000 L, the saving potential gains 0.64 percentage points, while the days of full supply rise from 93.9% to 96.6%, and the days of shortage fall almost by half. When the design criterion is comfort of use -- rainwater never running out at the points that use it -- and not the saving on the bill, reading the same table leads to another choice.

4.4.6. What we have so far

The tutorial adopts 4,000 L and moves on. It is the ideal volume by the hydraulic criterion with the 1 %/m³ threshold, and it is the value the project carries up to section 4.9, where it will be questioned again, this time with money and carbon included in the analysis.

Table 110 -- State of the tutorial project after choosing the volume

Project so far	
series	Florianópolis, 30 years
catchment area	150 m²
total demand	600 L/day
suppliable fraction	28%
tank volume	**4,000 L**
saving potential	27.12%
days of full supply	96.6%

One hydraulic decision remains before money can be discussed: does this water need to go up to an elevated tank, or can it be pressurised directly from the cistern? That is the subject of section 4.5.

4.5. Is an upper tank needed?

> By the end of this section you will be able to answer the question with a number, not with habit. The answer usually surprises.

The classic Brazilian rainwater harvesting arrangement uses two tanks: a buried cistern, which receives the water from the roof, and an elevated water tank, which feeds the points of use by gravity; between the two, a pump lifts the water. The arrangement is so widespread that almost every project adopts it without examining why, and this section examines it.

4.5.1. What the upper tank is for

The upper tank is not for storing more water; it is for delivering it. The distinction seems subtle and decides the whole analysis, because everything that follows depends on not confusing storage capacity with distribution capacity.

The cistern is buried, and the points of use -- toilet, utility sink, garden tap -- are above it. Something has to overcome that difference in level, and there are two ways of doing it with a pump.

- **Direct pressurisation.** A small pump with a pressure switch starts whenever a point is opened. There is no upper tank.
- **Pumping to an elevated tank.** The pump fills the tank from time to time, and gravity does the rest.

When the cistern is above the points of use, there is a third one: **gravity**, with no pump. Step **4. Reservoir & Costs** asks which of the three arrangements the project adopts, and it is this answer that defines the energy and the pump sets in the cost and in the carbon.

Table 111 -- The three supply arrangements and what each one adds to the project

Arrangement	Energy calculated	Upper tank and pump sets
pumping to an upper tank	volume pumped, at the manometric head, plus the start-ups	enter the cost and the carbon
direct pressurisation	all the rainwater consumed, at the working pressure	only the pump sets
gravity	none	none

In direct pressurisation, step 4 asks for the model of the pressure pump, chosen from a catalogue of six models with the manufacturer's flow x pressure curve, the working pressure and the flow at the point of use. The pressure pump starts at each flush, 5.05 per person per day according to the end-use survey of Mayer et al. (1999), and stays on while the cistern of the toilet fills, at the pace of its float valve. The energy is that time multiplied by the electrical power. In the tutorial, the smallest model of the catalogue would use 40.5 kWh per year, five times the 7.7 kWh of pumping to a tank; section 2.6 explains why, and shows that the model chosen weighs more than the pressure.

Pumping to an elevated tank has three real practical advantages. The pump starts a few times a day, instead of dozens, which extends the life of the equipment; the pressure at the points of use is constant, because it comes from the height of the water column and not from the pump running; and there is a reserve available during a power cut. None of these three advantages is a water saving, and that is precisely the confusion this section clears up.

4.5.2. The fair comparison

Comparing a 4,000 L system without an elevated tank with another of 4,000 L plus a 1,000 L tank is a methodological trap, because the second has more total storage and would win for that reason, not because of the arrangement. To isolate the effect of the arrangement, the storage must be kept constant: so 5,000 L in a single tank is compared with 4,000 L below plus 1,000 L above.

Table 112 -- Single tank compared with two tanks, keeping the same total storage

Same storage of 5,000 L	One tank only	4,000 below + 1,000 above
saving potential	27.47%	27.30%
days of full supply	97.9%	97.3%
volume pumped	--	59.8 m³/year
pump starts	--	68 per year
pump hours	--	12.6 h/year
energy	--	7.7 kWh/year
energy cost	--	R$ 5.37/year
potential difference		**-0.17 points**

Keeping the same storage, the arrangement with an elevated tank delivers about a sixth of a percentage point less saving than the single tank. The difference is not a rounding error, and it has a cause: **the cistern overflows while the upper tank still has room**.

Rain enters only through the cistern, and the overflow is calculated there before the day's consumption. Pumping, however, only happens when there is consumption and the upper tank falls below the trigger level. So, on a rainy day with the cistern full, the water exceeding the 4,000 L below is lost even if the tank above has hundreds of litres free. A single 5,000 L tank would keep that water.

That is also why a higher trigger level yields more: with it, the pump fills the tank earlier, the cistern is emptier when the rain arrives, and it overflows less. The reserve below the trigger level is not locked away, because it is consumed right after each pumping and when the cistern runs dry.

> **The upper tank does not increase the water saving. It costs.** In our project it costs the tank, it costs the pump, it costs 7.7 kWh per year of energy, and it costs a sixth of a point of potential compared with a single tank of the same total volume.
>
> That does not make it wrong. It makes the decision **hydraulic**, not one of water yield: adopt it if the project needs gravity pressure or a reserve against power cuts, and not expecting it to yield more water.

4.5.3. The trigger level

If you adopt the tank, there is a parameter to adjust: the reserve volume that is not consumed, below which the pump starts.

Table 113 -- Effect of the pump trigger level on the saving potential and on the number of starts

Trigger level	Potential	Starts per year
0 L	27.24%	62
100 L	27.30%	68
300 L	27.33%	87
500 L	27.39%	121

A higher trigger level means **more potential and more starts**, for the reason above. From 0 to 500 L, the potential rises by 0.15 points and the starts almost double, from 62 to 121 per year.

The trade-off is between a marginal water gain and the wear of the pump. Residential pump sets cope well with a few starts a day; what kills them is continuous on-off cycling. One hundred and twenty-one starts a year is one every three days, and none of the values in the table comes close to being a problem.

> **Where the trigger level really matters** is in buildings with concentrated demand, where the tank empties quickly. There it can exceed a thousand starts a year, and it is better to size the tank than to adjust the trigger level.

4.5.4. The energy, and why it is small

Seven point seven kilowatt-hours a year. Five reais and thirty-seven centavos.

It looks too little, and it is worth understanding why. Lifting one cubic metre of water twelve metres costs, in hydraulic energy, about 0.033 kWh. With the efficiency of the pump and that of the motor, and adding the pump start-ups, it comes to about 0.13 kWh per cubic metre pumped. Sixty cubic metres a year give the 7.7 kilowatt-hours of the table.

Water is heavy, but twelve metres is little. That is why the energy cost of rainwater harvesting is negligible compared with the saving on the water and sewage bill -- and that is also why the argument that «pumping water uses more energy than it saves» does not hold in small buildings.

> **Warning about the nameplate power.** The program asks for the efficiency **of the pump** and the efficiency **of the motor** separately. The nameplate power is the mechanical power at the shaft, and the motor efficiency is what separates what the shaft delivers from what the meter records. Left blank, the program adopts Table 13 of ABNT NBR 17094-2: 60.5% for half a horsepower. Section 2.6 gives the details.

4.5.5. The tutorial's decision

We adopt pumping to a 1,000 L upper tank, with a 100 L trigger level.

The decision is hydraulic, not one of water yield. The elevated tank gives constant pressure at the points of use without depending on the pump at each flush, keeps water available during a power cut and makes the pump start 68 times a year, instead of the more than seven thousand starts of a pressure pump triggered at each flush by four residents, which would use 40 to 99 kWh per year depending on the model. The price is what the table shows: the tank, the pumps, 7.7 kWh per year and a sixth of a point of potential compared with a single 5,000 L tank. There are two pumps, the duty pump and the standby, as item 6.5.11.2 of ABNT NBR 5626:2020 requires for pumping and pressurisation systems.

Compared with the 4,000 L cistern on its own, the potential rises from 27.12% to 27.30%, because the tank adds 1,000 L of storage. From here on, the cost, the carbon and the tutorial's decision include the tank, the two pump sets and the pumping energy.

Table 114 -- State of the tutorial project after the decision on the upper tank

Project so far	
lower tank volume	4,000 L
arrangement	pumping to an upper tank
upper tank	1,000 L, 100 L trigger level
saving potential	27.30%

Now we know how much water the system delivers. It remains to know **what water** it is.

4.6. What quality the use requires

> By the end of this section the project will have a treatment line defined, and you will be able to tell what the standard requires from what the program recommends.

Up to this point the tutorial has treated rainwater only as volume: how much is collected, how much is stored, how much is consumed. Water, however, is also a quality, and quality costs money -- in equipment at installation and in consumables every month. This section defines the quality class of the project and sizes the corresponding treatment line.

4.6.1. What the standard says, and what it does not say

ABNT NBR 15527 is the Brazilian standard for rainwater harvesting for non-potable uses. In the 2019 edition, its Table 1 sets three minimum parameters: *E. coli* below 200 per 100 mL, turbidity below 5.0 NTU and pH between 6.0 and 9.0 (item 4.6.3). Disinfection is at the designer's discretion (4.6.4); if chlorine is used, the standard recommends a free residual between 0.5 and 2.0 mg/L, with a maximum of 5 mg/L (4.6.5), and asks for monitoring at least every six months (4.6.7). Apparent colour and total coliforms, which the 2007 edition limited, left the table.

What the current standard does **not** do is divide non-potable uses into classes with different requirements. That division existed in the 2007 version and did not survive the 2019 revision, which chose a single set of parameters applicable to all non-potable uses.

The program, in turn, offers four classes, and the reason is practical: toilet flushing and garden watering do not expose the user in the same way, even if the standard does not distinguish them (Figure 57).

> **Reuse is not harvesting.** ABNT NBR 15575-1:2025, in Table 26 of item 18.4.2, has much stricter parameters for non-potable uses -- absence of total and thermotolerant coliforms, residual chlorine of 0.5 to 3.0 mg/L, turbidity below 2.0 NTU and colour below 15 Hazen units --, but they apply to the **reuse** of wastewater. Rainwater is not reuse: NBR 15527:2019 distinguishes the two terms (items 3.5 and 3.18), and it is its Table 1 that applies to rainwater harvesting. The European standard EN 16941-1:2024 sets no quality table: it lets the use and a risk assessment decide whether treatment beyond the filter is needed.

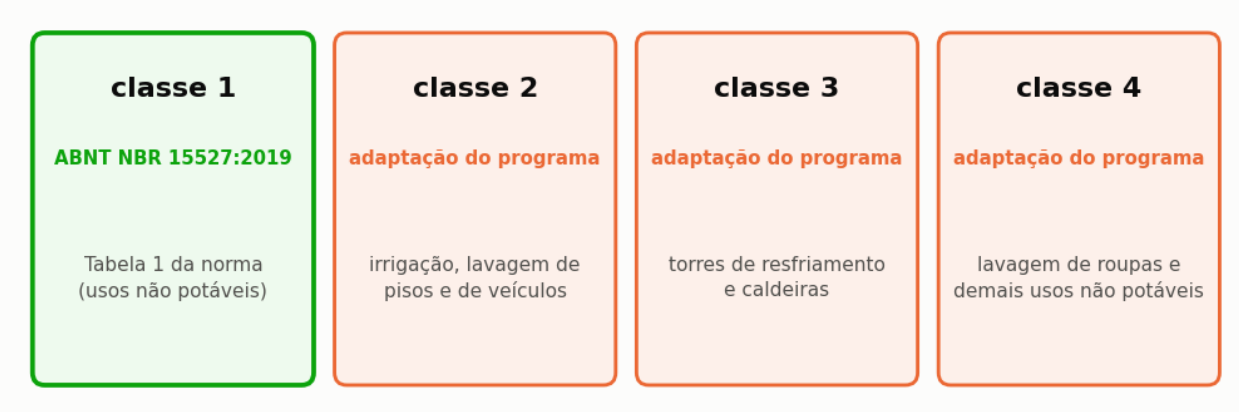


Figure 57 -- The program's four quality classes and the normative status of each one. Only class 1 corresponds to the parameters of ABNT NBR 15527:2019; classes 2 to 4 are adaptations made by the program, suitable for comparing treatments and unsuitable for presentation as a normative requirement.

> **Classes 2 to 4 are adaptations made by the program, not by the standard.** Netuno itself says so on the screen and in the report. In a published study, do not present them as a normative requirement: present them as the criterion adopted, and cite the source.

4.6.2. The class recommends, you decide

Choosing a class imposes no equipment on the project. The class expresses a recommendation of the program, based on the typical requirement of the intended use, and the decision to adopt it or not remains with whoever signs the project.

Table 115 -- Cost of the treatment line according to the disinfection technology adopted

Disinfection	CAPEX	Annual running cost	Annual CO2e
Tablet chlorination	R$ 1,547	R$ 43	0.2 kg
Chlorination + ultraviolet	R$ 2,747	R$ 259	6.4 kg
Chlorination + ozonation	R$ 5,047	R$ 243	15.2 kg

Table 116 -- Fine-filtration recommendation and cost of the line by quality class

Target class	Fine filtration recommended?	CAPEX without	CAPEX with
class 1	no	R$ 1,547	R$ 1,927
class 2	yes	R$ 1,547	R$ 1,927
class 3	yes	R$ 1,547	R$ 1,927
class 4	yes	R$ 1,547	R$ 2,097

The second table reveals the point: without the polishing filter, the four classes cost exactly the same. The target class changes a single thing, which is whether the program recommends fine filtration or not, and the corresponding filter only enters the budget when the user ticks it explicitly.

When the class recommends polishing and the user declines it, the program records the refusal in the report, instead of imposing the equipment silently or letting the decision go unrecorded. It is the right conduct in any design program: whoever signs decides, and the decision is documented for whoever comes later.

4.6.3. Disinfection

Disinfection is the stage on which the microbiological safety of the stored water depends, and the program offers three technologies. The difference between them is large, both in cost and in behaviour, and the criterion that separates them is not the effectiveness of disinfection -- all three disinfect -- but the persistence of the effect.

Tablet chlorination. A feeder with calcium hypochlorite tablets in the line. It is the default option. For the 82 m³ of our project, the treatment line with chlorination costs R$ 1,547 of investment and R$ 47 per year of running cost, values that include the first flush sized in the same step -- R$ 617 of investment and R$ 40 per year, as section 2.5 details -- and the tablets. It leaves a **residual** in the water, which protects the tank and the internal network against recontamination.

Ultraviolet. A reactor with a UV lamp. The line rises to R$ 2,747 of investment and R$ 263 per year of running cost, with the lamp and the reactor's energy. **It leaves no residual**: it disinfects what passes through it, and nothing protects the water afterwards.

Ozonation. The line rises to R$ 5,047 of investment and R$ 247 per year of running cost. Also without residual.

> **Warning.** In the program, ultraviolet and ozone enter as an **additional** barrier to chlorination, not as substitutes. It is the right practice: without a residual, the treated water becomes contaminated again on the way to the point of use. That is why the UV and ozone lines add the cost of chlorine. Anyone who expected that replacing chlorine with UV would **reduce** the cost will see the opposite.

The carbon impact follows the same order: 0.4 kg CO2e per year with chlorination, 6.5 with ultraviolet and 15.4 with ozonation, for the volume treated in this project. The whole difference is in the electricity consumed by the lamp and the ozone generator, since the chemical consumable of chlorination has a modest manufacturing emission by comparison.

4.6.4. The inlet filter

Before any disinfection comes the mechanical inlet filter, which removes leaves and coarse solids from the water before it reaches the tank. The program sizes it by the roof area, not by the intended quality class, because what this filter has to cope with is the peak flow of the roof -- a quantity that does not depend on the final destination of the water.

Table 117 -- Mechanical inlet filter sized by the roof area

Roof	Filter	Mesh
up to 200 m²	self-cleaning vortex DN 100	350 µm
up to 500 m²	self-cleaning vortex DN 150	400 µm
above	basket or cyclonic DN 200	500 µm

Our 150 m² roof is in the first range. These filters retain particles well below the 1 mm that EN 16941-1:2024 asks of the pre-treatment.

4.6.5. The incremental cost rule

The investment in the treatment line is not added in full to the total investment of the system, and the reason is to avoid double counting. The budget of a rainwater system already sets aside a generic treatment allowance within the accessories item, because no designer prices a system without some filtering; adding the detailed line on top of that allowance would charge treatment twice. The program deducts the generic allowance and adds only the increment, as Equation 31 formalises in section 2.5.

The same goes for the running cost: chlorination is already paid in the chemical consumables line of the cash flow, and the treatment line deducts that part so as not to charge the hypochlorite twice.

And in the life cycle assessment, the treatment line **replaces** the chlorination term instead of adding to it. Section 2.5 details the rule.

4.6.6. The tutorial's decision

Class 1, tablet chlorination, no polishing filter.

The water goes to the toilet and the outdoor area. Class 1, which reproduces Table 1 of ABNT NBR 15527:2019 (*E. coli* below 200 per 100 mL, turbidity below 5.0 NTU and pH between 6.0 and 9.0), covers both, which are on the list of item 4.1.7 of the standard. A project that adds the washing machine goes beyond that list, and item 4.1.8 asks the responsible professional to define and declare the quality criterion for it, as section 4.2 records. Chlorination leaves a residual, and its cost is the lowest of the three. An incremental investment of R$ 1,197 over the allowance already planned, of which R$ 617 is the first flush, and R$ 43.45 per year of running cost: R$ 3.45 of tablets and R$

40 of first-flush maintenance.

Table 118 -- State of the tutorial project after defining the quality class

Project so far	
volume	4,000 L
quality class	1, normative
disinfection	tablet chlorination
polishing filter	no
incremental treatment investment, with the first flush	R$ 1,197

We have volume, arrangement and quality. Now the question that decides whether the project leaves the drawing board.

4.7. What it costs and how fast it pays back

> By the end of this section you will have the twenty-year cash flow, and you will know why the **tariff regime of your municipality** decides the feasibility more than any parameter of the project.

> **International Edition.** The tutorial is priced in Brazilian reais, with the CASAN tariff of Florianópolis. In the International Edition the costs start at zero and the tariff is typed in, in the currency of the project; the calculation that follows is the same, and so is the lesson about tariff structures.

4.7.1. The saving is the difference between two bills

The most common mistake in the economic analysis of rainwater systems is to multiply the volume saved by an average tariff. The result is almost never right, and the reason lies in the structure of Brazilian tariffs -- and of many tariffs elsewhere --, which are progressive by consumption block, as section 3.3 details: the cubic metre that is no longer consumed is the last one on the bill, the most expensive, not the average one.

Netuno avoids the problem by calculating two complete bills -- the one the building would pay without the system and the one it pays with it -- and taking the difference between the two.

Table 119 -- Make-up of the water and sewage bill before and after the rainwater system

The CASAN bill	Consumption	Value
without the system	18.26 m³/month	R$ 390.75
with the system	13.28 m³/month	R$ 248.06
saving	4.99 m³/month	**R$ 142.69**

Table 120 -- Economic indicators of the tutorial project over twenty years of analysis

20-year analysis	Value
initial investment	R$ 14,061.19
annual saving on water	R$ 1,712.26
annual running cost	R$ 48.82
net present value	R$ 2,745.40
internal rate of return	7.54% per year
discounted payback	16.3 years

The result deserves attention. The system saves 4.99 m³ per month and the bill falls by R$ 142.69, which is R$ 28.60 per cubic metre saved -- well above the R$ 14.31 of the block in which consumption fell. The difference has two origins acting together: progressivity, which makes the saved cubic metre worth the price of the highest block, and sewage, which in Florianópolis is charged at 100% of the water value and therefore doubles the value of each cubic metre saved.

> **Sewage is a percentage of the water part, not of the total bill.** One hundred per cent of the water means that the sewage part equals the water part, and so sewage is half of the bill. Saving water saves both parts, and that is why the effective value of the saved cubic metre almost doubles.

4.7.2. The CASAN tariff regime

Florianópolis uses the availability-charge regime, in which the utility charges a fixed monthly fee of R$ 46.34 that adds to the metered consumption, and charges the progressive blocks from the first cubic metre consumed.

For a rainwater harvesting project, this is the most favourable configuration there is: each cubic metre saved produces a discount from the first one, because the fixed fee is charged for making the network available, not for an included volume. It is still paid, and it does not prevent the saving on consumption.

Not every municipality adopts this regime, and the difference between them decides the feasibility of entire projects. The program distinguishes six.

Table 121 -- The six tariff regimes and the effect of each one on the saving

Regime	What happens when saving	Municipalities in the database
availability charge	each m³ saved is deducted	122
exponential formula	each m³ saved is deducted, at the base price raised to the exponent	34
no minimum	likewise	165
allowance in the minimum	saving below the allowance **deducts nothing**	329
retroactive allowance	beyond the allowance, everything from zero is charged	2
payment floor	the bill does not go below the floor	0

> **Warning, and it is the point that brings down most studies.** In a municipality with an **allowance in the minimum** -- most of the database, 329 of the 652 -- the utility charges a fixed value that already includes, say, 10 m³. A house that consumes 12 m³ and saves 3 goes down to 9, enters the allowance, and **the bill stops falling**. The system saves water and does not save money.
>
> Running that case with the Florianópolis tariff would produce a payback that does not exist. Section 3.3 teaches how to build the structure of your municipality correctly.

4.7.3. The investment

R$ 14,061.19, split as follows:

Table 122 -- Make-up of the initial investment of the tutorial project

Item	Value
4,000 L concrete cistern	R$ 5,304
1,000 L polyethylene upper tank	R$ 460
two half-hp pump sets, duty and standby	R$ 600
labour and excavation	R$ 3,500
piping	R$ 1,200
accessories	R$ 1,800
treatment line, increment	R$ 580
first flush: 300 L chamber and leaf separator	R$ 617

Tank prices come from the program's database, which has values by volume and by material, with a declared source. For volumes outside the table, the program interpolates; it only extrapolates beyond it if the regression has

an R² above 0.90, and warns when it does not.

The first-flush chamber. The first flush is not only the depth subtracted from the rain: it is a physical component, with a cost. Step **3. Quality & Treatment** sizes it by default, and in the tutorial the 2 mm depth over 150 m² needs a 300 L chamber, which with the leaf separator costs R$ 616.90 of investment and R$ 40 per year of running cost. Without it, the investment would be R$ 13,444.29, the net present value R$ 4,032 and the discounted payback 14.8 years: leaving out the first flush brings the payback forward by a year and a half, and that is why a study that leaves it out looks more feasible than the system that gets built. Section 2.5 details the sizing.

4.7.4. The result

Table 123 -- Economic indicators of the tutorial project over twenty years

Indicator	Value
net present value	R$ 2,745
internal rate of return	7.54% per year
discounted payback	16.3 years

The project pays for itself, but only just. It returns 7.5% per year against a minimum rate of 6.2%, and pays back the investment in the seventeenth year of twenty. What sustains it is the Florianópolis tariff, which is high, and sewage at 100%; what squeezes it is the 28% fraction, which limits the saving to less than five cubic metres a month, and the concrete cistern, which is the most expensive per litre after stainless steel. Section 4.8 shows that changing the material changes the account.

The running cost is small: R$ 48.82 per year, of which R$ 3.45 of chlorine tablets, R$ 40 of first-flush maintenance and R$ 5.36 of pumping energy, the 7.7 kWh per year of section 4.5. The energy comes from the same function that feeds the environmental assessment, according to the arrangement declared in step **4. Reservoir & Costs**: pumping to an upper tank, direct pressurisation or gravity.

The result of an isolated volume says little without its neighbours. The economic tab repeats the calculation for all the volumes of the sweep and plots the curve, and it is there that you can see whether the technical optimum and the economic optimum fall in the same place (Figure 58).

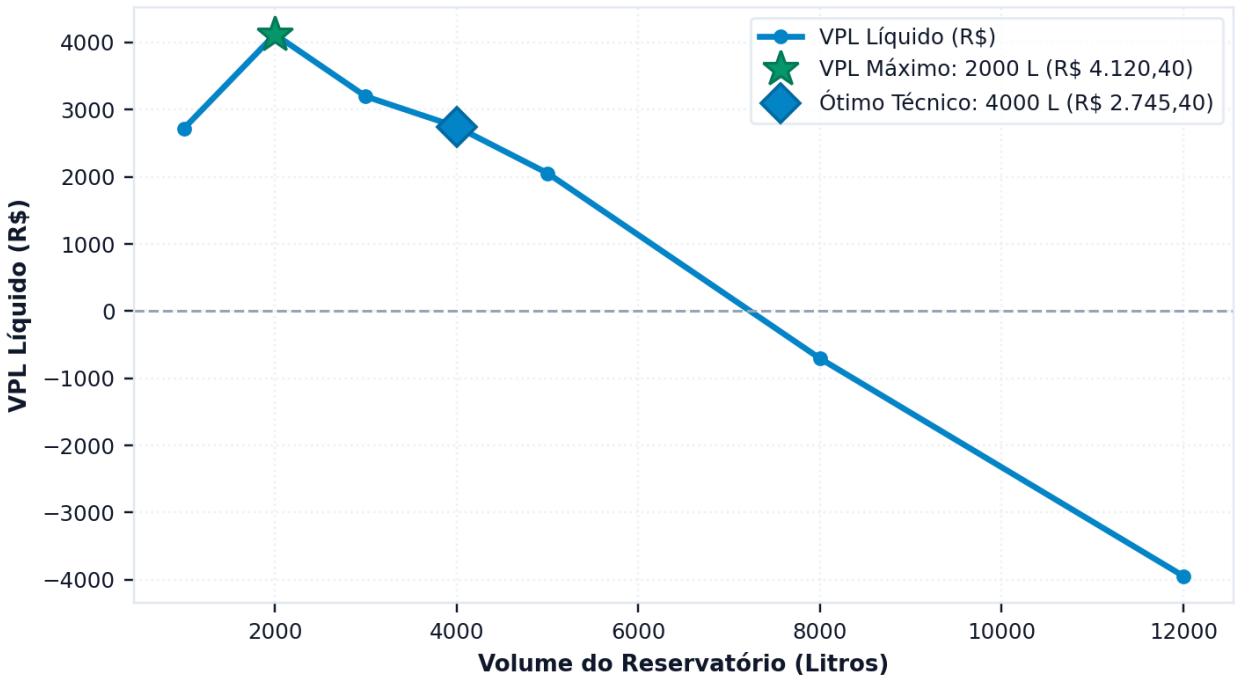


Figure 58 -- The chart of the economic tab: the net present value of each volume of the sweep, with a concrete cistern, with the economic maximum and the technical optimum marked. When the two coincide, the program marks them with a single symbol. Figure generated by the program itself.

4.7.5. What the projection assumes

Twenty years is a long time, and the projection depends on assumptions worth declaring:

Table 124 -- Assumptions of the economic projection and the weight of each one in the result

Assumption	Value adopted	Where it weighs
minimum attractive rate of return	0.5% per month	discounts the flow; the higher, the worse the NPV
cost inflation	0.35% per month	applies only to the running cost
water tariff escalation	5% per year	applies to the saving
sewage tariff escalation	5% per year	likewise, and indexed separately
escalation interval	12 months	in steps, not continuously
analysis period	20 years	minimum design life of the plumbing system, Table 25 of ABNT NBR 15575-1:2025

> **The water and sewage escalations are separate**, not a single index on the aggregate bill. The two parts are adjusted by different acts of the utility, and treating them together indexes half of the bill by the wrong index.

The tariff escalation is the most fragile assumption of the list. Five per cent a year for twenty years multiplies the tariff by 2.65. If the real tariff rises more than that, the project returns more; if it rises less, it returns less. A serious study declares the value adopted and shows the sensitivity -- section 3.7 teaches how.

4.7.6. What we have so far

Table 125 -- State of the tutorial project after the economic analysis

Project so far	
volume	4,000 L, concrete
arrangement	pumping to a 1,000 L upper tank
investment	R$ 14,061
annual saving	R$ 1,712
NPV over 20 years	R$ 2,745
payback	16.3 years

Money says yes, with no margin.

It remains to hear the carbon. And it will disagree.

4.8. How much carbon it avoids

> By the end of this section you will have the carbon balance of the system and a surprise: with the cistern we chose, the project **emits more than it avoids** over its service life.

4.8.1. The calculation

A rainwater harvesting system avoids emissions because each cubic metre it delivers is no longer abstracted, treated and pumped by the utility, and those three operations consume energy. The same system, however, emits carbon on three fronts of its own: the manufacture of its components, the energy consumed by internal pumping and the treatment consumables applied to the water.

The environmental balance of the system is the difference between what it avoids and what it emits, and the question that organises this section is in how many years the net annual credit pays off the debt incurred in manufacture. That period is the **carbon payback**, and it plays in the environmental dimension the same role that the payback period plays in the economic one.

For our project, a 4,000 L concrete cistern pumping to a 1,000 L upper tank:

Table 126 -- Carbon balance of the tutorial project with a concrete cistern

Part	Value
carbon embodied in the system	858 kg CO2e
emission avoided per year	20.9 kg CO2e
annual emission of pumping and treatment	0.9 kg CO2e
net annual balance	20.0 kg CO2e
carbon payback	**42.8 years**
balance at the end of 20 years	**-457 kg CO2e**

It takes almost forty-three years to pay off the carbon debt, against an adopted service life of twenty years. The system therefore never pays for itself environmentally: when it is decommissioned, it will have emitted 457 kg CO2e more than it avoided over its whole operation.

And that is the result of a project that, financially, pays for itself -- which shows, with uncomfortable clarity, why the dimensions of analysis must be kept separate until the moment of decision.

> **International Edition.** The emission factors of this section are Brazilian: the grid factor of the Ministry of Science, Technology and Innovation (MCTI) and the potable water factor by region. In the International Edition they start empty, and the environmental assessment is not calculated until the factors of the project's country are entered (appendix C).

4.8.2. Why the carbon saving is so small

Twenty-one kilograms of CO2e per year seems little for 60 cubic metres of water saved, and it is: the result is equivalent to 0.35 kg CO2e per cubic metre replaced.

The explanation lies in the nature of the product replaced. Treated water is not a carbon-intensive good. Abstracting, treating and distributing water costs, in Brazil, between 0.34 and 0.46 kg CO2e per cubic metre delivered, depending on the region and on the make-up of the local electricity mix, and that range is modest compared with

that of concrete, steel or fuels. Replacing a low-carbon product with another that requires the manufacture of a tank therefore does not produce the environmental gain intuition suggests.

> **The boundary matters.** The program's factor is per cubic metre **delivered** to the building, not per cubic metre produced at the plant. Distribution losses in Brazil range from 33.9% in the Southeast to 46.9% in the North: producing one cubic metre delivers between 0.53 and 0.66. Using the plant-outlet factor would underestimate the avoided emission by up to a third. Section 2.8 gives the details.

The environmental tab shows the same balance in graphic form, with the emission parts and the credit of the avoided water separated (Figure 59).

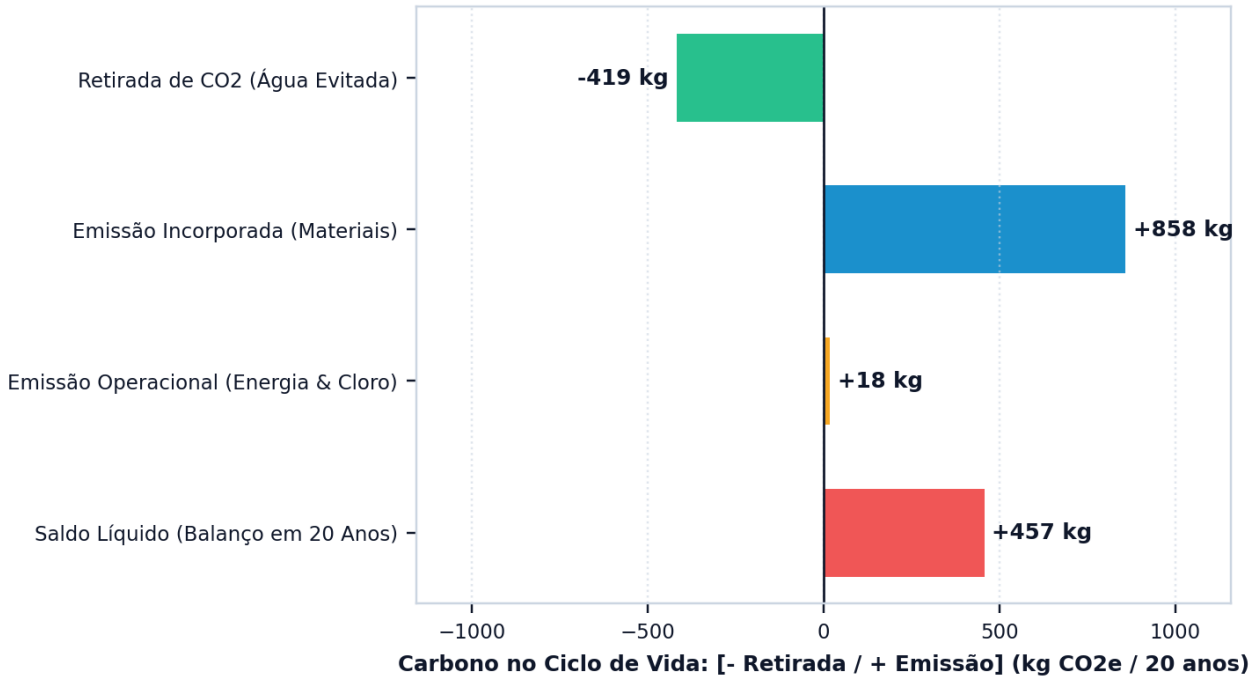


Figure 59 -- The chart of the environmental tab for the same project, with a concrete cistern: the parts of the carbon balance over twenty years. The sign follows the convention adopted throughout this chapter: emissions positive, avoided water negative. Figure generated by the program itself.

4.8.3. The material decides the sign

The carbon debt is almost all in the cistern. And a 4,000 L cistern can weigh 88 kg or 4,800 kg, depending on the material. The table below and Figure 60 show the path of the net emissions over the service life, and Figure 61 the final result of each material.

Throughout this chapter, **positive emits and negative removes**. It is the convention of the balance chart of the program's environmental tab, and the one that keeps the reading of an emissions chart intuitive: the line going down is the one improving.

Table 127 -- Net carbon emissions of the tutorial project according to the tank material

Cistern material	Mass	Embodied carbon	Carbon payback	Net emissions over 20 years	Price
Polyethylene (HDPE)	88 kg	327 kg CO2e	16.3 years	-74 kg	R$ 2,526
Reinforced concrete	4,800 kg	858 kg CO2e	42.8 years	457 kg	R$ 5,304
Glass fibre	57 kg	371 kg CO2e	18.5 years	-30 kg	R$ 3,410
Stainless steel	152 kg	748 kg CO2e	37.3 years	347 kg	R$ 9,093

Table 128 -- Effect of the concrete cistern volume on the net carbon emissions

Volume in concrete	Carbon payback	Net emissions over 20 years
2,000 L	27.4 years	141 kg CO2e
3,000 L	35.0 years	296 kg CO2e
4,000 L	42.8 years	457 kg CO2e
8,000 L	74.6 years	1,119 kg CO2e

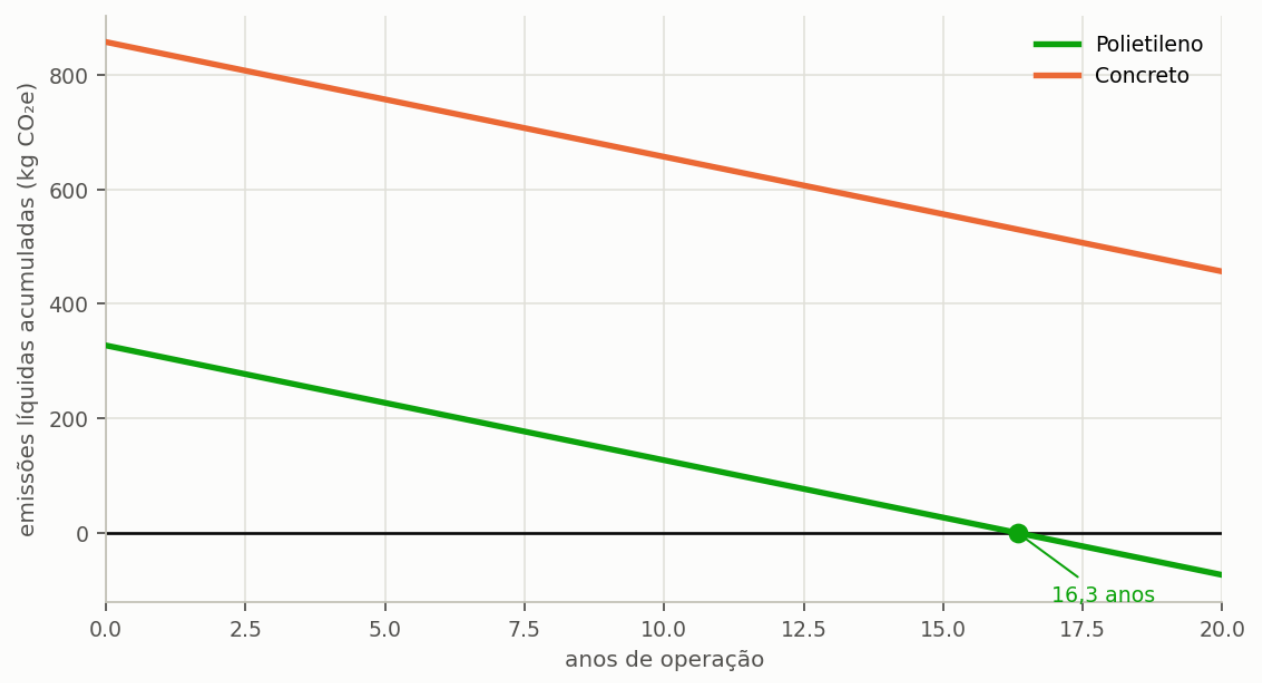


Figure 60 -- Cumulative net emissions of the 4,000 L cistern over the service life, for polyethylene and concrete. The curve starts from the carbon embodied in manufacture, above zero, and goes down with the annual credit of the avoided potable water; polyethylene crosses zero at 16.3 years and ends the service life removing carbon, and concrete does not cross it in the twenty years adopted.

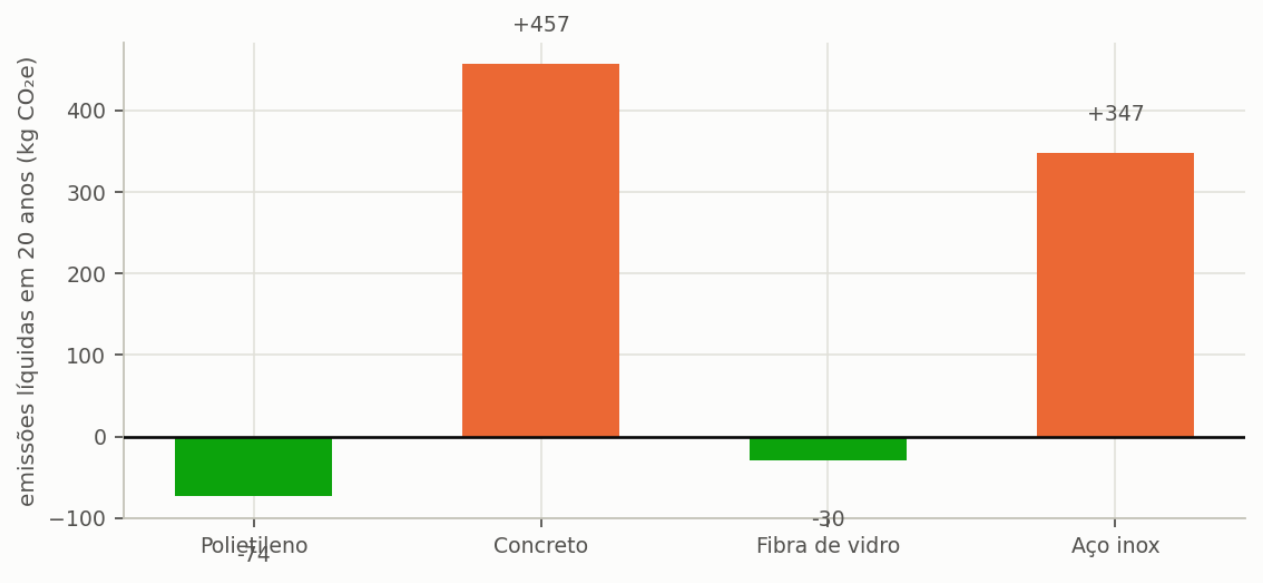


Figure 61 -- Net emissions over twenty years of the 4,000 L cistern, by material. In green, a negative value, that is, a net removal; in orange, positive: two of the four materials end their service life having emitted more than they avoided.

The choice of material reverses the result.

Polyethylene pays off the debt in 16.3 years and ends the twenty years having removed 74 kg. Concrete takes 42.8 years and ends having emitted 457 kg. Stainless steel also ends up emitting, 347 kg. With a 28% suppliable fraction, little water is saved, and only the light materials manage to pay for their own manufacture.

And the second table shows the other side of the same coin: in concrete, **the larger the cistern, the worse the carbon**. The 2,000 L one closes the twenty years having emitted 141 kg; the 8,000 L one, 1,119 kg. Each cubic metre of concrete adds 168 kg CO2e and yields less and less water.

4.8.4. There is an environmentally optimal volume

The observation above holds for concrete, but it is a particular case of something more general. The embodied carbon grows almost linearly with the volume; the avoided water grows with the supply, which saturates. The sum of the two parts has an **interior minimum**: there is a volume beyond which each additional litre of storage emits more than it avoids. Figure 62 shows the two curves.

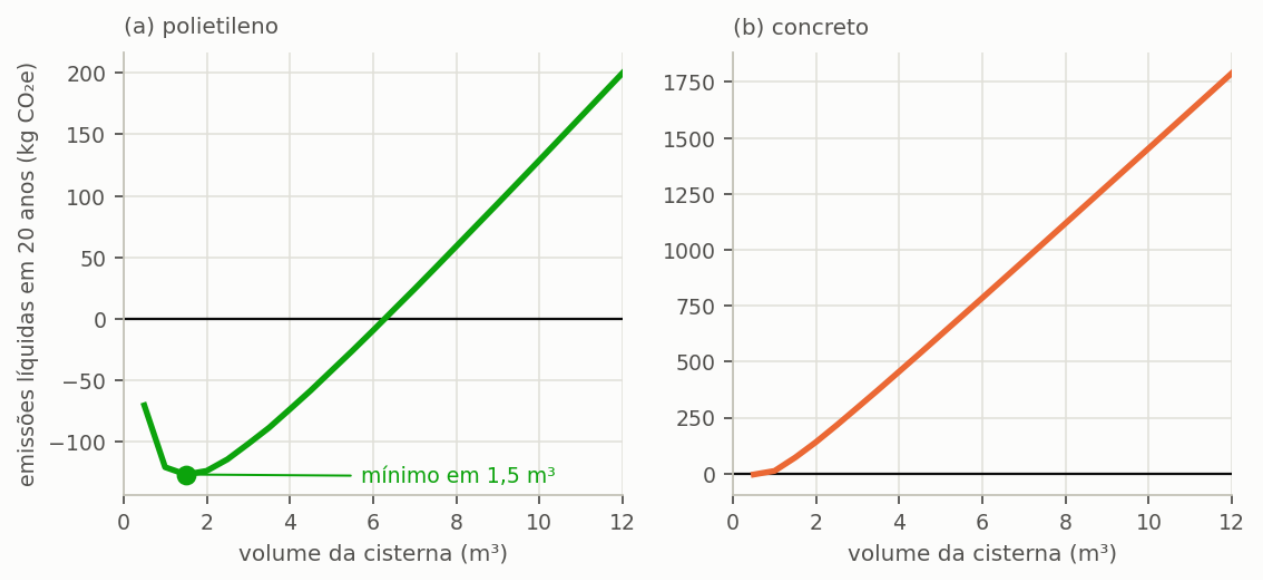


Figure 62 -- Net emissions over twenty years against the cistern volume, for (a) polyethylene and (b) concrete, in the tutorial project. In polyethylene the curve has an interior minimum at 1.5 m³ and crosses zero again at around 6 m³: above that volume the system emits more than it avoids. In concrete there is no interior minimum within the range, and the curve is positive throughout.

More storage is not monotonically better for the climate. It is the opposite of what common intuition suggests, and it follows from a simple property: the saturation of supply. The volume that minimises emissions is not the same as the one that maximises the water saving -- and section 4.9 deals with how to decide between the two.

> **The cheapest is also the best for carbon.** Polyethylene costs R$ 2,526 against R$ 5,304 for concrete, weighs fifty-five times less, and is the material in the table that closes the twenty years with the widest positive margin.
>
> This is not luck. Mass is carbon, and reinforced concrete is by far the most massive material on the list.

4.8.5. When concrete still makes sense

The table does not condemn concrete. It says that it needs a justification.

- **Longer service life.** The program adopts 20 years for all materials, the minimum design life that Table 25 of ABNT NBR 15575-1:2025 sets for the plumbing system. It is a floor, not a forecast: the standard admits higher values, as long as they are stated in the project (item 14.2.2 of NBR 15575-1). A well-built concrete cistern lasts 40 or 50 years; over a 50-year horizon, the 43-year payback is paid off, although without margin. If you adopt a different service life, **declare** it.
- **Large volumes.** Above 20 m³, polyethylene stops being practical and cast-in-place concrete becomes the only feasible option.
- **Space restrictions.** A buried concrete cistern adapts to the plot; a polyethylene one comes in fixed shapes.
- **Existing structure.** A concrete tank reused from another purpose has **zero** embodied carbon for the purposes of this project.

4.8.6. The tutorial's decision

We switch to polyethylene.

It is what a designer would do faced with this table, and it is what makes this tutorial honest: a study changes course when the analysis shows the course was wrong.

The switch improves the financial and the environmental result at the same time, without changing the saving potential, which is rare:

Table 129 -- Effect of switching from concrete to polyethylene on the three dimensions of analysis

	Concrete	Polyethylene
cistern price	R$ 5,304	R$ 2,526
total investment	R$ 14,061	R$ 11,283
financial payback	16.3 years	12.8 years
NPV over 20 years	R$ 2,745	R$ 5,524
carbon payback	42.8 years	16.3 years
net emissions over 20 years	+457 kg	-74 kg

Table 130 -- State of the tutorial project after the environmental analysis

Project so far	
volume	4,000 L
material	**polyethylene**
investment	R$ 11,283
NPV	R$ 5,524
financial payback	12.8 years
carbon payback	16.3 years

4.8.7. What to declare

Netuno's life cycle assessment is a **screening** one, not a complete inventory. All the default factors have a declared source, but some are approximations, such as that of the pump set and that of the hypochlorite, and a factor typed in without a source receives a screening label, which the program displays. In a published study:

- declare the **boundary** adopted, water delivered and not produced;
- declare the **service life** and justify it;
- declare which factors are approximations and which were entered without a source;
- declare the **electricity grid basis** and the year.

Appendix C has the source of each factor, and section 2.8 explains the seven stages of the inventory.

We now have three answers. The potential says one volume, money says another, carbon says a third. It is time to choose.

4.9. Which volume to choose, after all

> By the end of this section you will have a recommended volume, and you will know that it depends on weights that **you** declared -- not on a truth the program discovered.

4.9.1. Three criteria, three answers

The previous sections of this tutorial asked the same project three different questions -- how much water it delivers, how much money it returns and how much carbon it avoids -- and received three answers that do not coincide. This disagreement is not a defect of the analysis or an inconsistency in the data: it shows that the three criteria measure different things and behave differently with respect to the tank volume.

Table 131 -- Multi-criteria decision matrix: each sweep volume assessed in the three dimensions of analysis

Volume	Potential	NPV	Payback	Carbon balance over 20 years	Investment
1,000 L	23.30%	R$ 3,228	13.9 years	121 kg	R$ 9,217
2,000 L	25.89%	R$ 5,374	12.4 years	124 kg	R$ 9,897
3,000 L	26.80%	R$ 5,453	12.7 years	102 kg	R$ 10,809
4,000 L	27.30%	R$ 5,524	12.8 years	74 kg	R$ 11,283
5,000 L	27.58%	R$ 5,354	12.9 years	42 kg	R$ 11,757
8,000 L	27.94%	R$ 4,245	14.6 years	-59 kg	R$ 13,257
12,000 L	28.00%	R$ 2,737	16.5 years	-200 kg	R$ 14,831

Table 132 -- Winning volume under each criterion taken alone

Criterion alone	Winning volume
largest saving potential	12,000 L
shortest discounted payback, NPV as tie-breaker	2,000 L
largest carbon balance	2,000 L
compromise, equal weights	**4,000 L**

The first table is the portrait of the problem. Notice how the columns disagree:

- The **potential** grows monotonically: the larger the cistern, the more water. Maximum at 12,000 L.
- The **NPV** rises, flattens and starts to fall: maximum at 4,000 L, practically tied with 2,000 and 3,000 L, and falling from 5,000 L on, because the additional investment no longer pays for itself.
- The **carbon** rises and falls much earlier: maximum at 2,000 L, practically tied with 1,000 L. After that, each cubic metre of polyethylene adds more debt than credit.

There is no volume that wins on all three criteria at once. Each one has its optimum at a different point of the curve, and the choice between them is not a technical question with a single answer: it is a question of priority, which must be declared by whoever decides and cannot be discovered by the program. That is exactly the gap that multi-criteria methods fill.

4.9.2. How TOPSIS solves it

The program uses TOPSIS over **seven criteria**:

Table 133 -- The seven multi-criteria criteria and the direction of preference of each one

Criterion	Direction
saving potential	maximise
overflow	minimise
net present value	maximise
discounted payback	minimise
internal rate of return	maximise
CO2e at the end of life	minimise
potable water consumption	minimise

The method works in four linked steps. First it normalises the decision matrix, so that criteria measured in incomparable units -- percentage points, currency, kilograms of carbon dioxide equivalent and years -- come to occupy the same scale. Then it applies the declared weights. It then calculates, for each volume alternative, the Euclidean distance to the ideal solution, made up of the best values of each criterion, and to the anti-ideal solution, made up of the worst. Finally it ranks the alternatives by their relative closeness to the ideal, as formulated in section 2.10.

Applied to the tutorial project with equal weights across the three dimensions, split equally among the criteria of each one, the method points to the 4,000 L volume.

> **Three statements a published study must make.**
>
> The program's normalisation is a **variant** of Hwang and Yoon: it shifts by `x - min(x)` before the vector normalisation, to admit a negative NPV. It is a deliberate choice, and presenting it as canonical TOPSIS would be incorrect.
>
> SMAA-VIKOR, available alongside, draws weights and returns the probability of each volume taking each rank. It is the right answer when there is no consensus on the weights.
>
> Opricovic's conditions C1 and C2, of acceptable advantage and of stability, **are not implemented**. Section 2.10 gives the details.

The decision tab shows the score of each candidate volume (Figure 63).

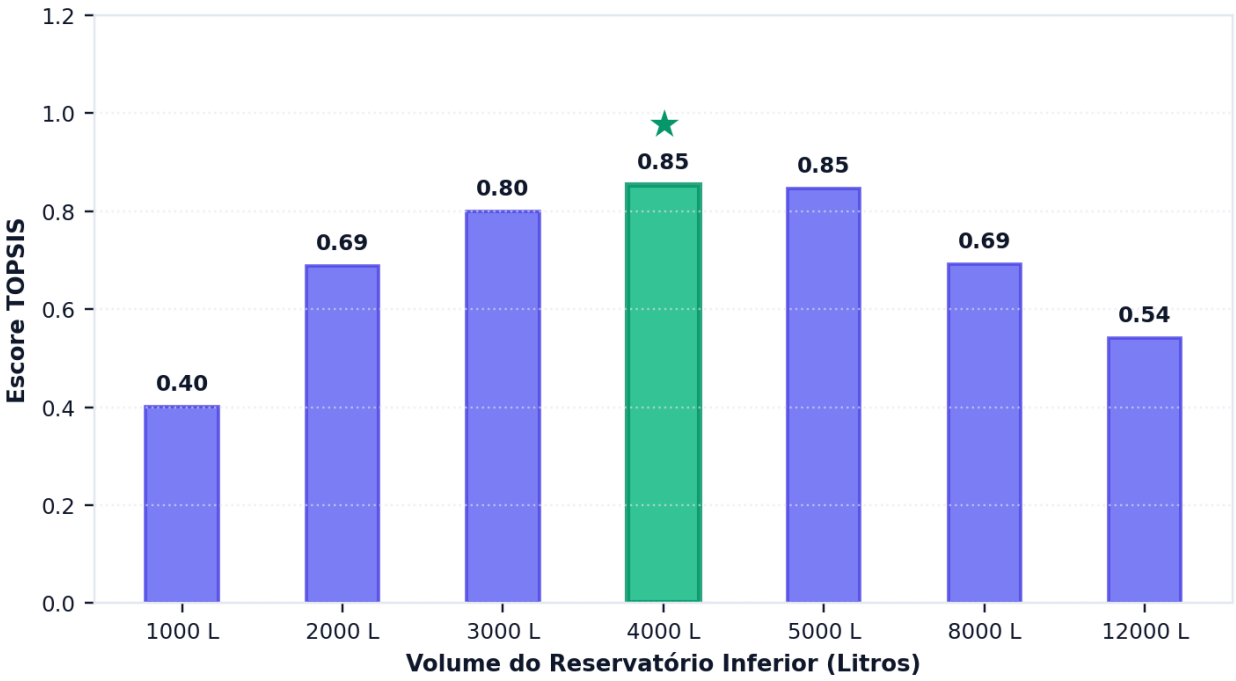


Figure 63 -- The chart of the decision tab: the closeness-to-ideal score of each candidate volume, with the winner highlighted. Scores close to one another mean technically equivalent alternatives, and in that case the choice comes to depend on the declared weight. Figure generated by the program itself.

4.9.3. The weight changes the answer

The table below shows what happens when the weight is shifted.

Table 134 -- Volume chosen by TOPSIS according to the dimension given the dominant weight, and the runner-up

Dominant weight	Volume	Runner-up
equal weights	4,000 L	5,000 L
technical, 60%	5,000 L	4,000 L
economic, 60%	4,000 L	3,000 L
environmental, 60%	4,000 L	5,000 L

Favouring the technical criterion moves the recommendation to 5,000 L, which was expected: the saving potential grows monotonically with the volume, and a dominant weight on that dimension pushes the choice up. The shift is small because the potential is already close to the ceiling, and the extra cubic metre yields little. The economic and environmental criteria, in turn, both converge to 4,000 L.

That convergence, however, is not a general rule, and reading it as one would wrongly generalise a particular case. It happens here because the 4,000 L volume sits at the shallow maximum of the net present value curve and still ends its service life removing carbon. In a municipality with a lower tariff, the economic optimum would move down and the divergence between the two criteria would show clearly.

> **Declare the weights.** «Optimal volume by TOPSIS» without the weights is an impossible statement to reproduce. Two papers with different weights reach different volumes and both are right.

A compromise solution gives up something, and it is worth knowing what. The program unfolds the single score into seven, one per criterion, and shows the distance separating the chosen alternative from the best value reached in each of them (Figure 64).

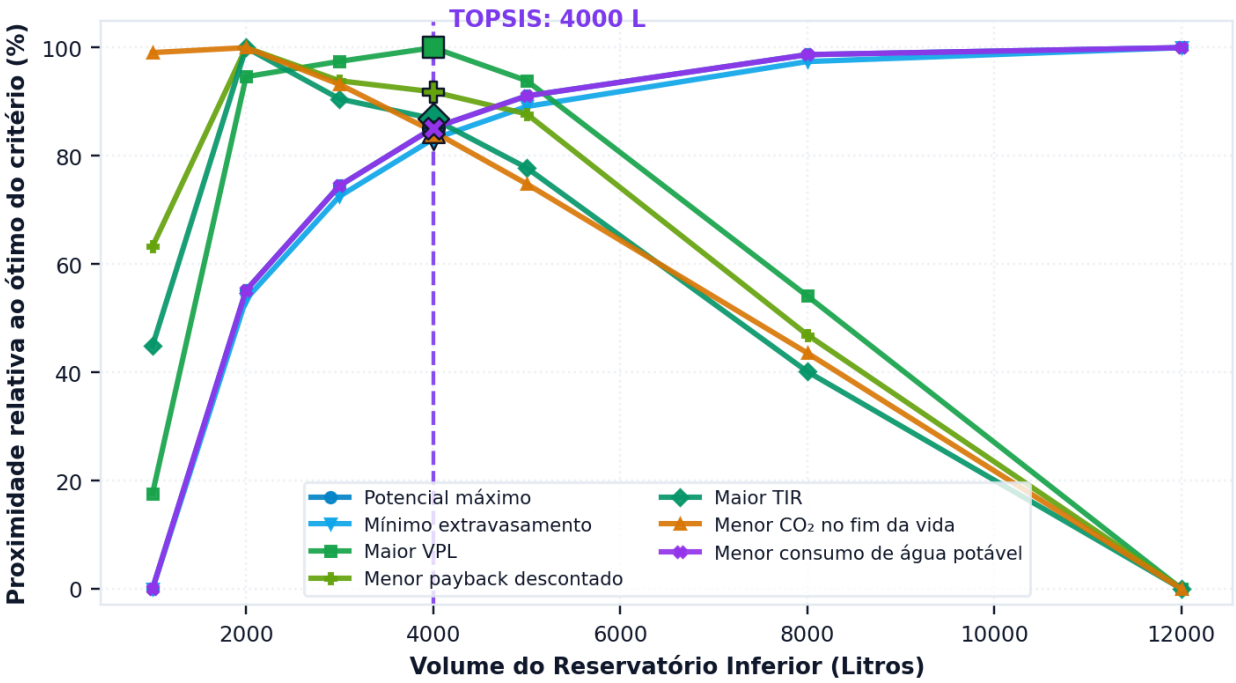


Figure 64 -- The second chart of the decision tab: how close the chosen volume comes to the best possible value on each of the seven criteria. It is the figure that shows what the compromise solution is giving up, and in which dimension. Figure generated by the program itself.

4.9.4. The tutorial's recommendation

The tutorial's recommendation is the **4,000 L polyethylene** tank, and the reason to adopt it is not only that it wins under equal weights. It also wins under a dominant economic weight and under a dominant environmental weight, and comes second by a narrow margin under a dominant technical weight. A recommendation that survives four different weightings is robust precisely because it does not depend on any of them.

It is also worth recording that this is the same volume the hydraulic criterion of section 4.4 had pointed to, with a marginal-gain threshold of 1 %/m³. Two methodologically independent routes -- one that looks only at the yield curve and another that weighs seven criteria from three dimensions -- reached the same result, and that agreement increases confidence in the recommendation.

The agreement, however, is not guaranteed in any other case, and that is why the tutorial took the long road instead of stopping at section 4.4. The purpose was not to reach the volume faster, but to know that the two routes agree in **this** project, instead of assuming they always agree.

4.9.5. The project, closed

Table 135 -- Summary of the tutorial project, with all the parameters and results

Netuno 5 . single-family house in Florianópolis	
precipitation series	Florianópolis, 30 years, 1,795 mm/year
catchment area	150 m²
runoff coefficient	0.85
first flush	2.0 mm, every event
residents	4
total demand	600 L/day
fraction suppliable by rain	28%
lower tank	4,000 L, polyethylene
supply arrangement	pumping to an upper tank
upper tank	1,000 L, 100 L trigger level
quality class	1, tablet chlorination
first-flush chamber	300 L
potable water saving potential	**27.30%**
days of full supply	97.3%
water saved	59.8 m³/year
pumping energy	7.7 kWh/year
investment	R$ 11,283
net present value over 20 years	**R$ 5,524**
discounted payback	12.8 years
net emissions over 20 years	**-74 kg CO2e**
carbon payback	16.3 years

One step remains, which is to communicate the result: a study that is not reported in a verifiable way is indistinguishable, for whoever reads it, from a well-presented guess. That is the subject of the next section.

4.10. The report, and what to declare in a paper

> By the end of this section you will have the report issued and, if you intend to publish, the list of what must appear in the method of the paper. That list is the shortest and the most important part of the manual.

4.10.1. The three reports

The program issues three different reports, and the difference is not one of length but of audience and purpose. Choosing the wrong report for the right reader is the most common way of producing a document nobody reads: a client does not go through a life cycle inventory item by item, and a certification assessor is not satisfied with a four-page conclusion.

The **executive report** takes four to six pages and brings together the technical result, the investment, the return and an objective conclusion. It is the document handed to the client, and its virtue is that it fits in a single reading, without requiring the reader to know the method.

The **environmental and ESG report** presents the life cycle inventory broken down by component, the carbon balance over the service life and how the project fits the five frameworks dealt with in section 2.9 -- LEED, AQUA-HQE, Selo Casa Azul, EDGE and GRI 303 (in the International Edition, LEED, EDGE and GRI 303). It is the document attached to a certification process, and the breakdown by component exists precisely because an assessor checks item by item.

The **research report** has the complete method, the declared assumptions, the series used and the results with the precision a paper requires. It is the basis on which a publication is written, and it is the only one of the three that includes the analytical figures.

Besides the three, step **10. Export** generates the schematic drawing in DXF, the BIM model in JSON and spreadsheets with the complete daily series and with the volume sweep, as section 3.9 details.

4.10.2. What the report does not decide for you

The report carries the numbers; the context that makes them reproducible is yours. No automatically generated document can declare why that precipitation series was chosen, whether the per capita consumption adopted was measured or taken from the literature, or what justifies the service life considered in the environmental assessment. It is that information that allows a third party to repeat the study and reach the same result, and it is what the rest of this section deals with.

4.10.3. The declaration list

A study carried out in Netuno that fails to declare any of the items below is not reproducible by third parties and, which is usually worse, is not comparable with another study done in the same program -- because two simulations with different assumptions produce numbers that look comparable and are not. The list is organised in the order in which the decisions appear in the work, and can be followed as a script for writing the method.

4.10.3.1. Program and version

- **Netuno version**, with the full number, and the **edition** (Brazil or International). Calculation fixes between versions shift results, and without the declared version there is no way of knowing which calculation produced the published number.

4.10.3.2. Precipitation series

- **Source**: station, national database, or climate model.
- If the national database: **model, scenario and time window**.
- **Trimming**: start and end dates, and number of years.
- **Treatment of gaps** and the criterion for dropping an incomplete year.

4.10.3.3. Building and demand

- Catchment area and **how it was obtained** -- horizontal projection.
- Runoff coefficient and its origin.
- Per capita consumption and whether it is **measured or a reference value**.
- **Fraction suppliable by rainwater** and the end uses that make it up. It is the ceiling of the result, and without it the potential cannot be interpreted.
- Whether any use is outside the list of item 4.1.7 of ABNT NBR 15527:2019, such as the washing machine, and which quality criterion was adopted for it.

4.10.3.4. Supply arrangement

- **Arrangement**: pumping to an upper tank, direct pressurisation or gravity. It defines the energy and the pump sets in the cost and in the carbon.
- With pumping to a tank: tank volume, trigger level and manometric head.
- With direct pressurisation: working pressure and starts per person per day.

4.10.3.5. First flush

- Depth adopted and **mode** among the three.
- **The divergence from Netuno 4**, if the study compares the versions: Netuno 4 treats the depth as an event threshold, Netuno 5 subtracts it, and the difference is worth 2.3 to 2.7 percentage points of saving over 4,050 reference cases.

4.10.3.6. Volume and choice criterion

- If it came from a sweep: **the marginal-gain threshold adopted**. Without it, «ideal volume» means nothing.
- If it came from multi-criteria: **the weights of the seven criteria** and the method, TOPSIS or SMAA-VIKOR.
- That the TOPSIS normalisation is a **variant** with a shift by `x - min(x)`.

4.10.3.7. Economic analysis

- **Municipality and utility**, with the effective date of the tariff table.
- **Minimum-charge regime** among the six. It is what changes results most between municipalities.
- Sewage percentage, and that it applies to the **water part** of the bill.
- Minimum attractive rate of return, inflation, analysis period.
- **Tariff escalations** for water, sewage and energy, with the interval.

4.10.3.8. Life cycle assessment

- **Boundary**: water delivered, not produced.
- **Service life** adopted and its justification.
- **Material** of the tank. It reverses the sign of the result, as section 4.8 shows.
- Basis of the electricity grid emission factor, **with the year**.
- Which factors are **approximations** and which carry a **screening label**.

4.10.3.9. Certifications

- That quality classes 2 to 4 are **adaptations made by the program**, not by NBR 15527.
- For LEED, the **version** of the framework: v4/v4.1 or v5.
- For EDGE, that the program assesses **only the water bar**, and that the base case used is the declared one or the modelled demand.

4.10.4. The ready-made paragraph

For our project, the method would read like this:

> The simulation was carried out in Netuno 5.0.0, Brazil Edition, with a daily-step behavioural model (overflow calculated before the demand is withdrawn, with the day's rain available for the day's consumption, an order between YAS and YBS), on a 30-year precipitation series for Florianópolis (10,950 days, mean of 1,795 mm/year). A catchment area of 150 m² in horizontal projection was adopted, with a runoff coefficient of 0.85, taken from the collection efficiency eta that ABNT NBR 15527:2019 recommends in item 4.1.6, and a 2.0 mm first flush on every event, subtracted from the daily depth -- a deliberate divergence from Netuno 4, which treats the value as an event threshold. The total demand adopted was 600 L/day (4 residents, 150 L/person.day), with a fraction suppliable by rainwater of 28%, corresponding to the toilet and the outdoor uses, within the list of item 4.1.7 of ABNT NBR 15527:2019. The 4,000 L volume resulted from the marginal-gain criterion with a 1 %/m³ threshold and was confirmed by a TOPSIS multi-criteria analysis with seven criteria and equal weights across the technical, economic and environmental dimensions, with normalisation in a variant of Hwang and Yoon shifted by x - min(x). The points of use are supplied by pumping to a 1,000 L upper tank, with a 100 L trigger level, a 0.5 hp pump, a manometric head of 12 m and the motor efficiency of Table 13 of ABNT NBR 17094-2. The economic analysis used the CASAN tariff structure in force since 01/04/2026, an availability-charge regime, sewage at 100% of the water value, a minimum attractive rate of return of 0.5% per month, inflation of 0.35% per month, escalations of 5% per year at 12-month intervals and a 20-year period. The life cycle assessment adopted the boundary of water delivered, a polyethylene tank and a service life of 20 years.

That is about two hundred and fifty words, and they are the difference between a study another researcher can repeat and a loose number nobody can check. The length is modest compared with what is gained: a method written this way answers in advance most of the questions a reviewer would ask.

4.10.5. The end of the tutorial

Ten sections, a single project followed from start to finish, and a complete study: it started from a house without a system and arrived at a recommendation of volume, material, quality class and economic and environmental feasibility, with each decision justified by the number that supported it. The table of section 4.9 brings together the consolidated result.

The way forward depends on what you intend to do next:

- **Solve a case different from the tutorial** -- chapter 3, the task guides.
- **Understand where each number comes from** -- chapter 2, the method.
- **Not make the mistakes the program does not prevent** -- section 5.1.
- **Defend the study before a reviewer** -- chapter 5.

5. Mistakes, limitations and reliability

5.1. Ten mistakes the program does not prevent

> These are not defects of the program. They are **engineering** mistakes it accepts without complaint, because it has no way of knowing. Each one gives the mistake, why it is tempting, what it does to the result, and how to avoid it.

5.1.1. Using the sloping area of the roof

The mistake. Entering the area of the roof surface instead of the horizontal projection.

Why it is tempting. It is the area that appears in the roof tile quantities, and the one the builder knows.

What it does to the result. It overestimates the catchment in the ratio `1 ÷ cos(slope)`. On a roof with a 30% pitch, 4.4% of water that does not exist; at 60%, 16.6%.

How to avoid it. Item 3.6 of NBR 15527:2019 defines the catchment area as the horizontal projection; EN 16941-1:2024 does the same. Do not use the contributing area of NBR 10844, which adds half the height of the roof and the walls to size gutters. Rain falls vertically, and what it sees is the shadow of the roof on the ground. If you have the sloping area, multiply it by `cos(slope)`, as in Figure 65.

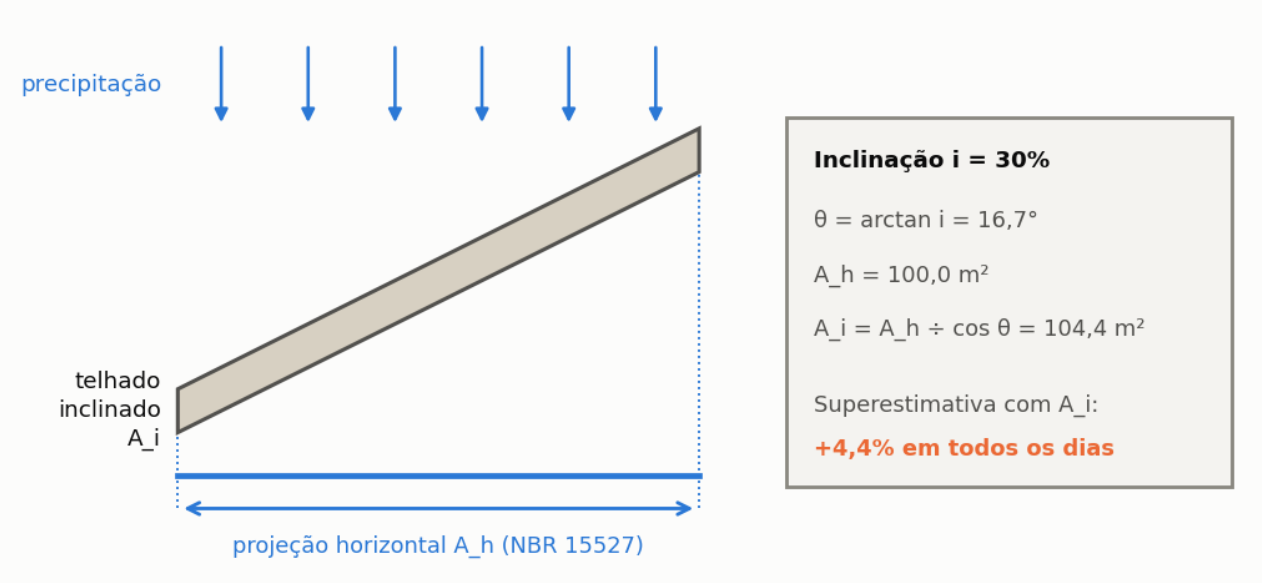


Figure 65 -- Catchment area as the horizontal projection of the roof, according to item 3.6 of ABNT NBR 15527:2019. For a 30% pitch, θ = 16.7° and the sloping area is 4.4% larger than the horizontal projection; using the sloping area overestimates the volume collected in that proportion on every day of the series.

5.1.2. Sizing by the mean monthly rainfall

The mistake. Using the mean monthly precipitation in a spreadsheet calculation, instead of the daily series.

Why it is tempting. The monthly mean is the figure published in the climatological normals, and the spreadsheet fits on one page.

What it does to the result. It systematically overestimates. The mean hides that rain arrives concentrated: a month with 200 mm spread over five events fills the cistern and makes it overflow five times, while the mean calculation assumes a continuous flow the cistern would absorb entirely.

How to avoid it. It is this program's reason for existing. Section 2.1 shows the overflow happening day by day; a calculation by means does not see it.

5.1.3. Confusing the first-flush depth with the volume of the device

The mistake. Entering in the first-flush field the volume of the self-cleaning chamber, in litres.

Why it is tempting. The device is sized in litres, and the field asks for a number.

What it does to the result. The field expects **millimetres**. Entering 300, thinking of the device's 300 litres, produces a first flush of 300 mm -- larger than almost any daily rainfall ever recorded in Brazil. The system collects **practically nothing**.

How to avoid it. `d_f` is a depth. The volume of the device is `A x d_f`: in the tutorial project, 150 m² x 2 mm = 300 litres. They are the same decision expressed in two units, and the program wants the first.

5.1.4. Ignoring the minimum-charge regime

The mistake. Building the tariff table with the progressive blocks and forgetting how the utility treats minimum consumption.

Why it is tempting. The blocks are what is published prominently. The minimum-charge regime is in a footnote, when it is there at all.

What it does to the result. In a municipality with an **allowance within the minimum**, which are 329 of the 652 in the database, saving below the allowance **discounts nothing**. A study that assumes a linear discount produces a payback that does not exist.

How to avoid it. Section 3.3. And the practical test: simulate two consumptions, one inside and one outside the minimum block, and compare them with a real bill of the building.

5.1.5. Simulating a school or hospital with a residential tariff

The mistake. Using the residential tariff of the catalogue for a non-residential building.

Why it is tempting. The catalogue has 652 municipalities ready to use, almost all of them only with the residential category, and the program does not prevent it.

What it does to the result. It underestimates the saving. Non-residential categories are more expensive: at CASAN in April 2026, the first block costs R$ 6.81/m³ for commercial against R$ 3.08/m³ for residential.

How to avoid it. The program **warns** when the building type is not residential, and states the direction of the error. Choose the building's category where the catalogue has it, today in the CORSAN municipalities; elsewhere, enter the structure of the right category in custom mode. Section 3.5 deals with the subject. In the International Edition the tariff is always typed in, so type the one for the building's category.

5.1.6. Expecting the upper tank to increase the saving

The mistake. Adopting the elevated tank expecting more water.

Why it is tempting. It is the classic Brazilian arrangement, and «more storage» sounds like «more water».

What it does to the result. With the same total storage, the arrangement with an elevated tank delivers **less**: the cistern overflows while the tank above still has room, because pumping only happens when there is consumption. In the tutorial, about a sixth of a percentage point less, plus 7.7 kWh a year of energy.

How to avoid it. Decide on the tank by a **hydraulic** criterion -- gravity pressure, a reserve against power cuts -- not a water-saving one. Section 4.5 makes the fair comparison.

5.1.7. Adopting too short a series

The mistake. Simulating with two or three years of series.

Why it is tempting. It is what can be obtained quickly from a nearby station.

What it does to the result. Two effects added together. The **empty-tank bias**, because the simulation starts with the cistern dry and the first months penalise the result; and the **dependence on the stretch**, because a short series may fall in an atypically dry or wet period.

How to avoid it. Thirty years is comfortable, ten is the minimum defensible. Below that, declare the limitation and show the result as a range.

5.1.8. Choosing the quality class by cost

The mistake. Adopting class 1 because it is the cheapest, for a use that would require more.

Why it is tempting. The program allows it, and the cost difference is visible immediately.

What it does to the result. It does not change the number; it changes the **validity** of the project. A system feeding a washing machine with water treated to the normative minimum is a quality decision, not a budget one.

How to avoid it. Choose by the **requirement of the use**. And remember that classes 2 to 4 are adaptations of the program, not requirements of NBR 15527: use them as a declared criterion, not as a standard.

5.1.9. Comparing with Netuno 4 without declaring the divergence

The mistake. Presenting results of the two versions side by side as if they were comparable.

Why it is tempting. They are the same program, in different versions.

What it does to the result. Netuno 5 estimates, on average, **2.3 to 2.7 percentage points less** saving than Netuno 4 over 4,050 reference cases, and about **half** of that difference comes from the treatment of the first flush, which changed deliberately.

How to avoid it. Declare the divergence, and compare with a first flush of zero, where the two rules coincide. Sections 3.8 and 6.1 go into detail.

5.1.10. Publishing the volume with the highest saving as the recommendation

The mistake. Choosing from the sweep the volume with the highest potential.

Why it is tempting. It is the column rainwater harvesting studies report, and more is better.

What it does to the result. In the tutorial, 20,000 L give the highest potential, the same 12,000 L already gave. They also cost **3.9 times more** than the 4,000 L and carry **4.1 times more embodied carbon**, to deliver 0.88 percentage points more. And, in concrete, the larger volume is environmentally **worse** in absolute terms.

How to avoid it. Declare the selection criterion. If it is hydraulic, the marginal-gain threshold; if it is multi-criteria, the weights. And report the winners of each criterion alone, because the divergence between them **is** the result.

5.1.11. The underlying mistake

The ten above have a common root: **treating the program as a calculator instead of a model**.

A calculator receives numbers and returns numbers. A model receives assumptions and returns the consequences of those assumptions. Every Netuno result is conditional on what was declared, and the quality of the study lies in the

quality and explicitness of those declarations.

That is why section 4.10 gives the list of what to declare, and why section 5.2 gives the known limitations. A study that does not declare its assumptions is not reproducible, however correct the calculation.

5.2. Known limitations

> Where the program does not serve, and why. A program that declares its own limits is more trustworthy than one that seems to serve for everything, because the second leaves users to discover them for themselves.

This section derives from the issue register the research group keeps, and reflects the state of version 5.0.0. A limitation leaves this list when it is actually resolved, not when it stops being mentioned -- a distinction that seems obvious and that the documentation of scientific programs often does not observe.

The limitations are grouped by origin, because the remedy each group admits is different. Those of the model follow from representation choices and only change with a revision of the model itself. Those of the data depend on surveying, and can be supplied by the user. The methodological ones are declared departures from a reference formulation. And those of implementation are pending work, with no conceptual obstacle.

5.2.1. Limitations of the model

There is no hourly sub-step. A 40 mm rain concentrated in two hours and the same depth spread over twenty-four hours produce the same result. For tanks very small relative to the catchment area, this underestimates the overflow.

There are no losses in the tank. Evaporation and leakage are not modelled.

The initial condition is an empty tank. In series of less than a year, this penalises the result noticeably.

The runoff coefficient is single and constant. It does not vary with rainfall intensity or with the time since the last event, when in reality the first event after a dry spell has a lower coefficient.

The first flush is a depth, not a device. The program calculates the volume and cost of the device, `A x d_f`, but not its hydraulic arrangement, which requires a design of its own.

5.2.2. Limitations of the tariff data

These limitations concern the Brazil Edition catalogue. In the International Edition there is no catalogue: the tariff is typed in, and its accuracy is the user's.

The tariff catalogue covers almost only the residential category. The 652 municipalities of the database carry the residential structure; the non-residential categories only enter with the provider's official table, and are currently transcribed in the CORSAN municipalities -- commercial and public in 34, industrial in 30. Elsewhere, the commercial, industrial and public categories, which exist at every utility and are usually more expensive, have to be entered by hand by the user, as section 3.3 describes. Until they are, a study of a school, shop or factory calculates the saving with the residential tariff, and therefore **underestimates** it: the error is conservative and its direction is known, but it must be declared. Appendix D shows, municipality by municipality, what has already been collected.

Five BRK Tocantins municipalities with a decreasing tariff in the first block, which underestimates the bill in the first block.

Fourteen municipalities with sewage by blocks of their own, represented by a single percentage. It is a declared approximation.

No social tariff and no water tariff flags. The low-income class is poorly represented. At CASAN, the social tariff is R$ 8.63 against R$ 46.34 for the ordinary availability charge -- a difference that would invert the conclusion of a study of social housing.

5.2.3. Limitations of the life cycle assessment

It is a screening assessment, not an inventory conforming to ISO 14040.

A single impact category. Carbon only. There is no eutrophication, acidification, land use or toxicity.

The manufacturing emission of the pump set is an approximation by mass. The program multiplies the catalogue mass of each power rating by the intensity per kilogram of an environmental product declaration of a pump of another class. The order of magnitude is defensible; the value for each power rating is not a measurement.

Direct pressurisation is modelled by mean values. The energy uses a single working pressure, the mean of the pressure switch range, and not the on-off cycle of each use; and no start-up energy or standby consumption is added, which a pressure pump with a variable-frequency drive would have. Old projects without an upper tank open with the arrangement «not stated» and receive no energy until one is chosen.

There is no end of life. Neither credit for recycling nor burden for disposal.

There is no maintenance. Component replacements over the service life do not enter.

5.2.4. Limitations of water quality

Classes 2 to 4 are adaptations of the program, not of NBR 15527. The matrix of parameters by use is a feature of the 2007 version of the standard and did not survive the 2019 revision.

The constants of the roof wash-off model are not calibrated. The resulting water quality is an **assumption**, not a measurement. The program does not replace laboratory testing.

5.2.5. Declared methodological limitations

The R95p percentile is calculated over the whole series, not over the fixed ETCCDI base period 1961-1990. Series of different lengths are not directly comparable on this index.

Opricovic's conditions C1 and C2 are not implemented in VIKOR. The compromise solution should be read with the weight sensitivity analysis alongside.

The TOPSIS normalisation is a variant with translation by `x - min(x)`, to admit a negative net present value.

The tariff adjustment is a single percentage over the whole analysis period, which makes the twenty-year projection fragile. It is the assumption that most moves the economic result.

5.2.6. Limitations of implementation

Six fields are saved in the file and never read. Among them the **installation month**, so the cash flow always starts in month 1.

Importing a Netuno 4 project does not recognise variable demand, and silently downgrades it to constant. Section 3.8 shows how to check.

The pump start-up time, of 108 s per start, represents the priming time of a pump whose suction line has drained, and is applied to **every** start. In the reference case it accounts for 19% of the pump's annual consumption. It is conservative.

There is no option to treat the first flush as an event threshold, as Netuno 4 did, although the verification protocol mentions it. To compare the two versions, use a first flush of zero, where the two rules coincide.

5.2.7. What to do with this list

In a published work, cite the limitations that affect your case -- not all of them, which would turn the method section into a defensive list, but those that actually bear on the conclusions presented. A declared limitation is a caveat; an omitted limitation, when discovered, is an error.

Three examples guide the selection. A study of a school needs to cite the tariff category issue, because it shifts the whole economic result. A study of environmental performance needs to cite the single impact category and the approximation by mass of the pump set emission factor. And a study of social housing needs to cite the absence of a social tariff, which in that context can invert the conclusion.

A reviewer who finds one of these limitations left unsaid is entirely right to raise it, and raising it usually costs more time than declaring it would have cost.

5.3. How the program is verified

> This section describes Netuno's verification suite, so that a reviewer has what they need to judge how far a result produced by the program can be trusted.

A scientific program whose verification is not declared is like a claim without evidence: it may be right, but whoever uses it has no way of knowing, and whoever reviews work done with it has no way of judging. This section is the evidence behind the numbers presented in the rest of the manual, and it was written to be read by those who need to decide whether to trust them.

5.3.1. The suite

The program's verification consists of 1,544 automated tests, spread over 135 files and organised into five complementary families (Figure 66): unit tests, invariant tests, integration tests, graphical interface tests and the test that replicates the results of Netuno 4. The separation is not merely organisational: each family catches a distinct class of defect, and the absence of any of them would leave a flank exposed.

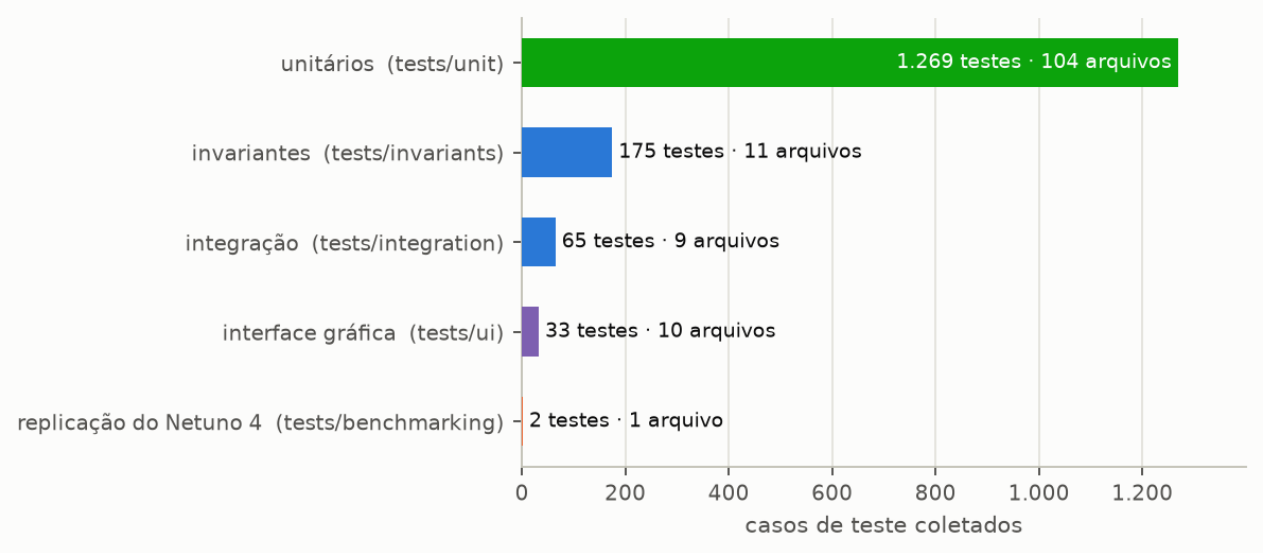


Figure 66 -- Number of tests and of files in each family of the verification suite, as collected in this version. Unit tests account for more than four fifths of the total, because they run fast and locate the error close to its origin.

The whole suite runs at every change to the code, and no change is merged while a test is failing. That discipline is what allows us to say that a fixed defect does not come back: the fix comes with the test that watches over it.

The verification layers of the program form a hierarchy, and each one catches a kind of error the others would not reach. A unit test does not detect a model error, because it checks the function against what is expected of it; and a manual check, which would detect that error, does not cover a thousand input combinations (Figure 67).

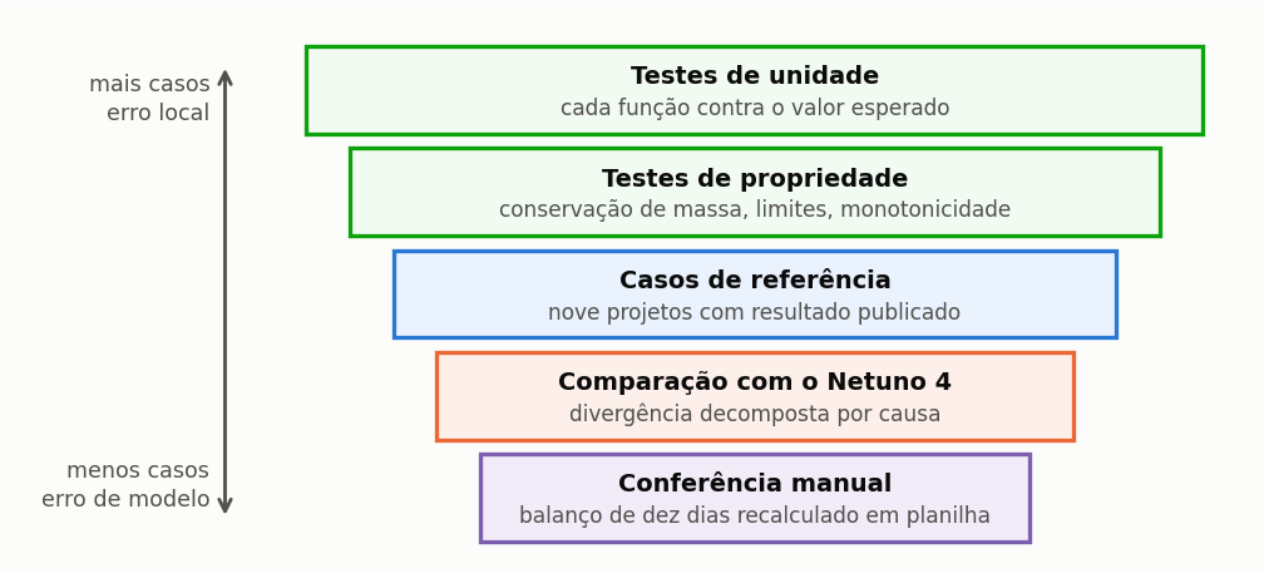


Figure 67 -- The five verification layers of the program, from the unit tests, numerous and aimed at local errors, to the manual check, occasional and aimed at model errors. No layer replaces another: it is the overlap of the five that supports the published result.

5.3.2. The invariants

The invariants are the most important family of the suite, and the reason lies in the nature of what they check. A value test checks whether the program produces a specific number for a specific input; an invariant checks whether a **property** holds for any admissible input. The first detects that a calculation has changed; the second detects that the model is no longer physically consistent.

Mass conservation requires that, over ten-year series, the volume collected equals the sum of the volume overflowed, consumed and left in storage. The residual measured is of the order of 10-¹■ per cent of the volume collected, that is, the rounding error of floating-point arithmetic itself. **Closure of the demand balance** does the same on the consumption side: the total demand must be exactly the sum of three parts, which are the rainwater consumed, the potable water bought and the unmet demand.

The **range of the potential** checks that the saving potential stays between zero and one hundred per cent for every combination of volume, runoff coefficient and area, which is checked over sixty combinations. The **ceiling of the suppliable fraction** is more specific and more revealing: the potential can never exceed the fraction of the demand rain can supply, and the check extends to volumes of one hundred cubic metres, far beyond any plausible building application.

Parity between engines compares the compiled core with the reference loop and requires both to return the same number, for single and double tanks, with and without the irrigation backup supply, within a tolerance of 10-■. **Monotonicity**, finally, requires that more volume never reduces the potential and that more demand never increases it -- two properties any correct implementation satisfies and a defective implementation violates conspicuously.

5.3.3. Analytical cases

A separate set gathers **closed-form** cases, that is, situations in which the correct result is known by deduction, with no need to simulate: constant rain, constant demand, a tank of infinite capacity and a tank of zero capacity. They are the strongest test available, because they do not depend on any reference value produced by the program itself. A test that compares today's output with yesterday's detects a regression, but perpetuates an error that already existed in both; an analytical case cannot do that.

5.3.4. Path tests

This family was born from the most persistent defect this program has shown over its history: a value calculated correctly that does not reach whoever should consume it. It is a class of error that escapes formula tests entirely, because each part on its own is correct -- the source calculates well, the destination would calculate well if it received the value, and what fails is the stretch between the two. Only a path test catches it: change the input on the screen and check that the final result actually moves.

Five real defects found by this family, and now watched by it, illustrate the pattern: the choice of disinfection technology did not reach the life cycle assessment; unticking the polishing filter had no effect in classes 2 to 4; the four tax fields were zeroed by the screen itself when it was closed; the irrigation backup supply option did not reach the compiled cores; and thirty of the eighty-seven fields of the time did not survive saving and reopening the project.

5.3.5. The numbers of the manual

With the writing of this manual a verification family of its own was created, meant to keep the document and the program from diverging: **each printed value has a matching assertion**. The numbers of section 2.1, those of the whole tutorial and each row of the ten-day table are covered, as is the requirement that certain sections declare certain caveats. When the program changes, the test fails and points to the page to correct.

It is the mechanism that keeps a whole manual from ageing silently, which is the usual fate of the documentation of scientific programs.

5.3.6. Verification against Netuno 4

The verification extends to the comparison with the previous version of the program, over four thousand and fifty reference cases, with the difference broken down by attributable cause. Section 6.1 presents the result of that comparison and the two deliberate methodological divergences that explain it.

5.3.7. What the verification does not prove

The distinction between verification and validation is demanded by any competent reviewer, and rightly so, because the two answer different questions and only one of them is within the reach of a test suite.

Verification answers whether the program correctly solves the equations it implements, and that is exactly what the suite described in this section tests. **Validation** answers whether the equations implemented adequately represent physical reality, and that requires field measurement: systems installed, monitored over time and compared with what the simulation predicted.

Netuno is verified. The validation of the daily behavioural model rests on the literature of the field, not on measurement of its own carried out by this group. A study that presented the program's results as if they were measurements would be making a methodological error.

5.3.8. The verification report in the interface

The **Help -> Verification Report** menu opens, inside the program, the verification report of the version: the distinction between verification and validation, the kernels exercised, the invariants, the analytical cases and the comparison with Netuno 4. It is the reference to cite when a study needs to declare how the program was verified.

5.3.9. Who runs the suite, and when

The suite comes with the program's source code, not with the installation: it is run by the research group at every change, before any version is released. What this means for those using the installed program is that the version on their machine has passed the 1,544 tests described here, including those that check each number printed in this manual.

> **IN DEPTH . CHECKING FOR YOURSELF**
> The source code is public, under the GPL v3 licence, and the suite runs with a single command in the development environment. A reviewer who wants to check the invariants -- mass closure, monotonicity, parity between the two calculation engines -- finds the test files organised by subject, and appendix B links each process of the program to what verifies it.

6. Fixes, bugs and versioning

6.1. Comparison with Netuno 4

> This section presents the measurement against 4,050 reference cases and, more importantly, the **breakdown** of the difference by cause.

6.1.1. The role of the comparison

It is worth starting by making clear what this comparison is **not**. Netuno 4 is not a fidelity target for Netuno 5. The current version adopts different modelling decisions, deliberate and documented, and it would be a cause for concern, not satisfaction, if the results coincided exactly.

The value of Netuno 4 in this verification is another, and a greater one: it offers an independent set of results, produced by another implementation of the same conceptual model, by other hands and in another language. Without an external reference of this kind, verification would be limited to internal properties of the program -- necessary, as section 5.3 describes, and unable to detect a systematic error that respected them all.

6.1.2. The reference data

The comparison between the two versions rests on four thousand and fifty cases exported from Netuno 4 and preserved since then, byte for byte, precisely so that the reference would not change during development. They are spread over two sets, with different cities and parameter ranges.

Table 136 -- The sets of reference cases exported from Netuno 4

Set	Cases	Cities
Floripa/Brasília	1,458	Florianópolis, Brasília
Manaus/São Paulo	2,592	Manaus, São Paulo

6.1.3. The total divergence

Table 137 -- Divergence measured between the two versions, by set of cases

Set	Mean absolute error	Bias	Maximum error	<= 5 pp
Floripa/Brasília	2.29 pp	-2.29 pp	7.43 pp	97.4%
Manaus/São Paulo	2.71 pp	-2.71 pp	10.60 pp	88.6%

The bias is systematically negative in both sets, that is, Netuno 5 estimates less saving than Netuno 4 in practically every case. The result is expected, and the next subsection breaks the difference down into its attributable causes.

6.1.4. The breakdown, which is the number that matters

Comparing the potential at the optimal volume of one program directly with that of the other mixes three differences of different kinds, and the resulting number does not say which one accounts for what. The breakdown separates them by comparing the two programs in three successively more controlled configurations, over the 2,583 cases of Manaus and São Paulo.

Table 138 -- Divergence in three successively more controlled configurations

Configuration	Mean absolute error	Bias	<= 5 pp
1. Netuno 5 optimum against Netuno 4 optimum	2.71 pp	-2.71 pp	88.6%
2. evaluated at the volume Netuno 4 declared optimal	1.60 pp	-1.60 pp	96.4%
3. likewise, adopting Netuno 4's threshold first flush	**0.25 pp**	**-0.25 pp**	**100.0%**

The three configurations use the same cases, excluding the nine with a null reference dealt with below. In each one, Netuno 5 estimates less than Netuno 4 in almost every case, and that is why the mean absolute error practically coincides with the bias.

The resulting attribution:

Table 139 -- Attribution of the divergence to its three sources

Source of the divergence	Contribution	Nature
optimal volume selection criterion	1.11 pp (41%)	design difference
treatment of the first flush	1.36 pp (50%)	deliberate methodological difference
residual of the water balance	**0.25 pp (9%)**	--

> **With the two known differences neutralised, the Netuno 5 engine reproduces Netuno 4 with a mean absolute error of 0.25 percentage points and 100% of the cases within 5 points.**
>
> The residual is consistent with the two-decimal precision of the reference spreadsheet itself. It is **this** number that verifies the engine. The total bias of 2.71 pp does not measure error: it measures the sum of two design decisions, both deliberate.

Figure 68 breaks the total divergence down into the three sources of the table.

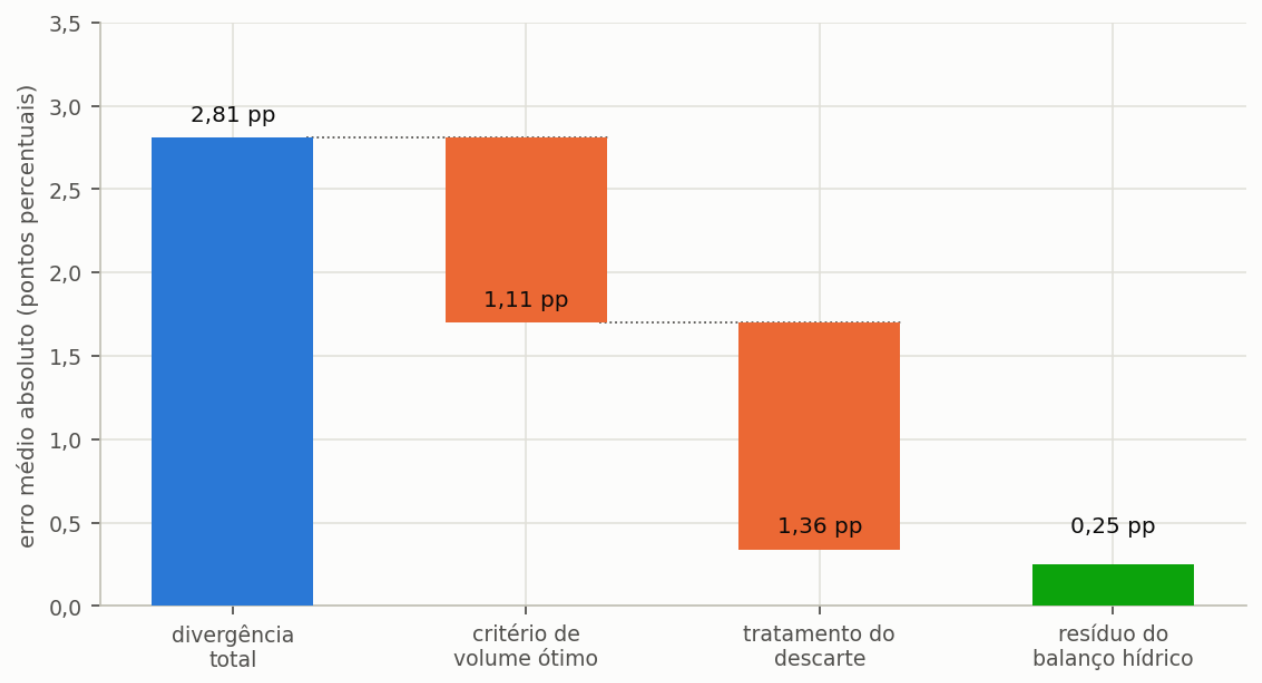


Figure 68 -- The mean divergence relative to Netuno 4, broken down. The first bar is the total observed; the two orange ones are declared method differences, one of design criterion and the other of first-flush treatment; the last is the residual of the water balance, which is the number that actually verifies the engine.

6.1.5. The two differences, in detail

6.1.5.1. The first flush

Netuno 4 treats the first-flush depth as an event threshold: when the day's precipitation is below it, nothing is collected; when it exceeds it, the whole rain is collected, with no subtraction at all. The Netuno 4 manual is explicit about it, stating that a daily precipitation below that value makes the program consider that there was no precipitation.

Netuno 5 subtracts the depth from the day's precipitation, a physically more faithful treatment because the water diverted to the self-cleaning device does not reach the cistern. The change accounts for half of the total divergence measured between the two versions.

6.1.5.2. The optimal volume criterion

Both programs choose the ideal volume by marginal-gain criteria, but the implementations and the search grids differ, so each one elects a different volume and the comparison comes to include that difference of choice. Evaluating the two programs at the same volume -- the one Netuno 4 declared optimal -- the bias falls from 2.71 to 1.60 percentage points, which attributes 41% of the total difference to the selection criterion.

6.1.6. How to measure the difference without changing the engine

The emulation of the old rule is done by transforming the input precipitation series, not by changing the water balance code. The methodological distinction is essential: measuring the divergence with modified code would validate a program that is not the one distributed to the user, which would defeat the purpose of the verification.

$$P' = P + d_{\mathrm{f}} \mathrm{se} P \geq d_{\mathrm{f}};\ P' = 0\ \mathrm{caso\ \ contrário} \tag{53}$$

Table 140 -- Variables of Equation 53

Symbol	Meaning	Unit
`P′`	transformed precipitation, to be supplied to Netuno 5	mm
`P`	precipitation observed on the day	mm
`d_f`	first-flush depth	mm

With this transformation applied, the Netuno 5 calculation `P_ef = max(0, P′ - d_f)` reproduces the threshold semantics exactly. The measurement does not run the risk of validating code that is not the one distributed.

> **A documentation caveat.** The internal verification protocol says to use an automatic option for this emulation. That option **is not implemented**. Use the series transformation, or compare with a first flush of zero, where the two rules coincide.

6.1.7. The null-reference cases

In some of the rows, Netuno 4 exported zero while Netuno 5 calculates a positive value. The incidence differs greatly between sets: 0.35% in Manaus and São Paulo against 34.98% in Floripa and Brasília.

The disparity points to a feature of the **reference data** -- probably Netuno 4 did not find an optimal volume under its own criterion -- and not to an error of Netuno 5, which never returns zero in these cases. They are kept apart from the statistics so as not to dominate them.

6.1.8. What to declare in a comparative study

- the **two numbers**: the total divergence and the residual after the breakdown;
- the two deliberate differences, named;
- whether the comparison was made with a zero first flush, with threshold emulation, or without neutralisation;
- the version of each program.

Presenting only the total divergence, without breaking it down, suggests an engine error that does not exist.

6.2. Error messages and what to do

> What each message means, what triggered it, and how to resolve it.

The program follows a deliberate policy towards impossible inputs: it rejects them instead of silently correcting them. The reason is that a negative catchment area is not a user preference to be accommodated, but a typing error that needs to be seen by whoever made it. Silently correcting it would produce a plausible result from a wrong input, which is the worst possible combination in a simulation program.

> **The messages of the calculation core are in Portuguese, in both editions.** The interface is translated, but the checks that run inside the calculation raise their messages in the language they were written in. The tables below quote each message as the program prints it, in Portuguese, followed by its meaning in English in brackets, so that you can recognise it on the screen.

6.2.1. Physical validation of the input

The messages below are raised before any calculation starts, and all of them name the field responsible and the value received, so that the correction does not require a search.

Table 141 -- Physical input validation messages and the action for each one

Message	What triggered it	What to do
«Volume do reservatório inferior negativo» (negative lower tank volume)	volume below zero	correct it; **zero is valid** and represents the no-system scenario
«Volume do reservatório superior negativo» (negative upper tank volume)	likewise, in the elevated tank	correct it; zero means no upper tank
«Área de captação negativa» (negative catchment area)	area below zero	correct it; use the horizontal projection in m²
«Coeficiente de escoamento fora de [0, 1]» (runoff coefficient outside [0, 1])	negative coefficient or above 1	above 1 the roof would deliver more water than the rain brings; usual roofs are between 0.80 and 0.95
«Lâmina de descarte negativa» (negative first-flush depth)	first flush below zero	the first flush removes water from the catchment, it does not add any
«Demanda diária negativa» (negative daily demand)	demand below zero	correct it
«Fração atendível por água pluvial fora de [0, 1]» (rainwater-suppliable fraction outside [0, 1])	fraction above 1	enter 0.4 for 40%, not 40
«... não é um número finito» (... is not a finite number)	field that reached the calculation empty, as text or infinite	fill in the named field with a number
«Modo de descarte desconhecido» (unknown first-flush mode)	mode other than 1, 2 or 3	choose one of the three modes; for no first flush, enter a depth of zero
«Intervalo de descarte inválido» (invalid first-flush interval)	mode 3 with an interval below 1 day	enter the interval in whole days
«Nível de acionamento da motobomba negativo» (negative pump start level)	level below zero	correct it; zero means starting with the tank empty

> The suppliable fraction is the most frequent message, and the rule that avoids it is a single one: where you type, a percentage from 0 to 100; where it is stored, a fraction from 0 to 1. Section 4.2 justifies the split. When the message appears, the value arrived as 40 in a place that stores 0.4 -- typically a hand-edited parameter file or an automation call.

Table 142 -- Suppliable fraction messages and the action for each one

Message	What triggered it	What to do
«... parece estar em percentual» (... seems to be a percentage)	fraction between 1 and 100 in the file, in the weekly, monthly or daily vector, or in automation	enter 0.4 for 40%
«... fora de [0, 100]» (... outside [0, 100])	negative percentage or above 100 on the screen, in the scenario table, in the spreadsheet or in `--share`	correct it; 40 means 40%
orange field, or the question «Simulate anyway?»	percentage between 0 and 1	if you meant 40%, type 40; if you really meant less than 1%, confirm

6.2.2. Precipitation series

The program does not distinguish a dry day from a day without a record, and so it does not guess: a gap in the series stops the calculation, saying where it is. Filling the gap, or trimming the series to a period without gaps, is a decision for whoever conducts the study, and the decision must be stated in the study.

Table 143 -- Precipitation series error messages and the action for each one

Message	Cause	What to do
«A série de precipitação está vazia» (the precipitation series is empty)	file with no numeric line	check that the file has a column of daily values, one per line
«... dia(s) sem valor numérico, o primeiro no dia N» (... day(s) without a numeric value, the first on day N)	record gap: empty line or text in the middle of the series	fill the gaps or trim the series, and declare the treatment
«... valor(es) negativo(s), o primeiro de X mm no dia N» (... negative value(s), the first of X mm on day N)	missing-data sentinel, such as -999	treat it as a gap: fill or trim

A header line before the data, such as «Date;Precipitation», is not an error: it is ignored and reading starts at the first numeric line.

6.2.3. Sweep

Table 144 -- Volume sweep error messages

Message	Cause	Solution
«O volume máximo da varredura deve ser finito e não negativo» (the maximum sweep volume must be finite and non-negative)	empty or negative field	enter the maximum volume in litres
«O passo de varredura deve ser finito e maior que zero» (the sweep step must be finite and greater than zero)	step zero or empty	enter the step; it will be rounded to the multiple of 250 L

There is also a warning that is not an error and should not be mistaken for one: when the sweep step is adjusted to the multiple of 250 L, or when the limit of forty volumes forces it to be widened, the program records the

adjustment and carries on normally. All that warning requires of the user is to declare in the study the step actually used, not the step originally requested.

6.2.4. Chlorination

Table 145 -- Chlorination parameter error messages

Message	Cause	Solution
«Pureza deve ficar em (0, 1]» (purity must be in (0, 1])	purity entered as a percentage	enter 0.65 for 65%, not 65
«A dose não pode ser negativa» (the dose cannot be negative)	dose below zero	correct it
«O fator de emissão não pode ser negativo» (the emission factor cannot be negative)	emission factor of the product below zero	correct it; the factor is per kilogram of commercial product

6.2.5. Economic analysis

Table 146 -- Economic analysis error messages and the action for each one

Message	Cause	Solution
«... deve ser um número finito e não negativo» (... must be a finite, non-negative number)	investment, tariff, cost, power or volume empty, negative or infinite	the message names the field; fill it in with a number equal to or greater than zero
«O período de análise deve ser um número inteiro de anos» (the analysis period must be a whole number of years)	fractional, zero or negative period	enter the period in whole years
«... deve ser um número finito e maior que -100%» (... must be a finite number greater than -100%)	adjustment, inflation or discount rate of -100% or less	check the sign of the rate
«... parece estar em percentual» (... seems to be a percentage)	annual rate entered as 5 in a function expecting a decimal fraction	enter 0.05 for 5%

> **Unit of the rates.** In the interface and in the project file, every rate is entered as a percentage, with the unit next to the field: the minimum attractive rate of return and inflation in % per month (p.m.) or in % per year (p.a.), according to the button beside them, and the tariff adjustments in % per adjustment period, whose length in months is in the neighbouring field. The project stores the minimum attractive rate and inflation in % per month. The conversion between month and year is compound, (1 + i)^12 - 1, not a multiplication by twelve: 1% a month is equivalent to 12.68% a year. In automation, the annual calculation functions receive the rates as a decimal fraction, 0.05 for 5%, and reject a value that looks like a percentage instead of reinterpreting it.

6.2.6. Multi-criteria decision

Table 147 -- Multi-criteria decision error messages and the action for each one

Message	Cause	Solution
«Informe 3 pesos de eixo ou 7 pesos de critérios TOPSIS» (enter 3 dimension weights or 7 TOPSIS criterion weights)	number of weights other than 3 or 7	enter one weight per dimension or one per criterion
«Os pesos TOPSIS devem ser finitos e não negativos» (TOPSIS weights must be finite and non-negative)	empty or negative weight	correct it; a weight of zero is valid and excludes the criterion
«Ao menos um peso TOPSIS deve ser maior que zero» (at least one TOPSIS weight must be greater than zero)	all weights zero	give weight to at least one criterion
«O critério TOPSIS ... da alternativa N não é finito» (TOPSIS criterion ... of alternative N is not finite)	undefined indicator in an alternative, typically a payback with no return	check the alternative indicated before deciding

6.2.7. Pump set

Table 148 -- Pump set parameter error messages

Message	Cause	Solution
«Rendimento deve ficar em (0, 1]» (efficiency must be in (0, 1])	efficiency as a percentage in the core	use 0.55 for 55%
«Rendimento do motor deve ficar em (0, 1]» (motor efficiency must be in (0, 1])	likewise, in the motor field	leave it blank for the program to adopt the NBR 17094-2 table

6.2.8. Files

Table 149 -- Error situations when reading and saving files

Situation	What happens	What to do
unreadable rainfall series	reading stops with a message	check the format: one reading per line, in mm, single column
`.netuno` file with invalid JSON	opening stops	the file may be truncated; use the backup copy
section missing in the file	the fields fall back to the class default	expected when opening a file from an earlier version; review the corresponding tab
climate database not installed	the tab offers only a file of your own	see section 1.2

6.2.9. Warnings that do not stop the calculation

The warnings gathered in this table do not prevent the calculation and do not indicate a defect: they are remarks the program makes about the conditions under which the result was produced. All of them need to be carried over into the text of the study, because all of them affect the interpretation of the number and none of them is evident in the result on its own.

Table 150 -- Warnings that do not stop the calculation and must be stated in the study

Warning	What it means
non-residential building type with a residential tariff	the economic result is **underestimated**; see section 3.5
the target class recommends fine filtration and it was declined	decision recorded in the report
emission factor with a screening label	the factor has no primary source; declare it
incomplete tariff structure	the municipality's table has a gap; check it against the resolution
sweep step adjusted	the effective step differs from the one requested
price extrapolation with low R²	the volume is outside the price table and the regression is not reliable

6.2.10. When the result looks wrong

Before suspecting a defect in the program, three checks resolve most of the cases in which the result is surprising, and each one corresponds to an input error with a signature of its own.

When the **potential comes out higher than expected**, the immediate suspect is the suppliable fraction: it is the ceiling of the whole result, and a too generous fraction inflates everything that comes after without anything in the report revealing the origin.

When the **potential comes out much lower**, check the first-flush depth. The corresponding mistake is described in section 5.1.3, and consists in entering the first flush in litres when the field asks for millimetres -- which discards, typically, the whole rain of every day.

When the **payback comes out absurd**, in either direction, check the municipality's tariff regime. It is the most frequent cause of an implausible economic result, and section 3.3 gives the test that identifies it in two minutes.

It is also worth going through the four input steps before accepting any result. The sidebar marks those still pending, and checking four screens takes less time than discovering the mistake after the report has been issued.

6.3. Version history and what changed

This section records what changed between versions of the program and, above all, **which changes alter results already published**. The distinction matters more than it seems: a fix to the interface affects no earlier study, whereas a fix to a calculation can shift every number of a paper already submitted. Anyone who has published with Netuno needs to know into which category each change falls, and that is why the history below separates the two explicitly.

The rule guiding this whole section is simple: a calculation fix obliges you to declare the version used; an interface fix does not.

6.3.1. The version you are using

The version number appears in the sidebar, in the footer and in the **About Netuno** menu. It is also saved inside each project file and printed in each report, so that an old study always carries with it the identification of the program that produced it. The redundancy is deliberate: reports circulate separately from files, and the question «which version was this done with?» usually comes up months later, when nobody remembers.

The numbering scheme follows the usual software engineering convention. The first number indicates the generation of the program, and changing it means a reformulation of the conceptual model; the second indicates a set of new features; and the third indicates fixes and adjustments that do not change the architecture. The move from Netuno 4 to Netuno 5, dealt with in section 6.1, is the only case of a change in the first number in the history of the program.

Version 5.0.0 is the first public release of Netuno 5. The numbers before it were development ones, and never circulated; no study was done with them.

6.3.2. How this history is kept

From now on, each version released gets an entry of its own in this section, with the release date and what changed, always split into two lists: what alters the results of studies already done and what does not. The most recent entry comes first.

Each entry follows the same model, so that two versions can be compared without rereading: a one-sentence summary, the new features, the fixes that alter results, those that do not, and what changed in the documentation. Sections with no content in that version simply do not appear.

The same list is distributed with the code, in the `CHANGELOG.md` file at the root of the repository, and feeds the versions page of the documentation site. It is not written three times: all three come from this section, and an automatic check refuses the build if they diverge. Those following the program through the repository, the site or the manual read the same record. The `CHANGELOG.md` is in Portuguese; this section is its translation.

Table 151 -- **Table 99** -- Overview of the published versions

Version	Date	What it brought	Fixes that change results
5.0.0	29 September 2026	First public release of Netuno 5: the simulation engine rewritten in Python and separated from the interface, with life cycle assessment, treatment line, multi-criteria decision, 3D piping layout and the documentation in three formats.	19

6.3.3. Version 5.0.0 -- 29 September 2026

> **In one sentence.** First public release of Netuno 5: the simulation engine rewritten in Python and separated from the interface, with life cycle assessment, treatment line, multi-criteria decision, 3D piping layout and the documentation in three formats.

The version was the result of a systematic review of the program, carried out with a specific aim: to find values that were calculated correctly in one module and discarded before reaching the next. This class of defect is particularly dangerous because it produces no visible error -- the program responds, the number looks plausible, and only the comparison between two points of the program itself reveals the inconsistency. The review went along five axes of propagation: from module to module, from the screen to the calculation core, between the interface and automation, between duplicated definitions of the same constant, and over time, that is, through saving and reopening a project.

6.3.3.1. New features

The program was rewritten in Python, with the simulation engine separated from the interface, and gained what Netuno 4 did not have: life cycle assessment of carbon, a sized treatment line, multi-criteria decision with stochastic acceptability analysis, a tariff database of 652 municipalities, an interface in four languages and export to CAD, BIM and spreadsheet. Section 6.1 compares the two programs number by number, and section 5.3 describes the verification suite that supports the comparison.

The demand now has two declared parts: the non-potable fraction of indoor consumption and the outdoor demand in litres per day, suspended on rainy days. The total suppliable fraction is derived from the two and appears in a field that cannot be edited. Section 4.2 describes the rule. A layer of tests opens each distributed project at both ends of the program, the interface and the command line, and requires them to agree, volume by volume, on the potential, the NPV, the payback, the carbon and the five optima.

The piping is now organised by the designer, run by run and in assembly order, from the connection points of the components (top, bottom and sides of the tank). The program checks whether each run reaches the declared destination, rejects pipes occupying the same space and generates, with the same data, the 3D plan in DXF, the 3D schematic with the real dimensions, the rotatable volumetric view and the glTF and HTML files for sharing, as well as the BIM manifest read by Revit.

The Brazil Edition (BRL) and the International Edition (INT) come from the same code. The Brazil Edition carries the audited Brazilian data. The International Edition is the neutral core: it opens in English, with tariffs, costs, currency and emission factors entered by the user. The differences are in section 1.2.

6.3.3.2. Fixes that alter results

The changes gathered in this table alter numbers relative to what the program calculated before each fix. As 5.0.0 is the first public release, no study depends on them; the table is kept because it describes what the program now calculates differently, and because it is the format the following versions will use.

Table 152 -- Fixes that alter the results of studies done with an earlier version

What was fixed	Effect on the result
the rated power of the pump set was treated as electrical input power, when it is shaft power	specific pumping consumption went from 0.0654 to 0.1080 kWh/m³
the start-up energy divided the rated power by the pump efficiency, although there is no flow at start-up	start-up share corrected, about 14% of the annual energy
the irrigation demand supplied by the mains did not appear in the summary	unmet demand is no longer underestimated by 10.85 L/day in the reference case
four tax fields were zeroed at each confirmation of the economic dialog	the water and energy bills now include the declared taxes
chlorination, ultraviolet and ozone were treated as mutually exclusive	the treatment line now combines stages, according to item 4.6.5 of ABNT NBR 15527:2019
the economic analysis only charged energy with an upper tank, and the environmental assessment charged pumping in any arrangement and lost chlorination without an upper tank	the supply arrangement is now declared in step 4, and cost and carbon read the same energy; projects without an upper tank no longer have implicit pumping in the carbon and now have chlorination
the hypochlorite emission factor was 1.85 kg CO2e/kg, with no declared process	it became 1.72 kg CO2e/kg, from Ecoinvent 3.8 on an active-chlorine basis
the interface did not restore the indoor fraction or the outdoor demand when opening a project, and always simulated with 35% and 0 L/day	the interface simulates with the project's fraction; the tutorial, which declares 28%, gave a 32.85% potential at 4,000 L on the screen and now gives 27.12%, as the manual and the command line do
saving from the interface wrote 35% and 50 L/day as indoor fraction and outdoor demand, whatever those on the screen were	the file saves the simulated fractions; old files that disagree are opened with the total fraction they declare, with a warning
in the weekly, monthly and daily modes, the fallback fraction was the total one, which already includes the outdoor demand	the fallback fraction is the indoor one, and the outdoor demand is no longer counted twice
the interface ignored the fixed volume declared in the project and simulated the technical optimum of the sweep	in fixed-volume mode, the interface simulates the declared volume, as the command line does
the environmental tab of the interface used its own chlorine dose and factor, not those of the project	the environmental assessment reads the chlorine declared in the water quality step
the command line charged the same cistern price throughout the sweep	the price follows the volume, as in the interface; the NPV no longer grows indefinitely with capacity
the command line left the first-flush device out of the embodied carbon, converted the pump power by another rule and rebuilt the piping with only the 50, 75, 100 and 150 mm sizes	the command line's environmental assessment reads the first flush, the pump set and the piping from the project, as the interface does
the command line used 5,000 draws in SMAA-VIKOR and weights 0.33/0.33/0.34 in TOPSIS; the interface, 20,000 draws and 0.333/0.333/0.334	both use 20,000 draws and the same weights
the no-cistern alternative entered the interface's multi-criteria analysis with a payback of zero, the best possible	an alternative that does not pay back enters without a payback, at both ends
the first-flush depth was 1.0 mm in the project file, in spreadsheet reading and in the engine, and 2.0 mm in the CAD exporter, the command line and the report	the fallback value became 2.0 mm on every path, according to item 4.1.5 of ABNT NBR 15527:2019; a project with no declared depth discards twice what it discarded, and the usable catchment falls in proportion to the days of light rain
the 34 CORSAN municipalities were stored as a straight line, `PB x C + basic service`, with sewage at 70% of the whole bill, although the utility charges `PB x C■ + basic service` and sewage per m³ consumed	the water follows the Table of Exponents (`n` from 1.00 up to 20 m³ to 1.13 between 101 and 150 m³) and sewage, R$ 6.40/m³ in ANEXO-5; in Alegrete, reducing consumption from 30 to 20 m³ now saves R$ 253.81 a month, not R$ 155.55

What was fixed	Effect on the result
the commercial, industrial, public and social categories were the residential tariff multiplied by fixed factors, with no provider table at all: x1.85 on commercial water, x2.30 on industrial, x1.65 on public and x0.50 on social	a category only enters with the provider's official table; currently, commercial and public in the 34 CORSAN municipalities and industrial in 30. Elsewhere, the category appears disabled and the study uses the custom structure; when the category the building type requires is not available, the program warns that the residential tariff underestimates the saving

6.3.3.3. Fixes that do not alter results

The changes below fix the behaviour of the program without touching any calculated value. No result changes because of them; it has only become harder to produce a wrong study.

Table 153 -- Behaviour fixes that do not alter any calculated result

What was fixed	What changes for you
the calendar demand modulation was not saved in the project file	a project with a weekly, monthly or daily profile reopens as it was saved
a JSON file of unknown schema was accepted and opened an entirely default project	opening stops with a message of its own
the standby pump checkbox forced two pumps in any situation	the quantity entered is respected
six fields were saved in the file and never read back	section 5.2 lists those that remain in that condition
A single name for the research group: Sustainable Environments Research Group, in Portuguese Grupo de Pesquisa em Ambientes Sustentáveis.	no calculated value changes
The Brasília and Manaus examples now use the official tariffs of each city, instead of those of Florianópolis.	no calculated value changes
A new example, `examples/exemplo_quatro_criterios.netuno`, in which the technical, economic, environmental and TOPSIS optima fall on four different volumes.	no calculated value changes
the spreadsheet scenario template cited rainfall series that are not distributed, and a scenario without its series was silently simulated with Florianópolis rain	the template uses the series in `data/rainfall`, and a missing series is an error naming the scenario and the file
the ANEEL tariff query had no overall timeout and wrote the file in use while downloading	the query gives up after three minutes, and an interrupted download does not leave the cache half written
the manual's equations lost the superscript: the NPV came out with `(1 + TMA) m`	exponents are superscript again
on the 3D schematic sheet, each accessory created a dimension plane and a two-line label, and the parts ended up on top of each other	the sheet became landscape A3, with dimension planes only at the main levels and the small parts identified by code, with type and level in a table alongside
the siphon overflow was drawn across the overflow pipe, with a gap on both sides	the siphon follows the direction of the pipe and touches the faces of the part
the volumetric view opened with building, ground, flow and shadows on, heavy on first opening	it opens with the labels only; the rest is turned on with the button
the text of the exponential formula widened the Tariff Manager column and narrowed the charts	the charts keep their width in any municipality
the Direct3D shader compiler warnings (X4122, X4000) appeared as «Qt warning»	they enter only the summary count; a compilation error still appears
the CAPEX of piping and accessories was an estimated allowance, and the LCA counted only the metres of pipe typed in; bends, fittings, parts and control cables of the 3D layout entered neither	in a new project, both now come from the layout, part by part, with the price and carbon of each item; old files keep the costs they declared
labour was fixed once, on generic allowances, and did not change with the volume in the sweep; the sweep also stopped charging the pressure pump when there was no upper tank	in a new project, labour is 30% of the materials of each volume, and the pump set enters every arrangement other than gravity, as in the economic engine
in the global optimiser, the currency symbol followed the language: in English, values in reais appeared as dollars	the symbol is that of the currency chosen in the project; in another currency, the 3D layout prices in reais and the automatic labour are no longer applied
the usual 3D layout was always the pumped one, with one pump set	the layout follows the project's supply (pumped, pressurised or gravity) and draws the standby pump set in parallel; the CAPEX and the LCA no longer charge upper-tank parts that do not exist and now charge those of the standby

6.3.3.4. Documentation

The documentation comes with the release in three formats, built from the same sources: the full manual in PDF, seven separate volumes for those who only need one part, and a documentation site with eighteen interactive figures, in which the reader changes the parameters and sees the result the program calculated. All three, plus the `CHANGELOG.md` distributed with the code, come from the same text files.

The manual now also exists in English, section by section, with the same numbers, figures, tables and equations; it is the manual the International Edition installs and opens. An automatic check refuses the build if the two versions diverge in structure. Three additions to the Portuguese text came with the translation: section 2.1 explains where the order of Netuno's balance stands between the YAS and YBS rules of the literature, which were previously given as equivalent to it; appendix D gained the blocks of each municipality by category, with the audit status and the date of each item of data; and the tariff material received is now registered in a control of its own, in `tarifas/fontes/REGISTRO.md`, which says what has already arrived and what remains to be collected.

> **IN DEPTH . WHY THE PUMPING FIX IS THE MOST RELEVANT**
> ABNT NBR 17094-2:2016 requires the motor efficiency to be marked on the plate **separately** from the power, which means that the rated power is the power available at the shaft, not the power drawn from the socket. Taking one for the other underestimates consumption by a factor equal to the motor efficiency, which for a half-hp two-pole machine is 60.5%. The effect is large in relative terms, though small in absolute terms, because the pumping energy of a building rainwater harvesting system is modest compared with the rest of the economic balance. Section 2.6 develops the full chain.

6.3.4. How to tell whether your version has a fix

The procedure is direct and does not depend on asking anyone: open one of the example files that come with the program, described in appendix J, run it, and compare the result obtained with the value printed in this manual. Each example case was generated by the same version of the program that generated this manual, and each published number has a matching automatic check, as section 5.3 describes. A divergence, therefore, indicates that the installed program is not that of this manual's version.

If the numbers match, your installation corresponds exactly to the manual you have in hand. If they do not, the difference is not a defect of your installation, but a version difference that needs to be identified before any study.

6.3.5. What to declare in a published work

Good practice in computational reproducibility requires a paper to identify unambiguously the program that produced the results. In Netuno's case, four items of information are enough, and all are available without effort: the version of the program, which appears in the report; the source of the precipitation series, with the period used; the tariff structure adopted, with its effective date; and a statement of which parameters were changed from the default values. Section 4.10 gives the full list, with a ready-made method paragraph to adapt.

Attaching the `.netuno` file to the supplementary material settles all four at once, because the file is self-contained and carries the rainfall series inside it. It is the most economical way of making a study verifiable by others, and costs an attachment of about 160 kB.

6.4. How to report a problem

Simulation programs have defects, and Netuno is no exception. The review carried out before this release found a dozen calculation defects, none of them accompanied by an error message, and most of them discovered not by reading the code but by comparing two points of the program itself that should agree and did not. The practical lesson is that the attentive user is the best source of detection a scientific program has, and that is why this section exists.

A good report is one that allows the problem to be reproduced without you being there. Everything asked for below serves that single aim.

6.4.1. What to send

A useful report has four components, and the absence of any of them usually turns the investigation into an exchange of messages that takes more of everyone's time than the fix itself.

The project file. It is the most important item, and on its own it replaces pages of description. The `.netuno` file is self-contained: it carries all the parameters and the whole rainfall series inside it, so whoever receives it reproduces your simulation exactly, on the same machine or any other. Without it, the investigation starts by trying to guess the inputs, and guessing inputs is the most efficient way of not finding a defect.

What you expected and what you got. This is the part only you can provide. Describe the number that appeared and the number that should have appeared, with the reason for the second -- be it a hand calculation, a result from an earlier version, a value from the literature or simply the order of magnitude experience suggests. A report that only says «the result is wrong» leaves the recipient to discover which of the forty numbers on the screen is the wrong one.

The version of the program. It appears in the sidebar, in the footer and in the **About Netuno** menu, and is already saved inside the project file and any exported report. State it anyway, because it is the first filter of the investigation: half the problems reported in scientific programs are defects already fixed in a version later than the one the user has installed. State the edition too, Brazil or International.

The error message, if there is one. Copy the full text, not a summary. Section 6.2 lists the messages the program raises deliberately, with the meaning of and the action for each one; if your message is there, the problem is probably one of input and has an immediate solution.

6.4.2. What to check before reporting

A considerable share of the results that look wrong are correct results of wrong inputs, and the check below takes a few minutes. It is not there to discourage the report, but so that your time and that of whoever investigates is spent on what really is a defect.

Start with section 5.1, which gathers the ten design mistakes the program has no way of preventing. Three of them account for most unexpected results: the catchment area entered as the sloping area of the roof instead of the horizontal projection, the first-flush depth entered in litres instead of millimetres, and the municipality's tariff regime treated as a minimum consumption when it is an availability charge -- or the reverse.

Then check whether the behaviour observed is not a declared limitation. Section 5.2 lists what the program admittedly does not do, and some of those limitations give exactly the impression of a defect. The most frequent case involves the tariff category: the catalogue covers the residential category of the 652 municipalities and the others only where the provider's official table has been transcribed, so that, in the other municipalities, a study of a school or a shop comes out with the residential tariff until the user enters that of the right category. The program warns about this instead of correcting it on its own, because the right table is public and specific to each utility.

6.4.3. Where to send it

Netuno is developed by the Sustainable Environments Research Group, in the Department of Civil Engineering of the Federal University of Santa Catarina. The group's contact is the reporting channel, and it serves for defects as well as for feature suggestions and requests for clarification of the method. Reports may be written in English.

Reports that come with the project file go straight into the verification suite described in section 5.3: a case that reproduces a defect becomes a test, and the test then keeps that defect from coming back. That is why the file matters so much, and it is also the most complete answer that can be given to whoever reported -- the problem was not only fixed, it is now watched.

> **IN DEPTH . WHAT MAKES A DEFECT HARD TO FIND**
>
> The defects that survive for years in a scientific program all have the same signature: they produce plausible numbers. An error that generates a negative value, or one a thousand times larger than expected, is found on the first run. But an efficiency factor applied twice, or a unit converted once more than necessary, produces a number that passes any visual inspection and survives years of use. That is why the program's verification rests on invariants -- the mass balance that must close, the monotonicity that must hold, the ceiling the curve cannot cross -- and not only on expected values. When you suspect a number, the quickest way to confirm the suspicion is to look for one of those identities it violates.

Appendices

Appendix A -- The project parameters

There are **116 parameters**. This is the complete list of what a project stores, with the exact name of each field inside the `.netuno` file.

The list was not transcribed by hand: it is extracted from the program itself for each version, and therefore cannot diverge from it.

The column **where it is entered** says on which screen the field appears; the column **who consumes it** says which calculation uses it. A field with neither is a field stored in the file and never read, and is listed in section 5.2.

Defaults in currency are the Brazil Edition ones, in reais (R$); the International Edition starts the costs and tariffs at zero, for the user to enter in the currency of the project.

A.1. Identification

3 parameters.

Table 154 -- Project parameters: identification

Parameter	Default	Unit	Where it is entered	Who consumes it
`project_name`	New Netuno project	--	Export, Main window, Reports	Automation, CAD and BIM, Reports, Spreadsheets
`project_id`	--	--	Main window, Reports	Automation
`created_at`	--	--	Main window	--

A.2. Rainfall

9 parameters.

Table 155 -- Project parameters: rainfall

Parameter	Default	Unit	Where it is entered	Who consumes it
`precip_file`	--	--	Main window, Project	Automation, Scenarios, Spreadsheets
`climate_source_mode`	national_db	--	--	--
`climate_city_uf`	SC	--	Project, Reports	--
`climate_city_name`	Florianópolis	--	Project, Reports	--
`climate_model`	INM-CM4	--	Project, Reports	--
`climate_scenario`	HIST	--	Project, Reports	--
`climate_start_date`	1980-01-01	--	Reports	--
`climate_end_date`	2013-12-31	--	Project, Reports	--
`start_date`	1980-01-01	--	Main window, Project, Tariffs and costs	Automation, Climate

Notes from the code:

- `climate_source_mode`: "national_db" (national database, Brazil Edition) or "custom_file" (own file)

A.3. Catchment

7 parameters.

Table 156 -- Project parameters: catchment

Parameter	Default	Unit	Where it is entered	Who consumes it
`discard`	--	mm	Charts, Documentation, Environmental, Main window, Project, Quality, Reports	Automation, CAD and BIM, LCA, Reports, Scenarios, Spreadsheets, Treatment line
`discard_mode`	1	code	Main window, Project, Sweep	Automation, Daily balance, Scenarios
`discard_days_interval`	7	days	Main window, Project, Sweep	Automation, Daily balance, Scenarios
`area`	100	m²	Building, Charts, Documentation, Environmental, Main window, Quality, Reports, Sweep	Automation, CAD and BIM, Daily balance, LCA, Reports, Scenarios, Spreadsheets, Treatment line
`irrigation_garden_area_m2`	0	m²	--	--
`runoff_coeff`	--	--	Building, Main window, Quality, Sweep	Automation, Daily balance, Scenarios, Spreadsheets
`first_flush_system_enabled`	no	--	Environmental, Quality	Automation, LCA, Treatment line

Notes from the code:

- `discard`: mm
- `discard_mode`: 1 = every event, 2 = first event after a dry spell, 3 = every X days
- `discard_days_interval`: interval for mode 3
- `area`: m²
- `irrigation_garden_area_m2`: Garden area in m² for the sizing wizard
- `runoff_coeff`: see hydrology_defaults

A.4. Building

2 parameters.

Table 157 -- Project parameters: building

Parameter	Default	Unit	Where it is entered	Who consumes it
`num_economias`	1	units	Building, Environmental	Automation, Economic
`building_template_id`	res_medio	--	Building	Automation, Tariffs

Notes from the code:

- `num_economias`: Number of dwelling units / apartments with individual metering

A.5. Demand

17 parameters.

Table 158 -- Project parameters: demand

Parameter	Default	Unit	Where it is entered	Who consumes it
`demand_type`	0	code	--	--
`demand_mode`	constant	--	Building, Environmental, Main window, Reports, Sweep	Automation, Daily balance
`demand_value`	200	L/day	Documentation, Economic, Main window	Automation, Daily balance
`demand_monthly`	--	--	Main window	Automation
`residents_type`	0	code	--	--
`residents_count`	4	residents	Economic, Environmental, Main window	Automation, Daily balance, Pump set
`residents_monthly`	--	--	Main window	Automation
`potable_indoor_share`	0.6	fraction	Building, Main window	Automation
`non_potable_indoor_share`	0.4	fraction	Building, Main window	Automation, Daily balance
`rainwater_share`	0.4	fraction	Documentation, Main window, Sweep	Automation, Daily balance, Scenarios, Spreadsheets
`weekly_demands`	--	--	Building, Main window, Sweep	Automation, Daily balance
`weekly_residents`	--	--	Building, Main window, Sweep	Automation, Daily balance
`weekly_shares`	--	--	Building, Main window, Sweep	Automation, Daily balance
`monthly_shares`	--	--	Building, Main window, Sweep	Automation, Daily balance
`daily_demands`	--	--	Building, Main window, Sweep	Automation, Daily balance
`daily_residents`	--	--	Building, Main window, Sweep	Automation, Daily balance
`daily_shares`	--	--	Building, Main window, Sweep	Automation, Daily balance

Notes from the code:

- `demand_mode`: "constant", "weekly", "monthly", "daily"
- `demand_value`: L/day, per resident or total
- `potable_indoor_share`: Potable fraction of indoor consumption; 1 - non_potable_indoor_share
- `non_potable_indoor_share`: Non-potable fraction of indoor consumption (toilets, washing machine); see fracao_atendivel
- `rainwater_share`: fraction from 0 to 1 (0.4 = 40%)

A.6. Irrigation

4 parameters.

Table 159 -- Project parameters: irrigation

Parameter	Default	Unit	Where it is entered	Who consumes it
`irrigation_liters_per_day`	0	L/day	Building, Environmental, Main window, Reports	Automation, Daily balance
`irrigation_rain_threshold_mm`	2	mm/day	Building, Main window	Automation, Daily balance
`irrigation_potable_backup`	yes	--	Building, Main window	Automation, Daily balance
`irrigation_rate_l_m2_day`	3	L/m².day	--	--

Notes from the code:

- `irrigation_liters_per_day`: Outdoor demand (watering, washing outdoor areas), L/day
- `irrigation_rain_threshold_mm`: Daily precipitation threshold that suppresses irrigation (mm/day)
- `irrigation_potable_backup`: If True, the mains supplies irrigation on dry days when the cistern runs dry
- `irrigation_rate_l_m2_day`: Daily watering rate (L/m²/day)

A.7. Water quality

9 parameters.

Table 160 -- Project parameters: water quality

Parameter	Default	Unit	Where it is entered	Who consumes it
`water_quality_target_class`	classe_1	--	Environmental, Quality	Automation, LCA, Treatment line
`water_quality_disinfection`	cloro	--	Environmental, Quality	Automation, LCA, Treatment line
`water_quality_polishing`	no	--	Environmental, Quality	Automation, LCA, Treatment line
`etap_enabled`	yes	--	Economic, Environmental, Optimiser	Automation, LCA, Treatment line
`chlorine_treatment_enabled`	yes	--	Economic, Environmental, Optimiser, Quality	Automation, Economic, LCA
`chlorine_dose_mg_l`	--	mg Cl2/L	Economic, Environmental, Optimiser, Quality	Automation, Economic, LCA, Reports, Treatment line
`chlorine_product_purity`	--	kg/kg	Environmental, Quality	Automation, Economic, LCA, Treatment line
`chlorine_cost_per_kg`	25	R$/kg	Economic, Optimiser, Quality	Automation, Economic, Prices, Treatment line
`chlorine_co2_kg_per_kg`	--	kg CO2e/kg	Environmental, Quality	Automation, LCA, Treatment line

Notes from the code:

- `water_quality_disinfection`: "cloro" (chlorine), "uv", "ozonio" (ozone), "nenhum" (none)
- `chlorine_dose_mg_l`: mg of active Cl2 per litre
- `chlorine_product_purity`: kg of active Cl2 per kg of product
- `chlorine_cost_per_kg`: price per kg of commercial PRODUCT

A.8. Tanks

12 parameters.

Table 161 -- Project parameters: tanks

Parameter	Default	Unit	Where it is entered	Who consumes it
`upper_tank_vol`	500	L	Economic, Environmental, Export, Main window, Optimiser, Pump wizard, Reports, Sweep, Tariffs and costs, Upper tank	Automation, CAD and BIM, Daily balance, Multi-criteria, Scenarios, Spreadsheets
`upper_tank_used`	1	code	Economic, Environmental, Main window, Optimiser, Reports, Sweep, Tariffs and costs	Automation, Daily balance, LCA, Pump set, Scenarios, Spreadsheets
`lower_tank_max`	10000	L	Main window, Sweep	Automation, Daily balance, Scenarios, Spreadsheets
`lower_tank_step`	250	L	Main window, Sweep	Automation, Daily balance, Scenarios, Spreadsheets
`lower_tank_mode`	1	code	Main window, Sweep	Automation
`lower_tank_fixed`	2000	L	Main window, Sweep, Tariffs and costs	Automation
`delta_ideal`	5	%/m³	Main window, Optimiser, Sweep	Automation, Scenarios
`econ_lower_tank_vol`	2000	L	Economic, Environmental, Sweep, Tariffs and costs	Automation
`lower_tank_material`	PEAD	--	Economic, Optimiser, Tariffs and costs	Automation, LCA
`upper_tank_material`	PEAD	--	Economic, Optimiser, Tariffs and costs	Automation, LCA
`cost_lower_tank`	0	R$	Economic, Sweep, Tariffs and costs	Automation, Economic
`cost_upper_tank`	0	R$	Economic, Sweep, Tariffs and costs	Automation, Economic

Notes from the code:

- `upper_tank_vol`: L
- `upper_tank_used`: 0 = no upper tank, 1 = with upper tank
- `lower_tank_max`: L
- `lower_tank_step`: L
- `lower_tank_mode`: 0 = fixed volume, 1 = sweep
- `lower_tank_fixed`: used in fixed-volume mode
- `delta_ideal`: marginal-gain threshold for the ideal volume, in %/m³
- `lower_tank_material`: "PEAD" (HDPE), "CONCRETO" (concrete), "FIBRA" (GRP), "ACO_INOX" (stainless steel)
- `upper_tank_material`: "PEAD" (HDPE), "CONCRETO" (concrete), "FIBRA" (GRP), "ACO_INOX" (stainless steel)

A.9. Pump set

18 parameters.

Table 162 -- Project parameters: pump set

Parameter	Default	Unit	Where it is entered	Who consumes it
pumping_threshold	50	L	Main window, Sweep	Automation, Daily balance, Scenarios, Spreadsheets
supply_arrangement	--	code	Economic, Environmental, Sweep	Automation, Economic, LCA, Pump set
pressurizer_pressure_m	18	m head	Sweep	Pump set
pressurizer_starts_per_person_day	5.05	1/person.day	Sweep	Pump set
pressurizer_model	komeco_tp820t	code	Sweep	Pump set
pressurizer_point_flow_l_min	9	L/min	Sweep	Pump set
pressurizer_power_kw	0.18	kW	Sweep	Pump set
pressurizer_flow_l_min	13.3	L/min	Sweep	Pump set
pressurizer_post_run_s	0	s	Sweep	Pump set
pump_count	2	units	Economic, Environmental, Optimiser, Reports, Tariffs and costs	Automation, Economic, LCA, Prices, Reports
pump_power_str	1/2	--	Reports	--
pump_power_kw	0.36775	kW	Charts, Economic, Optimiser, Reports, Sweep, Tariffs and costs	Automation, CAD and BIM, Economic, Prices, Pump set, Reports
pump_manometric_head_m	0	m head	Sweep	Automation, Pump set
pump_efficiency	0	%	Charts, Sweep, Tariffs and costs	Automation, Pump set
pump_flow_rate	0	L/h	Charts, Economic, Optimiser, Reports, Sweep, Tariffs and costs	Automation, Pump set
pump_startup_time_h	0.03	h	Charts, Economic, Optimiser, Tariffs and costs	Automation, Pump set
pump_motor_efficiency	0	%	Tariffs and costs	Automation, Pump set
pump_unit_cost	300	R$	Economic, Optimiser, Sweep, Tariffs and costs	Automation, Economic, Prices

Notes from the code:

- pumping_threshold: L, pump trigger level
- pump_count: duty + standby as the recommended initial configuration

- `pump_power_kw`: kW (1 hp (metric) = 0.7355 kW; 1/2 hp = 0.36775 kW)
- `pump_efficiency`: %, 0 = deduce from the size
- `pump_startup_time_h`: h
- `pump_motor_efficiency`: %
- `pump_unit_cost`: R$, Brazilian market reference for a 1/2 hp peripheral pump (zero in the International Edition)

A.10. Piping

1 parameter.

Table 163 -- Project parameters: piping

Parameter	Default	Unit	Where it is entered	Who consumes it
`pipes_list`	--	--	Economic, Tariffs and costs	Automation, LCA

A.11. Investment

3 parameters.

Table 164 -- Project parameters: investment

Parameter	Default	Unit	Where it is entered	Who consumes it
`cost_labor`	0	R$	Economic, Sweep, Tariffs and costs	Automation, Economic
`cost_piping`	0	R$	Economic, Sweep, Tariffs and costs	Automation, Economic
`cost_accessories`	0	R$	Economic, Sweep, Tariffs and costs	Automation, Economic, Treatment line

A.12. Operation

1 parameter.

Table 165 -- Project parameters: operation

Parameter	Default	Unit	Where it is entered	Who consumes it
`op_costs`	--	--	Economic, Optimiser, Sweep, Tariffs and costs	Automation, Economic

A.13. Tariffs

15 parameters.

Table 166 -- Project parameters: tariffs

Parameter	Default	Unit	Where it is entered	Who consumes it
`tariff_customized`	no	--	Project	--
`water_tariffs`	--	--	Project, Sweep, Tariffs, Tariffs and costs	Economic
`tariff_city_uf`	SC	--	Main window, Project, Reports, Tariffs, Tariffs and costs	--
`tariff_city_name`	Florianópolis	--	Main window, Project, Reports, Tariffs, Tariffs and costs	--
`tariff_category`	residencial	--	Economic, Tariffs, Tariffs and costs	Tariffs
`sewage_rate`	100	% of water	Project, Tariffs, Tariffs and costs	Automation, Economic, Tariffs
`sewage_mode`	1	code	Economic, Optimiser, Project, Tariffs, Tariffs and costs	Automation, Economic, Tariffs
`water_tax_fixed`	0	R$/month	Tariffs and costs	Automation, Economic
`water_tax_var`	0	%	Tariffs and costs	Automation, Economic
`tariff_escalation`	12	months	Economic, Optimiser, Sweep, Tariffs and costs	Automation, Economic
`water_tariff_escalation`	5	% per period	Economic, Optimiser, Sweep	Automation, Economic
`sewage_tariff_escalation`	5	% per period	Economic, Optimiser	Automation, Economic
`elec_tariff_escalation`	5	% per period	Economic, Optimiser, Sweep	Automation, Economic
`tariff_flag`	verde	--	Economic	Automation, Economic
`elec_tariff`	0.6957	R$/kWh	Charts, Economic, Optimiser, Project, Sweep, Tariffs and costs	Automation, Economic

Notes from the code:

- `tariff_category`: category of the tariff table applied; "personalizada" (custom) when typed in
- `sewage_mode`: 0 = fixed amount, 1 = percentage of the water part, 2 = amount per m³ consumed
- `water_tax_fixed`: currency
- `water_tax_var`: %
- `tariff_escalation`: months; length of the escalation period
- `water_tariff_escalation`: % per escalation period; with 12 months, % per year
- `sewage_tariff_escalation`: % per escalation period; with 12 months, % per year
- `elec_tariff_escalation`: % per escalation period; with 12 months, % per year
- `tariff_flag`: "verde" (green), "amarela" (yellow), "vermelha_1" (red 1), "vermelha_2" (red 2), "branca" (white); Brazil Edition
- `elec_tariff`: per kWh (Brazil Edition default: residential B1 tariff, CELESC, Florianópolis)

A.14. Energy

4 parameters.

Table 167 -- Project parameters: energy

Parameter	Default	Unit	Where it is entered	Who consumes it
`elec_city_uf`	SC	--	Main window, Project, Tariffs and costs	--
`elec_city_name`	Florianópolis	--	Main window, Project, Tariffs and costs	--
`elec_tax_fixed`	0	R$/month	Tariffs and costs	Automation, Economic
`elec_tax_var`	0	%	Tariffs and costs	Automation, Economic

Notes from the code:

- `elec_tax_fixed`: currency
- `elec_tax_var`: %

A.15. Incentives

3 parameters.

Table 168 -- Project parameters: incentives

Parameter	Default	Unit	Where it is entered	Who consumes it
`iptu_green_enabled`	no	--	Economic, Optimiser	Automation, Economic
`iptu_annual_value`	4000	R$/year	Economic, Optimiser	Automation, Economic
`iptu_discount_pct`	5	%	Economic, Optimiser	Automation, Economic

Notes from the code:

- `iptu_annual_value`: Annual municipal property tax (IPTU) of the building, per year; Brazil Edition
- `iptu_discount_pct`: Municipal green property-tax discount (IPTU Verde, %); Brazil Edition

A.16. Financial

4 parameters.

Table 169 -- Project parameters: financial

Parameter	Default	Unit	Where it is entered	Who consumes it
`inflation`	0.2	% per month	Documentation, Economic, Optimiser, Sweep, Tariffs and costs	Automation, Economic
`analysis_period_years`	20	years	Economic, Optimiser, Reports, Sweep, Tariffs and costs	Automation, Economic, LCA, Reports
`tma`	2	% per month	Economic, Optimiser, Sweep, Tariffs and costs	Automation, Economic, Spreadsheets
`install_month`	0	month	--	--

Notes from the code:

- `inflation`: % per month; inflation of O&M costs
- `analysis_period_years`: years
- `tma`: % per month; minimum attractive rate of return

A.17. Certifications

1 parameter.

Table 170 -- Project parameters: certifications

Parameter	Default	Unit	Where it is entered	Who consumes it
`edge_base_case_l_person_day`	0	L/person.day	Environmental	--

A.18. Other

3 parameters.

Table 171 -- Project parameters: other

Parameter	Default	Unit	Where it is entered	Who consumes it
`bim_layout`	--	--	Export, Main window	--
`capex_do_arranjo`	no	--	Main window, Sweep, Tariffs and costs	Treatment line
`mao_de_obra_automatica`	no	--	Main window, Sweep, Tariffs and costs	--

Appendix B -- The program's processes

There are **24 processes**, from the rainfall file to the report. Each one declares what triggers it, what it receives, the rules with their branches, what it produces and what happens when it fails.

It is the most technical appendix of the manual, and it serves two people: whoever reviews a study made with the program and needs to know exactly which calculation was done, and whoever found a strange result and wants to know where the number went through.

The names in `code` are the project file fields, listed in appendix A.

B.1. PROC-01 -- Load the precipitation series

Rainfall

Trigger. Project tab: choose a city in the national database or open a CSV file.

Inputs

- Daily CSV file, or a city of the CMIP6 database (652 cities, 19 models, historical plus SSP2-4.5 and SSP5-8.5)
- `climate_start_date` and `climate_end_date`, which trim the series
- `start_date`, the date of the first day of the series

Rules

1. The series becomes a `numpy` vector of millimetres per day; the vector index is the day, and the date only serves to know the weekday and the month.
2. An incomplete year is dropped on reading, so as not to bias annual totals.
3. The climate database ships embedded in the Brazil Edition, in `climbra.db`; in the International Edition, which does not include it, the series is always a user file.

Outputs

- `precip_data`: vector of mm/day, which is the input of **two** distinct processes -- PROC-04 (first flush and collection) and PROC-03 (irrigation suppression)

Exception. An unreadable file or one without a rainfall column stops loading with a message; the simulation does not run on an empty series.

B.2. PROC-02 -- Build the daily demand

Demand

Trigger. Start of the simulation, once per run.

Inputs

- `demand_mode`: constant, weekly, monthly or daily
- `demand_value` (L/day per person), `residents_count`
- `rainwater_share` or `non_potable_indoor_share`
- vectors `weekly_*`, `monthly_*`, `daily_*` when the mode uses them

Rules

1. Base demand of the day = `demand_value x residents_count`, with the multiplier of the weekday, month or day, depending on the mode.
2. The non-potable fraction splits the demand in two: indoor potable and indoor non-potable.

3. When the mode is weekly, monthly or daily, the fraction comes from the period vector itself; otherwise `non_potable_indoor_share` applies, and only then `rainwater_share`.
4. A null or zero value at any position of the vector falls back to the constant value.

Outputs

- `tot_arr`: total demand of the day
- `pot_in_arr`: indoor potable, which rain never supplies
- `non_pot_in_arr`: indoor non-potable, rain priority 1
- `pluv_arr` = indoor non-potable + active irrigation

Exception. An unreadable start date falls back to 01/01/1980 instead of stopping, because it only affects the weekday.

B.3. PROC-03 -- Suppress irrigation on rainy days

Demand

Trigger. Demand composition, one test per day of the series.

Inputs

- `irrigation_liters_per_day`
- `irrigation_rain_threshold_mm` (default 2 mm/day)
- `precip_data`, the **same** vector that feeds the collection

Rules

1. If the day's rain >= threshold, the day's irrigation is **zero** and the day is counted as suppressed.
2. Otherwise, the day's irrigation is the nominal volume.
3. It is the second reading of the rainfall file, independent of the first: a day can suppress irrigation and still yield collection.

Outputs

- `irr_arr`: irrigation of the day
- `irr_suppressed_mask`: suppressed days, which become `total_irrigation_days_suppressed` in the result

Exception. Without an irrigation volume, the vector is zero and the mask is false on every day.

B.4. PROC-04 -- Divert the first flush

Water balance

Trigger. Every day with rain, inside the daily loop.

Inputs

- `precip_mm` of the day
- `discard_mm`
- `discard_mode` (1, 2 or 3)
- `discard_days_interval`

Rules

1. **Mode 1**: diverts on every rain event.
2. **Mode 2**: diverts only on the first event after a dry spell, defined as a previous day with less than 1 mm.
3. **Mode 3**: diverts when `discard_days_interval` days have passed since the last diversion.

4. When the diversion applies: `p_eff = p - discard_mm` if `p > discard_mm`, otherwise **zero**: all the day's rain goes to the diverter.
5. When it does not apply: `p_eff = p`, the whole rain is usable.

Outputs

- `p_eff`: effective depth of the day, in mm, which goes **only** to PROC-05
- `eff_precip_mm` in the detailed daily series, when it is kept

Exception. A negative first-flush depth is rejected at the validation gate, before the loop.

B.5. PROC-05 -- Convert depth into collected volume

Water balance

Trigger. Every day, after the first flush.

Inputs

- `p_eff` from PROC-04
- `area` (m²)
- `runoff_coeff`

Rules

1. `v_ac = p_eff x area x runoff_coeff`, in litres.
2. This is where the rainfall file becomes volume, and the only place where it happens.
3. The coefficient is dimensionless and lies in [0, 1]; outside it the simulation does not start.

Outputs

- `v_ac`: volume collected in the day, which goes to PROC-06
- `harvested_liters` in the daily series

Exception. A negative area or a coefficient outside [0, 1] raises `ValueError` naming the field, in PROC-11.

B.6. PROC-06 -- Store and overflow

Water balance

Trigger. Every day, after collection.

Inputs

- `v_ac`
- volume left from the previous day
- `lower_tank_vol`

Rules

1. **Overflow before consumption**, as in the YAS rule; the day's rain, up to the capacity, already supplies the day's demand, which places the model between strict YAS and YBS.
2. `available = v_lower + v_ac`. If it exceeds the capacity, the surplus overflows and the tank is full.
3. The order matters: withdrawing the demand first would let the tank accept water that would physically have overflowed already, overestimating the yield.

Outputs

- `v_spill_today`, summed into `avg_spilled_l_day`, which becomes the overflow criterion in the multi-criteria decision

- `v_in_tank`, the volume available for supply

Exception. Zero volume is valid and represents the no-system scenario; a negative one is rejected.

B.7. PROC-07 -- Supply from a single tank

Water balance

Trigger. Every day, when there is no upper tank.

Inputs

- `v_in_tank`
- `pluvial_demand_per_day` from PROC-02

Rules

1. `v_c = min(rainwater demand, volume in the tank)`.
2. The remaining volume stays for the next day: `v_lower = v_in_tank - v_c`.

Outputs

- `v_c`: rainwater volume consumed in the day

Exception. None: the minimum already handles an empty tank and a null demand.

B.8. PROC-08 -- Supply with pumping to an upper tank

Water balance

Trigger. Every day, when `upper_tank_used` and the upper volume is positive.

Inputs

- `v_lower`, `v_upper`
- `upper_tank_vol`
- `pumping_threshold`, the reserve volume that is not consumed

Rules

1. Loop until the demand is met or the water runs out.
2. First consumes from the upper tank what lies **above** the pump trigger level.
3. If something is still missing and there is water below, starts the pump: transfers `min(v_lower, free space above)`, counts **one cycle** and records the batch.
4. After pumping, consumes again.
5. The trigger level is capped at the upper tank volume at the start of the process.

Outputs

- `v_c`
- `v_rec_today`, summed into `total_pumped_liters`
- `cycles_today`, summed into `total_pump_cycles`
- `max_pump_batch_liters`

Exception. A negative upper volume is rejected at validation. The loop has a guaranteed exit: if there is no water below or above, it stops.

B.9. PROC-09 -- Split the supply and top up with potable water

Water balance

Trigger. Every day, after supply.

Inputs

- `v_c`
- `non_pot_in_day`
- `irr_day`
- `pot_in_day`
- `irrigation_potable_backup`

Rules

1. **Priority 1**, indoor non-potable use: `c_pluv_in = min(v_c, indoor non-potable)`.
2. **Priority 2**, irrigation: `c_pluv_irr = min(v_c - c_pluv_in, irrigation of the day)`.
3. The potable water consumed is the sum of three parts: the indoor potable, the deficit of the indoor non-potable and -- only if the backup supply is on -- the irrigation deficit.
4. With the backup supply off, the garden is simply not watered on the dry day, and the mains is not used.

Outputs

- `potable_consumed_liters` of the day
- `avg_pluvial_indoor_consumed_l_day` and `avg_pluvial_irrigation_consumed_l_day`, which separate indoor from outdoor use for credits WEc1 and WEc2 of LEED v4 and Options 2 and 4 of WEc2 of LEED v5

Exception. None: all parts are minima and maxima with zero.

B.10. PROC-10 -- Classify the day's supply

Water balance

Trigger. Every day, at the end of supply.

Inputs

- `v_c`
- `pluvial_demand_per_day`

Rules

1. **Full** when the consumption equals the rainwater demand, with a tolerance of 10-■ L.
2. **Partial** when something was consumed, but less than the demand.
3. **Null** when nothing was consumed.

Outputs

- `pct_days_full_satisfaction`, `pct_days_partial_satisfaction` and `pct_days_null_satisfaction`, which are the system's time reliability

Exception. None.

B.11. PROC-11 -- Validate the physical input

Water balance

Trigger. First line of `run_single_simulation`, before any calculation.

Inputs

- lower and upper volume
- area

- runoff coefficient
- first-flush depth
- daily demand
- rainwater fraction

Rules

1. Negative volume, negative area, negative first flush and negative demand are refused.
2. Runoff coefficient and rainwater fraction must lie in [0, 1].
3. **Zero volume remains valid**: it is the «no system» reference of the sweep.
4. The message names the field and the value received, instead of silently correcting it.

Outputs

- No output: it either passes or raises `ValueError`

Exception. It is the handling itself. Before this gate, a volume of -1,000 L returned -166.67% saving, and the number went on to the economic analysis, the LCA and the report.

B.12. PROC-12 -- Aggregate the period and compute the indicators

Water balance

Trigger. End of the daily loop.

Inputs

- loop accumulators
- `n_days`

Rules

1. **Saving potential** = rainwater consumed in the period ÷ total demand in the period x 100.
2. The daily means divide the totals by the number of simulated days.
3. Conversions to a year use **365.25** days, not 365, to absorb the leap years of the series.
4. The potential always lies in [0, 100] -- an invariant the validation gate exists to guarantee.

Outputs

- `SingleTankSimulationResult`, with 21 fields, which is the object **all** downstream modules consume

Exception. A null total demand returns zero potential instead of dividing by zero.

B.13. PROC-13 -- Sweep volumes and find the ideal volume

Water balance

Trigger. Sweep tab: simulate button.

Inputs

- `lower_tank_max`
- `lower_tank_step`
- `delta_ideal` (%/m³)

Rules

1. The discrete grid rounds the step **up** to a multiple of 250 L, the commercial quantum of tanks, and limits it to 40 positive volumes.

2. The maximum volume entered is always kept as the last point, so that only the last interval can be smaller than the step.
3. Zero enters the grid as the reference alternative.
4. Runs PROC-04 to PROC-12 once per volume.
5. The **ideal volume** is the first whose marginal gain `deltapotential/deltam³` falls below the threshold: from there on, more storage does not pay.

Outputs

- `SweepSimulationResult`: one result per volume, the ideal volume, and the record that the step was adjusted
- The list of volumes is the list of multi-criteria **candidates**

Exception. A step larger than the maximum returns a one-point grid. The step adjustment is recorded in `step_adjusted`, not silenced.

B.14. PROC-14 -- Compute the rainfall indices

Rainfall

Trigger. Charts tab and research report.

Inputs

- `precip_data`
- dates of the series

Rules

1. Nine indices: PRCPTOT, R95p, RX1day, RX5day, SDII, R20mm, CDD, CWD and the Walsh and Lawler seasonality index.
2. The set is **not** pure ETCCDI: seasonality is not in the set, and R10mm, which is, is missing.
3. The 95th percentile is computed over the **whole loaded series**, not over the fixed 1961-1990 base of ETCCDI.
4. RX5day uses `min_periods=1`, so the first windows of the year have 1 to 4 days.

Outputs

- Nine annual series, for chart and report

Exception. A series shorter than one year returns the indices that can be computed, without stopping.

B.15. PROC-15 -- Size the treatment plant

Quality and treatment

Trigger. A change in the Water Quality tab, or the start of the economic analysis and the LCA.

Inputs

- `water_quality_target_class` (4 use classes)
- `water_quality_disinfection`: chlorine, ultraviolet or ozone
- `water_quality_polishing`, polishing filter
- `annual_pluvial_m3` from PROC-12
- `area`
- chlorine dose, purity, price and CO2 factor
- `discard` and `first_flush_system_enabled`, for the first-flush diverter

Rules

1. The target class defines the mandatory stages; the chosen disinfection defines the equipment and the running cost.
2. The polishing filter is applied **if and only if** ticked; when the class would recommend it and the user declines, this is recorded in `polishing_recommended_but_declined` instead of being imposed.
3. What is bought is the **product** -- tablets with 65% active chlorine -- not pure chlorine. Charging the mass of active chlorine, rather than that of the product, underestimates the cost by about 35%.
4. **Incremental rule**: the CAPEX of the line deducts the generic treatment allowance already included in «Accessories», and the running cost deducts chlorination, which the cash flow already pays. The rule lives in one place, consumed by **both** economic engines.
5. With `first_flush_system_enabled`, the first-flush diverter, of volume `discard x area`, enters the CAPEX and the running cost of the line.
6. In the LCA, the treatment line **replaces** the chlorination term; it never adds to it.
7. With `etap_enabled = False`, the treatment line enters no calculation: not the CAPEX, not the running cost, not the environmental assessment.

Outputs

- `total_etap_capex`, which enters the CAPEX
- `total_etap_annual_opex`, which enters the running cost
- `total_etap_co2e_kg_yr`, which replaces chlorination in the LCA
- `disinfection_stages`, shown in the report

Exception. A purity outside (0, 1] is refused by the validator of the chlorination chain.

B.16. PROC-16 -- Compute the pumping energy

Pump set

Trigger. Economic analysis and LCA, from the same simulation result.

Inputs

- `supply_arrangement`: pumping to an upper tank, direct pressurisation or gravity
- `total_pumped_liters` and `total_pump_cycles` from PROC-08
- `pressurizer_pressure_m`, `pressurizer_starts_per_person_day`, `residents_count`
- `pump_power_kw`, `pump_efficiency`, `pump_flow_rate`
- `pump_manometric_head_m`
- `pump_startup_time_h`

Rules

1. The arrangement decides what is pumped. **Pumping to an upper tank**: the volume sent to the upper tank. **Direct pressurisation**: all the rainwater consumed, with the working pressure in place of the head and no start-up energy. **Gravity**: nothing. It is the only source of kWh of the economic analysis and the LCA.
2. Specific energy `e = rhogH/(eta_pump.eta_motor)` per m³ pumped, or `P/(Q.eta_motor)` when the flow rate is declared.
3. Nameplate power is shaft power and **already includes** the pump efficiency; that is why `P x hours / eta_pump` counts that efficiency twice, and what is missing is to divide by the motor efficiency, from ABNT NBR 17094-2. Four copies of that old formula survived in the program and overestimated consumption by 1.68 times -- the worst of them ranked volumes in the multi-criteria decision.
4. Start-up energy is added separately, by the number of cycles.
5. Flow rate and efficiency not entered fall back to typical values **for the chosen power**, not to fixed constants.

Outputs

- `annual_kwh`, which becomes cost in the economic analysis and emission in the LCA
- `pump_hours_year`, `annual_cycles`

Exception. Gravity arrangement, an arrangement not stated in an old project, or zero power: the energy is zero.

B.17. PROC-17 -- Compute the water and sewage bill

Economic

Trigger. Twice per projected month: with and without harvesting.

Inputs

- monthly volume in litres
- `water_tariffs`: quadruples (minimum, maximum, value, code)
- `sewage_mode` and `sewage_rate`
- fixed and variable taxes
- `num_economias`

Rules

1. Consumption is evaluated **at the level of one dwelling unit** and multiplied by the number of units, so as not to distort the progressive blocks in a building with individual metering.
2. Code **1** or **0**: progressive block, charged only on the volume within it.
3. Code **-1**, availability charge: a fixed value that **adds** on top of the metered consumption.
4. Code **-2**, retroactive allowance: up to the allowance volume the fixed value is paid; beyond it, **all** the volume from zero is charged by the blocks.
5. Code **-3**, payment floor: it does not add, it **bounds from below**. With a floor of R$ 40 and a charged consumption of R$ 55, saving up to R$ 15 reduces nothing -- and that is exactly what a rainwater harvesting study needs to see.
6. Sewage is a percentage of the **water part**, not of the total bill. In Florianópolis it is 100% of the water, which is half the bill.
7. The six tariff regimes are named and audited in the database of 652 municipalities.

Outputs

- Value of the water and sewage bill per month, for the projection of PROC-18

Exception. An empty table falls back to the CASAN structure. An incomplete structure is flagged by the validation gate, which separates errors from warnings.

B.18. PROC-18 -- Project the cash flow

Economic

Trigger. Economic tab, after the technical simulation.

Inputs

- CAPEX of the five items plus pump and treatment line
- monthly water and sewage saving from PROC-17
- running costs: pumping, chlorination, treatment line and declared operating costs
- `tma`, `inflation`, `analysis_period_years`
- separate escalations for water, sewage and energy, and the interval between them

- green property-tax discount (IPTU Verde), when enabled

Rules

1. Water and sewage are escalated by **separate** indices, not by the aggregate bill.
2. Inflation applies only to running costs; tariff escalations apply only to the bill.
3. The escalation is applied every `tariff_escalation` months, in steps.
4. The tariff flag adjusts the electricity tariff: yellow, red 1, red 2, or the off-peak discount in the white tariff (Brazil Edition).
5. The green property-tax discount enters as an annual saving divided by twelve.
6. The payback is the first month in which the discounted cumulative cash flow crosses zero.

Outputs

- `vpl`, `tir_monthly_pct`, `tir_annual_pct`, `payback_months`
- `monthly_cash_flows`, month by month
- `annual_pump_kwh`, `annual_chlorine_kg`, `etap_capex_brl`

Exception. A negative monthly saving is truncated at zero before entering the cash flow. An IRR without a real root returns null instead of an invented number.

B.19. PROC-19 -- Assess the life cycle

LCA

Trigger. Environmental tab, after the technical simulation.

Inputs

- `tank_vol_liters`
- `annual_pluvial_m3` from PROC-12
- materials and masses of the two tanks
- lengths per pipe diameter
- `pump_count`
- grid electricity factor
- production route of the potable water
- treatment line result from PROC-15

Rules

1. Seven stages: lower cistern, upper tank, piping, pump set, pumping operation, treatment operation and avoided potable water.
2. Zero volume **without** rainwater volume is the no-system scenario, and receives no fixed impact for tank, pipe or pump.
3. The treatment line **replaces** the chlorination term; it does not add to it. Chlorination follows the treatment, not the upper tank.
4. Upper tank only in the pumping arrangement; pump sets in every arrangement with a pump; energy from PROC-16.
5. The avoided water is the water **delivered**, not produced: distribution losses in Brazil range from 33.9% in the Southeast to 46.9% in the North, so producing 1 m³ delivers 0.53 to 0.66 m³. The published factor of 0.236 kg CO2e/m³ is at the plant outlet; per m³ delivered it becomes 0.336 to 0.461 depending on the region. In the International Edition the user enters the factor for the delivered water.

6. The carbon payback is the embodied carbon divided by the net annual saving; without a positive net saving, it returns 999 years instead of infinity.

Outputs

- `total_embodied_co2_kg`, `net_annual_avoided_co2_kg`
- `lifetime_net_co2_saved_kg`, which is a multi-criteria criterion
- `environmental_payback_years`
- `inventory_items`, the item-by-item inventory

Exception. A factor without a primary source receives a screening label, and the label is shown -- the gap is declared, not hidden.

B.20. PROC-20 -- Map onto certifications

ESG and certifications

Trigger. Environmental tab and reports.

Inputs

- `potential_potable_savings_pct`
- annual rainwater and potable volumes
- indoor/outdoor split from PROC-09
- CO2 avoided over the life cycle
- `area`, for the catchment area of WEM14
- EDGE base case, when declared

Rules

1. **LEED v4**: WEc1 up to 2 points for outdoor use, WEc2 up to 6 points for indoor use, on the official scale. Without an outdoor demand, WEc1 is zero and says why.
2. **LEED v5 BD+C** (rating system of November 2025): WEc2 up to 8 points, by Option 1 (whole project, 30% to 65%, +1 with an alternative source) or by the sum of Options 2 (fixtures and fittings, 30% to 40%), 4 (outdoor, 50% and 100%) and 6 (on-site reuse, 2 points), new construction only. SSc3 recorded as not assessed; its reuse point does not exceed the cap of 3.
3. **AQUA-HQE Residential** (reference of 22/11/2024, Brazil Edition): items EAU.1.2 (HQE, 20% saving), EAU.1.3 (HQE 3, 40%) and EAU.1.5 (HQE 3, existence of a rainwater system).
4. **Casa Azul + CAIXA** (guide V023, 29/04/2026, Brazil Edition): criterion 3.6 met by the existence of the system; 2, 4 or 5 points depending on the scope of use, only when stated.
5. **EDGE**: **does not score points**. It measures a percentage against a local base case and compares it with 20%, in three independent bars. The advanced level raises the *energy* bar to 40%, not the water one. The program assesses only the water bar, by measure WEM14, and says that the other two are outside its reach.
6. Without a declared base case, EDGE uses the modelled demand and warns that the number is a **floor**.
7. **GRI 303**: avoided withdrawal and circularity rate.

Outputs

- `credit_details`: the credit matrix that feeds the on-screen table and the reports
- `esg_summary_text`
- cards for LEED, AQUA, Casa Azul, EDGE and GRI

Exception. A null total demand returns zero per cent without dividing by zero.

B.21. PROC-21 -- Rank volumes by multiple criteria

Multi-criteria

Trigger. Optimiser tab, with the sweep done.

Inputs

- one candidate per sweep volume, each with potential, overflow, NPV, discounted payback, IRR, life-cycle CO2 and potable consumption
- weights of the seven criteria

Rules

1. Seven criteria: **maximise** potential, NPV and IRR; **minimise** overflow, payback, CO2 and potable consumption.
2. Normalisation shifts by `x - min(x)` **before** the vector normalisation of Hwang and Yoon, to admit a negative NPV. It is a deliberate variant, and must be declared as a variant.
3. SMAA-VIKOR draws weights and returns the probabilities of each volume taking each rank.
4. Opricovic's conditions C1 and C2 are not implemented.

Outputs

- Ranking of the volumes, distance to the ideal per criterion, compromise volume

Exception. A single candidate returns a trivial ranking. A criterion constant across candidates does not contribute to the distance.

B.22. PROC-22 -- Save and reopen the project

Project file

Trigger. Save and open menu.

Inputs

- the whole `NetunoConfig`, with the fields of appendix A
- interface state
- current analysis

Rules

1. JSON format, in named sections.
2. Reading uses a default value field by field, so an old file keeps opening.
3. `created_at` and `precip_file` describe the act of saving, not the project: that is why they do not come back.
4. It is the fifth crossing of the program -- the one that crosses **time** rather than modules. Thirty of the 87 fields of the time did not survive the round trip, including the whole treatment-line choice, the manometric head, the tank materials, irrigation and the property tax.

Outputs

- `.netuno` file
- rebuilt `NetunoConfig`

Exception. A section missing from the file falls back to the class defaults. Invalid JSON stops opening with a message.

B.23. PROC-23 -- Issue reports and exports

Reports

Trigger. Reports tab; Export tab.

Inputs

- results of PROC-12, PROC-18, PROC-19, PROC-20 and PROC-21
- project data

Rules

1. Three reports: executive, environmental with ESG, and research.
2. Export to CAD and BIM from the sizing.
3. Spreadsheets with the daily series and the sweep.

Outputs

- HTML, PDF, DXF, JSON for BIM and XLSX

Exception. Conversion to PDF tries the Qt engine and falls back to the alternative generator when it is not available.

B.24. PROC-24 -- Automation for large-scale studies

Automation

Trigger. The `netuno` command line, described in section 3.6; it comes with the installation as a Python package, not with the Windows installer.

Inputs

- project file
- rainfall file

Rules

1. Calls the **same** calculation core as the window, without a screen.
2. Only part of the parameters can be reached through it; the others exist only in the interface, and the way is to build the project in the window and vary by automation only what changes between cases.
3. It serves factorial matrices of thousands of cases and uncertainty propagation by sampling, which are the instruments of the studies published by the group.

Outputs

- JSON and CSV with the results

Exception. A parameter out of range is refused before simulating, with the same validation gate as the interface.

Appendix C -- Emission factors

Every factor used by the life cycle assessment, with its provenance, boundary and degree of confidence. The full detail, with the sources item by item, is in `docs/procedencia_fatores_co2.md` (in Portuguese).

The material, treatment and pump factors below are the defaults of both editions. The water and grid factors are Brazilian, and the International Edition does not ship them: there, the user enters the emission factor of the local water supply and of the local grid.

C.1. Tank materials

Table 172 -- Emission factors of the tank materials, with mass and boundary

Material	Mass per useful m³	Factor	Boundary
high-density polyethylene	22 kg/m³	1.612 kg CO2e/kg	cradle to gate
reinforced concrete	1,200 kg/m³	0.140 kg CO2e/kg	cradle to gate
fibreglass	15 kg/m³	3.250 kg CO2e/kg	cradle to gate
stainless steel AISI 304	38 kg/m³	3.700 kg CO2e/kg	cradle to gate

The mass per useful cubic metre is what converts tank volume into mass of material. It is that, and not the factor, that makes concrete dominate the inventory: 1,200 kg per cubic metre against 22 for polyethylene.

> **Practical consequence, shown in section 4.8:** the material decides the **sign** of the environmental result. With concrete and a twenty-year service life, the tutorial system does not pay off its carbon debt.

C.2. Mains potable water

Nineteen production routes available, by macro-region and type of source.

The boundary is per cubic metre delivered to the building, not per cubic metre produced at the works. The conversion uses the distribution losses of each macro-region (Figure 69):

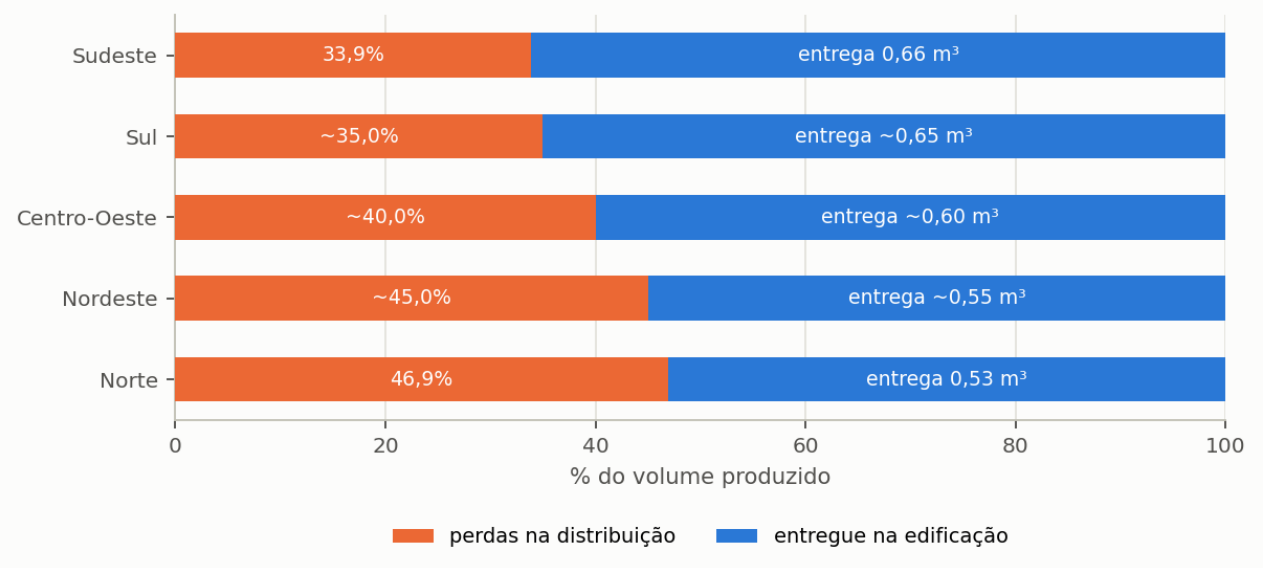


Figure 69 -- How much of each cubic metre produced actually reaches the building, by Brazilian macro-region. It is this fraction that converts an emission factor at the outlet of the works into a factor per cubic metre delivered, and ignoring it underestimates the avoided emission by up to a third.

C.3. Electricity grid

The emission factor of the Brazilian grid changes **every year**, and a study without the year declared cannot be reproduced. The program carries five reference databases and records which one was used.

C.4. Treatment consumables

Table 173 -- Emission factors of the treatment consumables

Consumable	Factor
calcium hypochlorite tablet at 65%	1.72 kg CO2e/kg of product
ultraviolet lamp	grid energy
ozone generator	grid energy

The mass is of **product**, not of active chlorine. Tablets have about 65% active chlorine.

C.5. Pump set

There is no Environmental Product Declaration for a Brazilian residential pump set. The program builds the value from two sources it has read: the mass of each power rating, from the Dancor catalogue, and the intensity of 4.77 kg CO2e per kilogram of pump, from the Grundfos EPD EPD-GRU-20250340-CBA1-EN (2025, modules A1 to A3).

Table 174 -- Manufacturing emission of the pump set by power rating, with the catalogue mass

Power	Mass	Manufacturing emission
1/4 hp	6.1 kg	29.1 kg CO2e
1/2 hp	6.4 kg	30.5 kg CO2e
3/4 hp	13.5 kg	64.4 kg CO2e
1 hp	15.3 kg	73.0 kg CO2e
1 1/2 hp	17.5 kg	83.5 kg CO2e
2 hp	20.7 kg	98.8 kg CO2e

> **Declare it in the study.** The EPD is for a European circulator with on-board electronics, and the reference mass includes the packaging. The intensity per kilogram is an order-of-magnitude approximation, not an inventory of the installed pump; where there is an EPD from the manufacturer used, it prevails. The previous values, from 12 to 55 kg CO2e, were 1.7 to 2.7 times below these, and four of them were arithmetic extrapolations.

C.6. Factors with a screening label

Not every parameter has a primary source, and those without one are marked in the report: today, the masses per useful cubic metre of polyethylene, concrete and stainless steel, which are geometric scenarios, and any factor typed in without a source. In the September 2026 revision, the last default emission factor without provenance, that of hypochlorite, received it:

Table 175 -- Provenance of the hypochlorite emission factor

Step	Value
database and model	Ecoinvent 3.8, *cutoff*
process	*sodium hypochlorite production, product in 15% solution state*, geography RoW (rest of the world)
method	EF 3.1, *Climate change* category
result	2.522 kg CO2e per kg of pure NaOCl
conversion	0.9525 kg of active chlorine per kg of NaOCl, hence 2.65 kg CO2e per kg of active chlorine
factor adopted	2.65 x 0.65 = **1.72 kg CO2e per kg of tablet**

> **Declare the approximation.** Ecoinvent 3.8 has no calcium hypochlorite, and the factor is that of sodium hypochlorite brought to an active-chlorine basis. Factors entered by the user without a source still receive the screening label, and an inventory that presents all factors with the same degree of confidence hides where the weakness lies.

C.7. What to consult before publishing

1. `docs/procedencia_fatores_co2.md`, for the source item by item;
2. the Documentation tab, life cycle assessment chapter, for the justifications shown to the user;
3. the inventory of the environmental report, which lists each item with the factor applied.

Appendix D -- Tariff regimes

> **Brazil Edition.** The tariff catalogue is Brazilian and ships only with the Brazil Edition. In the International Edition the tariff is entered by the user, in one of the same six regimes, and this appendix serves as a description of them.

The database holds **652 municipalities** with tariff structure, sewage percentage, minimum-charge regime, effective date and source. This list is extracted from the program's own database, not transcribed.

D.1. The block codes

Each row of the tariff table is a quadruple `(minimum, maximum, value, code)`.

The six codes and the effect of each on the bill are drawn in section 2.7, and are not repeated here.

D.2. The six regimes

Table 176 -- Distribution of the catalogue municipalities across the six tariff regimes

Regime	Municipalities	What happens when saving
No minimum	165	Each m³ saved reduces the bill.
Availability charge	122	A fixed value that **adds** on top of the metered consumption. Each m³ saved is deducted.
Allowance in the minimum	329	The fixed value already includes a volume. Saving **below** the allowance deducts nothing.
Retroactive allowance	2	Up to the allowance the fixed value is paid; beyond it, **everything from zero** is charged.
Payment floor	0	The bill does **not** go below the floor. It does not add: it bounds.
Exponential formula	34	Basic service plus `PB x C■`, with `n` set by the month's consumption (CORSAN). Each m³ saved is deducted.

> The payment floor is implemented, and no municipality in the database had been classified under it as of this version.

D.3. Distribution by state

The catalogue coverage follows the population, not the territory: the state capitals and the metropolitan regions are almost all registered, and the interior of the North and Centre-West regions concentrates the municipalities still missing (Figure 70).

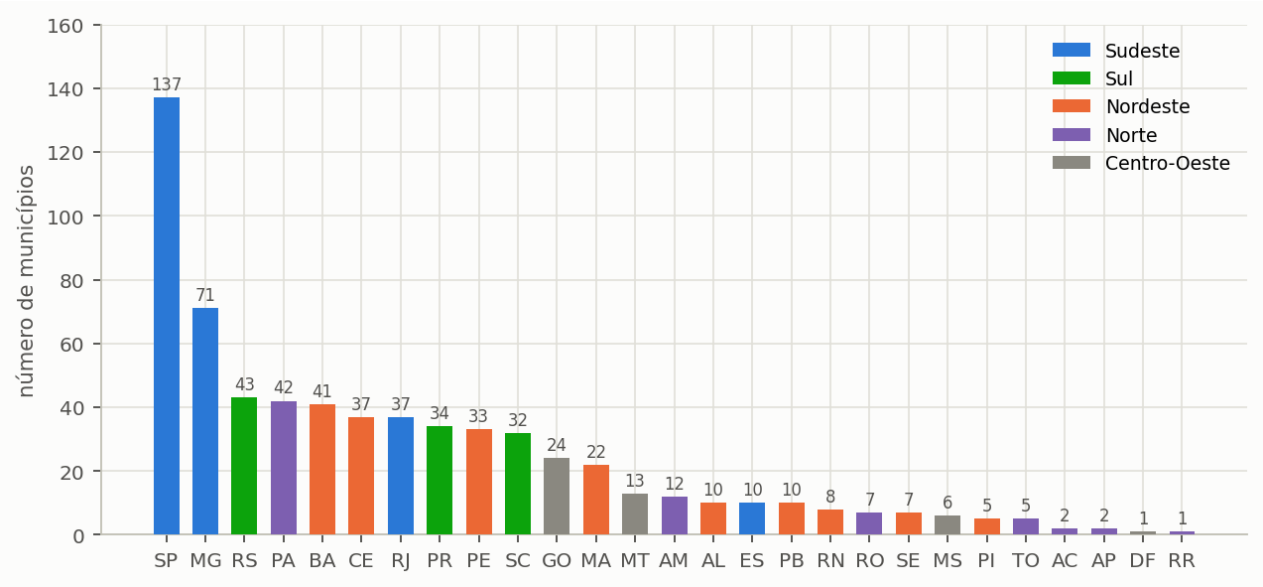


Figure 70 -- The distribution of the 652 catalogued municipalities across the Brazilian states, with colours identifying the macro-region. São Paulo and Minas Gerais alone account for almost a third of the catalogue.

D.4. Updating and auditing the database

The tariff database is maintained by the group with an update routine run outside the interface, before each version. The routine has three components.

Electricity. The residential tariffs of subgroup B1 are read from the open database of approved distributor tariffs published by the Brazilian Electricity Regulatory Agency (ANEEL), and associated with the distributor of each catalogue municipality.

Water and sewage. A register of primary evidence associates each provider with the regulatory agency, the official tariff page and the type of act that publishes them. Providers with a standard portal are read automatically; the others are flagged for manual checking of the decree or resolution, and the resulting structure goes through the same regime validation described in this appendix.

Change detection. Each monitored official page is recorded with the SHA-256 signature of its content and the `ETag` and `Last-Modified` headers. A new check compares the signatures and points out the pages changed since the last collection, which are the candidates for a tariff revision.

The routine can simulate the update without writing the database, run only the change detection, restrict itself to one state or one municipality, and separate electricity from sanitation. Each run writes an audit report in Markdown and an HTML panel with the status of each municipality.

D.5. Confidence grade

Each record carries the confidence grade of the collection:

Table 177 -- Confidence grade of the tariff records in the catalogue

Grade	Records
A -- high	580
B -- good	63
C -- structure to be checked	5
D -- incomplete structure	4

> **Always check the effective date** of your municipality's record against the utility's resolution. Tariffs are adjusted every year, and section 3.3 teaches the test that identifies a wrong structure.

D.6. The blocks by municipality and category

The table below gives the price of water, in R$/m³, in each monthly consumption interval, for each structure of the catalogue. The columns are the finest split the catalogue needs: the union of the block limits of all providers, so that within each column no provider changes the price. Municipalities with the same table form one row, identified by the number in the first column, which repeats in the following parts.

The residential category covers the 652 municipalities; the others appear where the provider's official table has been collected. The table is generated from the database for each version and grows with it. The fixed part adds on top of consumption; a zero price in the first blocks is the volume already included in the fixed value (allowance in the minimum).

The colour of the row says how far the data can be trusted: **green** (🟢), audited against the primary source, the provider's resolution or official table; **yellow** (🟡), collected automatically or flagged for review, and so to be confirmed against the current resolution; **red** (🔴), incomplete structure, not to be used without correction.

There are 652 municipalities and five categories, 3260 combinations. The first table says, for each category, in how many municipalities the tariff table is audited, to be confirmed, incomplete or not yet collected; each municipality appears in full in the following tables, and the categories not yet collected are listed by provider at the end of the section.

Table 178 -- Coverage of the tariff catalogue: municipalities by category and status of the data

Category	🟢 audited	🟡 to be confirmed	🔴 incomplete	🔴 not collected	Municipalities
Residential	363	285	4	0	652
Commercial	34	0	0	618	652
Public	34	0	0	618	652
Industrial	30	0	0	622	652
Social	0	0	0	652	652

Table 179 -- The tariff structures of the catalogue, numbered for the price-by-block tables that follow

No.	Provider: municipalities	Category	Fixed (R$/month)	Sewage
1	BRK Alagoas: Maceió (AL)	Residential	--	100%
2	BRK Alagoas: Marechal Deodoro (AL)	Residential	--	80%
3	BRK Alagoas: Rio Largo (AL)	Residential	--	80%
4	BRK Ambiental Cachoeiro: Cachoeiro de Itapemirim (ES)	Residential	45.00	80%
5	BRK Ambiental Limeira: Limeira (SP)	Residential	37.10	100%
6	BRK Ambiental Porto Ferreira: Porto Ferreira (SP)	Residential	42.00	90%
7	BRK Ambiental Sumaré: Sumaré (SP)	Residential	36.35	80%
8	BRK Ambiental Uruguaiana: Uruguaiana (RS)	Residential	--	70%
9	BRK Tocantins: Araguaína (TO), Gurupi (TO), Palmas (TO), Paraíso do Tocantins (TO), Porto Nacional (TO)	Residential	--	80%
10	CAEMA: Açailândia (MA), Bacabal (MA), Balsas (MA), Barra do Corda (MA), Barreirinhas (MA), Buriticupu (MA), Caxias (MA), Chapadinha (MA), Codó (MA), Coroatá (MA), Grajaú (MA), Imperatriz (MA), Itapecuru-Mirim (MA), Paço do Lumiar (MA), Pinheiro (MA), Santa Inês (MA), Santa Luzia (MA), São José de Ribamar (MA), São Luís (MA), Timon (MA), Tutóia (MA), Viana (MA)	Residential	33.58	100%
11	CAER: Boa Vista (RR)	Residential	39.92	80%
12	CAERD: Ariquemes (RO), Jaru (RO), Ji-Paraná (RO), Porto Velho (RO), Rolim de Moura (RO)	Residential	57.51	70%
13	CAERN: Assu (RN), Caicó (RN), Ceará-Mirim (RN), Extremoz (RN), Macaíba (RN), Mossoró (RN), Parnamirim (RN)	Residential	56.96	70%
14	CAERN: Natal (RN)	Residential	51.40	70%
15	CAESB: Brasília (DF)	Residential	11.62	100%
16	CAESC - Companhia de Água, Esgoto e Saneamento de Coari: Coari (AM)	Residential	32.50	80%
17	CAGECE: Acaraú (CE), Aquiraz (CE), Aracati (CE), Barbalha (CE), Beberibe (CE), Boa Viagem (CE), Brejo Santo (CE), Camocim (CE), Canindé (CE), Cascavel (CE), Caucaia (CE), Crateús (CE), Crato (CE), Eusébio (CE), Fortaleza (CE), Granja (CE), Horizonte (CE), Icó (CE), Iguatu (CE), Itaitinga (CE), Itapipoca (CE), Juazeiro do Norte (CE), Limoeiro do Norte (CE), Maracanaú (CE), Maranguape (CE), Morada Nova (CE), Pacajus (CE), Pacatuba (CE), Quixadá (CE), Quixeramobim (CE), Russas (CE), Sobral (CE), São Gonçalo do Amarante (CE), Tauá (CE), Tianguá (CE), Trairi (CE), Viçosa do Ceará (CE)	Residential	--	80%
18	CAGEPA: Bayeux (PB), Cabedelo (PB), Cajazeiras (PB), Campina Grande (PB), Guarabira (PB), João Pessoa (PB), Patos (PB), Santa Rita (PB), Sapé (PB), Sousa (PB)	Residential	54.82	80%
19	CASAL: Arapiraca (AL)	Residential	--	80%

No.	Provider: municipalities	Category	Fixed (R$/month)	Sewage
20	CASAN: Araranguá (SC), Biguaçu (SC), Camboriú (SC), Canoinhas (SC), Caçador (SC), Chapecó (SC), Concórdia (SC), Criciúma (SC), Florianópolis (SC), Imbituba (SC), Indaial (SC), Itapema (SC), Içara (SC), Mafra (SC), Navegantes (SC), Palhoça (SC), Rio do Sul (SC), São Francisco do Sul (SC), São José (SC), Tijucas (SC), Videira (SC), Xanxerê (SC)	Residential	46.34	100%
21	CEDAE/serviço local - validar área: Guapimirim (RJ), Itaperuna (RJ), Macaé (RJ), Valença (RJ)	Residential	--	100%
22	CESAMA Juiz de Fora: Juiz de Fora (MG)	Residential	15.01	80%
23	CESAN: Cariacica (ES), Guarapari (ES), Serra (ES), Vila Velha (ES), Vitória (ES)	Residential	57.39	80%
24	CODAU Uberaba: Uberaba (MG)	Residential	--	95%
25	COHASB / Prefeitura de Humaitá - prestação local: Humaitá (AM)	Residential	32.50	80%
26	Companhia Ituana de Saneamento (CIS): Itu (SP)	Residential	36.96	100%
27	Companhia Águas de Joinville: Joinville (SC)	Residential	34.93	80%
28	COMPESA: Abreu e Lima (PE), Araripina (PE), Arcoverde (PE), Belo Jardim (PE), Bezerros (PE), Buíque (PE), Cabo de Santo Agostinho (PE), Camaragibe (PE), Carpina (PE), Caruaru (PE), Escada (PE), Garanhuns (PE), Goiana (PE), Gravatá (PE), Igarassu (PE), Ipojuca (PE), Jaboatão dos Guararapes (PE), Limoeiro (PE), Moreno (PE), Olinda (PE), Ouricuri (PE), Palmares (PE), Paudalho (PE), Paulista (PE), Pesqueira (PE), Petrolina (PE), Recife (PE), Salgueiro (PE), Santa Cruz do Capibaribe (PE), Serra Talhada (PE), Surubim (PE), São Lourenço da Mata (PE), Vitória de Santo Antão (PE)	Residential	61.77	100%
29	COMUSA Novo Hamburgo: Novo Hamburgo (RS)	Residential	43.21	70%
30	COPASA: Alfenas (MG), Araxá (MG), Barbacena (MG), Belo Horizonte (MG), Betim (MG), Bom Despacho (MG), Campo Belo (MG), Caratinga (MG), Cataguases (MG), Congonhas (MG), Conselheiro Lafaiete (MG), Contagem (MG), Coronel Fabriciano (MG), Curvelo (MG), Divinópolis (MG), Esmeraldas (MG), Extrema (MG), Frutal (MG), Guaxupé (MG), Ibirité (MG), Ipatinga (MG), Itajubá (MG), Itaúna (MG), Janaúba (MG), Januária (MG), João Monlevade (MG), Lagoa Santa (MG), Lavras (MG), Leopoldina (MG), Montes Claros (MG), Nova Lima (MG), Nova Serrana (MG), Paracatu (MG), Patos de Minas (MG), Pedro Leopoldo (MG), Pouso Alegre (MG), Ribeirão das Neves (MG), Sabará (MG), São Francisco (MG), São Sebastião do Paraíso (MG), Teófilo Otoni (MG), Timóteo (MG), Três Corações (MG), Ubá (MG), Unaí (MG), Varginha (MG), Vespasiano (MG)	Residential	25.77	74%
31	COSAMA: Tabatinga (AM)	Residential	32.50	80%

No.	Provider: municipalities	Category	Fixed (R$/month)	Sewage
32	COSANPA: Abaetetuba (PA), Acará (PA), Alenquer (PA), Altamira (PA), Ananindeua (PA), Baião (PA), Barcarena (PA), Belém (PA), Benevides (PA), Bragança (PA), Breves (PA), Cametá (PA), Canaã dos Carajás (PA), Capanema (PA), Capitão Poço (PA), Castanhal (PA), Dom Eliseu (PA), Igarapé-Miri (PA), Itaituba (PA), Juruti (PA), Marabá (PA), Marituba (PA), Moju (PA), Monte Alegre (PA), Novo Repartimento (PA), Oriximiná (PA), Paragominas (PA), Parauapebas (PA), Portel (PA), Redenção (PA), Rondon do Pará (PA), Santa Izabel do Pará (PA), Santarém (PA), São Félix do Xingu (PA), São Miguel do Guamá (PA), Tailândia (PA), Tomé-Açu (PA), Tucuruí (PA), Vigia (PA), Viseu (PA), Xinguara (PA), Óbidos (PA)	Residential	--	80%
33	CSA Equatorial: Macapá (AP), Santana (AP)	Residential	42.91	100%
34	DAAE Araraquara: Araraquara (SP)	Residential	22.70	100%
35	DAAE Rio Claro: Rio Claro (SP)	Residential	35.36	100%
36	DAE Americana: Americana (SP)	Residential	up to 6 m³: 14.30 (retroactive)	100%
37	DAE Bauru: Bauru (SP)	Residential	up to 6 m³: 19.37 (retroactive)	65%
38	DAE Jundiaí: Jundiaí (SP)	Residential	40.84	69%
39	DAE Sant'Ana do Livramento: Santana do Livramento (RS)	Residential	--	80%
40	DAE Valinhos: Valinhos (SP)	Residential	23.73	100%
41	DAE/VG: Várzea Grande (MT)	Residential	31.20	80%
42	DAEB Bagé: Bagé (RS)	Residential	53.37	50%
43	DAEP Penápolis: Penápolis (SP)	Residential	7.31	70%
44	DAEPA: Patrocínio (MG)	Residential	--	80%
45	DAESC Coruripe: Coruripe (AL)	Residential	--	80%
46	DAMAE São João del-Rei: São João del-Rei (MG)	Residential	--	80%
47	DEMSUR: Muriaé (MG)	Residential	--	80%
48	DMAE Porto Alegre: Porto Alegre (RS)	Residential	--	100%
49	DMAE Poços de Caldas: Poços de Caldas (MG)	Residential	--	85%
50	DMAE Uberlândia: Uberlândia (MG)	Residential	--	80%
51	DMAES Ponte Nova: Ponte Nova (MG)	Residential	--	80%
52	EMASA Balneário Camboriú: Balneário Camboriú (SC)	Residential	44.95	100%

No.	Provider: municipalities	Category	Fixed (R$/month)	Sewage
53	EMBASA: Alagoinhas (BA), Barra (BA), Barreiras (BA), Bom Jesus da Lapa (BA), Brumado (BA), Caetité (BA), Camaçari (BA), Campo Formoso (BA), Candeias (BA), Casa Nova (BA), Conceição do Coité (BA), Cruz das Almas (BA), Dias d'Ávila (BA), Euclides da Cunha (BA), Eunápolis (BA), Feira de Santana (BA), Guanambi (BA), Ilhéus (BA), Ipirá (BA), Irecê (BA), Itaberaba (BA), Itabuna (BA), Itamaraju (BA), Itapetinga (BA), Jacobina (BA), Jequié (BA), Juazeiro (BA), Lauro de Freitas (BA), Luís Eduardo Magalhães (BA), Paulo Afonso (BA), Porto Seguro (BA), Ribeira do Pombal (BA), Salvador (BA), Santo Amaro (BA), Santo Antônio de Jesus (BA), Santo Estêvão (BA), Senhor do Bonfim (BA), Serrinha (BA), Simões Filho (BA), Teixeira de Freitas (BA), Vitória da Conquista (BA)	Residential	44.77	80%
54	GS Inima SAMAR: Araçatuba (SP)	Residential	28.48	80%
55	Iguá Sergipe: Aracaju (SE), Estância (SE), Itabaiana (SE), Lagarto (SE), Nossa Senhora do Socorro (SE), São Cristóvão (SE), Tobias Barreto (SE)	Residential	49.26	80%
56	Ourinhos Saneamento: Ourinhos (SP)	Residential	--	50%
57	Prefeitura de Birigui - Diretoria de Água e Esgoto: Birigui (SP)	Residential	23.90	100%
58	Prestador municipal/local - Iranduba: Iranduba (AM)	Residential	32.50	80%
59	Prestador municipal/local - Manicoré: Manicoré (AM)	Residential	32.50	80%
60	Prestador municipal/local - São Gabriel da Cachoeira: São Gabriel da Cachoeira (AM)	Residential	32.50	80%
61	Prestador municipal/local - Tefé: Tefé (AM)	Residential	32.50	80%
62	Prolagos: Cabo Frio (RJ), São Pedro da Aldeia (RJ)	Residential	--	100%
63	RIC Ambiental: Marília (SP)	Residential	19.71	75%
64	Rio+ Saneamento: Itaguaí (RJ), Rio das Ostras (RJ), Seropédica (RJ)	Residential	--	100%
65	SAAE / prestação municipal de Manacapuru - confirmar: Manacapuru (AM)	Residential	32.50	80%
66	SAAE Angra dos Reis: Angra dos Reis (RJ)	Residential	--	100%
67	SAAE Aracruz / CESAN conforme área: Aracruz (ES)	Residential	57.39	80%
68	SAAE Atibaia: Atibaia (SP)	Residential	52.97	100%
69	SAAE Barra Mansa: Barra Mansa (RJ)	Residential	--	100%
70	SAAE Cacoal: Cacoal (RO)	Residential	17.58	60%
71	SAAE Capivari: Capivari (SP)	Residential	33.11	90%
72	SAAE Cruzeiro: Cruzeiro (SP)	Residential	42.00	90%
73	SAAE Formiga: Formiga (MG)	Residential	21.40	50%
74	SAAE Ibitinga: Ibitinga (SP)	Residential	--	100%
75	SAAE Indaiatuba: Indaiatuba (SP)	Residential	35.10	90%
76	SAAE Itabira: Itabira (MG)	Residential	--	80%
77	SAAE Itabirito: Itabirito (MG)	Residential	22.73	60%
78	SAAE Itacoatiara: Itacoatiara (AM)	Residential	32.50	80%
79	SAAE Itapira: Itapira (SP)	Residential	22.71	100%

No.	Provider: municipalities	Category	Fixed (R$/month)	Sewage
80	SAAE Jaboticabal: Jaboticabal (SP)	Residential	28.91	100%
81	SAAE Jacareí: Jacareí (SP)	Residential	42.00	90%
82	SAAE Lagoa da Prata: Lagoa da Prata (MG)	Residential	41.70	65%
83	SAAE Lençóis Paulista: Lençóis Paulista (SP)	Residential	42.00	90%
84	SAAE Linhares: Linhares (ES)	Residential	45.00	80%
85	SAAE Lucas do Rio Verde: Lucas do Rio Verde (MT)	Residential	--	80%
86	SAAE Manhuaçu: Manhuaçu (MG)	Residential	--	80%
87	SAAE Mariana: Mariana (MG)	Residential	29.82	100%
88	SAAE Maués: Maués (AM)	Residential	32.50	80%
89	SAAE Mogi Mirim: Mogi Mirim (SP)	Residential	20.25	140%
90	SAAE Nova Mutum: Nova Mutum (MT)	Residential	24.82	80%
91	SAAE Parintins: Parintins (AM)	Residential	32.50	80%
92	SAAE Passos: Passos (MG)	Residential	--	80%
93	SAAE Pirapora: Pirapora (MG)	Residential	43.06	50%
94	SAAE Salto: Salto (SP)	Residential	32.22	100%
95	SAAE Sete Lagoas: Sete Lagoas (MG)	Residential	30.96	80%
96	SAAE Sorocaba: Sorocaba (SP)	Residential	22.00	92%
97	SAAE São Carlos: São Carlos (SP)	Residential	--	100%
98	SAAE São Mateus: São Mateus (ES)	Residential	--	80%
99	SAAE Três Pontas: Três Pontas (MG)	Residential	--	80%
100	SAAE Três Rios: Três Rios (RJ)	Residential	--	100%
101	SAAE Vilhena: Vilhena (RO)	Residential	57.51	70%
102	SAAE Viçosa: Viçosa (MG)	Residential	--	50%
103	SAAE Volta Redonda: Volta Redonda (RJ)	Residential	--	100%
104	SAAEB Ambiental: Bebedouro (SP)	Residential	34.62	80%
105	SAAEB Barretos: Barretos (SP)	Residential	28.90	50%
106	SAAEJA: Jaguariúna (SP)	Residential	22.81	100%
107	SAAET: Taquaritinga (SP)	Residential	42.00	90%
108	SABESP: Amparo (SP)	Residential	29.47	90%
109	SABESP: Artur Nogueira (SP), Assis (SP), Avaré (SP), Batatais (SP), Boituva (SP), Botucatu (SP), Campo Limpo Paulista (SP), Caçapava (SP), Cosmópolis (SP), Fernandópolis (SP), Franca (SP), Hortolândia (SP), Ibiúna (SP), Itapetininga (SP), Itapeva (SP), Itatiba (SP), Itupeva (SP), Lins (SP), Lorena (SP), Mococa (SP), Mogi das Cruzes (SP), Monte Mor (SP), Nova Odessa (SP), Paulínia (SP), Piedade (SP), Pindamonhangaba (SP), Porto Feliz (SP), Registro (SP), Santa Bárbara d'Oeste (SP), Suzano (SP), São José dos Campos (SP), São João da Boa Vista (SP), São Roque (SP), Tatuí (SP), Taubaté (SP), Tremembé (SP), Tupã (SP), Votorantim (SP), Várzea Paulista (SP)	Residential	40.42	80%

No.	Provider: municipalities	Category	Fixed (R$/month)	Sewage
110	SABESP: Arujá (SP), Barueri (SP), Caieiras (SP), Cajamar (SP), Carapicuíba (SP), Cotia (SP), Diadema (SP), Embu das Artes (SP), Embu-Guaçu (SP), Ferraz de Vasconcelos (SP), Francisco Morato (SP), Franco da Rocha (SP), Guarulhos (SP), Itapecerica da Serra (SP), Itapevi (SP), Itaquaquecetuba (SP), Jandira (SP), Mairiporã (SP), Mauá (SP), Osasco (SP), Poá (SP), Ribeirão Pires (SP), Santa Isabel (SP), Santana de Parnaíba (SP), Santo André (SP), São Bernardo do Campo (SP), São Paulo (SP), Taboão da Serra (SP), Vargem Grande Paulista (SP)	Residential	40.42	100%
111	SABESP: Bertioga (SP), Caraguatatuba (SP), Cubatão (SP), Guarujá (SP), Itanhaém (SP), Mongaguá (SP), Peruíbe (SP), Praia Grande (SP), Santos (SP), São Sebastião (SP), São Vicente (SP), Ubatuba (SP)	Residential	40.42	100%
112	SABESP: Bragança Paulista (SP)	Residential	40.42	100%
113	SABESP: Olímpia (SP)	Residential	25.34	80%
114	SABESP: Presidente Prudente (SP)	Residential	32.27	80%
115	SAE Araguari: Araguari (MG)	Residential	33.89	80%
116	SAE Ituiutaba: Ituiutaba (MG)	Residential	18.12	70%
117	SAE Louveira: Louveira (SP)	Residential	27.93	85%
118	SAEC Catanduva: Catanduva (SP)	Residential	24.79	100%
119	SAECIL Leme: Leme (SP)	Residential	35.33	85%
120	SAEG: Guaratinguetá (SP)	Residential	30.06	80%
121	SAEMA Araras: Araras (SP)	Residential	25.71	80%
122	SAEMAS Sertãozinho: Sertãozinho (SP)	Residential	25.31	80%
123	SAEP Pirassununga: Pirassununga (SP)	Residential	29.91	100%
124	SAERP Ribeirão Preto: Ribeirão Preto (SP)	Residential	14.06	128%
125	SAERP São José do Rio Pardo: São José do Rio Pardo (SP)	Residential	29.99	80%
126	SAESA: São Caetano do Sul (SP)	Residential	42.00	90%
127	SAEV Ambiental: Votuporanga (SP)	Residential	28.22	100%
128	SAMAE Blumenau / BRK Ambiental Blumenau: Blumenau (SC)	Residential	47.43	112%
129	SAMAE Brusque: Brusque (SC)	Residential	42.38	80%
130	SAMAE Caxias do Sul: Caxias do Sul (RS)	Residential	38.30	80%
131	SAMAE Gaspar: Gaspar (SC)	Residential	50.58	80%
132	SAMAE Jaraguá do Sul: Jaraguá do Sul (SC)	Residential	30.16	80%
133	SAMAE Mogi Guaçu: Mogi Guaçu (SP)	Residential	18.91	100%
134	SAMAE São Bento do Sul: São Bento do Sul (SC)	Residential	31.40	80%
135	SAMAE Tangará da Serra: Tangará da Serra (MT)	Residential	--	80%
136	SANASA Campinas: Campinas (SP)	Residential	53.65	123%
137	SANEACRE: Cruzeiro do Sul (AC), Rio Branco (AC)	Residential	38.37	80%

No.	Provider: municipalities	Category	Fixed (R$/month)	Sewage
138	SANEAGO: Anápolis (GO), Aparecida de Goiânia (GO), Caldas Novas (GO), Catalão (GO), Cidade Ocidental (GO), Cristalina (GO), Formosa (GO), Goianira (GO), Goianésia (GO), Goiânia (GO), Inhumas (GO), Itumbiara (GO), Jataí (GO), Luziânia (GO), Mineiros (GO), Morrinhos (GO), Novo Gama (GO), Planaltina (GO), Rio Verde (GO), Santo Antônio do Descoberto (GO), Senador Canedo (GO), Trindade (GO), Valparaíso de Goiás (GO), Águas Lindas de Goiás (GO)	Residential	17.46	100%
139	Saneaqua: Mairinque (SP)	Residential	--	100%
140	SANEAR Colatina: Colatina (ES)	Residential	--	80%
141	SANEAR Rondonópolis: Rondonópolis (MT)	Residential	36.50	80%
142	SANEBAVI Vinhedo: Vinhedo (SP)	Residential	40.20	100%
143	Saneouro: Ouro Preto (MG)	Residential	25.72	95%
144	SANEP Pelotas: Pelotas (RS)	Residential	28.96	80%
145	SANEPAR - Curitiba: Curitiba (PR)	Residential	55.09	85%
146	SANEPAR - Demais localidades: Almirante Tamandaré (PR), Apucarana (PR), Arapongas (PR), Araucária (PR), Cambé (PR), Campo Largo (PR), Campo Mourão (PR), Castro (PR), Cianorte (PR), Colombo (PR), Fazenda Rio Grande (PR), Foz do Iguaçu (PR), Francisco Beltrão (PR), Guarapuava (PR), Ibiporã (PR), Irati (PR), Londrina (PR), Marechal Cândido Rondon (PR), Maringá (PR), Medianeira (PR), Paranaguá (PR), Paranavaí (PR), Pato Branco (PR), Pinhais (PR), Piraquara (PR), Ponta Grossa (PR), Rolândia (PR), Sarandi (PR), São José dos Pinhais (PR), Telêmaco Borba (PR), Toledo (PR), Umuarama (PR), União da Vitória (PR)	Residential	55.09	80%
147	Sanessol: Mirassol (SP)	Residential	42.00	90%
148	SANESUL: Corumbá (MS), Dourados (MS), Naviraí (MS), Ponta Porã (MS), Três Lagoas (MS)	Residential	16.50	70%
149	SEMAE Piracicaba: Piracicaba (SP)	Residential	31.99	100%
150	SeMAE Rio Preto: São José do Rio Preto (SP)	Residential	26.63	100%
151	SEMAE São Leopoldo: São Leopoldo (RS)	Residential	61.29	80%
152	SEMASA Itajaí: Itajaí (SC)	Residential	41.70	80%
153	SEMASA Lages: Lages (SC)	Residential	44.00	80%
154	Serviço local - Barra do Piraí: Barra do Piraí (RJ)	Residential	--	100%
155	São Gabriel Saneamento S.A.: São Gabriel (RS)	Residential	33.89	80%
156	Tubarão Saneamento: Tubarão (SC)	Residential	--	100%
157	Verde Alagoas: União dos Palmares (AL)	Residential	--	80%
158	Águas Alta Floresta / Norte Saneamento: Alta Floresta (MT)	Residential	--	80%
159	Águas Cuiabá (Iguá): Cuiabá (MT)	Residential	--	90%
160	Águas da Imperatriz: Teresópolis (RJ)	Residential	--	90%
161	Águas das Agulhas Negras: Resende (RJ)	Residential	--	100%
162	Águas de Andradina: Andradina (SP)	Residential	42.00	90%

No.	Provider: municipalities	Category	Fixed (R$/month)	Sewage
163	Águas de Barra do Garças (Aegea): Barra do Garças (MT)	Residential	--	80%
164	Águas de Jahu: Jaú (SP)	Residential	42.00	90%
165	Águas de Juturnaíba: Araruama (RJ)	Residential	--	100%
166	Águas de Manaus: Manaus (AM)	Residential	--	80%
167	Águas de Matão: Matão (SP)	Residential	45.32	100%
168	Águas de Niterói: Niterói (RJ)	Residential	--	100%
169	Águas de Nova Friburgo: Nova Friburgo (RJ)	Residential	--	100%
170	Águas de Pará de Minas: Pará de Minas (MG)	Residential	--	80%
171	Águas de Primavera (Aegea): Primavera do Leste (MT)	Residential	--	90%
172	Águas de Sinop (Aegea): Sinop (MT)	Residential	--	100%
173	Águas de Sorriso: Sorriso (MT)	Residential	--	80%
174	Águas de Teresina: Teresina (PI)	Residential	46.83	80%
175	Águas de Valadares (Aegea): Governador Valadares (MG)	Residential	14.67	70%
176	Águas do Imperador: Petrópolis (RJ)	Residential	--	100%
177	Águas do Pantanal (autarquia municipal): Cáceres (MT)	Residential	--	50%
178	Águas do Paraíba: Campos dos Goytacazes (RJ)	Residential	--	100%
179	Águas do Piauí: Floriano (PI), Parnaíba (PI), Picos (PI), Piripiri (PI)	Residential	52.89	80%
180	Águas do Rio: Belford Roxo (RJ), Duque de Caxias (RJ), Japeri (RJ), Mesquita (RJ), Nilópolis (RJ), Nova Iguaçu (RJ), Queimados (RJ), São João de Meriti (RJ)	Residential	--	100%
181	Águas do Rio: Cachoeiras de Macacu (RJ), Itaboraí (RJ), Magé (RJ), Maricá (RJ), Rio Bonito (RJ), Saquarema (RJ), São Gonçalo (RJ)	Residential	--	100%
182	Águas do Rio / Iguá / Rio+ Saneamento: Rio de Janeiro (RJ)	Residential	--	100%
183	Águas do Sertão: Delmiro Gouveia (AL), Palmeira dos Índios (AL)	Residential	--	80%
184	Águas do Sertão: Penedo (AL)	Residential	--	80%
185	Águas do Sertão: São Miguel dos Campos (AL)	Residential	--	80%
186	Águas Guariroba: Campo Grande (MS)	Residential	19.87	70%
187	Águas Pontes e Lacerda / Norte Saneamento: Pontes e Lacerda (MT)	Residential	--	90%

Table 180 -- Price of water by monthly consumption interval, in R$/m³, by municipality and category (part 1 of 3: 0-5 to 20-25 m³)

No.	0-5	5-6	6-7	7-10	10-12	12-13	13-15	15-17	17-20	20-25
1	7.64	7.64	7.64	7.64	14.61	14.61	14.61	16.89	16.89	18.04
2	6.82	6.82	6.82	6.82	12.70	12.70	12.70	14.27	14.27	15.46
3	7.64	7.64	7.64	7.64	14.61	14.61	14.61	16.89	16.89	18.04
4	0.00	0.00	0.00	0.00	6.10	6.10	6.10	6.10	6.10	9.80
5	0.00	0.00	0.00	0.00	5.11	5.11	5.11	10.14	10.14	10.14
6	0.00	0.00	0.00	0.00	5.60	5.60	5.60	5.60	5.60	9.80
7	0.00	0.00	0.00	0.00	3.64	3.64	3.64	3.64	3.64	4.01
8	10.20	10.20	10.20	10.20	10.70	10.70	10.70	10.70	10.70	15.34
9	8.76	3.05	3.05	3.05	8.66	8.66	8.66	11.07	11.07	13.53
10	0.00	0.00	0.00	0.00	6.80	6.80	6.80	6.80	6.80	10.77
11	0.00	0.00	0.00	0.00	5.40	5.40	5.40	5.40	5.40	8.90
12	0.00	0.00	0.00	0.00	6.52	6.52	6.52	7.17	7.17	8.61
13	0.00	0.00	0.00	0.00	6.34	6.34	6.34	7.51	7.51	8.45
14	0.00	0.00	0.00	0.00	5.73	5.73	5.73	6.77	6.77	7.64
15	4.30	4.30	4.30	5.15	5.15	5.15	10.21	10.21	10.21	14.81
16	0.00	0.00	0.00	0.00	4.50	4.50	4.50	4.50	4.50	7.20
17	6.77	6.77	6.77	6.77	8.78	8.78	8.78	9.49	9.49	16.29
18	0.00	0.00	0.00	0.00	7.08	7.08	7.08	7.08	7.08	9.32
19	7.28	7.28	7.28	7.28	13.90	13.90	13.90	16.07	16.07	17.17
20	3.08	3.08	3.08	3.08	14.31	14.31	14.31	14.31	14.31	14.31
21	4.50	4.50	4.50	4.50	6.50	6.50	6.50	6.50	6.50	10.50
22	1.65	4.46	4.46	4.46	5.58	5.58	5.58	6.32	6.32	8.55
23	2.26	2.26	2.26	2.26	4.52	4.52	4.52	8.14	8.14	9.77
24	3.91	3.91	3.91	3.91	5.20	5.20	5.20	5.20	5.20	5.74
25	0.00	0.00	0.00	0.00	4.50	4.50	4.50	4.50	4.50	7.20
26	0.00	0.00	0.00	0.00	4.34	4.34	4.34	4.34	4.34	6.63
27	1.49	1.49	1.49	1.49	9.89	9.89	9.89	9.95	9.95	9.95
28	0.00	0.00	0.00	0.00	7.09	7.09	7.09	7.09	7.09	8.42
29	9.23	9.23	9.23	9.23	9.23	9.23	9.23	9.23	9.23	9.23
30	2.69	5.33	5.33	5.33	8.06	8.06	8.06	10.83	10.83	13.66
31	0.00	0.00	0.00	0.00	4.50	4.50	4.50	4.50	4.50	7.20
32	4.91	4.91	4.91	4.91	7.00	7.00	7.00	7.00	7.00	9.38
33	0.00	0.00	0.00	0.00	4.49	4.49	4.49	4.49	4.49	5.06
34	0.00	0.00	0.00	0.00	4.09	4.09	4.09	4.09	4.09	5.83
35	0.00	0.00	0.00	0.00	7.19	7.19	7.19	7.19	7.19	10.77
36	--	--	3.49	3.49	3.49	3.49	3.49	3.49	3.49	3.49
37	--	--	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23

No.	0-5	5-6	6-7	7-10	10-12	12-13	13-15	15-17	17-20	20-25
38	0.00	0.00	0.00	0.00	5.18	5.18	5.18	7.65	7.65	11.01
39	2.37	2.37	2.37	2.37	2.37	2.37	2.37	2.37	2.37	2.37
40	0.00	0.00	0.00	0.00	4.40	4.40	4.40	4.40	5.69	5.69
41	0.00	0.00	0.00	0.00	4.20	4.20	4.20	4.20	4.20	6.80
42	4.26	4.26	4.26	4.26	5.11	5.11	5.11	5.11	5.11	6.84
43	3.76	3.76	3.76	3.76	3.76	3.76	3.76	3.76	3.76	4.89
44	4.10	4.10	4.10	4.10	5.95	5.95	5.95	5.95	5.95	8.60
45	7.28	7.28	7.28	7.28	9.80	9.80	9.80	9.80	9.80	14.50
46	4.10	4.10	4.10	4.10	5.95	5.95	5.95	5.95	5.95	8.60
47	4.10	4.10	4.10	4.10	5.95	5.95	5.95	5.95	5.95	8.60
48	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50
49	3.70	3.70	3.70	3.70	4.44	4.44	4.44	5.22	5.22	6.74
50	3.85	3.85	3.85	3.85	5.60	5.60	5.60	5.60	5.60	8.20
51	4.10	4.10	4.10	4.10	5.95	5.95	5.95	5.95	5.95	8.60
52	0.66	0.66	0.66	0.66	4.70	4.70	4.70	4.70	4.70	4.70
53	0.00	0.00	1.77	1.77	12.53	12.53	12.53	13.41	13.41	15.08
54	0.00	0.00	0.00	0.00	0.00	3.50	3.50	3.50	3.50	6.46
55	0.00	0.00	0.00	0.00	11.02	11.02	11.02	11.02	11.02	16.75
56	3.11	3.11	3.11	3.11	6.17	6.17	6.17	6.17	6.17	7.77
57	0.00	0.00	0.00	0.00	2.90	2.90	2.90	2.90	2.90	3.34
58	0.00	0.00	0.00	0.00	4.50	4.50	4.50	4.50	4.50	7.20
59	0.00	0.00	0.00	0.00	4.50	4.50	4.50	4.50	4.50	7.20
60	0.00	0.00	0.00	0.00	4.50	4.50	4.50	4.50	4.50	7.20
61	0.00	0.00	0.00	0.00	4.50	4.50	4.50	4.50	4.50	7.20
62	17.04	17.04	17.04	17.04	22.32	22.32	22.32	35.74	35.74	35.74
63	0.00	2.31	2.31	2.31	2.31	2.31	2.31	6.24	6.24	6.24
64	6.98	6.98	6.98	6.98	6.98	6.98	6.98	15.37	15.37	15.37
65	0.00	0.00	0.00	0.00	4.50	4.50	4.50	4.50	4.50	7.20
66	4.50	4.50	4.50	4.50	6.50	6.50	6.50	6.50	6.50	10.50
67	0.00	0.00	0.00	0.00	4.52	4.52	4.52	8.14	8.14	9.77
68	0.00	0.00	0.00	0.00	7.97	7.97	7.97	7.97	7.97	13.13
69	4.50	4.50	4.50	4.50	6.50	6.50	6.50	6.50	6.50	10.50
70	0.00	0.00	0.00	0.00	2.11	2.11	2.11	2.43	2.43	2.79
71	0.00	0.00	0.00	0.00	5.33	5.33	5.33	5.33	5.33	6.65
72	0.00	0.00	0.00	0.00	5.60	5.60	5.60	5.60	5.60	9.80
73	0.60	0.85	0.85	0.85	2.82	2.82	2.82	4.29	4.29	6.26
74	2.41	2.41	2.41	2.41	2.41	2.41	2.41	2.99	2.99	3.59
75	0.00	0.52	0.52	0.52	4.70	4.70	4.70	4.70	4.70	6.16
76	4.10	4.10	4.10	4.10	5.95	5.95	5.95	5.95	5.95	8.60

No.	0-5	5-6	6-7	7-10	10-12	12-13	13-15	15-17	17-20	20-25
77	1.66	1.95	1.95	1.95	3.84	3.84	3.84	5.64	5.64	6.19
78	0.00	0.00	0.00	0.00	4.50	4.50	4.50	4.50	4.50	7.20
79	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7.29	7.29	8.27
80	0.00	0.00	0.00	0.00	3.48	3.48	3.48	3.48	3.48	4.38
81	0.00	0.00	0.00	0.00	5.60	5.60	5.60	5.60	5.60	9.80
82	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.58	3.58	3.76
83	0.00	0.00	0.00	0.00	5.60	5.60	5.60	5.60	5.60	9.80
84	0.00	0.00	0.00	0.00	6.10	6.10	6.10	6.10	6.10	9.80
85	2.43	2.43	2.43	2.43	3.12	3.12	3.12	3.12	3.12	4.15
86	4.10	4.10	4.10	4.10	5.95	5.95	5.95	5.95	5.95	8.60
87	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
88	0.00	0.00	0.00	0.00	4.50	4.50	4.50	4.50	4.50	7.20
89	0.00	0.00	0.00	0.00	2.22	2.22	2.22	9.09	9.09	9.17
90	0.00	0.00	0.00	0.00	5.22	5.22	5.22	5.22	5.22	7.54
91	0.00	0.00	0.00	0.00	4.50	4.50	4.50	4.50	4.50	7.20
92	4.10	4.10	4.10	4.10	5.95	5.95	5.95	5.95	5.95	8.60
93	1.01	1.30	1.30	1.30	2.44	2.44	2.44	4.16	4.16	6.45
94	0.00	0.00	0.00	0.00	3.60	3.60	3.60	3.92	3.92	4.31
95	0.00	0.00	--	--	--	--	--	--	--	--
96	0.00	0.00	0.00	0.00	3.30	3.30	3.30	4.81	4.81	6.98
97	2.41	2.41	2.41	2.41	6.05	6.05	6.05	6.05	6.05	7.77
98	3.87	3.87	3.87	3.87	6.11	6.11	6.11	7.44	7.44	8.21
99	4.10	4.10	4.10	4.10	5.95	5.95	5.95	5.95	5.95	8.60
100	4.50	4.50	4.50	4.50	6.50	6.50	6.50	6.50	6.50	10.50
101	0.00	0.00	0.00	0.00	7.85	7.85	7.85	7.85	7.85	11.90
102	2.82	2.93	2.93	2.93	3.14	3.14	3.14	7.03	7.03	7.03
103	4.05	4.05	4.05	4.05	4.85	4.85	4.85	4.85	4.85	8.20
104	0.00	0.00	0.00	0.00	--	--	--	--	--	--
105	0.00	0.00	0.00	0.00	5.99	5.99	5.99	6.71	6.71	7.58
106	0.00	0.00	0.00	0.00	2.76	2.76	2.76	3.22	3.22	3.96
107	0.00	0.00	0.00	0.00	5.60	5.60	5.60	5.60	5.60	9.80
108	0.00	0.00	0.00	0.00	0.00	4.17	4.17	4.17	4.17	13.78
109	0.00	0.00	0.00	0.00	5.69	5.69	5.69	5.69	5.69	8.74
110	0.00	0.00	0.00	0.00	6.40	6.40	6.40	6.40	6.40	15.95
111	0.00	0.00	0.00	0.00	5.69	5.69	5.69	5.69	5.69	8.74
112	0.00	0.00	0.00	0.00	5.69	5.69	5.69	5.69	5.69	7.53
113	0.00	0.00	0.00	0.00	5.09	5.09	5.09	5.09	5.09	5.16
114	0.00	0.00	0.00	0.00	4.53	4.53	4.53	4.53	4.53	6.95
115	0.00	0.00	0.00	0.00	2.34	2.34	2.34	2.34	2.34	2.57

No.	0-5	5-6	6-7	7-10	10-12	12-13	13-15	15-17	17-20	20-25
116	1.67	2.88	2.88	2.88	4.08	4.08	4.08	5.04	5.04	5.77
117	0.00	0.00	0.00	0.00	3.78	3.78	3.78	3.78	3.78	4.68
118	0.00	0.00	0.00	0.00	3.25	3.25	3.25	3.25	3.25	7.75
119	0.00	0.00	0.00	0.00	4.03	4.03	4.03	4.03	4.03	5.15
120	0.00	0.00	0.00	0.00	4.12	4.12	4.12	4.12	4.12	6.91
121	0.00	0.00	0.00	0.00	3.21	3.21	3.21	3.21	3.21	4.13
122	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.05	2.05	4.81
123	0.00	0.00	0.00	0.00	--	--	--	--	--	--
124	0.00	0.00	0.00	0.00	2.48	2.48	2.48	3.93	3.93	5.39
125	0.00	0.00	0.00	0.00	--	--	--	--	--	--
126	0.00	0.00	0.00	0.00	5.60	5.60	5.60	5.60	5.60	9.80
127	0.00	0.00	0.00	0.00	4.87	4.87	4.87	4.87	4.87	8.31
128	0.00	0.00	0.00	0.00	8.63	8.63	8.63	8.63	8.63	8.63
129	0.00	0.00	0.00	0.00	7.80	7.80	7.80	10.01	10.01	11.37
130	0.00	8.77	8.77	8.77	10.54	10.54	10.54	16.56	16.56	21.75
131	0.00	0.00	0.00	0.00	7.81	7.81	7.81	7.81	7.81	7.81
132	1.59	1.80	1.80	1.80	8.05	8.05	8.05	10.23	10.23	11.37
133	0.00	0.00	0.00	0.00	2.29	2.29	2.29	2.82	2.82	7.52
134	2.36	2.55	2.55	2.55	8.83	8.83	8.83	9.40	9.40	10.02
135	5.85	5.85	5.85	5.85	7.90	7.90	7.90	7.90	7.90	10.80
136	0.00	0.00	0.00	0.00	9.96	9.96	9.96	10.18	10.18	10.43
137	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.68	3.68	4.57
138	5.77	5.77	5.77	5.77	6.51	6.51	6.51	7.45	7.45	8.45
139	4.71	4.71	4.71	4.71	6.50	6.50	6.50	6.50	6.50	10.04
140	2.85	2.85	2.85	2.85	5.92	5.92	5.92	8.50	8.50	9.36
141	0.00	0.00	0.00	0.00	5.10	5.10	5.10	5.10	5.10	8.20
142	0.00	0.00	0.00	0.00	4.56	4.56	4.56	5.73	5.73	8.00
143	3.67	3.67	3.67	3.67	10.06	10.06	10.06	12.46	12.46	14.30
144	6.13	6.13	6.13	6.13	7.05	7.05	7.05	7.05	7.05	9.32
145	0.00	1.70	1.70	1.70	9.49	9.49	9.49	9.54	9.54	9.62
146	0.00	1.70	1.70	1.70	9.49	9.49	9.49	9.54	9.54	9.62
147	0.00	0.00	0.00	0.00	5.60	5.60	5.60	5.60	5.60	9.80
148	5.92	5.92	5.92	5.92	7.45	7.45	7.45	9.15	9.15	11.80
149	0.00	0.00	0.00	0.00	5.49	5.49	5.49	7.88	7.88	11.28
150	0.00	0.00	0.00	0.00	3.84	3.84	3.84	3.84	3.84	5.25
151	5.15	5.15	5.15	5.15	5.15	5.15	5.15	11.45	11.45	11.45
152	0.00	0.00	0.00	0.00	7.48	7.48	7.48	7.48	7.48	7.48
153	0.00	0.00	0.00	0.00	7.48	7.48	7.48	7.48	7.48	7.48
154	4.50	4.50	4.50	4.50	6.50	6.50	6.50	6.50	6.50	10.50

No.	0-5	5-6	6-7	7-10	10-12	12-13	13-15	15-17	17-20	20-25
155	6.74	6.74	6.74	6.74	6.74	6.74	6.74	6.74	6.74	6.74
156	5.25	5.25	5.25	5.25	9.68	9.68	9.68	9.68	9.68	11.11
157	5.58	5.58	5.58	5.58	10.25	10.25	10.25	11.14	11.14	12.22
158	5.85	5.85	5.85	5.85	7.90	7.90	7.90	7.90	7.90	10.80
159	4.92	4.92	4.92	4.92	6.01	6.01	6.01	6.01	6.01	10.06
160	5.28	5.28	5.28	5.28	5.28	5.28	5.28	11.62	11.62	11.62
161	4.50	4.50	4.50	4.50	6.50	6.50	6.50	6.50	6.50	10.50
162	0.00	0.00	0.00	0.00	5.60	5.60	5.60	5.60	5.60	9.80
163	4.33	4.33	4.33	4.33	6.50	6.50	6.50	6.50	6.50	10.83
164	0.00	0.00	0.00	0.00	5.60	5.60	5.60	5.60	5.60	9.80
165	13.79	13.79	13.79	13.79	17.71	17.71	17.71	26.45	26.45	26.45
166	6.42	6.42	6.42	6.42	12.43	12.43	12.43	12.43	12.43	18.98
167	0.00	0.00	0.00	0.00	4.53	4.53	4.53	4.53	4.53	6.04
168	5.65	5.65	5.65	5.65	5.65	5.65	5.65	14.13	14.13	14.13
169	4.25	4.25	4.25	4.25	7.10	7.10	7.10	7.10	7.10	7.10
170	4.10	4.10	4.10	4.10	5.95	5.95	5.95	5.95	5.95	8.60
171	5.76	5.76	5.76	5.76	8.64	8.64	8.64	8.64	8.64	14.40
172	6.01	6.01	6.01	6.01	8.54	8.54	8.54	8.54	8.54	14.32
173	3.06	3.06	3.06	3.06	4.59	4.59	4.59	4.59	4.59	7.66
174	0.00	0.00	0.00	0.00	8.72	8.72	8.72	8.72	8.72	8.72
175	1.79	1.83	1.83	1.83	5.24	5.24	5.24	5.57	5.57	5.62
176	4.80	4.80	4.80	4.80	6.80	6.80	6.80	6.80	6.80	11.50
177	3.27	3.27	3.27	3.27	5.72	5.72	5.72	5.72	5.72	8.87
178	4.85	4.85	4.85	4.85	7.20	7.20	7.20	7.20	7.20	11.50
179	0.00	0.00	0.00	0.00	9.81	9.81	9.81	10.92	10.92	12.80
180	7.52	7.52	7.52	7.52	7.52	7.52	7.52	16.54	16.54	16.54
181	7.44	7.44	7.44	7.44	7.44	7.44	7.44	16.36	16.36	16.36
182	6.98	6.98	6.98	6.98	6.98	6.98	6.98	15.37	15.37	15.37
183	7.39	7.39	7.39	7.39	14.10	14.10	14.10	16.30	16.30	17.41
184	6.14	6.14	6.14	6.14	10.68	10.68	10.68	12.17	12.17	13.72
185	5.49	5.49	5.49	5.49	10.48	10.48	10.48	12.11	12.11	12.94
186	8.03	8.03	8.03	8.03	10.28	10.28	10.28	10.46	10.46	11.57
187	3.03	3.03	3.03	3.03	4.55	4.55	4.55	4.55	4.55	7.57

Table 181 -- Price of water by monthly consumption interval, in R$/m³, by municipality and category (part 2 of 3: 25-30 to 70-75 m³)

No.	25-30	30-35	35-40	40-45	45-50	50-55	55-60	60-65	65-70	70-75
1	18.04	18.63	18.63	18.86	18.86	18.99	18.99	18.99	18.99	18.99
2	15.46	15.97	15.97	15.92	15.92	15.14	15.14	15.14	15.14	15.14
3	18.04	18.63	18.63	18.86	18.86	18.99	18.99	18.99	18.99	18.99
4	9.80	9.80	9.80	9.80	9.80	14.80	14.80	14.80	14.80	14.80
5	10.14	13.18	13.18	13.18	13.18	13.18	13.18	16.48	16.48	16.48
6	9.80	9.80	9.80	9.80	9.80	14.50	14.50	14.50	14.50	14.50
7	4.01	5.64	5.64	5.64	5.64	7.57	7.57	7.57	7.57	7.57
8	15.34	15.85	15.85	15.85	15.85	16.18	16.18	16.18	16.18	16.18
9	15.67	17.50	17.50	17.50	17.50	17.50	17.50	17.50	17.50	17.50
10	10.77	13.68	13.68	13.68	13.68	21.48	21.48	21.48	21.48	21.48
11	8.90	8.90	8.90	8.90	8.90	8.90	8.90	8.90	8.90	8.90
12	9.87	11.82	11.82	11.82	11.82	16.50	16.50	16.50	16.50	16.50
13	8.45	9.74	9.74	9.74	9.74	16.80	16.80	16.80	16.80	16.80
14	7.64	8.79	8.79	8.79	8.79	16.80	16.80	16.80	16.80	16.80
15	14.81	22.22	22.22	22.22	28.88	28.88	28.88	28.88	28.88	28.88
16	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
17	16.29	16.29	16.29	16.29	16.29	28.77	28.77	28.77	28.77	28.77
18	9.32	12.67	12.67	12.67	12.67	12.67	12.67	12.67	12.67	12.67
19	17.17	17.72	17.72	17.96	17.96	18.07	18.07	18.07	18.07	18.07
20	19.14	19.14	19.14	19.14	19.14	24.08	24.08	24.08	24.08	24.08
21	10.50	10.50	10.50	10.50	10.50	15.00	15.00	15.00	15.00	15.00
22	8.55	8.55	8.55	11.16	11.16	11.16	11.16	11.16	11.16	11.16
23	9.77	11.24	11.24	11.24	11.24	12.36	12.36	12.36	12.36	12.36
24	5.74	6.13	6.13	6.13	6.13	6.88	6.88	6.88	6.88	6.88
25	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
26	6.63	6.63	6.63	6.63	6.63	7.94	7.94	7.94	7.94	7.94
27	13.18	13.18	13.63	13.63	13.63	13.63	13.63	13.63	13.63	13.63
28	8.42	11.59	11.59	11.59	11.59	13.74	13.74	13.74	13.74	13.74
29	9.23	9.23	9.23	9.23	9.23	9.23	9.23	9.23	9.23	9.23
30	13.66	13.66	13.66	16.54	16.54	16.54	16.54	16.54	16.54	16.54
31	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
32	9.38	10.58	10.58	14.66	14.66	19.04	19.04	19.04	19.04	19.04
33	5.06	6.58	6.58	6.58	6.58	9.31	9.31	9.31	9.31	9.31
34	5.83	7.64	7.64	9.07	9.07	10.82	10.82	10.82	10.82	10.82
35	10.77	10.77	10.77	10.77	10.77	15.55	15.55	15.55	15.55	15.55
36	3.49	3.49	3.49	3.49	3.49	3.49	3.49	3.49	3.49	3.49
37	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23

No.	25-30	30-35	35-40	40-45	45-50	50-55	55-60	60-65	65-70	70-75
38	11.01	11.01	11.01	11.01	11.01	11.01	11.01	11.01	11.01	11.01
39	2.37	2.37	2.37	2.37	2.37	2.37	2.37	2.37	2.37	2.37
40	7.77	7.77	10.69	10.69	10.69	22.18	22.18	22.18	22.18	22.18
41	6.80	6.80	6.80	6.80	6.80	6.80	6.80	6.80	6.80	6.80
42	6.84	8.10	8.10	8.94	8.94	11.17	11.17	11.17	11.17	11.17
43	4.89	5.26	5.26	5.64	5.64	6.01	6.01	6.01	6.01	6.01
44	8.60	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
45	14.50	14.50	14.50	14.50	14.50	19.80	19.80	19.80	19.80	19.80
46	8.60	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
47	8.60	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
48	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50
49	6.74	8.37	8.37	9.93	9.93	11.47	11.47	13.08	13.08	14.61
50	8.20	11.50	11.50	11.50	11.50	11.50	11.50	11.50	11.50	11.50
51	8.60	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
52	7.83	7.83	7.83	7.83	7.83	7.83	7.83	7.83	7.83	7.83
53	16.82	18.49	18.49	20.28	20.28	24.40	24.40	24.40	24.40	24.40
54	6.46	8.82	8.82	8.82	8.82	12.69	12.69	12.69	12.69	12.69
55	16.75	23.49	23.49	23.49	23.49	32.67	32.67	32.67	32.67	32.67
56	7.77	10.04	10.04	10.04	10.04	15.77	15.77	15.77	15.77	15.77
57	3.34	3.70	3.70	5.06	5.06	6.46	6.46	6.46	6.46	6.46
58	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
59	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
60	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
61	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
62	42.88	42.88	51.46	51.46	63.18	63.18	80.25	80.25	91.26	91.26
63	6.24	8.41	8.41	8.41	8.41	10.20	10.20	10.20	10.20	10.20
64	15.37	20.94	20.94	20.94	41.88	41.88	41.88	55.84	55.84	55.84
65	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
66	10.50	10.50	10.50	10.50	10.50	15.00	15.00	15.00	15.00	15.00
67	9.77	11.24	11.24	11.24	11.24	12.36	12.36	12.36	12.36	12.36
68	13.13	13.13	13.13	13.13	13.13	16.09	16.09	16.09	16.09	16.09
69	10.50	10.50	10.50	10.50	10.50	15.00	15.00	15.00	15.00	15.00
70	3.21	3.69	3.69	3.69	3.69	4.06	4.06	4.06	4.06	4.06
71	6.65	9.53	9.53	14.27	14.27	15.67	15.67	15.67	15.67	15.67
72	9.80	9.80	9.80	9.80	9.80	14.50	14.50	14.50	14.50	14.50
73	8.23	9.71	9.71	9.96	9.96	10.01	10.01	10.01	10.01	10.01
74	4.11	4.66	5.33	6.07	6.92	7.88	7.88	7.88	7.88	8.99
75	6.16	6.83	6.83	9.65	9.65	9.65	9.65	11.45	11.45	11.45
76	8.60	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00

No.	25-30	30-35	35-40	40-45	45-50	50-55	55-60	60-65	65-70	70-75
77	6.83	7.49	7.49	8.29	8.29	9.09	9.09	9.09	9.09	9.09
78	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
79	8.27	8.37	8.50	8.67	8.67	10.44	10.44	10.44	10.44	10.44
80	4.38	7.21	7.21	8.37	8.37	11.88	11.88	13.03	13.03	14.36
81	9.80	9.80	9.80	9.80	9.80	14.50	14.50	14.50	14.50	14.50
82	3.93	4.31	4.31	4.51	4.51	4.72	4.72	4.72	4.72	4.72
83	9.80	9.80	9.80	9.80	9.80	14.50	14.50	14.50	14.50	14.50
84	9.80	9.80	9.80	9.80	9.80	14.80	14.80	14.80	14.80	14.80
85	5.96	5.96	7.03	7.03	7.03	8.20	8.20	8.20	8.20	8.20
86	8.60	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
87	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
88	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
89	9.17	9.56	9.56	10.05	10.05	10.66	10.66	10.66	10.66	10.66
90	7.54	9.28	9.28	11.59	11.59	13.92	13.92	13.92	13.92	13.92
91	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
92	8.60	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
93	8.17	9.88	9.88	10.17	10.17	10.23	10.23	10.23	10.23	10.23
94	4.77	5.27	6.34	8.07	8.07	8.07	8.07	10.21	10.21	10.21
95	--	--	--	--	--	--	--	--	--	--
96	7.69	8.05	8.05	8.47	8.47	8.90	8.90	8.90	8.90	8.90
97	7.77	9.81	9.81	11.65	11.65	12.78	12.78	14.16	14.16	14.16
98	8.21	9.40	9.40	9.40	9.40	9.40	9.40	9.40	9.40	9.40
99	8.60	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
100	10.50	10.50	10.50	10.50	10.50	15.00	15.00	15.00	15.00	15.00
101	11.90	11.90	11.90	11.90	11.90	16.50	16.50	16.50	16.50	16.50
102	7.03	7.03	7.03	9.74	9.74	9.74	9.74	10.33	10.33	10.33
103	8.20	8.20	8.20	8.20	8.20	12.50	12.50	12.50	12.50	12.50
104	--	--	--	--	--	--	--	--	--	--
105	8.64	9.94	11.54	13.50	13.50	15.91	15.91	15.91	15.91	15.91
106	4.73	5.65	6.85	6.85	8.21	8.21	8.21	9.78	9.78	9.78
107	9.80	9.80	9.80	9.80	9.80	14.50	14.50	14.50	14.50	14.50
108	13.78	13.78	13.78	13.78	13.78	20.37	20.37	20.37	20.37	20.37
109	8.74	8.74	8.74	8.74	8.74	10.46	10.46	10.46	10.46	10.46
110	15.95	15.95	15.95	15.95	15.95	17.57	17.57	17.57	17.57	17.57
111	8.74	8.74	8.74	8.74	8.74	10.46	10.46	10.46	10.46	10.46
112	7.53	7.53	7.53	7.53	7.53	10.20	10.20	10.20	10.20	10.20
113	5.16	7.09	7.09	7.09	7.09	11.82	11.82	11.93	11.93	12.03
114	6.95	6.95	6.95	6.95	6.95	8.32	8.32	8.32	8.32	8.32
115	2.57	2.81	2.81	3.10	3.10	3.46	3.46	3.46	3.46	3.46

No.	25-30	30-35	35-40	40-45	45-50	50-55	55-60	60-65	65-70	70-75
116	6.49	6.97	6.97	7.45	7.45	7.69	7.69	7.69	7.69	7.69
117	4.68	6.43	6.43	7.25	7.25	10.21	10.21	10.21	10.21	15.08
118	7.79	8.34	8.34	9.77	9.77	11.47	11.47	11.47	11.47	11.47
119	5.15	6.62	6.62	6.90	6.90	9.23	9.23	9.23	9.23	9.23
120	6.91	6.91	6.91	6.91	6.91	11.31	11.31	11.31	11.31	11.31
121	4.13	5.58	5.58	5.58	5.58	7.82	7.82	7.82	7.82	7.82
122	5.39	5.83	5.93	6.46	6.46	7.02	7.02	7.67	7.67	9.92
123	--	--	--	--	--	--	--	--	--	--
124	7.18	9.33	9.33	12.57	12.57	15.77	15.77	15.77	15.77	15.77
125	--	--	--	--	--	--	--	--	--	--
126	9.80	9.80	9.80	9.80	9.80	14.50	14.50	14.50	14.50	14.50
127	8.31	8.25	8.25	8.25	8.25	9.94	9.94	9.94	9.94	9.94
128	8.63	11.00	11.00	11.00	11.00	11.00	11.00	11.00	11.00	11.00
129	12.54	12.54	12.54	12.54	12.54	13.18	13.18	13.18	13.18	13.18
130	21.75	22.66	22.66	22.66	22.66	24.36	24.36	24.36	24.36	24.36
131	10.30	10.30	10.30	12.49	12.49	14.31	14.31	14.31	14.31	14.31
132	12.49	14.79	14.79	14.79	14.79	14.79	14.79	14.79	14.79	14.79
133	7.52	7.58	7.58	9.34	9.34	10.74	10.74	10.74	10.74	10.74
134	10.34	10.34	10.34	10.34	10.34	10.51	10.51	10.51	10.51	10.51
135	10.80	14.20	14.20	14.20	14.20	14.20	14.20	14.20	14.20	14.20
136	12.82	13.65	13.65	13.65	13.65	20.94	20.94	20.94	20.94	20.94
137	5.46	6.35	7.24	8.12	9.01	9.90	9.90	9.90	9.90	9.90
138	9.54	10.89	10.89	12.33	12.33	14.06	14.06	14.06	14.06	14.06
139	10.04	10.04	10.04	10.04	10.04	11.91	11.91	11.91	11.91	11.91
140	9.36	9.98	9.98	9.98	9.98	10.41	10.41	10.41	10.41	10.41
141	8.20	8.20	8.20	8.20	8.20	8.20	8.20	8.20	8.20	8.20
142	8.00	8.00	11.63	11.63	11.63	16.92	16.92	16.92	16.92	16.92
143	14.30	14.30	14.30	22.98	22.98	22.98	22.98	22.98	22.98	22.98
144	9.32	10.74	10.74	10.74	10.74	12.33	12.33	12.33	12.33	12.33
145	9.62	16.28	16.28	16.28	16.28	16.28	16.28	16.28	16.28	16.28
146	9.62	16.28	16.28	16.28	16.28	16.28	16.28	16.28	16.28	16.28
147	9.80	9.80	9.80	9.80	9.80	14.50	14.50	14.50	14.50	14.50
148	11.80	15.60	15.60	15.60	15.60	15.60	15.60	15.60	15.60	15.60
149	11.28	11.28	11.28	11.28	11.28	11.28	11.28	11.28	11.28	11.28
150	5.25	6.99	6.99	8.64	8.64	13.14	13.14	13.14	13.14	13.14
151	11.45	16.03	16.03	16.03	16.03	16.03	16.03	16.03	16.03	16.03
152	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38
153	10.20	10.20	10.20	10.20	10.20	10.20	10.20	10.20	10.20	10.20
154	10.50	10.50	10.50	10.50	10.50	15.00	15.00	15.00	15.00	15.00

No.	25-30	30-35	35-40	40-45	45-50	50-55	55-60	60-65	65-70	70-75
155	6.74	6.74	6.74	6.74	6.74	6.74	6.74	6.74	6.74	6.74
156	11.11	13.59	13.59	13.59	13.59	16.29	16.29	16.29	16.29	16.29
157	12.22	12.64	12.64	13.29	13.29	13.92	13.92	13.92	13.92	13.92
158	10.80	14.20	14.20	14.20	14.20	14.20	14.20	14.20	14.20	14.20
159	10.06	12.31	12.31	12.31	12.31	16.29	16.29	16.29	16.29	16.29
160	11.62	15.85	15.85	15.85	31.70	31.70	31.70	42.27	42.27	42.27
161	10.50	10.50	10.50	10.50	10.50	15.00	15.00	15.00	15.00	15.00
162	9.80	9.80	9.80	9.80	9.80	14.50	14.50	14.50	14.50	14.50
163	10.83	14.29	14.29	22.95	22.95	22.95	22.95	22.95	22.95	22.95
164	9.80	9.80	9.80	9.80	9.80	14.50	14.50	14.50	14.50	14.50
165	33.10	33.10	42.43	42.43	51.93	51.93	66.02	66.02	80.27	80.27
166	18.98	25.86	25.86	29.84	29.84	29.84	29.84	34.02	34.02	34.02
167	6.04	6.04	6.04	6.04	6.04	6.04	6.04	6.04	6.04	6.04
168	14.13	17.53	17.53	17.53	33.92	33.92	33.92	45.23	45.23	45.23
169	7.10	11.80	11.80	11.80	11.80	15.50	15.50	15.50	15.50	15.50
170	8.60	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
171	14.40	19.01	19.01	26.90	26.90	26.90	26.90	26.90	26.90	26.90
172	14.32	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93
173	7.66	11.48	11.48	18.04	18.04	18.04	18.04	18.04	18.04	18.04
174	15.06	15.06	15.06	15.06	15.06	15.06	15.06	15.06	15.06	15.06
175	5.62	5.62	5.62	10.37	10.37	10.37	10.37	10.37	10.37	10.37
176	11.50	11.50	11.50	11.50	11.50	17.20	17.20	17.20	17.20	17.20
177	8.87	11.37	11.37	15.88	15.88	15.88	15.88	15.88	15.88	15.88
178	11.50	11.50	11.50	11.50	11.50	16.80	16.80	16.80	16.80	16.80
179	12.80	12.80	12.80	12.80	12.80	12.80	12.80	12.80	12.80	12.80
180	16.54	22.56	22.56	22.56	45.12	45.12	45.12	60.16	60.16	60.16
181	16.36	22.31	22.31	22.31	44.62	44.62	44.62	59.49	59.49	59.49
182	15.37	20.94	20.94	20.94	41.88	41.88	41.88	55.84	55.84	55.84
183	17.41	17.98	17.98	18.21	18.21	18.33	18.33	18.33	18.33	18.33
184	13.72	14.83	14.83	15.57	15.57	15.95	15.95	15.95	15.95	15.95
185	12.94	13.36	13.36	13.54	13.54	13.62	13.62	13.62	13.62	13.62
186	14.23	20.40	20.40	20.40	20.40	20.40	20.40	20.40	20.40	20.40
187	7.57	9.99	9.99	16.04	16.04	16.04	16.04	16.04	16.04	16.04

Table 182 -- Price of water by monthly consumption interval, in R$/m³, by municipality and category (part 3 of 3: 75-80 to 5,000 m³)

No.	75-80	80-90	90-99	99-100	100-150	150-200	200-300	300-500	500-5,000	> 5,000
1	18.99	18.99	19.09	19.09	19.09	19.10	19.10	19.10	19.10	19.10
2	15.14	15.14	16.96	16.96	16.96	19.88	19.88	19.88	19.88	19.88
3	18.99	18.99	19.09	19.09	19.09	19.10	19.10	19.10	19.10	19.10
4	14.80	14.80	14.80	14.80	14.80	14.80	14.80	14.80	14.80	14.80
5	16.48	16.48	16.48	16.48	20.60	20.60	20.60	20.60	20.60	20.60
6	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50
7	7.57	7.57	7.57	7.57	7.57	7.57	7.57	7.57	7.57	7.57
8	16.18	16.18	16.18	16.18	16.18	16.18	16.18	16.18	16.18	16.18
9	17.50	17.50	17.50	17.50	17.50	17.50	17.50	17.50	17.50	17.50
10	21.48	21.48	21.48	21.48	21.48	21.48	21.48	21.48	21.48	21.48
11	8.90	8.90	8.90	8.90	8.90	8.90	8.90	8.90	8.90	8.90
12	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.50
13	16.80	16.80	16.80	16.80	16.80	16.80	16.80	16.80	16.80	16.80
14	16.80	16.80	16.80	16.80	16.80	16.80	16.80	16.80	16.80	16.80
15	28.88	28.88	28.88	28.88	28.88	28.88	28.88	28.88	28.88	28.88
16	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
17	28.77	28.77	28.77	28.77	28.77	28.77	28.77	28.77	28.77	28.77
18	12.67	12.67	12.67	12.67	12.67	12.67	12.67	12.67	12.67	12.67
19	18.07	18.07	18.18	18.18	18.18	18.19	18.19	18.19	18.19	18.19
20	24.08	24.08	24.08	24.08	24.08	24.08	24.08	24.08	24.08	24.08
21	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00
22	11.16	11.16	11.16	11.16	11.16	11.16	11.16	11.16	11.16	11.16
23	12.36	12.36	12.36	12.36	12.36	12.36	12.36	12.36	12.36	12.36
24	6.88	6.88	6.88	6.88	7.03	7.03	7.28	7.28	7.28	7.28
25	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
26	7.94	7.94	7.94	7.94	7.94	7.94	7.94	7.94	7.94	7.94
27	13.63	13.63	13.63	13.63	13.63	13.63	13.63	13.63	13.63	13.63
28	13.74	13.74	26.40	26.40	26.40	26.40	26.40	26.40	26.40	26.40
29	9.23	9.23	9.23	9.23	9.23	9.23	9.23	9.23	9.23	9.23
30	16.54	16.54	16.54	16.54	16.54	16.54	16.54	16.54	16.54	16.54
31	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
32	19.04	19.04	19.04	19.04	19.04	19.04	19.04	19.04	19.04	19.04
33	9.31	9.31	9.31	9.31	9.31	9.31	9.31	9.31	9.31	9.31
34	10.82	10.82	10.82	10.82	12.76	12.76	15.21	15.21	15.21	15.21
35	15.55	15.55	15.55	15.55	17.91	17.91	17.91	17.91	17.91	17.91
36	3.49	3.49	3.49	3.49	3.49	3.49	3.49	3.49	3.49	3.49
37	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23

No.	75-80	80-90	90-99	99-100	100-150	150-200	200-300	300-500	500-5,000	> 5,000
38	11.01	11.01	11.01	11.01	11.01	11.01	11.01	11.01	11.01	11.01
39	2.37	2.37	2.37	2.37	2.37	2.37	2.37	2.37	2.37	2.37
40	22.18	22.18	22.18	22.18	22.18	22.18	22.18	22.18	22.18	22.18
41	6.80	6.80	6.80	6.80	6.80	6.80	6.80	6.80	6.80	6.80
42	11.17	11.17	11.17	11.17	11.17	11.17	11.17	11.17	11.17	11.17
43	6.39	6.39	6.39	6.39	6.76	6.76	6.76	6.76	6.76	6.76
44	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
45	19.80	19.80	19.80	19.80	19.80	19.80	19.80	19.80	19.80	19.80
46	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
47	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
48	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50
49	14.61	14.61	14.61	14.61	14.61	14.61	14.61	14.61	14.61	14.61
50	11.50	11.50	11.50	11.50	11.50	11.50	11.50	11.50	11.50	11.50
51	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
52	7.83	7.83	7.83	7.83	7.83	7.83	7.83	7.83	7.83	7.83
53	24.40	24.40	24.40	24.40	24.40	24.40	24.40	24.40	24.40	24.40
54	12.69	12.69	12.69	12.69	16.79	16.79	16.79	16.79	16.79	16.79
55	32.67	32.67	32.67	32.67	42.08	42.08	42.08	42.08	42.08	42.08
56	15.77	15.77	15.77	15.77	15.77	15.77	15.77	15.77	15.77	15.77
57	6.46	6.46	6.46	6.46	8.12	8.12	9.62	9.62	9.62	9.62
58	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
59	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
60	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
61	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
62	91.26	91.26	91.26	91.26	91.26	91.26	91.26	91.26	91.26	91.26
63	10.20	10.20	10.20	10.20	11.63	11.63	11.63	11.63	11.63	11.63
64	55.84	55.84	55.84	55.84	55.84	55.84	55.84	55.84	55.84	55.84
65	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
66	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00
67	12.36	12.36	12.36	12.36	12.36	12.36	12.36	12.36	12.36	12.36
68	16.09	16.09	16.09	16.09	16.09	16.09	16.09	16.09	16.09	16.09
69	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00
70	4.06	4.06	4.06	4.06	4.06	4.06	4.06	4.06	4.06	4.06
71	15.67	15.67	15.67	15.67	15.67	15.67	15.67	15.67	15.67	15.67
72	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50
73	10.06	10.06	10.06	10.06	10.11	10.11	10.16	10.16	10.16	10.16
74	8.99	8.99	8.99	8.99	10.22	10.22	10.22	10.22	10.22	10.22
75	11.45	23.32	23.32	23.32	23.32	23.32	23.32	23.32	23.32	23.32
76	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00

No.	75-80	80-90	90-99	99-100	100-150	150-200	200-300	300-500	500-5,000	> 5,000
77	9.99	9.99	9.99	9.99	10.99	10.99	12.09	12.09	12.09	12.09
78	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
79	10.44	10.44	10.44	10.44	10.44	10.44	10.44	10.44	10.44	10.44
80	14.36	15.93	17.47	17.47	19.22	19.22	19.22	19.22	19.22	19.22
81	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50
82	5.11	5.11	5.11	5.11	5.54	5.54	5.72	5.72	5.72	5.72
83	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50
84	14.80	14.80	14.80	14.80	14.80	14.80	14.80	14.80	14.80	14.80
85	8.20	8.20	8.20	8.20	8.20	8.20	8.20	8.20	8.20	8.20
86	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
87	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
88	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
89	11.67	11.67	11.67	11.67	12.75	12.75	12.75	12.75	12.75	12.75
90	13.92	13.92	13.92	13.92	13.92	13.92	13.92	13.92	13.92	13.92
91	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20	7.20
92	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
93	10.29	10.29	10.29	10.29	10.34	10.34	10.40	10.40	10.40	10.40
94	10.21	13.58	13.58	13.58	13.58	13.58	13.58	13.58	13.58	13.58
95	--	--	--	--	--	--	--	--	--	--
96	9.12	9.12	9.12	9.12	10.95	10.95	13.13	15.75	15.75	15.75
97	14.16	14.16	14.16	14.16	14.16	14.16	14.16	14.16	14.16	14.16
98	9.40	9.40	9.40	9.40	9.40	9.40	9.40	9.40	9.40	9.40
99	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
100	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00
101	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.50
102	10.33	10.33	10.33	10.33	10.33	10.33	10.33	10.95	10.95	10.95
103	12.50	12.50	12.50	12.50	12.50	12.50	12.50	12.50	12.50	12.50
104	--	--	--	--	--	--	--	--	--	--
105	15.91	15.91	15.91	15.91	17.18	17.18	17.18	17.18	17.18	17.18
106	9.78	9.78	9.78	9.78	11.75	11.75	11.75	11.75	11.75	11.75
107	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50
108	20.37	20.37	20.37	25.49	25.49	25.49	25.49	25.49	25.49	25.49
109	10.46	10.46	10.46	10.46	10.46	10.46	10.46	10.46	10.46	10.46
110	17.57	17.57	17.57	17.57	17.57	17.57	17.57	17.57	17.57	17.57
111	10.46	10.46	10.46	10.46	10.46	10.46	10.46	10.46	10.46	10.46
112	10.20	10.20	10.20	10.20	10.20	10.20	10.20	10.20	10.20	10.20
113	12.03	12.03	12.03	12.03	12.03	12.03	12.03	12.03	12.03	12.03
114	8.32	8.32	8.32	8.32	8.32	8.32	8.32	8.32	8.32	8.32
115	3.46	3.46	3.46	3.46	3.46	3.46	3.46	3.46	3.46	3.46

No.	75-80	80-90	90-99	99-100	100-150	150-200	200-300	300-500	500-5,000	> 5,000
116	7.93	7.93	7.93	7.93	8.17	8.17	8.41	8.41	8.41	8.41
117	15.08	15.08	15.08	15.08	15.08	15.08	15.08	15.08	15.08	15.08
118	11.47	11.47	11.47	11.47	13.46	13.46	13.46	13.46	13.46	13.46
119	9.23	9.23	9.23	9.23	10.87	10.87	10.87	10.87	10.87	10.87
120	11.31	11.31	11.31	11.31	11.31	11.31	11.31	11.31	11.31	11.31
121	7.82	7.82	7.82	7.82	11.34	11.34	11.34	11.34	17.02	26.39
122	9.92	12.03	12.81	12.81	13.20	13.20	13.20	13.20	13.20	13.20
123	--	--	--	--	--	--	--	--	--	--
124	15.77	19.02	19.02	19.02	21.87	21.87	21.87	21.87	21.87	21.87
125	--	--	--	--	--	--	--	--	--	--
126	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50
127	9.94	9.94	9.94	9.94	9.94	9.94	9.94	9.94	9.94	9.94
128	11.00	11.00	11.00	11.00	11.00	11.00	11.00	11.00	11.00	11.00
129	13.18	13.18	13.18	13.18	13.18	13.18	13.18	13.18	13.18	13.18
130	24.36	24.36	24.36	24.36	28.35	28.35	28.35	28.35	28.35	28.35
131	14.31	14.31	14.31	14.31	14.31	14.31	14.31	14.31	14.31	14.31
132	14.79	14.79	14.79	14.79	14.79	14.79	14.79	14.79	14.79	14.79
133	10.74	10.74	10.74	10.74	10.74	10.74	10.74	10.74	10.74	10.74
134	10.51	10.51	10.51	10.51	10.51	10.51	10.51	10.51	10.51	10.51
135	14.20	14.20	14.20	14.20	14.20	14.20	14.20	14.20	14.20	14.20
136	20.94	20.94	20.94	20.94	20.94	20.94	20.94	20.94	20.94	20.94
137	9.90	9.90	9.90	9.90	9.90	9.90	9.90	9.90	9.90	9.90
138	14.06	14.06	14.06	14.06	14.06	14.06	14.06	14.06	14.06	14.06
139	11.91	11.91	11.91	11.91	11.91	11.91	11.91	11.91	11.91	11.91
140	10.41	10.41	10.41	10.41	10.41	10.41	10.41	10.41	10.41	10.41
141	8.20	8.20	8.20	8.20	8.20	8.20	8.20	8.20	8.20	8.20
142	22.82	22.82	22.82	22.82	22.82	22.82	22.82	22.82	22.82	22.82
143	22.98	22.98	22.98	22.98	22.98	22.98	22.98	22.98	22.98	22.98
144	12.33	12.33	12.33	12.33	14.18	14.18	14.18	14.18	14.18	14.18
145	16.28	16.28	16.28	16.28	16.28	16.28	16.28	16.28	16.28	16.28
146	16.28	16.28	16.28	16.28	16.28	16.28	16.28	16.28	16.28	16.28
147	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50
148	15.60	15.60	15.60	15.60	15.60	15.60	15.60	15.60	15.60	15.60
149	11.28	11.28	11.28	11.28	11.28	11.28	11.28	11.28	11.28	11.28
150	13.14	13.14	13.14	13.14	14.59	14.59	14.59	14.59	14.59	14.59
151	16.03	16.03	16.03	16.03	16.03	16.03	16.03	16.03	16.03	16.03
152	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38
153	10.20	10.20	10.20	10.20	10.20	10.20	10.20	10.20	10.20	10.20
154	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00

No.	75-80	80-90	90-99	99-100	100-150	150-200	200-300	300-500	500-5,000	> 5,000
155	6.74	6.74	6.74	6.74	6.74	6.74	6.74	6.74	6.74	6.74
156	16.29	16.29	16.29	16.29	16.29	16.29	16.29	16.29	16.29	16.29
157	13.92	13.92	13.96	13.96	13.96	13.97	13.97	13.97	13.97	13.97
158	14.20	14.20	14.20	14.20	14.20	14.20	14.20	14.20	14.20	14.20
159	16.29	16.29	16.29	16.29	16.29	16.29	16.29	16.29	16.29	16.29
160	42.27	42.27	42.27	42.27	42.27	42.27	42.27	42.27	42.27	42.27
161	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00
162	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50
163	22.95	22.95	22.95	22.95	22.95	22.95	22.95	22.95	22.95	22.95
164	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50	14.50
165	80.27	80.27	80.27	80.27	80.27	80.27	80.27	80.27	80.27	80.27
166	34.02	34.02	34.02	34.02	34.02	34.02	34.02	34.02	34.02	34.02
167	6.04	6.04	6.04	6.04	6.04	6.04	6.04	6.04	6.04	6.04
168	45.23	45.23	45.23	45.23	45.23	45.23	45.23	45.23	45.23	45.23
169	15.50	15.50	15.50	15.50	15.50	15.50	15.50	15.50	15.50	15.50
170	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
171	26.90	26.90	26.90	26.90	26.90	26.90	26.90	26.90	26.90	26.90
172	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93
173	18.04	18.04	18.04	18.04	18.04	18.04	18.04	18.04	18.04	18.04
174	15.06	15.06	15.06	15.06	15.06	15.06	15.06	15.06	15.06	15.06
175	10.37	10.37	10.37	10.37	10.37	10.37	10.37	10.37	10.37	10.37
176	17.20	17.20	17.20	17.20	17.20	17.20	17.20	17.20	17.20	17.20
177	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88
178	16.80	16.80	16.80	16.80	16.80	16.80	16.80	16.80	16.80	16.80
179	12.80	12.80	12.80	12.80	12.80	12.80	12.80	12.80	12.80	12.80
180	60.16	60.16	60.16	60.16	60.16	60.16	60.16	60.16	60.16	60.16
181	59.49	59.49	59.49	59.49	59.49	59.49	59.49	59.49	59.49	59.49
182	55.84	55.84	55.84	55.84	55.84	55.84	55.84	55.84	55.84	55.84
183	18.33	18.33	18.44	18.44	18.44	18.45	18.45	18.45	18.45	18.45
184	15.95	15.95	16.01	16.01	16.01	16.01	16.01	16.01	16.01	16.01
185	13.62	13.62	13.70	13.70	13.70	13.71	13.71	13.71	13.71	13.71
186	20.40	20.40	20.40	20.40	20.40	20.40	20.40	20.40	20.40	20.40
187	16.04	16.04	16.04	16.04	16.04	16.04	16.04	16.04	16.04	16.04

The categories below do not yet have the provider's official table in the catalogue. In those municipalities the program does not offer them, and the way forward is to enter the table in Custom mode.

Table 183 -- Tariff categories not yet collected, by provider

Provider: municipalities	Categories not collected
BRK Alagoas: Maceió (AL), Marechal Deodoro (AL), Rio Largo (AL)	Commercial, Public, Industrial, Social
BRK Ambiental Cachoeiro: Cachoeiro de Itapemirim (ES)	Commercial, Public, Industrial, Social
BRK Ambiental Limeira: Limeira (SP)	Commercial, Public, Industrial, Social
BRK Ambiental Porto Ferreira: Porto Ferreira (SP)	Commercial, Public, Industrial, Social
BRK Ambiental Sumaré: Sumaré (SP)	Commercial, Public, Industrial, Social
BRK Ambiental Uruguaiana: Uruguaiana (RS)	Commercial, Public, Industrial, Social
BRK Tocantins: Araguaína (TO), Gurupi (TO), Palmas (TO), Paraíso do Tocantins (TO), Porto Nacional (TO)	Commercial, Public, Industrial, Social
CAEMA: Açailândia (MA), Bacabal (MA), Balsas (MA), Barra do Corda (MA), Barreirinhas (MA), Buriticupu (MA), Caxias (MA), Chapadinha (MA), Codó (MA), Coroatá (MA), Grajaú (MA), Imperatriz (MA), Itapecuru-Mirim (MA), Paço do Lumiar (MA), Pinheiro (MA), Santa Inês (MA), Santa Luzia (MA), São José de Ribamar (MA), São Luís (MA), Timon (MA), Tutóia (MA), Viana (MA)	Commercial, Public, Industrial, Social
CAER: Boa Vista (RR)	Commercial, Public, Industrial, Social
CAERD: Ariquemes (RO), Jaru (RO), Ji-Paraná (RO), Porto Velho (RO), Rolim de Moura (RO)	Commercial, Public, Industrial, Social
CAERN: Assu (RN), Caicó (RN), Ceará-Mirim (RN), Extremoz (RN), Macaíba (RN), Mossoró (RN), Natal (RN), Parnamirim (RN)	Commercial, Public, Industrial, Social
CAESB: Brasília (DF)	Commercial, Public, Industrial, Social
CAESC - Companhia de Água, Esgoto e Saneamento de Coari: Coari (AM)	Commercial, Public, Industrial, Social
CAGECE: Acaraú (CE), Aquiraz (CE), Aracati (CE), Barbalha (CE), Beberibe (CE), Boa Viagem (CE), Brejo Santo (CE), Camocim (CE), Canindé (CE), Cascavel (CE), Caucaia (CE), Crateús (CE), Crato (CE), Eusébio (CE), Fortaleza (CE), Granja (CE), Horizonte (CE), Icó (CE), Iguatu (CE), Itaitinga (CE), Itapipoca (CE), Juazeiro do Norte (CE), Limoeiro do Norte (CE), Maracanaú (CE), Maranguape (CE), Morada Nova (CE), Pacajus (CE), Pacatuba (CE), Quixadá (CE), Quixeramobim (CE), Russas (CE), Sobral (CE), São Gonçalo do Amarante (CE), Tauá (CE), Tianguá (CE), Trairi (CE), Viçosa do Ceará (CE)	Commercial, Public, Industrial, Social
CAGEPA: Bayeux (PB), Cabedelo (PB), Cajazeiras (PB), Campina Grande (PB), Guarabira (PB), João Pessoa (PB), Patos (PB), Santa Rita (PB), Sapé (PB), Sousa (PB)	Commercial, Public, Industrial, Social
CASAL: Arapiraca (AL)	Commercial, Public, Industrial, Social
CASAN: Araranguá (SC), Biguaçu (SC), Camboriú (SC), Canoinhas (SC), Caçador (SC), Chapecó (SC), Concórdia (SC), Criciúma (SC), Florianópolis (SC), Imbituba (SC), Indaial (SC), Itapema (SC), Içara (SC), Mafra (SC), Navegantes (SC), Palhoça (SC), Rio do Sul (SC), São Francisco do Sul (SC), São José (SC), Tijucas (SC), Videira (SC), Xanxerê (SC)	Commercial, Public, Industrial, Social
CEDAE/serviço local - validar área: Guapimirim (RJ), Itaperuna (RJ), Macaé (RJ), Valença (RJ)	Commercial, Public, Industrial, Social
CESAMA Juiz de Fora: Juiz de Fora (MG)	Commercial, Public, Industrial, Social
CESAN: Cariacica (ES), Guarapari (ES), Serra (ES), Vila Velha (ES), Vitória (ES)	Commercial, Public, Industrial, Social
CODAU Uberaba: Uberaba (MG)	Commercial, Public, Industrial, Social
COHASB / Prefeitura de Humaitá - prestação local: Humaitá (AM)	Commercial, Public, Industrial, Social

Provider: municipalities	Categories not collected
Companhia Ituana de Saneamento (CIS): Itu (SP)	Commercial, Public, Industrial, Social
Companhia Águas de Joinville: Joinville (SC)	Commercial, Public, Industrial, Social
COMPESA: Abreu e Lima (PE), Araripina (PE), Arcoverde (PE), Belo Jardim (PE), Bezerros (PE), Buíque (PE), Cabo de Santo Agostinho (PE), Camaragibe (PE), Carpina (PE), Caruaru (PE), Escada (PE), Garanhuns (PE), Goiana (PE), Gravatá (PE), Igarassu (PE), Ipojuca (PE), Jaboatão dos Guararapes (PE), Limoeiro (PE), Moreno (PE), Olinda (PE), Ouricuri (PE), Palmares (PE), Paudalho (PE), Paulista (PE), Pesqueira (PE), Petrolina (PE), Recife (PE), Salgueiro (PE), Santa Cruz do Capibaribe (PE), Serra Talhada (PE), Surubim (PE), São Lourenço da Mata (PE), Vitória de Santo Antão (PE)	Commercial, Public, Industrial, Social
COMUSA Novo Hamburgo: Novo Hamburgo (RS)	Commercial, Public, Industrial, Social
COPASA: Alfenas (MG), Araxá (MG), Barbacena (MG), Belo Horizonte (MG), Betim (MG), Bom Despacho (MG), Campo Belo (MG), Caratinga (MG), Cataguases (MG), Congonhas (MG), Conselheiro Lafaiete (MG), Contagem (MG), Coronel Fabriciano (MG), Curvelo (MG), Divinópolis (MG), Esmeraldas (MG), Extrema (MG), Frutal (MG), Guaxupé (MG), Ibirité (MG), Ipatinga (MG), Itajubá (MG), Itaúna (MG), Janaúba (MG), Januária (MG), João Monlevade (MG), Lagoa Santa (MG), Lavras (MG), Leopoldina (MG), Montes Claros (MG), Nova Lima (MG), Nova Serrana (MG), Paracatu (MG), Patos de Minas (MG), Pedro Leopoldo (MG), Pouso Alegre (MG), Ribeirão das Neves (MG), Sabará (MG), São Francisco (MG), São Sebastião do Paraíso (MG), Teófilo Otoni (MG), Timóteo (MG), Três Corações (MG), Ubá (MG), Unaí (MG), Varginha (MG), Vespasiano (MG)	Commercial, Public, Industrial, Social
CORSAN: Canoas (RS), Erechim (RS), Esteio (RS), Santa Cruz do Sul (RS)	Industrial
CORSAN: Alegrete (RS), Alvorada (RS), Bento Gonçalves (RS), Cachoeira do Sul (RS), Cachoeirinha (RS), Camaquã (RS), Campo Bom (RS), Canoas (RS), Capão da Canoa (RS), Carazinho (RS), Cruz Alta (RS), Erechim (RS), Esteio (RS), Farroupilha (RS), Gravataí (RS), Guaíba (RS), Ijuí (RS), Lajeado (RS), Montenegro (RS), Parobé (RS), Passo Fundo (RS), Rio Grande (RS), Santa Cruz do Sul (RS), Santa Maria (RS), Santa Rosa (RS), Santo Ângelo (RS), Sapiranga (RS), Sapucaia do Sul (RS), São Borja (RS), Taquara (RS), Tramandaí (RS), Vacaria (RS), Venâncio Aires (RS), Viamão (RS)	Social
COSAMA: Tabatinga (AM)	Commercial, Public, Industrial, Social
COSANPA: Abaetetuba (PA), Acará (PA), Alenquer (PA), Altamira (PA), Ananindeua (PA), Baião (PA), Barcarena (PA), Belém (PA), Benevides (PA), Bragança (PA), Breves (PA), Cametá (PA), Canaã dos Carajás (PA), Capanema (PA), Capitão Poço (PA), Castanhal (PA), Dom Eliseu (PA), Igarapé-Miri (PA), Itaituba (PA), Juruti (PA), Marabá (PA), Marituba (PA), Moju (PA), Monte Alegre (PA), Novo Repartimento (PA), Oriximiná (PA), Paragominas (PA), Parauapebas (PA), Portel (PA), Redenção (PA), Rondon do Pará (PA), Santa Izabel do Pará (PA), Santarém (PA), São Félix do Xingu (PA), São Miguel do Guamá (PA), Tailândia (PA), Tomé-Açu (PA), Tucuruí (PA), Vigia (PA), Viseu (PA), Xinguara (PA), Óbidos (PA)	Commercial, Public, Industrial, Social
CSA Equatorial: Macapá (AP), Santana (AP)	Commercial, Public, Industrial, Social
DAAE Araraquara: Araraquara (SP)	Commercial, Public, Industrial, Social
DAAE Rio Claro: Rio Claro (SP)	Commercial, Public, Industrial, Social
DAE Americana: Americana (SP)	Commercial, Public, Industrial, Social
DAE Bauru: Bauru (SP)	Commercial, Public, Industrial, Social
DAE Jundiaí: Jundiaí (SP)	Commercial, Public, Industrial, Social

Provider: municipalities	Categories not collected
DAE Sant'Ana do Livramento: Santana do Livramento (RS)	Commercial, Public, Industrial, Social
DAE Valinhos: Valinhos (SP)	Commercial, Public, Industrial, Social
DAE/VG: Várzea Grande (MT)	Commercial, Public, Industrial, Social
DAEB Bagé: Bagé (RS)	Commercial, Public, Industrial, Social
DAEP Penápolis: Penápolis (SP)	Commercial, Public, Industrial, Social
DAEPA: Patrocínio (MG)	Commercial, Public, Industrial, Social
DAESC Coruripe: Coruripe (AL)	Commercial, Public, Industrial, Social
DAMAE São João del-Rei: São João del-Rei (MG)	Commercial, Public, Industrial, Social
DEMSUR: Muriaé (MG)	Commercial, Public, Industrial, Social
DMAE Porto Alegre: Porto Alegre (RS)	Commercial, Public, Industrial, Social
DMAE Poços de Caldas: Poços de Caldas (MG)	Commercial, Public, Industrial, Social
DMAE Uberlândia: Uberlândia (MG)	Commercial, Public, Industrial, Social
DMAES Ponte Nova: Ponte Nova (MG)	Commercial, Public, Industrial, Social
EMASA Balneário Camboriú: Balneário Camboriú (SC)	Commercial, Public, Industrial, Social
EMBASA: Alagoinhas (BA), Barra (BA), Barreiras (BA), Bom Jesus da Lapa (BA), Brumado (BA), Caetité (BA), Camaçari (BA), Campo Formoso (BA), Candeias (BA), Casa Nova (BA), Conceição do Coité (BA), Cruz das Almas (BA), Dias d'Ávila (BA), Euclides da Cunha (BA), Eunápolis (BA), Feira de Santana (BA), Guanambi (BA), Ilhéus (BA), Ipirá (BA), Irecê (BA), Itaberaba (BA), Itabuna (BA), Itamaraju (BA), Itapetinga (BA), Jacobina (BA), Jequié (BA), Juazeiro (BA), Lauro de Freitas (BA), Luís Eduardo Magalhães (BA), Paulo Afonso (BA), Porto Seguro (BA), Ribeira do Pombal (BA), Salvador (BA), Santo Amaro (BA), Santo Antônio de Jesus (BA), Santo Estêvão (BA), Senhor do Bonfim (BA), Serrinha (BA), Simões Filho (BA), Teixeira de Freitas (BA), Vitória da Conquista (BA)	Commercial, Public, Industrial, Social
GS Inima SAMAR: Araçatuba (SP)	Commercial, Public, Industrial, Social
Iguá Sergipe: Aracaju (SE), Estância (SE), Itabaiana (SE), Lagarto (SE), Nossa Senhora do Socorro (SE), São Cristóvão (SE), Tobias Barreto (SE)	Commercial, Public, Industrial, Social
Ourinhos Saneamento: Ourinhos (SP)	Commercial, Public, Industrial, Social
Prefeitura de Birigui - Diretoria de Água e Esgoto: Birigui (SP)	Commercial, Public, Industrial, Social
Prestador municipal/local - Iranduba: Iranduba (AM)	Commercial, Public, Industrial, Social
Prestador municipal/local - Manicoré: Manicoré (AM)	Commercial, Public, Industrial, Social
Prestador municipal/local - São Gabriel da Cachoeira: São Gabriel da Cachoeira (AM)	Commercial, Public, Industrial, Social
Prestador municipal/local - Tefé: Tefé (AM)	Commercial, Public, Industrial, Social
Prolagos: Cabo Frio (RJ), São Pedro da Aldeia (RJ)	Commercial, Public, Industrial, Social
RIC Ambiental: Marília (SP)	Commercial, Public, Industrial, Social
Rio+ Saneamento: Itaguaí (RJ), Rio das Ostras (RJ), Seropédica (RJ)	Commercial, Public, Industrial, Social
SAAE / prestação municipal de Manacapuru - confirmar: Manacapuru (AM)	Commercial, Public, Industrial, Social
SAAE Angra dos Reis: Angra dos Reis (RJ)	Commercial, Public, Industrial, Social
SAAE Aracruz / CESAN conforme área: Aracruz (ES)	Commercial, Public, Industrial, Social
SAAE Atibaia: Atibaia (SP)	Commercial, Public, Industrial, Social

Provider: municipalities	Categories not collected
SAAE Barra Mansa: Barra Mansa (RJ)	Commercial, Public, Industrial, Social
SAAE Cacoal: Cacoal (RO)	Commercial, Public, Industrial, Social
SAAE Capivari: Capivari (SP)	Commercial, Public, Industrial, Social
SAAE Cruzeiro: Cruzeiro (SP)	Commercial, Public, Industrial, Social
SAAE Formiga: Formiga (MG)	Commercial, Public, Industrial, Social
SAAE Ibitinga: Ibitinga (SP)	Commercial, Public, Industrial, Social
SAAE Indaiatuba: Indaiatuba (SP)	Commercial, Public, Industrial, Social
SAAE Itabira: Itabira (MG)	Commercial, Public, Industrial, Social
SAAE Itabirito: Itabirito (MG)	Commercial, Public, Industrial, Social
SAAE Itacoatiara: Itacoatiara (AM)	Commercial, Public, Industrial, Social
SAAE Itapira: Itapira (SP)	Commercial, Public, Industrial, Social
SAAE Jaboticabal: Jaboticabal (SP)	Commercial, Public, Industrial, Social
SAAE Jacareí: Jacareí (SP)	Commercial, Public, Industrial, Social
SAAE Lagoa da Prata: Lagoa da Prata (MG)	Commercial, Public, Industrial, Social
SAAE Lençóis Paulista: Lençóis Paulista (SP)	Commercial, Public, Industrial, Social
SAAE Linhares: Linhares (ES)	Commercial, Public, Industrial, Social
SAAE Lucas do Rio Verde: Lucas do Rio Verde (MT)	Commercial, Public, Industrial, Social
SAAE Manhuaçu: Manhuaçu (MG)	Commercial, Public, Industrial, Social
SAAE Mariana: Mariana (MG)	Commercial, Public, Industrial, Social
SAAE Maués: Maués (AM)	Commercial, Public, Industrial, Social
SAAE Mogi Mirim: Mogi Mirim (SP)	Commercial, Public, Industrial, Social
SAAE Nova Mutum: Nova Mutum (MT)	Commercial, Public, Industrial, Social
SAAE Parintins: Parintins (AM)	Commercial, Public, Industrial, Social
SAAE Passos: Passos (MG)	Commercial, Public, Industrial, Social
SAAE Pirapora: Pirapora (MG)	Commercial, Public, Industrial, Social
SAAE Salto: Salto (SP)	Commercial, Public, Industrial, Social
SAAE Sete Lagoas: Sete Lagoas (MG)	Commercial, Public, Industrial, Social
SAAE Sorocaba: Sorocaba (SP)	Commercial, Public, Industrial, Social
SAAE São Carlos: São Carlos (SP)	Commercial, Public, Industrial, Social
SAAE São Mateus: São Mateus (ES)	Commercial, Public, Industrial, Social
SAAE Três Pontas: Três Pontas (MG)	Commercial, Public, Industrial, Social
SAAE Três Rios: Três Rios (RJ)	Commercial, Public, Industrial, Social
SAAE Vilhena: Vilhena (RO)	Commercial, Public, Industrial, Social
SAAE Viçosa: Viçosa (MG)	Commercial, Public, Industrial, Social
SAAE Volta Redonda: Volta Redonda (RJ)	Commercial, Public, Industrial, Social
SAAEB Ambiental: Bebedouro (SP)	Commercial, Public, Industrial, Social
SAAEB Barretos: Barretos (SP)	Commercial, Public, Industrial, Social
SAAEJA: Jaguariúna (SP)	Commercial, Public, Industrial, Social
SAAET: Taquaritinga (SP)	Commercial, Public, Industrial, Social

Provider: municipalities	Categories not collected
SABESP: Amparo (SP), Artur Nogueira (SP), Arujá (SP), Assis (SP), Avaré (SP), Barueri (SP), Batatais (SP), Bertioga (SP), Boituva (SP), Botucatu (SP), Bragança Paulista (SP), Caieiras (SP), Cajamar (SP), Campo Limpo Paulista (SP), Caraguatatuba (SP), Carapicuíba (SP), Caçapava (SP), Cosmópolis (SP), Cotia (SP), Cubatão (SP), Diadema (SP), Embu das Artes (SP), Embu-Guaçu (SP), Fernandópolis (SP), Ferraz de Vasconcelos (SP), Franca (SP), Francisco Morato (SP), Franco da Rocha (SP), Guarujá (SP), Guarulhos (SP), Hortolândia (SP), Ibiúna (SP), Itanhaém (SP), Itapecerica da Serra (SP), Itapetininga (SP), Itapeva (SP), Itapevi (SP), Itaquaquecetuba (SP), Itatiba (SP), Itupeva (SP), Jandira (SP), Lins (SP), Lorena (SP), Mairiporã (SP), Mauá (SP), Mococa (SP), Mogi das Cruzes (SP), Mongaguá (SP), Monte Mor (SP), Nova Odessa (SP), Olímpia (SP), Osasco (SP), Paulínia (SP), Peruíbe (SP), Piedade (SP), Pindamonhangaba (SP), Porto Feliz (SP), Poá (SP), Praia Grande (SP), Presidente Prudente (SP), Registro (SP), Ribeirão Pires (SP), Santa Bárbara d'Oeste (SP), Santa Isabel (SP), Santana de Parnaíba (SP), Santo André (SP), Santos (SP), Suzano (SP), São Bernardo do Campo (SP), São José dos Campos (SP), São João da Boa Vista (SP), São Paulo (SP), São Roque (SP), São Sebastião (SP), São Vicente (SP), Taboão da Serra (SP), Tatuí (SP), Taubaté (SP), Tremembé (SP), Tupã (SP), Ubatuba (SP), Vargem Grande Paulista (SP), Votorantim (SP), Várzea Paulista (SP)	Commercial, Public, Industrial, Social
SAE Araguari: Araguari (MG)	Commercial, Public, Industrial, Social
SAE Ituiutaba: Ituiutaba (MG)	Commercial, Public, Industrial, Social
SAE Louveira: Louveira (SP)	Commercial, Public, Industrial, Social
SAEC Catanduva: Catanduva (SP)	Commercial, Public, Industrial, Social
SAECIL Leme: Leme (SP)	Commercial, Public, Industrial, Social
SAEG: Guaratinguetá (SP)	Commercial, Public, Industrial, Social
SAEMA Araras: Araras (SP)	Commercial, Public, Industrial, Social
SAEMAS Sertãozinho: Sertãozinho (SP)	Commercial, Public, Industrial, Social
SAEP Pirassununga: Pirassununga (SP)	Commercial, Public, Industrial, Social
SAERP Ribeirão Preto: Ribeirão Preto (SP)	Commercial, Public, Industrial, Social
SAERP São José do Rio Pardo: São José do Rio Pardo (SP)	Commercial, Public, Industrial, Social
SAESA: São Caetano do Sul (SP)	Commercial, Public, Industrial, Social
SAEV Ambiental: Votuporanga (SP)	Commercial, Public, Industrial, Social
SAMAE Blumenau / BRK Ambiental Blumenau: Blumenau (SC)	Commercial, Public, Industrial, Social
SAMAE Brusque: Brusque (SC)	Commercial, Public, Industrial, Social
SAMAE Caxias do Sul: Caxias do Sul (RS)	Commercial, Public, Industrial, Social
SAMAE Gaspar: Gaspar (SC)	Commercial, Public, Industrial, Social
SAMAE Jaraguá do Sul: Jaraguá do Sul (SC)	Commercial, Public, Industrial, Social
SAMAE Mogi Guaçu: Mogi Guaçu (SP)	Commercial, Public, Industrial, Social
SAMAE São Bento do Sul: São Bento do Sul (SC)	Commercial, Public, Industrial, Social
SAMAE Tangará da Serra: Tangará da Serra (MT)	Commercial, Public, Industrial, Social
SANASA Campinas: Campinas (SP)	Commercial, Public, Industrial, Social
SANEACRE: Cruzeiro do Sul (AC), Rio Branco (AC)	Commercial, Public, Industrial, Social

Provider: municipalities	Categories not collected
SANEAGO: Anápolis (GO), Aparecida de Goiânia (GO), Caldas Novas (GO), Catalão (GO), Cidade Ocidental (GO), Cristalina (GO), Formosa (GO), Goianira (GO), Goianésia (GO), Goiânia (GO), Inhumas (GO), Itumbiara (GO), Jataí (GO), Luziânia (GO), Mineiros (GO), Morrinhos (GO), Novo Gama (GO), Planaltina (GO), Rio Verde (GO), Santo Antônio do Descoberto (GO), Senador Canedo (GO), Trindade (GO), Valparaíso de Goiás (GO), Águas Lindas de Goiás (GO)	Commercial, Public, Industrial, Social
Saneaqua: Mairinque (SP)	Commercial, Public, Industrial, Social
SANEAR Colatina: Colatina (ES)	Commercial, Public, Industrial, Social
SANEAR Rondonópolis: Rondonópolis (MT)	Commercial, Public, Industrial, Social
SANEBAVI Vinhedo: Vinhedo (SP)	Commercial, Public, Industrial, Social
Saneouro: Ouro Preto (MG)	Commercial, Public, Industrial, Social
SANEP Pelotas: Pelotas (RS)	Commercial, Public, Industrial, Social
SANEPAR - Curitiba: Curitiba (PR)	Commercial, Public, Industrial, Social
SANEPAR - Demais localidades: Almirante Tamandaré (PR), Apucarana (PR), Arapongas (PR), Araucária (PR), Cambé (PR), Campo Largo (PR), Campo Mourão (PR), Castro (PR), Cianorte (PR), Colombo (PR), Fazenda Rio Grande (PR), Foz do Iguaçu (PR), Francisco Beltrão (PR), Guarapuava (PR), Ibiporã (PR), Irati (PR), Londrina (PR), Marechal Cândido Rondon (PR), Maringá (PR), Medianeira (PR), Paranaguá (PR), Paranavaí (PR), Pato Branco (PR), Pinhais (PR), Piraquara (PR), Ponta Grossa (PR), Rolândia (PR), Sarandi (PR), São José dos Pinhais (PR), Telêmaco Borba (PR), Toledo (PR), Umuarama (PR), União da Vitória (PR)	Commercial, Public, Industrial, Social
Sanessol: Mirassol (SP)	Commercial, Public, Industrial, Social
SANESUL: Corumbá (MS), Dourados (MS), Naviraí (MS), Ponta Porã (MS), Três Lagoas (MS)	Commercial, Public, Industrial, Social
SEMAE Piracicaba: Piracicaba (SP)	Commercial, Public, Industrial, Social
SeMAE Rio Preto: São José do Rio Preto (SP)	Commercial, Public, Industrial, Social
SEMAE São Leopoldo: São Leopoldo (RS)	Commercial, Public, Industrial, Social
SEMASA Itajaí: Itajaí (SC)	Commercial, Public, Industrial, Social
SEMASA Lages: Lages (SC)	Commercial, Public, Industrial, Social
Serviço local - Barra do Piraí: Barra do Piraí (RJ)	Commercial, Public, Industrial, Social
São Gabriel Saneamento S.A.: São Gabriel (RS)	Commercial, Public, Industrial, Social
Tubarão Saneamento: Tubarão (SC)	Commercial, Public, Industrial, Social
Verde Alagoas: União dos Palmares (AL)	Commercial, Public, Industrial, Social
Águas Alta Floresta / Norte Saneamento: Alta Floresta (MT)	Commercial, Public, Industrial, Social
Águas Cuiabá (Iguá): Cuiabá (MT)	Commercial, Public, Industrial, Social
Águas da Imperatriz: Teresópolis (RJ)	Commercial, Public, Industrial, Social
Águas das Agulhas Negras: Resende (RJ)	Commercial, Public, Industrial, Social
Águas de Andradina: Andradina (SP)	Commercial, Public, Industrial, Social
Águas de Barra do Garças (Aegea): Barra do Garças (MT)	Commercial, Public, Industrial, Social
Águas de Jahu: Jaú (SP)	Commercial, Public, Industrial, Social
Águas de Juturnaíba: Araruama (RJ)	Commercial, Public, Industrial, Social
Águas de Manaus: Manaus (AM)	Commercial, Public, Industrial, Social

Provider: municipalities	Categories not collected
Águas de Matão: Matão (SP)	Commercial, Public, Industrial, Social
Águas de Niterói: Niterói (RJ)	Commercial, Public, Industrial, Social
Águas de Nova Friburgo: Nova Friburgo (RJ)	Commercial, Public, Industrial, Social
Águas de Pará de Minas: Pará de Minas (MG)	Commercial, Public, Industrial, Social
Águas de Primavera (Aegea): Primavera do Leste (MT)	Commercial, Public, Industrial, Social
Águas de Sinop (Aegea): Sinop (MT)	Commercial, Public, Industrial, Social
Águas de Sorriso: Sorriso (MT)	Commercial, Public, Industrial, Social
Águas de Teresina: Teresina (PI)	Commercial, Public, Industrial, Social
Águas de Valadares (Aegea): Governador Valadares (MG)	Commercial, Public, Industrial, Social
Águas do Imperador: Petrópolis (RJ)	Commercial, Public, Industrial, Social
Águas do Pantanal (autarquia municipal): Cáceres (MT)	Commercial, Public, Industrial, Social
Águas do Paraíba: Campos dos Goytacazes (RJ)	Commercial, Public, Industrial, Social
Águas do Piauí: Floriano (PI), Parnaíba (PI), Picos (PI), Piripiri (PI)	Commercial, Public, Industrial, Social
Águas do Rio: Belford Roxo (RJ), Cachoeiras de Macacu (RJ), Duque de Caxias (RJ), Itaboraí (RJ), Japeri (RJ), Magé (RJ), Maricá (RJ), Mesquita (RJ), Nilópolis (RJ), Nova Iguaçu (RJ), Queimados (RJ), Rio Bonito (RJ), Saquarema (RJ), São Gonçalo (RJ), São João de Meriti (RJ)	Commercial, Public, Industrial, Social
Águas do Rio / Iguá / Rio+ Saneamento: Rio de Janeiro (RJ)	Commercial, Public, Industrial, Social
Águas do Sertão: Delmiro Gouveia (AL), Palmeira dos Índios (AL), Penedo (AL), São Miguel dos Campos (AL)	Commercial, Public, Industrial, Social
Águas Guariroba: Campo Grande (MS)	Commercial, Public, Industrial, Social
Águas Pontes e Lacerda / Norte Saneamento: Pontes e Lacerda (MT)	Commercial, Public, Industrial, Social

The second table gives, for the same rows, the effective date declared by the source and the date on which the official page was last checked by the database monitoring routine. A row without a date is a source the routine does not monitor yet. The effective dates are quoted as the sources state them, in Portuguese.

Table 184 -- Effective date and last check of the source of each tariff structure, with the status of the data

No.	Declared effective date	Source checked on	Status
1	Vigente desde dezembro de 2025	2026-08-31	audited
2	Vigente desde dezembro de 2025	2026-08-31	audited
3	Vigente desde dezembro de 2025	2026-08-31	audited
4	Resoluções ARSP / Concessões ES	2026-08-31	to be confirmed
5	Contas emitidas desde julho de 2026	2026-08-31	audited
6	Resoluções ARES-PCJ / ARSESP	2026-08-31	to be confirmed
7	Vigente desde fevereiro de 2026	2026-08-31	audited
8	Tabela vigente desde 30/08/2025; ainda vigente em 19/08/2026	2026-08-31	audited
9	Vigente desde 05/12/2025	2026-08-31	to be confirmed
10	Água vigente desde 05/01/2024; relação esgoto/água confirmada em 19/01/2026	2026-08-31	audited
11	Estrutura Tarifária CAER Oficial	2026-08-31	to be confirmed
12	Vigente em 2025/2026	2026-08-31	audited
13	Vigente desde 01/04/2025	2026-08-31	audited
14	Vigente desde 01/07/2025	2026-08-31	audited
15	Resolução ADASA nº 70	2026-08-31	audited
16	Estrutura Municipal Vigente	2026-08-31	to be confirmed
17	Vigente desde 05/11/2025	2026-08-31	to be confirmed
18	Vigente desde 30/06/2025	2026-08-31	audited
19	Vigente desde dezembro de 2025	2026-08-31	audited
20	Vigente desde 01/04/2026	2026-08-31	to be confirmed
21	Regulação Oficial	2026-08-31	to be confirmed
22	Res. ARISB-MG 332/2025; vigente em 19/08/2026	2026-08-31	audited
23	Tabela aplicável desde 01/08/2025 e vigente em 19/08/2026	2026-08-31	audited
24	Vigente desde março de 2026	2026-08-31	audited
25	Estrutura Municipal Vigente	2026-08-31	to be confirmed
26	Vigente desde maio de 2026	2026-08-31	audited
27	01/03/2026	2026-08-31	audited
28	Resolução ARPE nº 289	2026-08-31	audited
29	2026 vigente em 19/08/2026	2026-08-31	audited
30	22/01/2026-21/01/2027	2026-08-31	to be confirmed
31	Estrutura Municipal Vigente	2026-08-31	to be confirmed
32	Resoluções ARCON-PA 2024-2026	2026-08-31	audited
33	Vigente de 30/08/2025 a 29/08/2026	2026-08-31	audited
34	Vigente desde novembro de 2025	2026-08-31	audited
35	Vigente desde fevereiro de 2026	2026-08-31	audited
36	Vigente em 19/08/2026	2026-08-31	audited
37	Resolução nº 02, de 11/02/2026	2026-08-31	audited

No.	Declared effective date	Source checked on	Status
38	Res. 664/2025; vigente em 19/08/2026	2026-08-31	audited
39	Referência vigente pesquisada em 19/08/2026	2026-08-31	to be confirmed
40	Vigente desde abril de 2023; último ato listado pela ARES-PCJ	2026-08-31	audited
41	Estrutura Tarifária DAE/VG	2026-08-31	to be confirmed
42	2026 vigente em 19/08/2026	2026-08-31	audited
43	Vigente desde 01/03/2026	2026-08-31	audited
44	Resoluções ARISB-MG / Prestadores Locais	2026-08-31	to be confirmed
45	Resolução ARSAL Nº 233	2026-08-31	to be confirmed
46	Resoluções ARISB-MG / Prestadores Locais	2026-08-31	to be confirmed
47	Resoluções ARISB-MG / Prestadores Locais	2026-08-31	to be confirmed
48	2026 vigente em 19/08/2026	2026-08-31	audited
49	Vigente desde 01/06/2026	2026-08-31	audited
50	Estrutura Tarifária DMAE / ARESAN	2026-08-31	to be confirmed
51	Resoluções ARISB-MG / Prestadores Locais	2026-08-31	to be confirmed
52	Vigente desde 04/05/2026	2026-08-31	audited
53	Vigente desde 20/07/2026	2026-08-31	to be confirmed
54	Desde 01/03/2026	2026-08-31	audited
55	Vigente desde 01/08/2025	2026-08-31	audited
56	Vigente em 19/08/2026	2026-08-31	audited
57	Desde 03/2026	2026-08-31	to be confirmed
58	Estrutura Municipal Vigente	2026-08-31	to be confirmed
59	Estrutura Municipal Vigente	2026-08-31	to be confirmed
60	Estrutura Municipal Vigente	2026-08-31	to be confirmed
61	Estrutura Municipal Vigente	--	to be confirmed
62	Vigente desde 01/12/2025	2026-08-31	audited
63	Vigente desde abril de 2026	2026-08-31	audited
64	Deliberação AGENERSA	--	to be confirmed
65	Estrutura Municipal Vigente	2026-08-31	to be confirmed
66	Regulação Oficial	2026-08-31	to be confirmed
67	SAAE: efeitos desde 09/06/2026; Res. ARSP 103/2026	2026-08-31	to be confirmed
68	Vigente desde dezembro de 2025	2026-08-31	to be confirmed
69	Regulação Oficial	2026-08-31	to be confirmed
70	Vigente desde 28/12/2025	2026-08-31	to be confirmed
71	Res. 704/2026; desde 07/2026	2026-08-31	to be confirmed
72	Resoluções ARES-PCJ / ARSESP	2026-08-31	to be confirmed
73	Res. ARISB-MG 361/2025; vigente em 19/08/2026	2026-08-31	to be confirmed
74	Vigente em 19/08/2026	2026-08-31	to be confirmed
75	Res. ARES-PCJ 699/2026; vigente em 19/08/2026	2026-08-31	to be confirmed

No.	Declared effective date	Source checked on	Status
76	Resoluções ARISB-MG / Prestadores Locais	2026-08-31	to be confirmed
77	Vigente desde agosto de 2026	2026-08-31	to be confirmed
78	Estrutura Municipal Vigente	2026-08-31	to be confirmed
79	Vigente desde junho de 2026	2026-08-31	to be confirmed
80	Res. 713/2026; vigente em 19/08/2026	2026-08-31	to be confirmed
81	Resoluções ARES-PCJ / ARSESP	2026-08-31	to be confirmed
82	Res. ARISB-MG 373/2025; vigente em 19/08/2026	2026-08-31	to be confirmed
83	Resoluções ARES-PCJ / ARSESP	2026-08-31	to be confirmed
84	Resoluções ARSP / Concessões ES	2026-08-31	to be confirmed
85	Tabela oficial de 2026	2026-08-31	to be confirmed
86	Resoluções ARISB-MG / Prestadores Locais	2026-08-31	to be confirmed
87	Desde 01/2026	2026-08-31	to be confirmed
88	Estrutura Municipal Vigente	2026-08-31	to be confirmed
89	Vigente desde julho de 2026	2026-08-31	to be confirmed
90	Vigente desde abril de 2025; mantida como vigente pelo SAAE em 24/08/2026	2026-08-31	to be confirmed
91	Estrutura Municipal Vigente	2026-08-31	to be confirmed
92	Resoluções ARISB-MG / Prestadores Locais	2026-08-31	to be confirmed
93	Res. ARISB-MG 367/2025 c/ correção 377/2026; vigente em 19/08/2026	2026-08-31	to be confirmed
94	Vigente desde novembro de 2025	--	to be confirmed
95	Vigente desde 03/03/2026	2026-08-31	incomplete structure
96	Res. 703/2026; desde 07/2026	2026-08-31	to be confirmed
97	Vigente desde setembro de 2025	2026-08-31	to be confirmed
98	26/02/2026-25/02/2027; Res. ARSP 098/2026	2026-08-31	to be confirmed
99	Resoluções ARISB-MG / Prestadores Locais	2026-08-31	to be confirmed
100	Regulação Oficial	--	to be confirmed
101	Resoluções AGERO / CAERD	2026-08-31	to be confirmed
102	Resolução ARIS-MG nº 227/2026	2026-08-31	to be confirmed
103	Decreto Municipal	--	to be confirmed
104	Desde 03/2026	2026-08-31	incomplete structure
105	Vigente desde setembro de 2025; tabela sucessora ainda não aplicável na data de corte	2026-08-31	to be confirmed
106	Res. 696/2026; vigente em 19/08/2026	2026-08-31	to be confirmed
107	Resoluções ARES-PCJ / ARSESP	2026-08-31	to be confirmed
108	Res. 697/2026; vigente em 19/08/2026	2026-08-31	audited
109	Desde 01/01/2026	2026-08-31	audited
110	Desde 01/01/2026	2026-08-31	audited
111	Desde 01/01/2026	2026-08-31	audited

No.	Declared effective date	Source checked on	Status
112	Desde 01/01/2026	2026-08-31	audited
113	Res. 659/2025; vigente em 19/08/2026	2026-08-31	audited
114	Desde 01/01/2026	2026-08-31	audited
115	Res. ARISB-MG 368/2025; vigente em 19/08/2026	2026-08-31	audited
116	Res. ARISB-MG 371/2025; vigente em 19/08/2026	2026-08-31	audited
117	Resolução ARES-PCJ nº 687/2026	2026-08-31	audited
118	Vigente desde janeiro de 2026	2026-08-31	audited
119	Tabela da Resolução ARES-PCJ nº 680/2025 vigente em 2026	2026-08-31	audited
120	Vigente desde 15/03/2026	2026-08-31	audited
121	Vigente desde dezembro de 2025	2026-08-31	audited
122	Tabela da Resolução ARES-PCJ nº 678/2025 vigente em 2026	2026-08-31	audited
123	Res. 695/2026; vigente em 19/08/2026	2026-08-31	incomplete structure
124	Vigente desde maio de 2026	2026-08-31	audited
125	Res. 694/2026; vigente em 19/08/2026	2026-08-31	incomplete structure
126	Resoluções ARES-PCJ / ARSESP	2026-08-31	to be confirmed
127	Vigente no exercício de 2026	2026-08-31	audited
128	Água desde 01/01/2026; esgoto desde 01/04/2026	2026-08-31	audited
129	Tarifa de água vigente desde 17/05/2026; esgoto ainda não faturado no cenário atual	2026-08-31	audited
130	Vigente desde 01/07/2026	2026-08-31	audited
131	Tarifa de água vigente desde 01/07/2026; adicional de esgoto condicionado à unidade atendida	2026-08-31	audited
132	Tabela tarifária do exercício de 2026	2026-08-31	audited
133	Resolução ARES-PCJ nº 686/2026	2026-08-31	audited
134	01/08/2026	2026-08-31	audited
135	Resoluções AGER-MT / Concessões	2026-08-31	to be confirmed
136	Vigente desde fevereiro de 2026	2026-08-31	audited
137	Vigente desde julho de 2025	2026-08-31	audited
138	Desde 01/04/2026	2026-08-31	audited
139	Res. ARSESP 1787/2026; vigente em 19/08/2026	2026-08-31	audited
140	Vigente em 19/08/2026	2026-08-31	audited
141	Estrutura Tarifária SANEAR	2026-08-31	to be confirmed
142	Vigente desde agosto de 2026	2026-08-31	audited
143	Tabela jan/2026 - Res. ARISB-MG 369/2025	2026-08-31	audited
144	Vigente de 07/2026 a 06/2027	--	audited
145	Vigente em 19/08/2026	2026-08-31	to be confirmed
146	Vigente em 19/08/2026	2026-08-31	to be confirmed
147	Resoluções ARES-PCJ / ARSESP	2026-08-31	to be confirmed
148	Portaria AGEMS 315 / SANESUL	2026-08-31	audited

No.	Declared effective date	Source checked on	Status
149	Res. 638/2025 adotada no corte 19/08/2026	2026-08-31	audited
150	Vigente em agosto de 2026, após o prazo legal mínimo	2026-08-31	audited
151	Tabela tarifária do exercício de 2026	2026-08-31	audited
152	Vigente para faturas com vencimento desde 05/10/2025	2026-08-31	audited
153	Tabela tarifária do exercício de 2026	2026-08-31	audited
154	Regulação Oficial	2026-08-31	to be confirmed
155	2026 vigente em 19/08/2026	2026-08-31	to be confirmed
156	Vigente de 01/05/2026 a 30/04/2027	2026-08-31	audited
157	Vigente desde 08/03/2026	2026-08-31	audited
158	Resoluções AGER-MT / Concessões	2026-08-31	to be confirmed
159	Vigente desde 20/03/2025	2026-08-31	audited
160	Vigente desde 06/01/2026	2026-08-31	audited
161	Regulação Oficial	2026-08-31	to be confirmed
162	Resoluções ARES-PCJ / ARSESP	2026-08-31	to be confirmed
163	Vigente desde janeiro de 2026	2026-08-31	audited
164	Resoluções ARES-PCJ / ARSESP	2026-08-31	to be confirmed
165	Vigente desde 01/12/2025	2026-08-31	audited
166	Vigente desde fevereiro de 2026	2026-08-31	to be confirmed
167	Res. 665/2025; vigente em 19/08/2026	2026-08-31	audited
168	Deliberação AGENERSA	2026-08-31	to be confirmed
169	Deliberação AGENERSA	2026-08-31	to be confirmed
170	Resoluções ARISB-MG / Prestadores Locais	2026-08-31	to be confirmed
171	Vigente desde janeiro de 2026	2026-08-31	audited
172	Estrutura vigente após as parcelas/reajustes de junho de 2026	2026-08-31	audited
173	Vigente desde novembro de 2025	2026-08-31	audited
174	Vigente desde 28/06/2026	2026-08-31	audited
175	Vigente em 19/08/2026	2026-08-31	audited
176	Decreto Municipal	2026-08-31	to be confirmed
177	Tabela oficial consultada em 24/08/2026	2026-08-31	audited
178	Decreto Municipal	2026-08-31	to be confirmed
179	Vigente desde 01/03/2026	--	audited
180	Vigente desde dezembro de 2025	2026-08-31	to be confirmed
181	Vigente desde dezembro de 2025	2026-08-31	to be confirmed
182	Deliberação AGENERSA	2026-08-31	to be confirmed
183	Vigente desde 08/03/2026	2026-08-31	audited
184	Vigente desde 08/03/2026	2026-08-31	audited
185	Vigente desde 08/03/2026	2026-08-31	audited
186	Vigente desde 03/01/2026	2026-08-31	audited

No.	Declared effective date	Source checked on	Status
187	Estrutura tarifária 2026; aplicação publicada para março de 2026	2026-08-31	audited

The CORSAN municipalities charge by the exponential formula `PB x C■ + SB` (section 3.3), with no price per block; the exponent `n` is that of the category's column in the Exponent Table.

Table 185 -- The CORSAN exponential formula in each tariff table and collected category

Municipalities	Category	PB (R$/m³)	SB (R$/month)	Sewage (R$/m³)
Erechim (RS)	Residential	6.95	32.95	4.86
Erechim (RS)	Commercial	7.90	58.79	5.53
Erechim (RS)	Public	7.90	117.40	5.53
Santa Cruz do Sul (RS)	Residential	7.91	37.50	5.53
Canoas (RS), Esteio (RS)	Residential	8.68	41.20	6.07
Santa Cruz do Sul (RS)	Commercial	8.98	66.92	6.28
Santa Cruz do Sul (RS)	Public	8.98	133.64	6.28
Alegrete (RS), Alvorada (RS), Bento Gonçalves (RS), Cachoeira do Sul (RS), Cachoeirinha (RS), Camaquã (RS), Campo Bom (RS), Capão da Canoa (RS), Carazinho (RS), Cruz Alta (RS), Farroupilha (RS), Gravataí (RS), Guaíba (RS), Ijuí (RS), Lajeado (RS), Montenegro (RS), Parobé (RS), Passo Fundo (RS), Rio Grande (RS), Santa Maria (RS), Santa Rosa (RS), Santo Ângelo (RS), Sapiranga (RS), Sapucaia do Sul (RS), São Borja (RS), Taquara (RS), Tramandaí (RS), Vacaria (RS), Venâncio Aires (RS), Viamão (RS)	Residential	9.15	43.39	6.40
Canoas (RS), Esteio (RS)	Commercial	9.88	73.48	6.91
Canoas (RS), Esteio (RS)	Public	9.88	146.79	6.91
Alegrete (RS), Alvorada (RS), Bento Gonçalves (RS), Cachoeira do Sul (RS), Cachoeirinha (RS), Camaquã (RS), Campo Bom (RS), Capão da Canoa (RS), Carazinho (RS), Cruz Alta (RS), Farroupilha (RS), Gravataí (RS), Guaíba (RS), Ijuí (RS), Lajeado (RS), Montenegro (RS), Parobé (RS), Passo Fundo (RS), Rio Grande (RS), Santa Maria (RS), Santa Rosa (RS), Santo Ângelo (RS), Sapiranga (RS), Sapucaia do Sul (RS), São Borja (RS), Taquara (RS), Tramandaí (RS), Vacaria (RS), Venâncio Aires (RS), Viamão (RS)	Commercial	10.41	77.41	7.28
Alegrete (RS), Alvorada (RS), Bento Gonçalves (RS), Cachoeira do Sul (RS), Cachoeirinha (RS), Camaquã (RS), Campo Bom (RS), Capão da Canoa (RS), Carazinho (RS), Cruz Alta (RS), Farroupilha (RS), Gravataí (RS), Guaíba (RS), Ijuí (RS), Lajeado (RS), Montenegro (RS), Parobé (RS), Passo Fundo (RS), Rio Grande (RS), Santa Maria (RS), Santa Rosa (RS), Santo Ângelo (RS), Sapiranga (RS), Sapucaia do Sul (RS), São Borja (RS), Taquara (RS), Tramandaí (RS), Vacaria (RS), Venâncio Aires (RS), Viamão (RS)	Public	10.41	154.64	7.28
Alegrete (RS), Alvorada (RS), Bento Gonçalves (RS), Cachoeira do Sul (RS), Cachoeirinha (RS), Camaquã (RS), Campo Bom (RS), Capão da Canoa (RS), Carazinho (RS), Cruz Alta (RS), Farroupilha (RS), Gravataí (RS), Guaíba (RS), Ijuí (RS), Lajeado (RS), Montenegro (RS), Parobé (RS), Passo Fundo (RS), Rio Grande (RS), Santa Maria (RS), Santa Rosa (RS), Santo Ângelo (RS), Sapiranga (RS), Sapucaia do Sul (RS), São Borja (RS), Taquara (RS), Tramandaí (RS), Vacaria (RS), Venâncio Aires (RS), Viamão (RS)	Industrial	11.83	154.64	8.28

Appendix E -- Climate database



> **Brazil Edition only.** The International Edition does not ship this database; there, the user loads the rainfall series of the site as a file of their own (section 4.1).

E.1. What the database contains

Table 186 -- Contents of the national climate database

What the database holds	Quantity
Brazilian cities	652
CMIP6 climate models	19
emission scenarios	2, SSP2-4.5 and SSP5-8.5
simulations per city and model	3: historical, SSP2-4.5 and SSP5-8.5
total time series	37,164 series
original uncompressed format	~6.5 GB (CSVs)
compiled Netuno 5 format	compressed SQLite database (`climbra.db`)

Each series is a continuous daily record of precipitation, in millimetres, for a combination of city, model and simulation: 652 cities x 19 models x 3 simulations give the 37,164 series. The series come from **CLIMBra** (Ballarin et al., 2023, *Scientific Data / Nature*, doi:10.1038/s41597-023-01956-z; repository *Science Data Bank*, doi:10.57760/sciencedb.02316), bias-corrected by *Quantile Delta Mapping* against the daily grid of Xavier, King and Scanlon (2016), at a resolution of 0.25°. The historical series covers 1980 to 2013 (12,419 days) and the future ones, 2015 to 2100 (31,411 days); the study windows are free cuts of these series. All are in the Gregorian calendar. The HadGEM3-GC31-LL, KACE and UKESM1-0-LL models, which originally use 360-day years, have one day fewer at the end of each series, which does not affect the cut.

E.2. Shipping in the executable and optimised format

To avoid the slow handling of tens of thousands of text files and to allow direct distribution in the Netuno 5 installer and executable, the 37,164 series were compiled and indexed in a **high-efficiency SQLite** database (`climbra.db`), with binary compression per series via `zlib`.

- **Access speed:** indexed access to the database takes **less than 1 millisecond** per series, against tens of milliseconds to open and parse CSV files on disk.
- **Transparency:** the Netuno 5 climate data engine gives priority to the compiled database and falls back automatically to the folder of loose `.csv` files if the database is not found.
- **Portability:** it lets the Netuno 5 installer provide the national database ready to use, without requiring the user to download and unpack 6.5 GB by hand.

E.3. The scenarios

The two CMIP6 Shared Socioeconomic Pathways available in the database are drawn in section 3.2: SSP2-4.5 is the intermediate pathway and SSP5-8.5 assumes high emissions without mitigation. CLIMBra does not include strong mitigation scenarios, such as SSP1-2.6.

E.4. The historical series that come with the program

Independently of the CMIP6 database, four long historical series come with the installation, in the program's `data/rainfall` folder:

Table 187 -- Historical precipitation series distributed with the program

City	Files
Florianópolis	3
Brasília	3
Manaus	3
São Paulo	3

The Florianópolis series used in the tutorial has 86 years; the first thirty years add up to a mean of 1,795 mm a year, with rain on 46% of the days.

E.5. What to declare in a study

- the **model or models**, by name, and the selection criterion;
- the SSP **scenario**;
- the **time window**, with the years;
- the **historical series** used as reference;
- whether there was downscaling or bias correction, and which;
- the **provenance and version** of the database.

Section 3.2 deals with conducting the study with future scenarios, and warns against the mistake of choosing the model by the result.

Appendix F -- Files, folders and formats

Everything the program reads and writes, and where each thing is kept.

F.1. What the program reads

Table 188 -- Formats the program reads and where each one is loaded

Format	Extension	What it is	Where it is loaded
project	`.netuno`	a complete study, with the rain inside	Open Project
old project	`.json`	project from an earlier version, same content	Open Project
legacy project	`.csv`	project from Netuno 4 or the early 5.0 series	Open Project
rainfall series	`.csv`	one column of millimetres per day	step 1. Project & Rainfall
daily demand	`.csv`	measured consumption, day by day	step 2. Building & Demand
scenarios	`.xlsx`	one row per scenario	Scenarios & Tools menu

The rainfall series in CSV is the simplest format possible: one line per day, with the depth in millimetres. The program accepts a point or a comma as the decimal separator, and ignores a header line if there is one. An incomplete year is discarded on reading, so as not to bias the annual totals.

> **Legacy project in CSV.** The old format does not keep everything the current version uses: demand modulation, materials, multi-criteria weights and the rainfall series are left out. When opening such a file, the program warns and adopts the default values for what is missing. Section 3.8 shows what to check after importing.

F.2. What the program writes

Table 189 -- Formats the program writes and where each one is generated

Format	Extension	What it carries	Where it is generated
project	`.netuno`	all the parameters and the rainfall series	Save Project
report	`.pdf`	the complete study, ready to hand over	step 9. PDF Reports
spreadsheet	`.xlsx`	daily series, sweep and indicators	step 10. Export
table	`.csv`	the same data, as delimited text	step 10. Export
data	`.json`	the results, for reprocessing	step 10. Export
drawing	`.dxf`	the hydraulic schematic, for CAD	step 10. Export
drawing	`.svg`	the same schematic, as a vector image	step 10. Export

In CSV export you choose the separator: **semicolon** to open in Excel set to Portuguese or another language that uses the decimal comma, **comma** for the international standard. Choosing wrongly makes the spreadsheet open everything in a single column.

The drawing comes in three presentations: **A3 technical sheet**, **simple schematic** and **block library**, the last accompanied by a JSON file with the properties of the components, for BIM coordination. Section 3.9 deals with all of them.

F.3. The project file, inside

The `.netuno` file is a text file in JSON format. It can be opened in any text editor, and that is deliberate: a published study needs to be inspectable without the program.

It has seven sections:

Table 190 -- The seven sections of the project file and what each one holds

Section	What it holds
`metadata`	project name, identifier and dates
`rainfall`	the city, the period and the **whole daily series**
`building_and_demand`	area, coefficient, occupants, consumption and fraction
`demand_modulation`	the weekly, monthly and daily profiles, when there are any
`reservoirs_and_pumping`	volumes, thresholds and the pump set
`water_quality_and_etap`	target class, disinfection and treatment
`financial_and_economic`	tariffs, taxes, adjustments and period

Appendix A lists all the parameters, with the exact name of each field.

> **The rainfall series goes inside the file.** That is what makes the project reproducible: whoever receives the file obtains exactly your numbers, even without the climate database installed. On the other hand, a project with thirty years of series takes up about 160 kB -- little for a file and a lot for an e-mail attachment with twenty of them.

F.4. Where things are kept

Table 191 -- Location of the program's files and of the user's files

What	Where
the program	`C:\Program Files\Netuno 5`
the rainfall series that come with it	in the program folder, in `data\rainfall`
the tariff database of the 652 municipalities	inside the program (Brazil Edition)
the manual's cases	in the program folder, in `casos`
the three reference studies	in the program folder, in `exemplos`
the national climate database, if installed	wherever its installer indicates
your projects and reports	**wherever you choose when saving**

The program does not create a working folder or keep hidden files. Each save asks for the destination, and nothing you produce stays inside the installation folder -- which means that reinstalling or updating the program deletes nothing of yours.

F.5. Compatibility between versions

Table 192 -- Compatibility between versions of the project file

Situation	What happens
project from an earlier version	opens; new fields take the default
project from this version in an earlier version	may not open, or may lose fields
Netuno 4 project in CSV	opens with a warning; check what section 3.8 lists
JSON file from another program	**rejected**, with a message of its own

The last row deserves an explanation. A JSON file that is not a Netuno project could, in principle, be accepted and filled in with default values -- and the user would simulate 100 m² believing they were simulating the 180 m² the file describes. The program prefers to stop the opening with a message of its own, which is inconvenient and is correct: a project opened wrongly produces an entire study that is wrong.

Appendix G -- Glossary

Technical terms and acronyms, in English, with the Portuguese term in brackets where it appears in the Brazilian literature, in the program's Portuguese interface or in this manual's figures. Many acronyms are Portuguese; they are listed under the acronym itself, as the program prints it. The terminology guidelines of Netuno 5 close the appendix.

ABNT. *Associação Brasileira de Normas Técnicas*, the body that publishes the Brazilian standards. A standard is identified by the acronym NBR followed by the number and the year of the edition, as in ABNT NBR 15527:2019.

ANEEL. *Agência Nacional de Energia Elétrica*, the Brazilian electricity regulator, which approves the tariffs of the distribution companies. The Brazil Edition reads from them the price of pumping energy, and the tariffs are updated from the agency's open data.

AQUA-HQE. Environmental certification system for buildings, the Brazilian adaptation of the French HQE led by Fundação Vanzolini. In the 2024 residential scheme, water is dealt with in the «Water consumption» theme, in items classified into the levels PR, HQE, HQE 2 and HQE 3.

Availability charge (*tarifa de disponibilidade*). Tariff regime in which a fixed value is **added** on top of the metered consumption. Every cubic metre saved earns a discount.

Allowance within the minimum (*franquia no mínimo*). Tariff regime in which the fixed value already includes a volume of consumption. Saving **below** the allowance does not reduce the bill.

BD+C (*Building Design and Construction*). Family of LEED rating systems for the design and construction of new or substantially refurbished buildings. It is the family dealt with in this manual, in versions 4, 4.1 and 5.

BIM (*building information modelling*). Digital representation of the building in which each component carries properties beyond geometry, which allows the designs of different disciplines to be coordinated.

BRL. Acronym of the program's Brazil Edition, the one with the Brazilian data: the CLIMBra climate database, tariffs of 652 municipalities, ANEEL energy, costs in reais and national certifications. Section 1.2 compares the two editions.

CAD (*computer-aided design*). The program exports the hydraulic schematic in DXF, an exchange format accepted by CAD programs.

CAPEX (*capital expenditure*). Initial investment.

CASAN. *Companhia Catarinense de Águas e Saneamento*, the water and sewage utility of Florianópolis and of much of Santa Catarina. Its tariff structure is the one adopted in this manual's tutorial.

Catchment area (*área de captação*). **Horizontal** projection of the impermeable roof surface on which water is collected, according to item 3.6 of NBR 15527:2019 and to EN 16941-1:2024. It is not the sloping area of the roof, nor the contributing area of NBR 10844, which serves to size gutters.

Carbon payback (*payback de carbono*). Time for the net annual emission saving to pay off the carbon embodied in the manufacture of the system.

CDD (*consecutive dry days*). Longest sequence of consecutive days with rain below 1 mm.

CLIMBra (*Climate Change Dataset for Brazil*). High-resolution gridded daily meteorological dataset (0.25° x 0.25°, ~28 km) for the whole of Brazil, covering 19 CMIP6 global climate models and the SSP2-4.5 and SSP5-8.5 emission scenarios. Developed by Ballarin et al. (2023) in *Scientific Data* (Nature) (doi:10.1038/s41597-023-01956-z) and publicly available from the *Science Data Bank* (doi:10.57760/sciencedb.02316). The raw projections were bias-corrected by *Quantile Delta Mapping* (QDM) using the observational dataset of Xavier, King and Scanlon (2016) as reference. In Netuno 5, CLIMBra is the climate database for 652 Brazilian cities (historical series from 1980 to 2013 and continuous future projections from 2015 to 2100), shipped with the Brazil Edition only.

CMIP6. Sixth phase of the *Coupled Model Intercomparison Project*, the coordinated international exercise that produces the climate projections used by the IPCC reports. The program's climate database derives from it.

CORSAN. *Companhia Riograndense de Saneamento*, the water and sewage utility of much of Rio Grande do Sul. It bills water by the exponential formula `PB x C■` plus the basic service charge, not by consumption blocks.

CSV (*comma-separated values*). Text file format in which each line is a record and the columns are separated by a character, typically a comma or a semicolon. It is the format of the precipitation series.

CV (*cavalo-vapor*). Metric horsepower, a unit of power equal to 0.7355 kW, usual in the Brazilian pump trade. A half-hp pump set has 0.37 kW of shaft power. This manual writes it as «hp».

CWD (*consecutive wet days*). Longest sequence of consecutive days with rain equal to or above 1 mm.

Discounted payback (*payback descontado*). Time for the cumulative discounted cash flow to cross zero.

DN (*diâmetro nominal*, nominal diameter). Commercial designation of the diameter of a pipe or fitting, in millimetres. It corresponds exactly to neither the internal nor the external diameter: it is a reference number from which the real dimensions follow, tabulated by standard.

DXF (*drawing exchange format*). Exchange format for technical drawings, read by practically all CAD programs.

EAU. Prefix of the items of the «Water consumption» theme in the AQUA-HQE rating system, from the French *eau*, water. The items related to rainwater harvesting are EAU.1.2, EAU.1.3 and EAU.1.5.

EDGE (*excellence in design for greater efficiencies*). Certification system of the World Bank's International Finance Corporation. It measures the percentage saving against a local baseline in three independent dimensions and requires each to cross the threshold.

EN (*Europäische Norm*, European standard). Standard published by the European Committee for Standardization. In this manual, EN 15978, which defines the building life cycle modules, EN 15804, which governs the environmental product declarations of construction products, and EN 16941-1, on rainwater harvesting systems.

EPD (*Environmental Product Declaration*). Third-party verified document, according to ISO 14025 and, in construction, EN 15804, that publishes the life cycle impacts of a product per declared unit. It is the preferred source of emission factors for materials and equipment.

ESG (*environmental, social and governance*). Set of environmental, social and governance criteria used in assessing companies and developments. The program feeds environmental indicators of that set.

ETAP (*estação de tratamento de água pluvial*). Rainwater treatment plant. In the program, the set of inlet filter, disinfection and polishing filter.

ETCCDI (*Expert Team on Climate Change Detection and Indices*). International working group that standardised the indices of climate extremes, of which the program calculates eight.

First flush (*descarte de primeiras águas*). Initial depth of rain diverted from the system, which carries the dirt deposited on the roof (a minimum of 2 mm according to NBR 15527:2019). In Netuno 5 it is **subtracted** from the day's rain; in Netuno 4 it was treated as an event threshold.

GRI (*Global Reporting Initiative*). International standard for corporate disclosure of social and environmental performance. Its 303 series deals with water and effluents.

HQE (*Haute Qualité Environnementale*). French high environmental quality rating system, basis of AQUA-HQE. It also designates the levels of the rating system's items: HQE, HQE 2 and HQE 3, in increasing order of requirement, above the prerequisite (PR).

IFC. The acronym designates two different things. In certification, it is the World Bank's *International Finance Corporation*, which develops EDGE. In the BIM context, it is *Industry Foundation Classes*, an open exchange format the program does **not** export: the program's BIM export is a JSON file of properties.

Ideal volume (*volume ideal*, optimal tank volume). Smallest tank volume of the simulation grid from which the marginal gain in potable water saving ($\Delta E_{pot} / \Delta V$) falls below a declared threshold (typically $1.0\%/\text{m}^3$ or $0.5\%/\text{m}^3$). It avoids oversizing tanks, securing the best cost-benefit ratio.

Individual metering (*micromedição*). Metering per dwelling unit in a multi-family building. Entered in the program as the number of units (*economias*).

INT. Acronym of the program's International Edition: it opens in English and carries no Brazilian data. Tariffs, costs, currency and emission factors are entered by the user.

IPCC (*Intergovernmental Panel on Climate Change*). Intergovernmental panel that synthesises scientific knowledge on climate change and organises the assessment reports to which the climate projections refer.

IPTU verde (green property tax). Discount on the municipal property tax granted by some Brazilian municipalities to buildings that adopt sustainability measures, among them rainwater harvesting. The percentage and the conditions vary with municipal legislation.

IRR (*taxa interna de retorno*, TIR). Internal rate of return. The discount rate that makes the net present value zero.

ISO 14040. International standard that sets out the principles and framework of life cycle assessment. An inventory conforming to ISO 14040 requires traceability of each flow and external critical review, requirements a screening assessment does not meet.

JSON (*JavaScript object notation*). Structured text file format, readable by people and by programs. It is the internal format of the `.netuno` file.

LCA (*avaliação de ciclo de vida*, ACV). Life cycle assessment. Method that accounts for the environmental impacts of a product over its whole existence, from the extraction of raw materials to disposal. The program's is a **screening** one, not a full inventory conforming to ISO 14040.

LCOW (*levelized cost of water*). Ratio between the present value of all the life cycle costs of the system (CAPEX, replacements, O&M, energy and chemicals) and the discounted volume of rainwater consumed over the analysis horizon, in currency per m³. It allows the economic viability of the system to be compared directly with the utility's tariff.

LEED (*leadership in energy and environmental design*). Environmental certification system for buildings of the *U.S. Green Building Council*, the most widespread internationally. It deals with water efficiency in the *Water Efficiency* category: in version 4, by credits WEc1 and WEc2; in version 5, by credit WEc2, *Enhanced Water Efficiency*.

m.c.a. (*metro de coluna d'água*). Metre of water column, a unit of pressure equal to that exerted by a column of water one metre high, about 9.81 kPa. It is the usual unit in building installations; this manual writes it as «m».

Marginal gain (*ganho marginal*). Change in the saving potential per cubic metre added to the tank, in %/m³ ($\Delta E_{pot} / \Delta V$). Selection criterion of the ideal volume.

Motor efficiency (*rendimento do motor*). Ratio between the mechanical power at the shaft and the electrical power drawn from the mains. ABNT NBR 17094-2 requires it to be marked on the plate.

NBR (*Norma Brasileira*). Brazilian standard, published by ABNT. The most cited in this manual are NBR 15527, on rainwater harvesting, NBR 10844, on building stormwater installations, NBR 5626, on building cold and hot water systems, and NBR 17094-2, on induction motors.

Netuno 5 manual suite (*Manuais do Netuno 5*). The program's documentation, built from the same sources in three formats: the complete manual, seven separate volumes, each with a part of it, and the documentation site. The volumes serve those who need only one part, to consult or send:

1. **Complete manual**: all the chapters and appendices. !Cover of the Complete Manual

2. **Volume 1 -- Start here**: what the program does, how to install it and where to start depending on your goal. !Cover of Volume 1 -- Start here
3. **Volume 2 -- The method**: the equations, the criteria and the sources behind each number the program publishes. !Cover of Volume 2 -- The method
4. **Volume 3 -- Guides**: other cities, future climate, tariffs, variable demand, batch, sensitivity and export. !Cover of Volume 3 -- Guides
5. **Volume 4 -- Tutorial**: a house project in Florianópolis, from the first input to the recommended volume. !Cover of Volume 4 -- Tutorial
6. **Volume 5 -- Mistakes, limitations and reliability**: the most common mistakes, the limits of the model and how the program is verified. !Cover of Volume 5 -- Reliability
7. **Volume 6 -- Fixes, bugs and versioning**: what changed since Netuno 4, the error messages and how to report a problem. !Cover of Volume 6 -- Versions
8. **Volume 7 -- Appendices**: parameters, processes, emission factors, tariffs, climate database, formats, glossary and references. !Cover of Volume 7 -- Appendices

The covers shown are those of the Portuguese volumes, which are published in Portuguese only. The complete manual exists in English too: it is this one, and the International Edition opens it from step 9. Documentation.

Non-potable water (*água não potável*). Water not intended for direct human consumption, but which meets the physical, chemical and microbiological quality standards set by specific standards (such as Table 1 of ABNT NBR 15527:2019 and ABNT NBR 16783:2019) for the uses admitted: flushing toilets and urinals, washing vehicles, pavements and floors, landscape irrigation, laundry and cooling towers.

NPV (*valor presente líquido*, VPL). Net present value.

Overflow (*extravasamento*, spillage). Volume that exceeds the capacity of the tank and is led away by the overflow pipe when the tank is already full.

Payment floor (*piso de pagamento*). Tariff regime in which the bill **does not go** below a value. Different from the availability charge: the floor limits from below, the availability charge adds on top.

PB (*preço-base*). Base price, in R$/m³, of the CORSAN exponential formula, `PB x C■`, in which `C` is the month's consumption and `n` the exponent of the exponential table for that consumption.

PEAD (*polietileno de alta densidade*). High-density polyethylene (HDPE), the material of the most common plastic tanks on the Brazilian market.

Pluvial demand (*demanda pluvial*, D_{pl}). Part of the building's daily water demand that can be supplied by the harvesting system ($D_{pl,t} = D_{tot,t} \times f$). It corresponds to the non-potable points of use connected to the rainwater network.

Potable water (*água potável*). Water that fully meets the potability standards set by Ministry of Health Ordinance GM/MS No. 888/2021, in Brazil, intended for drinking, food preparation and personal hygiene.

Potable water saving potential (*potencial de economia de água potável*, E_{pot}). Fraction of the **total** demand of the building supplied by rainwater ($E_{pot} = \sum V_{con} / \sum D_{tot} \times 100\%$). It is the program's central indicator, bounded above by the suppliable fraction ($E_{pot} \le f$).

pp. Percentage points. Used to express the difference between two percentages: from 35% to 37% is two percentage points, not two per cent.

PR (*pré-requisito*). Prerequisite: in AQUA-HQE, the level of a mandatory item, below the HQE, HQE 2 and HQE 3 levels.

PRCPTOT. Total annual precipitation on wet days.

PRFV (*plástico reforçado com fibra de vidro*). Glass-fibre reinforced plastic, the material of medium and large tanks. It appears in the program as «fibreglass».

Pump efficiency (*rendimento da bomba*). Ratio between the hydraulic power delivered to the water and the mechanical power at the shaft. It does not include the motor.

PVC. Polyvinyl chloride, the material of building cold-water piping and of the first-flush diversion columns. The dimensions of solvent-weld pipes follow ABNT NBR 5648.

R95p. Sum of the precipitation of the days above the 95th percentile of the wet days.

Rainwater (*água de chuva*). Atmospheric precipitation collected directly on impermeable roof surfaces (roofs and waterproofed slabs), before it reaches the ground or public drains. It is the official technical and normative term of ABNT NBR 15527:2019.

Rainwater harvesting (*aproveitamento de água de chuva*, AAC). Use of the water falling on roofs for non-potable uses in buildings, regulated in Brazil by ABNT NBR 15527:2019. «Harvesting» is preferred to «reuse», since rainwater collected on the roof is a primary resource and not an effluent of earlier human use.

Rainwater harvesting system (*sistema de aproveitamento de água pluvial*, SAAP). Integrated set of building components -- catchment area, pipes, first flush, treatment plant (ETAP), lower cistern, upper tank, pump set or pressure pump and an independent distribution network -- designed to collect, treat, store and supply non-potable water in accordance with ABNT NBR 15527 and ABNT NBR 5626.

Representative rainfall series length (*comprimento representativo de série de precipitação*, N_{rep}). Shortest continuous period, in years of daily rainfall data, needed to stabilise the sizing and performance of the system statistically (reproducing the ideal volume and the saving potential of the full series with confidence $\ge 90\%$), according to the method of Geraldi and Ghisi (2017, 2018, 2019). The length needed is not a fixed geographical constant: it grows with the suppliable fraction and with the demand level of the building.

Retroactive allowance (*franquia retroativa*). Tariff regime in which consumption up to the allowance pays a fixed value and consumption above it is billed in full from 0 m³.

Runoff coefficient (*coeficiente de escoamento*). Fraction of the rain that actually reaches the tank, after losses to absorption, evaporation and initial wetting of the roof.

RX1day, **RX5day**. Largest precipitation in one and in five consecutive days, in the year.

SB (*serviço básico*). Basic service charge: the fixed monthly part of the CORSAN water bill, added to the result of the exponential formula.

SDII (*simple daily intensity index*). Mean precipitation of the wet days.

SI (*seasonality index*). Walsh and Lawler seasonality index. Zero when rain is uniform over the year.

SMAA (*stochastic multicriteria acceptability analysis*). Method that draws weights and returns the probability of each alternative winning.

SNIS (*Sistema Nacional de Informações sobre Saneamento*). Brazil's national sanitation information system, the official database of the sector's indicators, kept by the federal government. It is the origin of the sanitation energy consumption indicators used in the environmental assessment.

SSP (*shared socioeconomic pathways*). Coherent hypotheses of global development and the corresponding emissions. They are combined with the climate models to produce projections.

Stochastic demand (*demanda estocástica*). Modelling daily water consumption as a probabilistic variable that varies from day to day and between realisations (Silva and Ghisi, 2016), allowing the uncertainty range of the sizing and of the performance indicators to be assessed under real variability of use.

Stormwater (*água pluvial*, pluvial water). Rainwater after urban surface runoff, or when led by building drainage and storage pipes and systems (ABNT NBR 10844). In Netuno's Portuguese text, the term applies preferably to building infrastructure and engineering components: harvesting system, rainwater network, pluvial demand, rainwater consumption and overflow.

Suppliable fraction (*fração atendível por água pluvial*, non-potable demand share, f). Percentage of the building's total water demand (D_{tot}) that can be supplied by rainwater ($f = D_{pl} / D_{tot}$). It is the **insurmountable asymptotic ceiling** of the potable water saving potential ($E_{pot} \le f$): even with infinite storage and unlimited rain, the potable saving never exceeds the suppliable fraction, because the rest ($1 - f$) is reserved for potable uses. It is distinct from volumetric reliability ($C_v = E_{pot} / f$), which assesses the supply of the rainwater part.

Tariff flag (*bandeira tarifária*). Surcharge or discount on the Brazilian electricity tariff, according to generation conditions. Green, yellow, red 1, red 2, and the white mode.

TFDI (*tarifa fixa de disponibilidade da infraestrutura*). Fixed infrastructure availability tariff, the name some utilities give to the part of the bill charged regardless of consumption. See *Availability charge*.

Time reliability (*confiabilidade temporal*, C_t). Percentage of days on which the system fully ($V_{con,t} = D_{pl,t}$), partly or not at all supplied the rainwater demand.

TMA (*taxa mínima de atratividade*). Minimum attractive rate of return. The discount rate adopted.

TOPSIS (*technique for order of preference by similarity to ideal solution*). Multi-criteria ranking method of Hwang and Yoon.

UV. Ultraviolet radiation, used in water disinfection by inactivating micro-organisms. It acts only at the moment of passage through the reactor and leaves no residual, unlike chlorination.

VIKOR. Multi-criteria decision method proposed by Opricovic, which seeks a compromise solution between conflicting criteria. See also *SMAA*.

Volumetric reliability (*confiabilidade volumétrica*, C_v). Fraction of the **rainwater** demand the system supplied ($C_v = \sum V_{con} / \sum D_{pl} = E_{pot} / f$). Not to be confused with the saving potential, whose denominator is the total demand of the building.

WEM. Coding of the EDGE water efficiency measures. WEM14 is the rainwater harvesting system.

XLSX. Microsoft Excel workbook format, based on compressed XML. It is one of the export formats for spreadsheets and command-line results.

YAS (*yield after spillage*). Operating rule in which overflow is calculated **before** the demand is withdrawn and, in the strict form, the day's withdrawal comes only from the volume stored the day before. The program calculates overflow first, as YAS does, but lets the day's rain meet the day's demand; it lies between YAS and YBS (section 2.1).

YBS (*yield before spillage*). Operating rule in which the demand is withdrawn **before** overflow is calculated. It assumes consumption before the day's rain and overestimates the yield; the program does not adopt it, and its result lies between YAS and YBS.

G.1. Netuno's terminology and naming guidelines

Consistent terminology is essential to technical rigour in scientific papers, technical reports, the graphical interface and Netuno's documentation. The conventions below detail the distinction between related terms that are often confused, first in Portuguese, the language of the program's reference text, and then in English.

G.1.1. 1. Rainwater, stormwater and the colloquial form

Table 193 -- Terminology for rainwater and stormwater, in Portuguese and English

Term	Definition and context of use	Status in Netuno	Reference
água de chuva (rainwater)	Precipitation collected directly on the roof (roof or waterproofed slab), before it reaches the ground or drains. It refers to the **water source** and the environmental resource.	**Canonical and preferred** for the resource collected. In English, «rainwater».	ABNT NBR 15527:2019, title and item 3.1.
água pluvial (stormwater, pluvial water)	Rainwater after urban surface runoff, or when led by building drainage and storage pipes and systems. It applies to the **physical infrastructure** and the engineering components.	**Canonical and preferred** for systems, networks and the balance. In English, «rainwater» is used for these too («rainwater network», «rainwater demand»), and «stormwater» only for drainage.	ABNT NBR 10844:1989; ABNT NBR 5626:2020.
água da chuva	Colloquial, informal expression.	**Discouraged**. Not to be used in publications, normative texts, Netuno reports or code.	Practice of technical and scientific writing.

G.1.2. 2. Suppliable fraction, pluvial demand, potential and reliability

To avoid the mathematical and conceptual ambiguities often found in the literature:

Table 194 -- Fundamental quantities of the water balance and relations between fraction and potential

Quantity	Symbol / Code	Definition	Formula	Relation / Ceiling
Suppliable fraction	f / `rainwater_share`	Share of the building's total demand corresponding to non-potable points of use that can be connected to the harvesting system.	$f = D_{pl} / D_{tot}$	Design parameter ($0 \le f \le 1$). It is the **asymptotic ceiling** of E_{pot}.
Pluvial demand	D_{pl} / `rainwater_demand`	Volume of water required by the building's non-potable points of use over a time interval.	$D_{pl,t} = D_{tot,t} \times f$	In L/day or m³/year.
Rainwater volume consumed	V_{con} / `pluvial_consumed`	Volume of non-potable water actually supplied by the rainwater tank to the points of use.	Determined by the daily balance (section 2.1)	$V_{con,t} \le D_{pl,t}$ and $V_{con,t} \le S_{t-1} + C_t$.
Potable water saving potential	E_{pot} / `potential_potable_savings_pct`	Percentage of the building's **total** demand saved by rainwater harvesting.	$E_{pot} = \frac{\sum V_{con}}{\sum D_{tot}} \times 100\%$	**Central indicator**. Always $E_{pot} \le f$.
Volumetric reliability	C_v / `volumetric_reliability_pct`	Percentage of the **rainwater** demand supplied by the stored rainwater.	$C_v = \frac{\sum V_{con}}{\sum D_{pl}} \times 100\%$	$C_v = \frac{E_{pot}}{f}$. It can reach 100% even when $E_{pot} < 100\%$.
Time reliability	C_t / `temporal_reliability_pct`	Percentage of days of the series on which the rainwater demand was 100% supplied ($V_{con,t} = D_{pl,t}$).	$C_t = \frac{N_{full_days}}{N_{total}} \times 100\%$	It measures the security of daily supply without potable top-up.

G.1.3. 3. Recommended naming for papers and reports

- Use **«rainwater harvesting»** (not «rainwater reuse» -- rainwater is not an effluent; its use is *harvesting*, not *reuse*).
- Use **«catchment area»** for the horizontal plane of the roof (NBR 15527, EN 16941-1) and keep **«contributing area»** exclusively for the hydraulic sizing of gutters under wind (NBR 10844).
- Use **«first flush»** for the initial depth diverted (a minimum of 2 mm according to NBR 15527), not «first-flush threshold».
- Use **«suppliable fraction»** or «non-potable demand share» (not «relative demand» or «saving fraction») to express the share of non-potable uses in total consumption.
- Use **«representative series length»** for the minimum length in years needed for the statistical stabilisation of the simulation (Geraldi and Ghisi method).
- Use **«LCOW»** or **«levelized cost of water»** for the unit cost metric over the life cycle discounted at the minimum attractive rate of return.

Appendix H -- References

Standards, publications and databases that support the program's decisions. The full list, with the provenance of each citation, is in the program's bibliography module and in the Documentation tab. Works published in Portuguese are cited by their original titles, with a description in English.

H.1. Brazilian technical standards

ABNT NBR 15527 -- Água de chuva: aproveitamento de coberturas em áreas urbanas para fins não potáveis. Requisitos. [Rainwater: harvesting from roofs in urban areas for non-potable purposes. Requirements.] It sets the quality parameters, the first flush and maintenance.

ABNT NBR 10844:1989 -- Instalações prediais de águas pluviais. Procedimento. [Building stormwater installations. Procedure.] It defines the contributing area (item 3.2 and Figure 2), with the increments for slope and walls, for the design flow of gutters and downpipes; it is not the catchment area of the water balance, which follows NBR 15527:2019, item 3.6.

ABNT NBR 16783 -- Uso de fontes alternativas de água não potável em edificações. [Use of alternative sources of non-potable water in buildings.]

ABNT NBR 5626:2020 -- Sistemas prediais de água fria e água quente. Projeto, execução, operação e manutenção. [Building cold and hot water systems. Design, construction, operation and maintenance.] Second edition, corrected version of 30/09/2020.

ABNT NBR 15575-1:2025 -- Edificações habitacionais. Desempenho. Parte 1: Requisitos gerais. [Residential buildings. Performance. Part 1: General requirements.] Seventh edition, 05/12/2025.

ABNT NBR 15575-6:2021 -- Edificações habitacionais. Desempenho. Parte 6: Requisitos para os sistemas hidrossanitários. [Residential buildings. Performance. Part 6: Requirements for plumbing systems.] Fifth edition, 29/09/2021.

ABNT NBR 17094-2:2016 -- Máquinas elétricas girantes. Parte 2: motores de indução monofásicos. Requisitos. [Rotating electrical machines. Part 2: single-phase induction motors. Requirements.] Table 13 gives the minimum nominal full-load efficiency of the IR1 line, and Table 14 that of the IR2 line. The standard requires the efficiency to be marked on the rating plate, which establishes that the rated power is the mechanical power at the shaft.

Portaria GM/MS 888/2021 -- Procedimentos de controle e vigilância da qualidade da água para consumo humano. [Ministry of Health ordinance on the control and surveillance of the quality of water for human consumption.] It governs residual chlorine and coliforms.

H.2. Certification systems

LEED v4 / v4.1 BD+C -- USGBC. *Water Efficiency* category, credits WEc1 and WEc2. Registration of commercial projects until 30 Jun. 2027.

LEED v5 BD+C -- U.S. GREEN BUILDING COUNCIL. *LEED v5 Building Design and Construction: rating system*. Washington, DC: USGBC, Nov. 2025. Prerequisites WEp1 (*Water Metering and Reporting*) and WEp2 (*Minimum Water Efficiency*), credits WEc2 (*Enhanced Water Efficiency*) and SSc3 (*Rainwater Management*).

AQUA-HQE Residential -- FUNDAÇÃO VANZOLINI; CERQUAL QUALITEL. *Referencial de avaliação do desempenho de edifícios residenciais em construção e renovação* [Assessment scheme for the performance of residential buildings under construction and renovation]. Version 00. São Paulo: Fundação Vanzolini, 22 Nov. 2024. «Water consumption» theme, items EAU.1.2, EAU.1.3 and EAU.1.5.

Selo Casa Azul + CAIXA -- CAIXA ECONÔMICA FEDERAL. *Guia Selo Casa Azul + CAIXA*. V023. Brasília: CAIXA, 29 Apr. 2026. Criterion 3.6, rainwater harvesting.

Selo Casa Azul Uni -- CAIXA ECONÔMICA FEDERAL. *Guia Selo Casa Azul Uni*. V001. Brasília: CAIXA, Dec. 2025. Criterion 7, rainwater harvesting.

EDGE -- IFC, World Bank Group. *EDGE Certification Process*, and the WEM coding of the water efficiency measures. Administered by GBCI in more than 120 countries.

GRI 303 -- Global Reporting Initiative. Water and effluents, indicators 303-3 and 303-5.

H.3. Methods

Hwang, C.-L.; Yoon, K. (1981). *Multiple Attribute Decision Making: Methods and Applications*. The original formulation of TOPSIS.

Opricovic, S.; Tzeng, G.-H. The VIKOR method and the conditions C1 and C2 of acceptable advantage and stability.

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Walsh, R. P. D.; Lawler, D. M. (1981). The seasonality index.

ETCCDI -- *Expert Team on Climate Change Detection and Indices*, of the WMO. The indices of climate extremes.

Mayer, P. W.; DeOreo, W. B. et al. (1999). *Residential End Uses of Water*. Denver: AWWA Research Foundation. Frequency of fixture use: 5.05 flushes and 0.37 washing-machine cycles per person per day; the flushes are the basis of the starts of direct pressurisation.

Komeco. *Pressurizador de água TP 820T e TP 825T G2: manual do usuário* [TP 820T and TP 825T G2 water pressure pump: user manual]. Version of 15/05/2007. Data sheet and flow x pressure chart; the pressure switch range of the TP 820T, 12 to 24 m, is the basis of the default working pressure.

Cardal. *Pressurizador Potenza: manual de instruções* [Potenza pressure pump: instruction manual]. IM 311. Technical specifications and flow x pressure curve (figure 8).

Jacuzzi. *Pressurizador residencial AcquaHouse: manual de instruções* [AcquaHouse residential pressure pump: instruction manual]. March 2017. Flow x pressure curves tested at the Pump Labelling Laboratory of the Federal University of Itajubá.

Syllent (Eberle). *Pressurizador para água quente, linha Impulse Press: manual de instalação e operação* [Hot-water pressure pump, Impulse Press line: installation and operation manual]. Drawing 3I4743. Flow x head table and the 10 s switch-off delay of the electronic control.

Rowa. *Pressurizadores Max Press E: manual de instalação* [Max Press E pressure pumps: installation manual], and the data sheet of the Max Press 20E: maximum pressure of 20 m and 5,000 L/h at 7 m.

H.4. Life cycle assessment standards

ABNT NBR ISO 14040 -- Environmental management: life cycle assessment, principles and framework. Corresponding to ISO 14040:2006. The four phases of the assessment.

ABNT NBR ISO 14044 -- Environmental management: life cycle assessment, requirements and guidelines. Corresponding to ISO 14044:2006.

EN 15978:2011 -- Sustainability of construction works: assessment of environmental performance of buildings, calculation method. The life cycle modules A to D of the building.

EN 15804:2012+A2:2019 -- Sustainability of construction works: environmental product declarations, core rules for the product category of construction products.

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H.5. Sizing and performance of rainwater harvesting systems

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H.6. Future climate

BALLARIN, A. S. et al. (2023). CLIMBra: Climate Change Dataset for Brazil. *Scientific Data*, v. 10. doi:10.1038/s41597-023-01956-z. The origin of the program's climate database.

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H.7. Life cycle and energy of rainwater systems

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H.8. The predecessor program

Ghisi, E.; Cordova, M. M. (2014). *Netuno 4 -- Manual do Usuário* [Netuno 4 -- User Manual]. Federal University of Santa Catarina, Department of Civil Engineering, Laboratory of Energy Efficiency in Buildings. Florianópolis, June 2014.

Section 6.1 gives the quantitative comparison between the two versions and the breakdown of the differences.

H.9. Databases

CMIP6 -- *Coupled Model Intercomparison Project*, phase 6. Climate projections that feed the IPCC reports.

ANEEL -- Open data on the approved tariffs of Brazilian electricity distribution companies.

State and municipal sanitation regulatory agencies -- tariff resolutions, one per municipality of the database.

SNIS -- *Sistema Nacional de Informações sobre Saneamento* [Brazil's national sanitation information system]. Distribution loss indices by macro-region.

H.10. How to cite the program

When publishing a study done with Netuno 5, cite the program **with the version**, and the edition, and declare the items of section 4.10. A study that cites «Netuno» without a version is not reproducible: fixes between versions change results, and section 2.6 gives the example of the pump set's consumption, which changed by 65% between two consecutive versions.

Appendix I -- One-page summaries

Detachable summaries. Each sheet fits on one page and serves as a quick reference beside the computer.

I.1. Sheet 1 -- The flow of a complete study

Table 195 -- Reference sheet: the ten steps of a complete study and the section that deals with each one

#	Step	What is decided in it	Section
1	Project and rain	series, area, coefficient, first flush	4.1
2	Building	building type, demand, suppliable fraction	4.2
3	First simulation	a guessed volume, to see	4.3
4	Sweep	return curve, ideal volume	4.4
5	Arrangement	is an upper tank needed?	4.5
6	Quality	class, disinfection, filters	4.6
7	Economics	real tariff, CAPEX, cash flow	4.7
8	Carbon	inventory, material, payback	4.8
9	Decision	seven criteria, declared weights	4.9
10	Report	outputs and the list of what to declare	4.10

Before publishing: section 4.10, the list of what to declare. Section 5.1, the ten mistakes. Section 5.2, the limitations that affect your case.

I.2. Sheet 2 -- The pitfalls, one line each

Table 196 -- Reference sheet: the ten design pitfalls and where each one is dealt with

Pitfall	Where
sloping area instead of the horizontal projection	5.1.1
mean monthly rain instead of the daily series	5.1.2
first flush in **litres** when the field wants millimetres	5.1.3
tariff **minimum-charge regime** ignored	5.1.4
school or hospital with a **residential tariff**	5.1.5
expecting more **water** from the upper tank	5.1.6
too **short a series**	5.1.7
quality class chosen by **cost**	5.1.8
comparing with **Netuno 4** without declaring the divergence	5.1.9
publishing the volume with the **highest saving** as the recommendation	5.1.10

I.3. Sheet 3 -- The six tariff regimes

Table 197 -- Reference sheet: the six tariff regimes and the effect of saving

Regime	When saving	Municipalities
no minimum	every m³ earns a discount	165
availability charge	every m³ earns a discount; the fixed part remains	122
exponential formula (CORSAN)	every m³ earns a discount; the bill is PB x C■ + basic service	34
allowance within the minimum	**below the allowance there is no discount at all**	**329**
retroactive allowance	going past it, everything is billed from zero	2
payment floor	the bill does not go below the floor	0

Quick test: simulate two consumptions, one inside and one outside the minimum block, and compare them with a real bill. If only one matches, the regime is wrong.

The municipality counts are those of the Brazil Edition catalogue; in the International Edition, the regime is the one you enter.

I.4. Sheet 4 -- The indicators, and what each one answers

Table 198 -- Reference sheet: the indicators and the question each one answers

Indicator	Denominator	Answers
saving potential	**total** demand	how much potable water consumption falls
volumetric reliability	**rainwater** demand	how much of what was intended was supplied
full, partial, no-supply days	days of the series	how often it can be counted on
overflow	--	is the tank small?
marginal gain	--	is it worth enlarging?

```
Epot = volumetric reliability x suppliable fraction
```

I.5. Sheet 5 -- The pump set energy chain

```
pumping:   e = rho . g . H ÷ (eta_pump . eta_motor)       kWh/m³
start-ups: E = plate x time ÷ eta_motor              kWh
```

The plate gives the power **at the shaft**. The motor efficiency separates the shaft from the socket. If left blank, Table 13 of NBR 17094-2 applies: 60.5% at half a hp, 69.3% at one hp.

I.6. Sheet 6 -- The list of what to declare, summarised

Program: version and edition, with revision.

Rain: source, cut, years; if a projection, model, scenario and window.

Building: area and how it was obtained, coefficient, per capita consumption and its source, **suppliable fraction** and the uses that make it up.

First flush: depth and mode; the divergence from Netuno 4, if comparing versions.

Volume: selection criterion -- marginal-gain threshold, or multi-criteria weights.

Economics: municipality, utility, effective date of the table, **minimum-charge regime**, sewage percentage, minimum attractive rate of return, inflation, adjustments and period; in the International Edition, the currency.

Carbon: boundary (water delivered), service life, tank material, grid database with the year, **approximate** factors or those with a **screening label**.

Certification: that classes 2 to 4 are adaptations of the program; for LEED, the version; for EDGE, that only the water bar was assessed.

Appendix J -- Example files

The manual prints results; these files are what you open to reproduce them. They are in the program's `casos` folder, and each one carries its rainfall series embedded: they open without the original CSV, on any machine.

J.1. The nine cases of the manual

Table 199 -- The nine example cases and what each one demonstrates

File	Section	What it demonstrates
`01_tutorial_florianopolis.netuno`	4.1 to 4.10	the final tutorial project: a 4,000 L polyethylene cistern pumping to a 1,000 L upper tank
`02_tutorial_varredura.netuno`	4.4 and 2.2	the project before the upper tank, in sweep mode, from 0 to 20,000 L, in steps of 1,000 L
`03_dez_dias.netuno`	2.1	the ten-day teaching case, which fits in a spreadsheet
`04_reservatorio_superior.netuno`	4.5	4,000 L below plus 1,000 above, the comparison of section 4.5 against the 5,000 L of a single tank
`05_manaus.netuno`	3.1	the tutorial project taken to Manaus, where it rains more and yields less
`06_brasilia.netuno`	3.1	the tutorial project taken to Brasília, with a marked dry season
`07_saopaulo.netuno`	3.1	the tutorial project taken to São Paulo
`08_escola_semanal.netuno`	3.4 and 3.5	a school of 300 pupils with demand modulated by day of the week
`09_hotel_mensal.netuno`	3.4	a seaside hotel with monthly occupancy in phase with the rain

J.2. What each one produces

The values below are obtained by **rereading the saved file** and simulating what was reread, not the configuration that generated it. That is the difference that matters: a parameter that does not survive saving shows up here as a wrong number, instead of going unnoticed.

Table 200 -- Result each example case produces when reread and simulated

File	Mode	Volume	Potential	Rainwater consumed
`01_tutorial_florianopolis.netuno`	fixed volume	4,000 L	27.30%	163.8 L/day
`02_tutorial_varredura.netuno`	sweep	ideal: 4,000 L	27.12%	21 volumes
`03_dez_dias.netuno`	fixed volume	2,000 L	36.00%	72.0 L/day
`04_reservatorio_superior.netuno`	fixed volume	4,000 L	27.30%	163.8 L/day
`05_manaus.netuno`	fixed volume	4,000 L	20.96%	125.8 L/day
`06_brasilia.netuno`	fixed volume	4,000 L	15.45%	92.7 L/day
`07_saopaulo.netuno`	fixed volume	4,000 L	25.74%	154.4 L/day
`08_escola_semanal.netuno`	fixed volume	30,000 L	34.34%	3,095.8 L/day
`09_hotel_mensal.netuno`	fixed volume	25,000 L	32.11%	1,854.9 L/day

J.3. How to run them

File, Open Project, choose the file, and the program loads everything: the rain, the building, the costs and the configuration of the analyses. Go through the result steps in the sidebar to see each dimension.

The files are in the `casos` folder, inside the program's installation folder. If you want to modify them, **save them under another name**: that way the original case keeps reproducing the number printed in the manual.

The water results above are the same in both editions. The cases carry Brazilian tariffs and costs in reais; in the International Edition, replace them with your own before reading the economic results.

J.4. The three studies in `exemplos`

Besides the manual's cases, the program comes with three reference studies in `exemplos`: a single-family house in Florianópolis, an office building in Brasília and a public school in Manaus. They are larger and less didactic than the manual's, and serve as a starting point for real projects.

> **They were re-saved in this version.** They were in an earlier flat schema, which the program read without recognising: they opened as a default project, with 100 m² and three occupants, and nothing warned. Opening now rejects an unknown format with a message of its own, and the three were converted to the current schema, with the rain embedded.

J.5. What these files are not

They are not design templates. They are **demonstration cases**: the costs are those of the tutorial, the tariff is CASAN's and the category is residential in all of them, including the school and the hotel, because of the limitation section 5.2 describes. For a real study, replace the tariff, costs and rainfall series with those of your case.